Identification Problems in the Modelling of Social Phenomena

Econometric Seminar LECON 2603

Ernesto San Martín

Interdisciplinary Laboratory of Social Statistics LIES UC, Chile

Faculty of Mathematics, Pontificia Universidad Católica de Chile, Chile

LIDAM/CORE, UCLouvain, Belgium

International Francqui Professor

Santiago & Louvain-la-Neuve, Second Semester, Academic Year 2025-2026

Contents

1 Fundamental concepts **1**

1.1 The problems of statistics . . . 1

1.2 Statistical model . . . 2

1.3 Identification problem . . . 4

1.4 Concluding remarks . . . 7

2 The Conceptual Depth of the Law of Total Probabilities **8**

2.1 Introduction . . . 8

2.2 Scores of te SIMCE test . . . 10

2.3 Distribution function of scores by administrative dependency of schools . . . 13

2.4 Conditional probability as a random variable . . . 16

2.5 Conditioning on other variables . . . 17

2.6 Conditionalization to zero probability events . . . 19

2.7 Marginal-conditional decomposition . . . 22

3 What can we learn when data are missing? **24**

3.1 Structure of the problem . . . 24

3.2 Partial identification of means . . . 26

3.3 Parameters that respect stochastic dominance . . . 28

3.4 Combining Multiple Sampling Processes . . . 30

3.5 Example: The National Longitudinal Survey for Young . . . 32

3.6 Instrumental variables . . . 34

3.6.1 Preliminaries . . . 34

3.6.2 Distributional assumptions . . . 34

3.6.3 Consequences of the distributional assumptions . . . 35

3.6.4 Estimation with nonresponses weights . . . 37

3.7 Concluding remark . . . 38

4 Conditional Prediction with Missing Data **39**

4.1 Missing outcomes . . . 39

4.2 Jointly missing outcome and covariates . . . 39

4.3 Missing covariates . . . 41

4.4 General Missing-Data Pattern . . . 42

4.5 Examples . . . 44

4.5.1 Motivation . . . 44

4.5.2 Some findings . . . 46

5 Dissecting Chilean Surveys: The Missing Outcomes Case **47**

5.1 CADEM survey . . . 48

5.1.1 General objective and methodological information . . . 48

5.1.2 How are the missing responses treated? . . . 49

5.1.3 Dissecting the CADEM survey . . . 50

5.1.4 Discussion . . . 57

5.2 CASEN survey . . . 57

5.2.1 Treatment of missing outcomes in the CASEN survey . . . 58

5.2.2 Dissecting the CASEN survey . . . 58

5.2.3 Discussion . . . 65

5.3 The Araucanía citizen consultation . . . 66

5.3.1 Historical and economical context . . . 66

5.3.2 Recent political context . . . 67

5.3.3 Organization of the Araucanía consultation and results . . . 68

5.3.4 How these results were used? . . . 68

5.3.5 Dissecting the use of Araucanía citizen consultation . . . 73

5.3.6 Discussion . . . 78

5.4 Concluding remarks . . . 79

6 How to report the uncertainty due to missing data? **81**

6.1 Introduction . . . 81

6.2 Proposal of reporting . . . 83

7 Contaminated outcomes **86**

7.1 Motivation . . . 86

7.2 Partial identification of $P(y^* \mid z = 1)$ and $P(y^*)$ . . . 87
7.3 Contamination through imputation . . . 88

8 A Qualitative Analysis of Peru's 2021 Presidential Elections: A Partial Identification Strategy 90
8.1 Introduction . . . 90
8.2 Motivation . . . 90
8.3 Background on electoral fraud . . . 91
8.3.1 Measuring electoral fraud . . . 93
8.3.2 Other perspectives . . . 93
8.4 Data . . . 94
8.4.1 Share of valid votes . . . 95
8.4.2 Voter turnout . . . 95
8.5 Participation and voting in first and second round . . . 95
8.5.1 Partial identification of the joint distribution of participation . . . 96
8.5.2 Partial identification of the conditional probabilities of participation . . . 97
8.5.3 Partial identification of the distributions for each round . . . 98
8.5.4 Empirical analysis of participation and voting using partial identification . . . 99
8.6 Electoral fraud . . . 100
8.6.1 Electoral fraud as a contaminated data problem . . . 101
8.6.2 Assessing electoral fraud . . . 102
8.7 More than two candidates . . . 103
8.8 Discussion . . . 105

9 A Critical View on the NEAT Equating Design: Statistical Modelling and Identifiability Problems 110
9.1 Introduction . . . 110
9.2 Modelling problems underlying the NEAT design . . . 111
9.2.1 Equating under the NEAT design: a summary of the literature . . . 111
9.2.2 What we question: A statement of the problem . . . 113
9.3 Statistical specification of the NEAT design and the inherent identification problem . . . 114
9.3.1 The sample space in the NEAT design . . . 114
9.3.2 The random variable Z . . . 115
9.3.3 The anchor random variable A . . . 117
9.3.4 The random variables X and Y . . . 117

9.3.5 Consequences of the model specification . . . 118
9.4 Approaches to tackle the Identification Problem . . . 119
9.4.1 Strong ignorability condition . . . 119
9.4.2 Partial identification analysis . . . 120
9.4.3 Empirical illustration . . . 124
9.5 Improved partial identification intervals . . . 126
9.5.1 Motivation . . . 126
9.5.2 Partial identification of the parameters of interest . . . 128
9.6 Discussion . . . 132
9.7 A definition of random assignment in terms of bias . . . 134
9.8 Proof of the equivalence between (9.2.5) and (9.4.2) . . . 136
9.9 Proof of Theorem 9.3 . . . 136

10 Brokering the Evaluation of Public Policies 139
10.1 Introduction . . . 139
10.1.1 Public policy implementation and uncertainty . . . 139
10.1.2 The features of the EBP paradigm . . . 140
10.1.3 The consequences of the EBP paradigm . . . 140
10.1.4 Limitations of the EBP paradigm due to additional uncertainties . . . 141
10.1.5 Are policies based on scientific evidence or on scientific knowledge? . . . 142
10.2 How to construct a prescriptive role for science in policy decisions? . . . 143
10.2.1 The standard solution . . . 144
10.2.2 Optimistic policymaker . . . 145
10.2.3 Conservative policymaker . . . 148
10.2.4 What if the policy decision harms all the individuals? . . . 150
10.2.5 What if the policy decision benefits all individuals? . . . 151
10.2.6 Overview of the solutions . . . 152
10.2.7 A plausible policy interpretation of strong ignorability? . . . 153
10.3 A Decision Problem . . . 154
10.3.1 Basic framework of decision theory . . . 155
10.3.2 Minimax-regret for the FBC perspective . . . 158
10.3.3 Ethical issues of fractional treatment rules . . . 159
10.4 The role of science in policy . . . 159
10.4.1 The impact of the Honest Broker role at the methodological level . . . 160
10.4.2 Implications of an Honest Broker strategy . . . 160

10.4.3 The challenge of the proposed strategy . . . 161
10.5 Proofs . . . 162
10.5.1 Proof of inequalities (10.2.5) . . . 162
10.5.2 Proof of inequalities (10.2.12) . . . 162
10.5.3 Proof of (10.2.16) . . . 163

11 How good/how bad is integrated school leadership for school administration? The Chilean case 164
11.1 The Chilean National System of Quality Assurance of Education . . . 164
11.1.1 Integrated leadership model in the Chilean policy context . . . 165
11.1.2 Information and guidance: the inputs provided to schools by the Agency of Quality 166
11.2 The impact of school leadership on intra-school practices . . . 167
11.2.1 Relating school leadership and intra-school practices . . . 167
11.2.2 Summary of the observational evidence . . . 168
11.3 Policy Evaluation . . . 171

12 A case study of sentencing and recidivism 173
12.1 Context . . . 173
12.1.1 Judge's sentences . . . 174
12.1.2 Structure of the problem . . . 175
12.1.3 Identified and unidentified parameters . . . 176
12.2 Impact evaluation of the treatments . . . 176
12.2.1 Application . . . 178
12.3 Two models of treatment selection . . . 178
12.3.1 Motivation . . . 178
12.3.2 Outcome optimization model . . . 179
12.3.3 Skimming model . . . 180

13 Identification Problems underlying Randomized Control Trials 182
13.1 How to define a Randomized Trial? . . . 182
13.1.1 Kolmogorov's axioms of probability theory . . . 182
13.1.2 What intuition underlies the expression *randomized control trial*? . . . 185
13.1.3 A structural definition of bias . . . 186
13.2 Compliance with Treatment Assignment . . . 187
13.2.1 Intention-to-Treat . . . 188
13.2.2 The Average Effect of Treatment on Compliers . . . 189

14 Identification problems in the extrapolation of experimental results **192**
14.1 Extrapolation of the Perry Preschool Project 192
14.2 The mixing problem 194
14.2.1 Partial identification without prior information 195
14.2.2 Ordered outcomes 196
14.2.3 Treatment independent from outcomes 197
14.2.4 Optimization policies 197
14.2.5 Known treatment distribution 198
14.2.6 Application 199

A Appendix **200**
A.1 Fréchet inequality 200
A.1.1 Statement of the problem 201
A.1.2 Solution 201
A.1.3 Step 2 203

References **205**

Chapter 1

Fundamental concepts

The objective of this section is to realize that the meaning of a statistical model essentially depends on an identification analysis. This is due to the fact that identifiability is related to *model construction*, not to *statistical estimation*. This explains why some authors consider an *identification problem* as a *pre-statistical problem* (McHugh, 1956).

1.1 The problems of statistics

In 1922, R. A. Fisher pointed out that, in spite of the large amount of fruitful applications of statistics, its basic principles were still in a "state of obscurity" and that "during the recent rapid development of practical methods, fundamental problems have been ignored" (Fisher, 1922, p.310). This evaluation led Fisher to define the scope of statistical methods as the reduction of data. "No human mind is capable of grasping in its entirety the meaning of any considerable quantity of numerical data" (Fisher, 1973, p.6). Consequently, data should be replaced by a relatively few quantity of values which shall contain the *relevant information* provided by the original data.

Conceptually speaking, the reduction of data is accomplished by constructing a hypothetical infinite population, of which the actual data are considered a random sample. This hypothetical population is fully characterized by a probability distribution, which in turn is specified by relatively few parameters. The parameters "are sufficient to describe it exhaustively in respect of all qualities under discussion". Thus, any information provided by a sample, which is used to estimate those parameters, is relevant information. This conceptual scheme leads to distinguish three problems of statistics (Fisher, 1922, 1973, Chapter 1): *Problems of Specification*, *Problems of Estimation* and *Problems of Distribution*. Problems of Specification arise in the choice of the mathematical form of the probability distribution which characterizes the observed population. Its choice "is not arbitrary, but requires an understanding of the way in which the data are supposed to, or did in fact, originate" (Fisher, 1973, p.8).

How to make precise the way in which the data are generated? A quarter of century later, Koopmans and Reiersøl (1950) suggested a response, which leads to reformulate the specification problem as defined by Fisher. Such a reformulation, "appropriate to many applications of statistical methods", is based

on a distinction between population and structure:

> In many fields the objective of the investigator's inquisitiveness is not just a "population" in the sense of a distribution of observable variables, but a physical structure projected behind this distribution, by which the latter is thought to be generated. The word "physical" is used merely to convey that the structure concept is based on the investigator's ideas as to the "explanation" or "formation" of the phenomena studied, briefly, on his theory of these phenomena, whether they are classified as physical in the literal sense, biological, psychological, sociological, economic or otherwise (Koopmans & Reiersøl, 1950, p.165).

A structure is determined on the basis of a substantive theory, combined with systematically collected data for the relevant variables. The collected data are represented by observed variables, whereas (some) aspects of a substantive theory are represented by unobservable or latent ones. Thus, a structure is composed of a particular probability distribution of the latent variables, combined with a relationship between the observed and unobserved variables. Koopmans and Reiersøl's reformulation of the specification problem is accordingly defined as follows:

> The specification is therefore concerned with the mathematical forms of both the distribution of the latent variables and the relationships connecting observed and latent variables (p.168).

The distribution of the observed variables is regarded as produced by an operation performed on the distribution of the unobserved variables. The specification problem is, therefore, concerned with specifying a model which, by hypothesis, contains the structure generating the distribution of the observed variables. The result of this reformulation is the emergence of the identification problem: can the distribution of the observations, generated by a given structure, be generated by only one structure in that model? When the answer is affirmative, it is said that the structure is identified by the observations. In such a case, the structure has a statistical meaning in the sense that it is supported by the observations through the probability distribution of the observations (Koopmans, 1949; Hurwicz, 1950; Clifford, 1982; Manski, 1995; San Martín, 2015). The main characteristic of the so-called structural modeling is that it considers structures and/or parameters as endowed with a meaning that should be supported by the observations.

1.2 Statistical model

The specification problem leads to define a *statistical model*. Its definition becomes apparent after making explicit the components of a statistical model:

- *Sample space* $(S, \mathcal{S})$, where S is the set of all possible outcomes, and $\mathcal{S}$ is the σ-field of subsets of S: the elements of $\mathcal{S}$ are considered as the events of interest.
- A probability distribution P^θ defined on the sample space $(S, \mathcal{S})$, where θ is a parameter. P^θ is accordingly called *sampling distribution* or *sampling probability*.

- $\theta \in \Theta$, where the set Θ is called *parameter space*.

Thus, a statistical model can compactly be written as

$$\mathcal{E} = \{(S, \mathcal{S}), P^\theta : \theta \in \Theta\}.$$

When the statistical model is dominated (that is, when for all $\theta \in \Theta$, P^θ is absolutely continuous with respect to a σ-finite measure λ defined on $(S, \mathcal{S})$, the statistical model is equivalently described through the likelihood function p^θ, where $p^\theta = \frac{dP^\theta}{d\lambda}$:

$$\mathcal{E} = \{(S, \mathcal{S}) : p^\theta : \theta \in \Theta\}.$$

Remark 1.1 It is important to remark that this concept covers what it is commonly called parametric, semi-parametric and non-parametric models: the distinction depends on the structure of Θ. Thus, the parameter space Θ of a parametric model is contained in a finite-dimensional space; for non-parametric models, the parameter space Θ is an infinite-dimensional space; semi-parametric models can be described by saying that Θ is a cartesian product of the form $\boldsymbol{A} \times \boldsymbol{\Psi}$, where $\boldsymbol{A}$ is a finite-dimensional space and $\boldsymbol{\Psi}$ is an infinite-dimensional space.

Ejemplo 1.2.1 The concept of statistical model is understand the difference between a fixed effect model and a random effect model: it is enough to make explicit the data generating process. As an example, let us consider an ANOVA 1 Fixed Factor (balanced case):

$$Y_{ij} \overset{\text{indep.}}{\sim} \mathcal{N}(\alpha + \beta_j, \sigma^2), \quad i = 1, \ldots, n;\ j = 1, \ldots, J. \tag{1.2.1}$$

Let

$$\mathbf{Y}_j = (Y_{1i}, \ldots, Y_{nj})' \quad j = 1, \ldots, J; \quad \mathbf{Y} = (\mathbf{Y}_1', \ldots, \mathbf{Y}_J')'.$$

The mutual independence implies that

$$\mathbf{Y} \sim \mathcal{N}_{nJ}\left(\alpha \mathbb{1}_{nJ} + \begin{pmatrix} \beta_1 \mathbb{1}_n \\ \vdots \\ \beta_J \mathbb{1}_n \end{pmatrix},\ \sigma^2 I_{nJ}\right), \tag{1.2.2}$$

where $\mathbb{1}_n$ is a n-dimensional vector of 1's and I_{nJ} is the $nJ \times nJ$ identity matrix. This, (1.2.2) characterizes the fixed- effect model: the data are mutually independent and the means depend of j.

A random-effect model is specified through a marginal-conditional decomposition:

$$\text{(i) } (Y_{ij} \mid \beta_j) \overset{\text{c.ind}}{\sim} \mathcal{N}(\alpha + \beta_j, \sigma^2), \quad \text{(ii) } \beta_j \overset{\text{ind}}{\sim} \mathcal{N}(0, \tau^2), \quad i = 1, \ldots, n;\ j = 1, \ldots, J. \tag{1.2.3}$$

This specification deserves three remarks:

1. The specification (1.2.3) does not correspond to the statistical model because a statistical model describes the *observations*.

2. (1.2.3.i) is a conditional distribution, whereas (1.2.1) is a marginal distribution: therefore, both distributions are *not* related between them and consequently one specification can not be viewed as a kind of "re-interpretation" of the other (as it is typically viewed in econometrics, statistics, biometrics or psychometrics). Moreover, in (1.2.1) β_j is a constant, whereas in (1.2.3.i) it is a random variable.

3. c.ind means that the for each j, $\{Y_{ij} : i = 1, \ldots, n\}$ are mutually independent conditionally on β_j.

The specification (1.2.3) induces a statistical model: it is obtained after integrating out the β_j's. For each j, it follows that

$$\mathbf{Y}_j \sim \mathcal{N}_n \left(\alpha \mathbb{1}_n, \begin{pmatrix} \sigma^2 + \tau^2 & \tau^2 & \cdots & \tau^2 \\ \tau^2 & \sigma^2 + \tau^2 & \cdots & \tau^2 \\ \vdots & \vdots & \ddots & \vdots \\ \tau^2 & \tau^2 & \cdots & \sigma^2 + \tau^2 \end{pmatrix} \right). \tag{1.2.4}$$

This means that two observations Y_{ij} and $Y_{i'j}$ are correlated between them. Therefore, a random effect model is used when the scientist have reason to believe that the observations nested in the j-group are correlated in the sense that the j-group induces such a correlation. This is not the case in a fixed-effect model. The choice depends on substantive considerations related to the fact that the j-group induces correlation or not.

■

1.3 Identification problem

The definition of statistical model implicitly involves a relationship between the parameter space and the set of sampling distributions defined on $(S, \mathcal{S})$. If we denote this set as $\mathcal{P}(S, \mathcal{S})$, such a relationship corresponds to the mapping $\mathbf{\Phi} : \Theta \longrightarrow \mathcal{P}(S, \mathcal{S})$ such that $\mathbf{\Phi}(\theta) = P^\theta$ for $\theta \in \Theta$.

Our interest is focused on the sample space $(S, \mathcal{S})$: there we collect specific features of interest characterizing a particular population. For instance, the number of correct responses in a test of 40 items. A sampling probability P^θ is supposed to describe the behavior of those features. This behavior could be interpreted as the variability of such features across the population. The relevant point here is that P^θ fully characterizes the population that is analyzed with respect to those features of interest. If we accept that, then we focus our interest on the family $\{P^\theta : \theta \in \Theta\}$, assuming that θ is a characteristic of P^θ which in turn describes the units of the population in terms of those features of interest. As Fisher said, the parameters "are sufficient to describe [the population of interest] exhaustively in respect of all qualities under discussion" (Fisher, 1973).

Paraphrasing Koopmans (1949), statistical inference unsupported by a substantive theory applies to whatever statistical regularities and, consequently, stable relationships can be discerned in the data. Such

purely empirical relationships are likely to be due to the presence and persistence of underlying structural relationships. However, the direction of this deduction cannot be reversed (from the empirical to the structural relationships) expect possibly with the help of a substantive theory which specifies the form of the structural relationships. The possibility of this deduction necessarily implies to solve an *identification problem* at the model construction level. As Haavelmo (1997) puts it,

> The one type of problems, seemingly paradoxical, grew out of a rather intricate consequence of successful economic theory. Strangely enough, the fact is that if an economic theory, an economic relation, is a good theory, true to reality, it may not be possible to quantify it by using data from the economic of which that relation is a part. This is the so-called "problem of identification" (p.14).

These considerations are applicable to all social and health sciences, among many others. This is a requirement of a scientific inquiry that focuses on the explanation of the phenomena, and not (only) on the mere prediction with implicit assumptions of structural stability (as is the case with most of the recent data science techniques).

The previous considerations lead to conclude that the properties of a population under study are represented by a parameter. Consequently, if a property supposed to be a characteristic of a population can not be represented by a parameter of the sampling probability, such a property is not supported by the observations: this corresponds to an identification problem. Therefore, the statistical meaning of parameters becomes explicit by an identification analysis.

Ejemplo 1.3.1 [From Fariña, González, and San Martín (2019)] Let us illustrate these considerations with the ANOVA 1 Fixed factor as introduced in Example 1.2.1. For this model, one of the following two identifiability restrictions are typically considered: $\beta_1 = 0$ or $\sum_{j=1}^{J} \beta_j = 0$. Under the first restriction parameter identifiability follows from the identity $\alpha = \mu_1 = E(Y_{i1})$ and $\beta_j = \mu_j - \mu_1 = E(Y_{ij}) - E(Y_{i1})$. Thus, under the first restriction, α corresponds to the mean of the first population, whereas β_j represents the deviation of the jth population mean with respect to the mean of the first population. Under the second restriction, we have that $\alpha = \frac{1}{J}\sum_{j=1}^{J} \mu_j = \frac{1}{J}\sum_{j=1}^{J} E(Y_{ij})$ and $\beta_j = \mu_j - \frac{1}{J}\sum_{j=1}^{J} \mu_j = E(Y_{ij}) - \frac{1}{J}\sum_{j=1}^{J} E(Y_{ij})$. Thus, under the second restriction, α represents the overall mean, whereas β_j corresponds to the deviation of the jth population mean with respect to the overall mean. The situation is different for the random-effects version of the model (1.2.3): the identifiability follows from the induced statistical model (1.2.4). Here, $\alpha = E(\mu_j) = E(Y_{ij})$, that is, it represents a common mean for each observation.

■

Ejemplo 1.3.2 [From San Martín, González, and Tuerlinckx (2009) and San Martín and De Boeck (2015)] Let us illustrate the previous consideration with the so-called Rasch model, which is widely used in the test industry. Let $\{Y_{ij} : i = 1, \ldots, I;\ j = 1, \ldots, J\}$ be a sequence of mutually independent binary random variables, where

$$Y_{ih} = \begin{cases} 1, & \text{if person } i \text{ correctly answers item } j; \\ 0, & \text{in any other case;} \end{cases},$$

where $Y_{ij} \sim \text{Bern}\,(p_{ij})$. Rasch model focuses on *relative comparisons*. Therefore, we introduce the following reparameterization:

$$p_{ij} = \frac{\lambda_{ij}}{1+\lambda_{ij}}, \quad \lambda_{ij} \geqslant 0,$$

which is equivalent to

$$\lambda_{ij} = \frac{P(Y_{ij}=1)}{P(Y_{ij}=0)}.$$

Thus, if $\lambda_{ij} > 1$, then person i has a probability to correctly answer item j greater than answering it incorrectly. A similar interpretation can be done of $\lambda_{ij} < 1$ or $\lambda_{ij} = 1$. The model specification is completed by assuming that

$$\lambda_{ij} = \frac{\epsilon_i}{\eta_j} \quad i = 1, \ldots, I,\ j = 1, \ldots, J, \tag{1.3.5}$$

where ϵ_i is a characteristic of person i and η_j is a characteristic of item j. The question is whether these parameters can be written as a function of λ_{ij}'s. The answer is *not*. As a matter of fact, let $\{\epsilon_i, \eta_j\}$ be a set of parameters satisfying the equalities (1.3.5); and let $\{\epsilon_i', \eta_j'\}$ be other different set. It follows that

$$\frac{\epsilon_i}{\eta_j} = \frac{\epsilon_i'}{\eta_j'} \qquad \forall\,(i,j) \in \{1,\ldots,I\} \times \{1,\ldots,J\},$$

which is equivalent to

$$\frac{\epsilon_i}{\epsilon_i'} = \frac{\eta_j}{\eta_j'} \qquad \forall\,(i,j) \in \{1,\ldots,I\} \times \{1,\ldots,J\},$$

which necessarily implies that

$$\frac{\epsilon_i}{\epsilon_i'} = \frac{\eta_j}{\eta_j'} = \alpha \qquad \forall\,(i,j) \in \{1,\ldots,I\} \times \{1,\ldots,J\},$$

where α is a constant. Therefore, $\{\epsilon_i, \eta_j\}$ are not identified and, consequently, can not be statistically interpreted.

In order to remove the inherent indeterminacy, we choose an item, for instance $j = 1$, as a *standard item* in the sense that we fix a "unit". Thus, $\eta_1 = 1$ and therefore

$$\epsilon_i = \lambda_{i1} = \frac{P(Y_{i1}=1}{P(Y_{i1}=0)}.$$

Therefore, the parameter that characterizes person i corresponds to the betting odd of answering correctly the standard item 1. In particular,

$$\epsilon_i > \epsilon_{i'} \Longleftrightarrow P(Y_{i1}=1) > P(Y_{i'1}=1).$$

That is, persons i and i' can be compared between them through their answers to the standard item 1. This comparison explains why psychometricians call ϵ_i *ability of person* i, although any psychological phenomenon underlies ϵ_i.

Similarly, from (1.3.5) and $\epsilon_i = \lambda_{1i}$, it follows that

$$\eta_j = \frac{P(Y_{i1} = 1)}{P(Y_{i1} = 0)} \frac{P(Y_{ij} = 0)}{P(Y_{ij} = 1)}.$$

Then

1. $\eta_j > 1 = \eta_1 \Longleftrightarrow P(Y_{i1} = 1) > P(Y_{ij} = 1)$ for all person i.
2. $\eta_j > \eta_k \Longleftrightarrow P(Y_{ik} = 1) > P(Y_{ij} = 1)$ for all person i.

These inequalities explains why psychometricians call η_j *difficulty of item* j. Thus, for instance, item j is more difficult that item k if and only if the probability to correctly answer item j is smaller than the probability to correctly answer item k.

Finally, the Rasch model allows us to compare the individual characteristic ϵ_i and the item characteristic η_j:

$$\epsilon_i > \eta_j \Longleftrightarrow P(Y_{ij} = 1) > P(Y_{ij} = 0).$$

Thus, the ability of person i is greater than the difficulty of item j if and only if the probability that person i answers correctly item j is greater than the probability of answering incorrectly. This property is know as the simultaneous representation "person $\times$ items". For an application, see Claro et al. (2012).

■

1.4 Concluding remarks

Identification analysis is a relevant component of model construction: the empirical interpretation of the parameters of interest depend on this type of analysis. Furthermore, all property that a scientist assumes to be a characteristic of a population under study *is* actually a characteristic of it *only if is a function of the corresponding sampling distribution*; for more details, see also San Martín (2018). This aspect becomes relevant for empirical research because such a research *always* involves an identification problem. The way in which the identification problem is solved will constitute the so-called *maintained assumption*. When this type of assumptions is combined with empirical evidence, we reach scientific conclusions and thus we conform a corpus of scientific knowledge (Manski, 2013a). Based on this knowledge, science becomes useful for informing policy decisions. Nevertheless, we will emphasize that the maintained assumptions are not unique and, therefore, we will make explicit the subjective side of scientific knowledge. This is one of the aspects we will discuss in what follows.

Chapter 2

The Conceptual Depth of the Law of Total Probabilities

2.1 Introduction

Let us consider the scores that students obtained in the 2013 SIMCE Mathematics test (a Chilean national test): those students were at 6th grade of elementary school. In order to have a panoramic view of these data, we will begin by describing some general elements that reflect the way in which the Chilean Educational System is organized.

• Students belong to schools which are classified in three categories: Public schools (denoted by MN), subsidized schools (denoted bby PS) and private schools (denoted by PP). From the data we have

1. 39.39% of students belong to MN schools;
2. 52.96% of students belong to PS schools;
3. 7.65% of students belong to PP schools.

• Schools are also classified in terms of socioeconomic level. The Chilean Agency of Quality of Education defines 5 socioeconomic levels, which are constructed from the information collected in surveys applied to parents and guardians. Five socioeconomic levels are defined, from the lowest to the highest. From the data we have

1. 10.45% of students belong to schools of low socioeconomic level (denoted by 1).
2. 31.54% of students belong to schools of middle low socioeconomic level (denoted by 2).
3. 34.30% of students belong to schools of middle socioeconomic level (denoted by 3).

4. 15.65% of students belong to schools of middle high socioeconomic level (denoted by 4).
5. 8.07% of students belong to schools of high socioeconomic level (denoted by 5).

• As we have discussed elsewhere, relative frequencies satisfy the axioms of ? (?), provided that the total number of statistical units with respect to which such frequencies are calculated is fixed. Thus, relative frequency is a measure of probability. Thus, if we denote by D a variable indicating the administrative unit of the educational establishment to which a student belongs, then D takes three values, namely, MN, PS, PP.The variable D induces a partition over the sample space M corresponding to the levels associated to student. Thus,

1. $D^{-1}\{\text{MN}\}$ comprises all students who belong to MN schools.
2. $D^{-1}\{\text{PS}\}$ comprises all students who belong to PS schools.
3. $D^{-1}\{\text{PP}\}$ comprises all students who belong to PP schools.

Furthermore,

$$P(D = \text{MN}) = 0.3939; \qquad P(D = \text{PS}) = 0.5296; \quad P(D = \text{PP}) = 0.0765.$$

• Similarly, let S be the socioeconomic level of a school to which students belong. Then S takes the values $\{1, 2, 3, 4, 5\}$ and, therefore, induces also a partition on M. Consequently,

$$\begin{array}{lll} P(S = 1) = 0.1045; & P(S = 2) = 0.3254; & P(S = 3) = 0.3430; \\ P(S = 4) = 0.1565; & P(S = 5) = 0.0807. & \end{array}$$

• It is interesting to know how the variables D and S intersect between them. So, we generate the contingency table of relative frequencies, which corresponds to the joint probability distribution of the vector (D, S) (we express this table in percentage terms):

	$S = 1$	$S = 2$	$S = 3$	$S = 4$	$S = 5$
$D = 1$	8.28	21.43	8.80	0.88	0.00
$D = 2$	2.17	10.11	25.49	14.49	0.70
$D = 3$	0.00	0.00	0.01	0.27	7.37

Exercise: Describe the partitioned induced by both D y S on M.

• In terms of the biological sex of students (denoted by G), 49.04% are female and 50.96% are male. When we consider this information with the administrative dependency of schools, we get the following joint distribution of (G, D) (we express this table in percentage terms):

	$D = \text{MN}$	$D = \text{PS}$	$D = \text{PP}$
$G = \text{F}$	18.91	26.36	3.76
$G = \text{M}$	20.48	26.60	3.88

When we interact the biologial sex of students with the socioeconomic level of schools, we obtain the joint distribution of (G, S) (we express this table in percentage terms):

	$S = 1$	$S = 2$	$S = 3$	$S = 4$	$S = 5$
$G = \text{F}$	4.93	14.97	17.22	7.90	4.01
$G = \text{M}$	5.52	16.56	17.09	7.75	4.05

• Three different forms of the test were applied (denoted by F):forms A, B and C. Form A was applied to 48.59% of students; form B, to 47.82% of students; form C, to 3.59% of students.

2.2 Scores of te SIMCE test

Given that 3 forms were applied, it is relevant to compare the results of the students in each of these forms. To do so, it is relevant to start with the most "panoramic" description, which is obtained by means of the cumulative distribution function of these scores. Figure 2.1 shows the cumulative distributions for each of the forms: it can be seen that forms A and B have, globally, a similar behavior[1] but form C has a different behavior.

In order to express the differences between these distribution functions, we will use the quantile function, which corresponds to the generalized inverse of the cumulative distribution function: for $alpha \in [0, 1]$, the quantile function $q(\alpha)$ corresponding to the cumulative distribution function F is given by

$$q(\alpha) = \inf\{t : F(t) \geqslant \alpha\}.$$

Table 2.1 shows some quantiles corresponding to Forms A, B and C. It can be concluded, for example, that

1. 5% of students who answered form A scored at most 171.99 points, while 5% of those who answered form B scored at most 167.72 points. These scores contrast with at most 155.04 points obtained by the 5% who answered form C.

2. 50% of students who answered form A scored at most 250.82 points. 50% of students who answered form B scored at most 249.05 points. However, 50% of students who answered form C scored at most 232.42 points.

These comments are illustrated through the quantiles 0.25 and 0.5 in Figure 2.2.

[1]There are statistical procedures that allow us to decide whether this observed equality is statistically justifiable: they are called hypothesis tests,

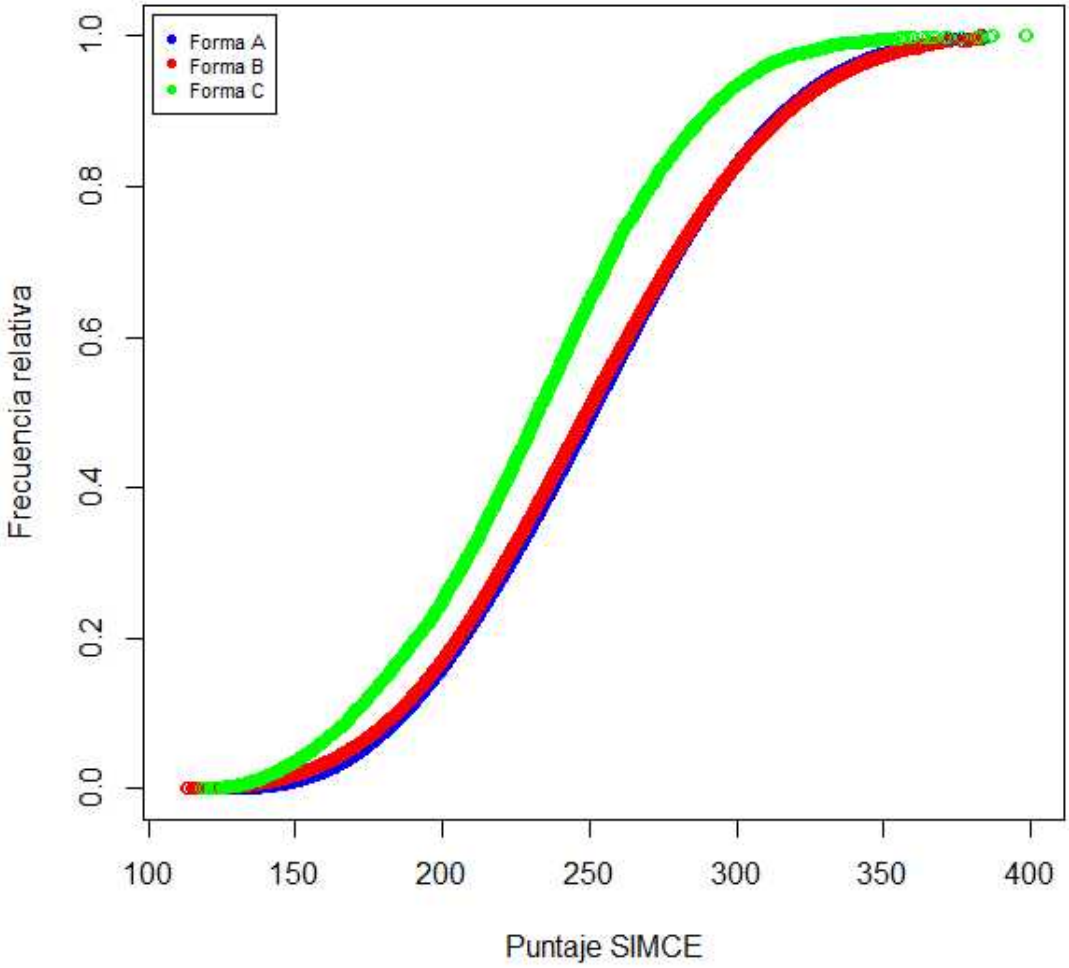


Figure 2.1: Distribution functions of the SIMCE Mathematics scores

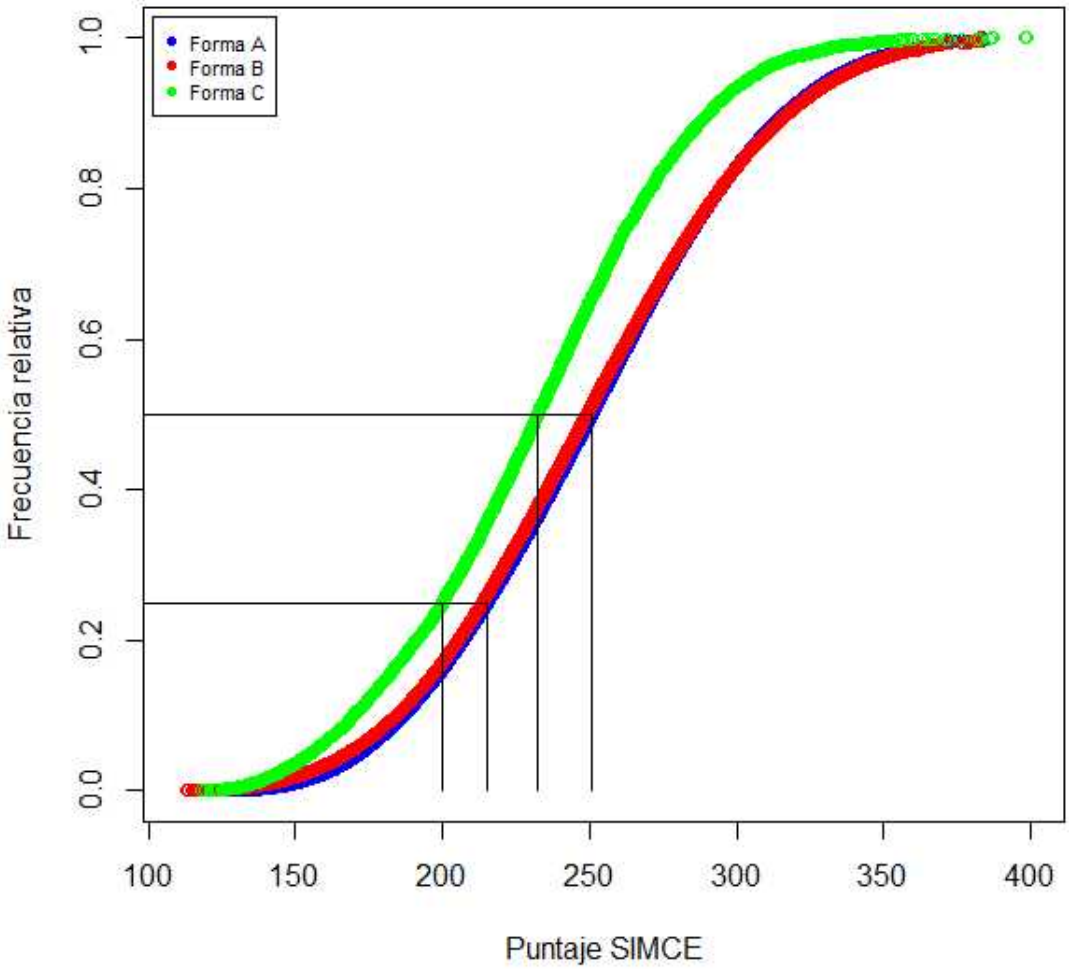


Figure 2.2: Cumulative distribution functions of SIMCE Mathematics scores

Table 2.1: Quantiles of the distributions of scores obtained in Forms A, B and C

Quantile	Form A	Form B	Form C
0.05	171.99	167.72	155.04
0.25	215.32	213.52	200.08
0.5	250.82	249.05	232.42
0.75	285.96	285.6	262.68
0.95	333.96	335.86	305.8

2.3 Distribution function of scores by administrative dependency of schools

Let us focus our attention on the distribution function of Form A scores. To plot the distribution function of the scores obtained in Form A, in the corresponding database we take the column containing the scores and draw the distribution.

Question: How do we graphically represent the cumulative distribution of scores of students belonging to MN schools? Although the question may seem naive, it is not, because the answer will allow us to understand a fundamental probabilistic concept. We report the answer provided by students after a brief moment's thought:

> It is enough to consider the scores of students belonging to MN schools: take those data set, and plot the distribution function.

Figure 2.3 shows the result of this response: the score distribution function of students belonging to MN schools. This distribution has been obtained *on one condition*, namely, that the students belong to a MN school. If we denote by X a random variable defined from M to the set N of possible scores (from 100 to 400 points that each student obtained in Form A), then we have represented graphically *the distribution function of scores conditional on the fact that students belong to a MN school*. This is written as

$$P(X \leqslant x \mid D = \text{MN}) \qquad x \in N.$$

Note that for obtaining this distribution function, we don't use the computation rule according to which

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

with A, B events.What is important is that a *condition* was introduced that allowed us to select certain data and thus obtain the conditional distribution of the scores.

Similarly, we can obtain the distributions of scores under the condition that students belong to PS schools, and the distribution of scores under the condition that students belong to PP schools. Figure 2.4 shows the three distributions. Thus, we have the graphical representation of

$$P(X \leqslant x \mid D = \text{MN}) \quad x \in N; \qquad P(X \leqslant x \mid D = \text{PS}) \quad x \in N; \qquad P(X \leqslant x \mid D = \text{PP}) \quad x \in N.$$

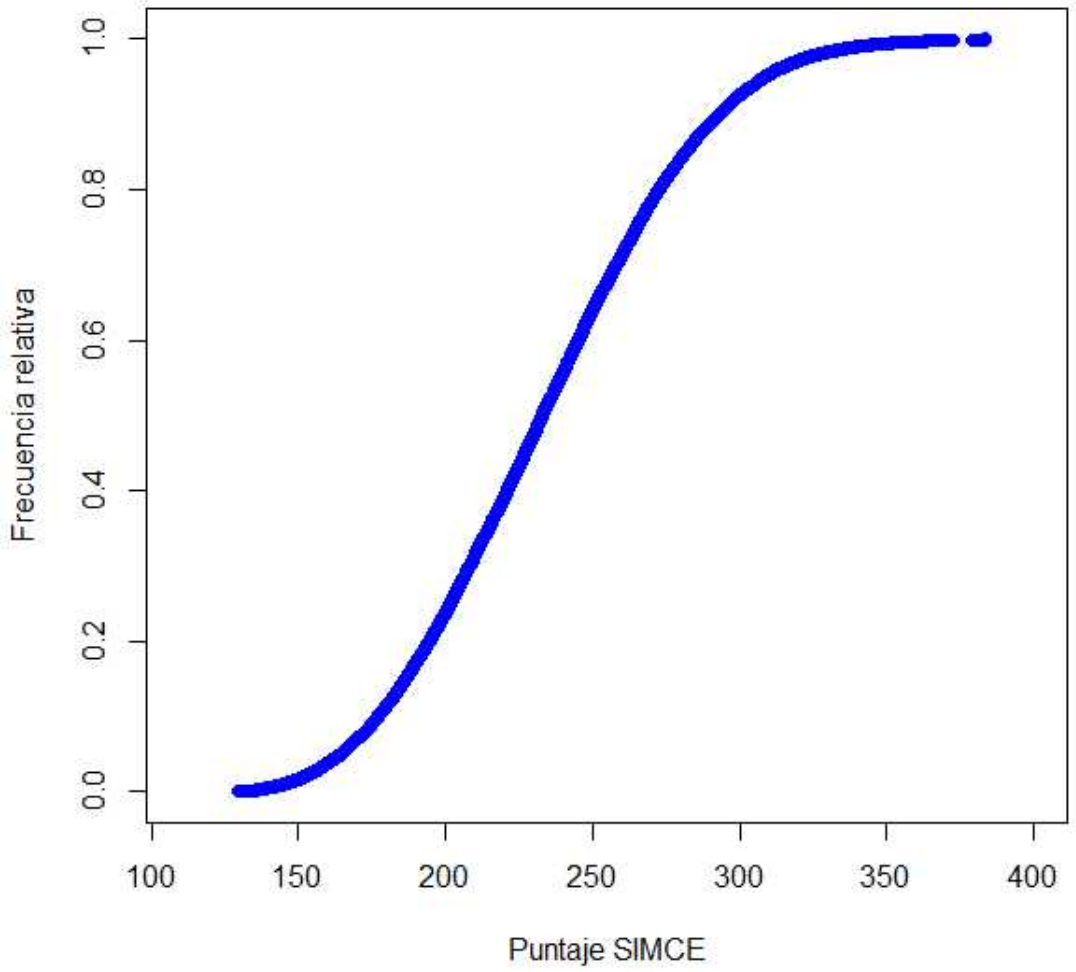


Figure 2.3: Cumulative distribution function of the SIMCE Mathematics scores for students in MN schools

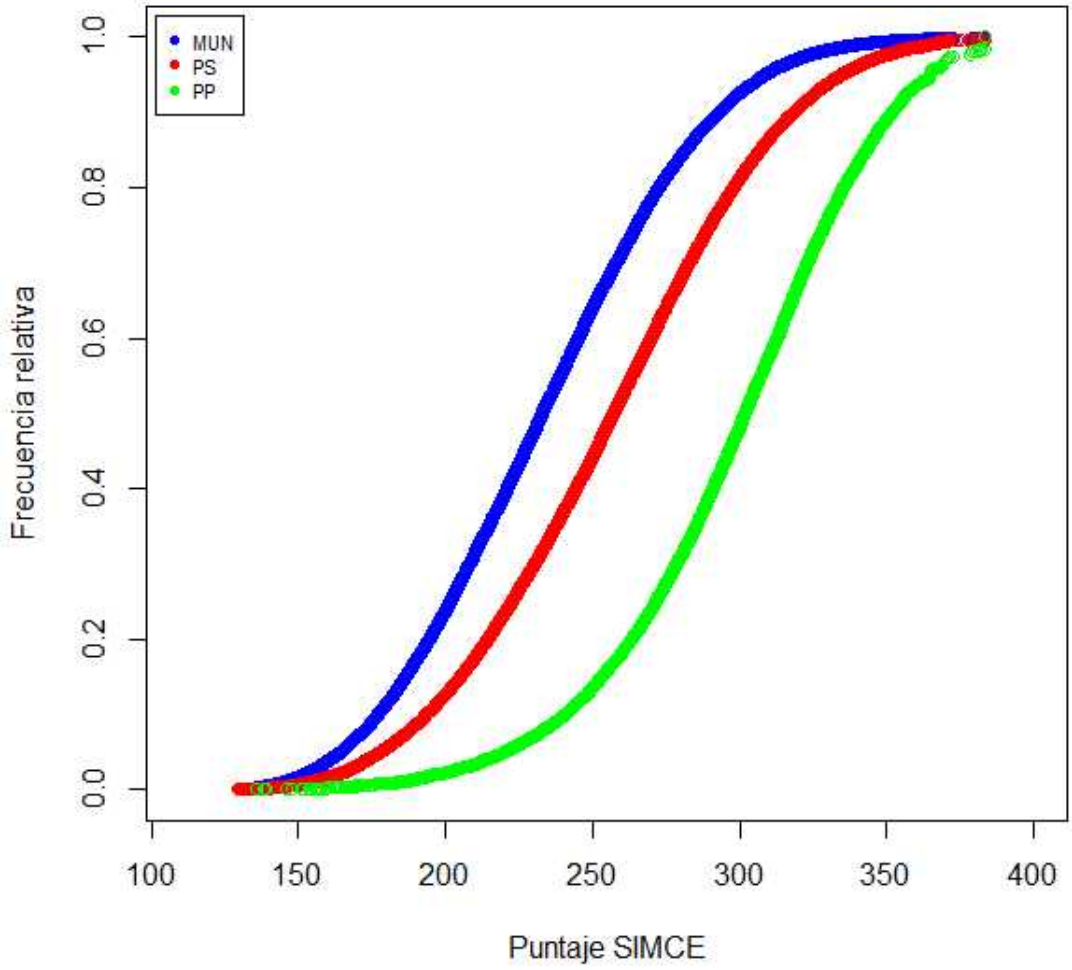


Figure 2.4: Cumulative distribution function of SIMCE Mathematics scores conditional on the type of school

Now, three probability distributions have been obtained conditionally to specific values taken by the random variable D, namely, $D = \text{MN}$, $D = \text{PS}$ y $D = \text{PP}$. Therefore, we have three values of a function of D: *the conditional distribution of X conditionally on D, denoted by $P(X \leqslant x \mid D)$ with $x \in N$, is a function of D and, hence, a random variable*. This function can be written as follows: for all $x \in N$,

$$\begin{aligned} P(X \leqslant x \mid D) &= P(X \leqslant x \mid D = \text{MN})\, \mathbb{1}_{\{D=\text{MN}\}} + P(X \leqslant x \mid D = \text{PS})\, \mathbb{1}_{\{D=\text{PS}\}} + \\ &\quad P(X \leqslant x \mid D = \text{PP})\, \mathbb{1}_{\{D=\text{PP}\}}, \end{aligned} \tag{2.3.1}$$

donde, por ejemplo,

$$\mathbb{1}_{\{D=\text{MN}\}}(m) = \begin{cases} 1, & \text{si } m = \text{MN}; \\ 0, & \text{si no.} \end{cases}$$

For instance,

$$P(X \leqslant x \mid D)(\text{PS}) = P(X \leqslant x \mid D = \text{PS}).$$

2.4 Conditional probability as a random variable

A conditional probability is, therefore, a random variable: it is in fact a transformation of the random variable on which it is conditioned. As such, it is possible to compute its expectation. Let us compute the expectation of $P(X \leqslant x \mid D)$ using (2.3.1) and recalling that, if $Y \sim \text{Bern}(p)$, then $E(Y) = p$: for all $x \in N$,

$$\begin{aligned} E\left[P(X \leqslant x \mid D)\right] &= E\Big[P(X \leqslant x \mid D = \text{MN})\, \mathbb{1}_{\{D=\text{MN}\}} + P(X \leqslant x \mid D = \text{PS})\, \mathbb{1}_{\{D=\text{PS}\}} + \\ &\quad P(X \leqslant x \mid D = \text{PP})\, \mathbb{1}_{\{D=\text{PP}\}}\Big] \\ &= P(X \leqslant x \mid D = \text{MN})\, E\left[\mathbb{1}_{\{D=\text{MN}\}}\right] + P(X \leqslant x \mid D = \text{PS})\, E\left[\mathbb{1}_{\{D=\text{PS}\}}\right] + \\ &\quad P(X \leqslant x \mid D = \text{PP})\, E\left[\mathbb{1}_{\{D=\text{PP}\}}\right] \\ &= P(X \leqslant x \mid D = \text{MN})\, P(D = \text{MN}) + P(X \leqslant x \mid D = \text{PS})\, P(D = \text{PS}) + \\ &\quad P(X \leqslant x \mid D = \text{PP})\, P(D = \text{PP}) \\ &= P(X \leqslant x), \end{aligned}$$

where the last equality follows from the Law of Total Probability.

From these equalities we can conclude that

1. The marginal distribution $P(X \leqslant x)$ corresponds to combine the conditional distributions $P(X \leqslant x \mid D = \text{MN})$, $P(X \leqslant x \mid D = \text{PS})$ y $P(X \leqslant x \mid D = \text{PP})$, taking into account the relative weights $P(D = \text{MN})$, $P(D = \text{PS})$ y $P(D = \text{PP})$. The relationship of this distribution to the conditional ones is graphically illustrated in Figure 2.5.

2. Moreover, the marginal distribution $P(X \leqslant x)$ corresponds to the expectation of the random variable $P(X \leqslant x \mid D)$.

3. The equality
$$P(X \leqslant x) = E\left[P(X \leqslant x \mid D)\right] \tag{2.4.2}$$
allows us to *define* the conditonal probability in the following terms: the random variable D induces a partition on the sample space M, which is denoted as
$$M = A_1 \cup A_2 \cup A_3,$$
where $A_i \cap A_j = \varnothing$ for $i \neq j$. Here $A_1 = D^{-1}\{\text{MN}\}$, $A_2 = D^{-1}\{\text{PS}\}$ y $A_3 = D^{-1}\{\text{PP}\}$. On this partition a random variable Y can be defined as follows:
$$Y(m) = a_i, \qquad \text{if } m \in A_i \quad \text{con } i = 1, 2, 3.$$
Taking into account that $\{A_1, A_2, A_3\}$ is a partition of M, a_1, a_2 y a_3 are different real numbers. For the random variable Y, the expectation is given by
$$E(Y) = \sum_i a_i P(A_i).$$
Then a random variable such that for each set A_i assumes the value $P(X \leqslant x \mid A_i)$ is called *conditional probability of* $\{X \leqslant x\}$ *conditionally on the partition induced by* D. The expectation of this random variable is precisely equal to $P(X \leqslant x)$.

   This definition constitutes one of the relevant contributions made by ? (?, Chapter I, §6). It shows that the Law of Total Probabilities corresponds to the definition of the conditional probability as a random variable.

2.5 Conditioning on other variables

We can also describe the behavior of the scores as a function of socioeconomic group S, a variable that takes 5 values. Figure 2.6 shows this evolution. We can also show the behavior of the distribution of scores as a function of the sex G of the students. Figure **??** shows this relationship.

Taking into account what has been discussed so far, we can say that the distribution of $P(X \leqslant x)$ is the conditional expectation of three different random variables:

$$\begin{aligned} P(X \leqslant x) &= E[P(X \leqslant x \mid D)] \\ &= E[P(X \leqslant x \mid S)] \\ &= E[P(X \leqslant x \mid G)] \qquad \forall\, x \in N. \end{aligned}$$

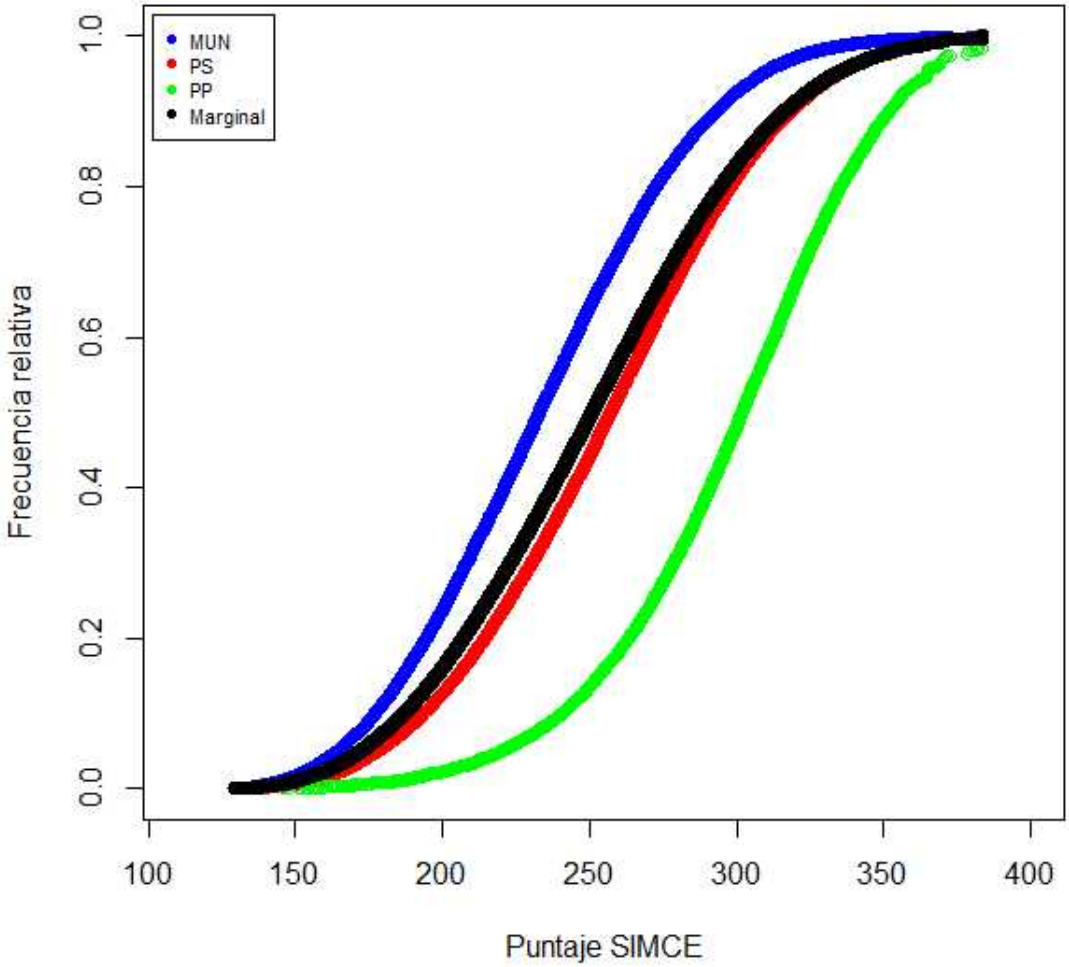


Figure 2.5: Función de distribución acumulada de los puntajes SIMCE Matemáticas condicionalmente a la dependencia administrativa del establecimiento educacional

This means that a phenomenon of interest, which is described by means of the probability distribution, can be considered as the marginalization of several conditional distributions. In the example at hand we can say that the distribution of scores in the SIMCE Mathematics test corresponds to the expected value of the distributions conditional on the administrative type of school; or to the expected value of the socioeconomic group of the school; or to the expected value of the biological sex of the students. Each of these variables, D, S, G, is related to the score. And there may be hundreds. In fact, the Law of Total Probabilities can be written as follows:

$$P(X \leqslant x) = E[P(X \leqslant x \mid W)] \quad \forall\, x \in N,$$

for *all* random variables W defined on M.

Conditional probability is used to model relationships between random variables, a matter already explained by Laplace (see below); therefore, the above equality tells us that a marginal phenomenon can be seen as the expected value of many conditional phenomena: *the data do not provide a criterion for choosing variables on which to condition*. To get a criterion, it is imperative to turn to and collaborate with experts in the fields from which the data come. The task of statisticians is to model the contextual knowledge they have. A purely exploratory analysis (as it is done by Data Science, following Gauss and Tukey) can be spurious whenever it pretends to replace the intuitions of the experts of the phenomena under study.

2.6 Conditionalization to zero probability events

A conditional probability has been characterized through the equality

$$P(B) = E[P(B \mid \mathcal{A}],$$

where $\mathcal{A} = \{A_q : q = 1, \ldots, r\}$. The numbers $P(B \mid A_q)$ can be computed using the computation rule

$$P(B \mid A_q) = \frac{P(B \cap A_q)}{P(A_q)}, \quad \text{si } P(A_q) \neq 0.$$

How can we define the conditional probability $P(B \mid mathcalA)$ if, for example, $P(A_1) = 0$? Let use the characterization of a conditional probability :

$$\begin{aligned} P(B) &= E[P(B \mid \mathcal{A})] \\ &= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \cdots + P(B \mid A_r)P(A_r). \end{aligned}$$

The quantity on the left side of the equality does not change whatever the value of $P(B \mid A_1) \in [0, 1]$ is, since that value multiplied by 0 (which is $P(A_1)$) will always give 0. Therefore, the number $P(B \mid A_1)$ is defined *arbitrarily*, without arbitrarily changing the random variable $P(B \mid mathcalA)$.

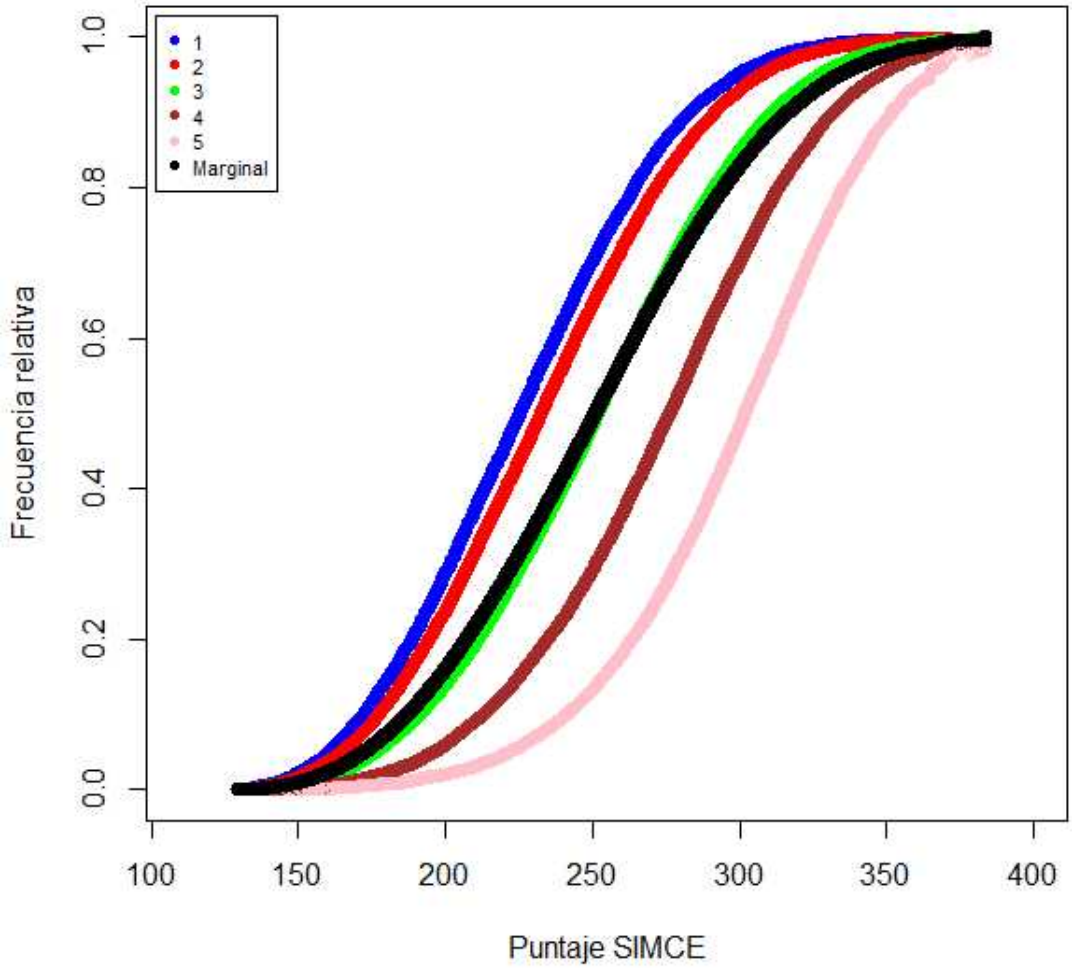


Figure 2.6: Cumulative distribution function of SIMCE Mathematics scores conditional on the socioeconomic level of schools

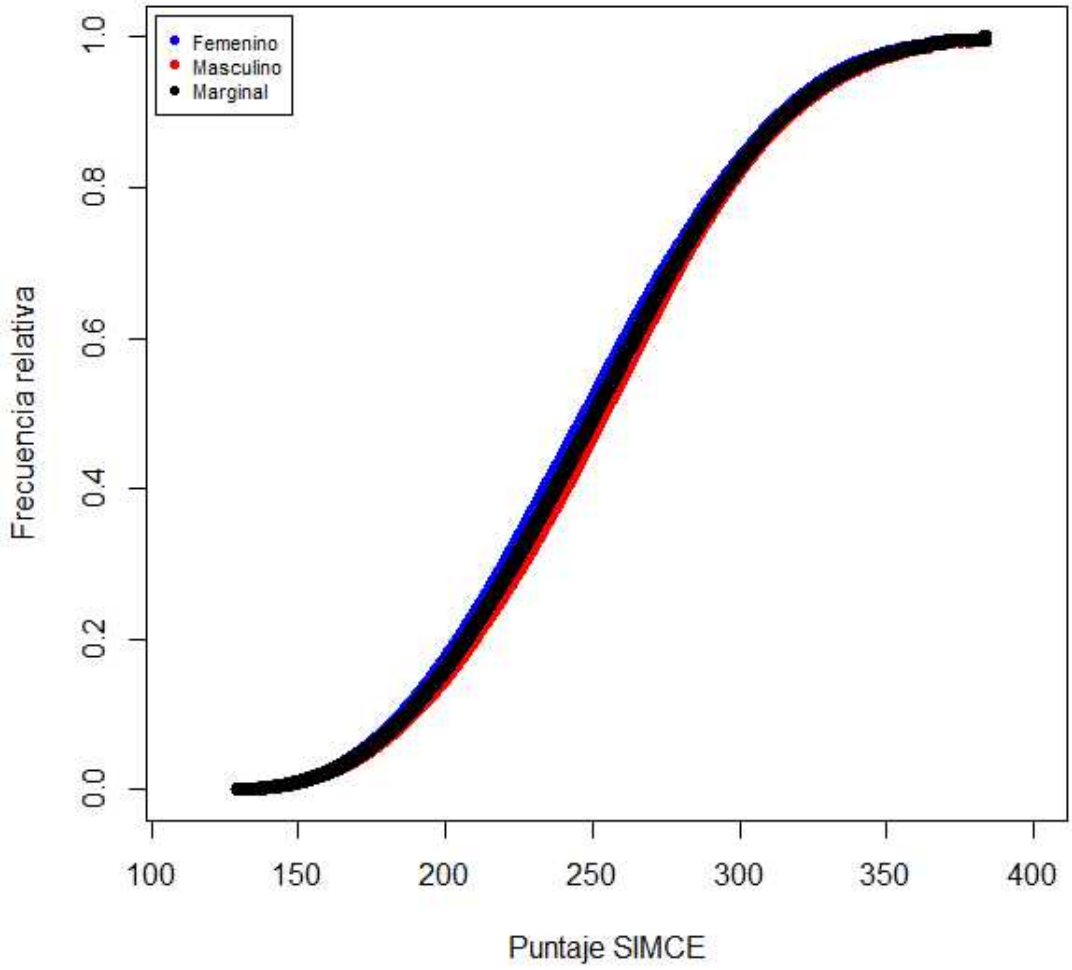


Figure 2.7: Cumulative distribution function of SIMCE Mathematics scores conditional on the biological sex of students

2.7 Marginal-conditional decomposition

The concept of conditional probability is intimately related to the statistical modeling of *dependent events*. This is specified through what we call marginal-conditional decomposition. In order to make such a decomposition explicit, we will use an argument due to Laplace (1820, pp. 181-182):

Suppose that the total number of possible cases that can occur is equal to p. Suppose further that

1. From the p possible cases, there are p' cases favorable to the occurrence of an event E_1.
2. From these p' cases, there are q cases favorable to the occurrence of event E_2.

These conditions imply that if event E_1 occurs, *not necessarily* event E_2 occurs. However, if event E_2 occurs, then *necessarily* event E_1 occurs. Therefore, events E_1 and E_2 are *dependent* in the sense that the occurrence of one of them depends on the occurrence of the other, although such dependence is not symmetric. More precisely, the conditions defined by one of the events determine the probability of occurrence of the other event.

As an example, let us consider that E_1 corresponds to the event "cloudy day in a given year", while E_2 corresponds to the event "rainy day in a given year". Thus, if it rains, then it is necessarily cloudy (the "being cloudy" is determined by the rainy day), but if it is cloudy it does not necessarily rain: raining is an event contingent to being cloudy.

Now, the probability of the occurrence of event E_1 is given by

$$\frac{p'}{p}.$$

In the context of the example, this corresponds to the proportion of cloudy days in a given year. The probability of the occurrence of event E_2 *under the condition that event* E_1 *occurred* is given by

$$\frac{q}{p'}.$$

In the context of the example, this corresponds to the proportion of rainy days among cloudy days –that is, a relative proportion. Finally, the probability of occurrence *simultaneously* of events E_1 and E_2 is given by

$$\frac{q}{p}.$$

In the context of the example, this would correspond to the proportion of rainy (and hence cloudy) days in a given year.

The above three probabilities are related to each other as follows:

$$\frac{q}{p} = \frac{p'}{p} \times \frac{q}{p'}; \tag{2.7.3}$$

that is, the probability of the compound event $E_1 \cap E_2$ is equal to the probability of event E_1, multiplied by the probability of event E_2, under the assumption that E_1 has occurred. In the context of the example, this means that the proportion of rainy (and therefore cloudy) days in a given year is equal to the proportion of cloudy days, multiplied by the proportion of rainy days among cloudy days.

Now, let us denote by $P(E_1 \cap E_2)$, the probability of the compound event $E_1 \cap E_2$; the probability of event E_1 we denote by $P(E_1)$; and the probability of occurrence of event E_2, under the condition determined by event E_1, we denote by $P(E_2 \mid E_1)$. Then the relation (2.7.3) is written as

$$P(E_1 \cap E_2) = P(E_1)\, P(E_2 \mid E_1). \tag{2.7.4}$$

The decomposition (2.7.4) corresponds to a marginal-conditional decomposition: it indicates to what extent the conditions defined by event E_1 determine the probability of occurrence of event E_2.

This decomposition can be extended to describe the joint occurrence of two events E_1 and E_2 under the condition determined by an event C:

$$P(E_1 \cap E_2 \mid C) = P(E_1 \mid C)\, P(E_2 \mid E_1 \cap C). \tag{2.7.5}$$

The decompositions (2.7.4) and (2.7.5) are thus useful for modeling dependent events.

Chapter 3

What can we learn when data are missing?

Typically, missing data are considered as an *estimation problem*. It is claimed that missing data cause estimation problem because they lead to use only the observable information. The statistical literature provides a plethora of methods leading to solve the problem in such a way that a data set with missing data is "transformed" into a "complete data set" (that is, a data set without missing data). However, this perspective is incorrect: the missing data is a property inherent to a specific data generating process and, therefore, it should be considered as a part of the specification problem. In this chapter, we will consider the *missing data problem* as a modelling problem, emphasizing the underlying identification problem. This chapter is based on Manski (2005b).

3.1 Structure of the problem

Let J denotes a population of statistical units. For each $j \in J$, we have an outcome y_j in a space Y. Thus, we have a probability space (J, Ω, P) and a random variable $y : J \longrightarrow Y$ with a probability distribution $P(y)$.

A sampling process draws statistical units at random from J. A realization of y may or may not be observable as indicated by the realization of a binary random variable z: y is observable if $z = 1$, and y is not observable if $z = 0$. Using the available information, is it possible to identify $P(y)$? If not, what can we learn about $P(y)$?

The structure of the missing data problem is displayed by the Law of Total Probability:

$$P(y) = P(y \mid z = 1)P(z = 1) + P(y \mid z = 0)P(z = 0). \tag{3.1.1}$$

The sampling process reveals the distribution of the observable outcomes $P(y \mid z = 1)$ and the distribution of observability $P(z)$. In other words, both $P(y \mid z = 1)$ and $P(z)$ are identified because such distributions are the parameters itself. However, the sampling process is not informative regarding the distribution of missing outcomes $P(y \mid z = 0)$. Hence the empirical evidence reveals that $P(y)$ lies in

the identification region

$$H[P(y)] \doteq \{P(y \mid z=1)P(z=1) + \gamma\, P(z=0), \;\; \gamma \in \Gamma_Y\}, \tag{3.1.2}$$

where Γ_Y denotes the space of all probability distributions on Y.

The feasible values of $P(y)$ are the mixtures of $P(y \mid z = 1)$ and all elements of Γ_Y, with mixing probabilities $P(z = 1)$ and $P(z = 0)$. Consequently, if we choose $\gamma_1 \neq \gamma_2$, then the induced $P(y)$'s are coherent with the empirical evidence. Therefore, $P(y)$ is *not* identified.

Note that

$$H[P(y)] \subset \Gamma_Y \quad \text{if } 0 < P(z=0) < 1;$$

and

$$H[P(y)] = \{P(y \mid z=1)\} \quad \text{if } P(z=0) = 0.$$

Hence $P(y)$ is *partially identified* when $0 < P(z = 0) < 1$ and it is *point-identified* when $P(z = 0) = 0$.

Ejemplo 3.1.1 A test of mathematics was intended to be applied to 1020 students. Each student voluntarily chooses to answer the test or not. Only 557 students answered the test. The test was scored between 1 and 7. If a student obtains at least 4, then she or he is considered to master the content of the test. The available empirical evidence reveals $P(z)$, namely

$$P(z=1) = \frac{557}{1020} = 0.55, \quad P(z=0) = \frac{114633}{670} = 0.45;$$

and the distribution $P(y \mid z = 1)$ of observable scores, which is depicted in Figure 3.1.

The identification region of $P(y)$ is plotted in Figure 3.2: it can be realized the uncertainty induced by the missing data, which means that there are a lot of possible probability distributions $P(y)$.

■

Distributional assumptions may have identifying power. One may assert that the distribution $P(y \mid z = 0)$ lies in some set $\Gamma_{0Y} \subset \Gamma_Y$. Then the identification region shrinks from $H[P(y)]$ to

$$H_1[P(y)] \doteq \{P(y \mid z=1)P(=1) + \gamma\, P(z=0), \;\; \gamma \in \Gamma_{0Y}\}. \tag{3.1.3}$$

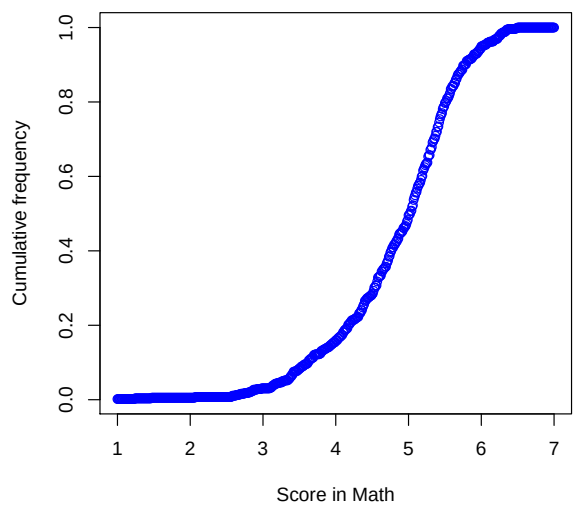
Or one may assert that the distribution of interest $P(y)$ lies in some set $H_0[P(y)] \subset \Gamma_Y$. Then the identification region shrinks from $H[P(y)]$ to

$$H_1[P(y)] \doteq H_0[P(y)] \cap H[P(y)].$$

Assumptions of the former and the latter types differ in that the former are necessarily non-refutable but the latter may be refutable. As a matter of fact,

1. An assumption that restricts $P(y \mid z = 0)$ is non-refutable because one does not observe the missing data.

Figure 3.1: Cumulative observable distribution $P(y \mid z = 1)$



2. An assumption that restricts $P(y)$ may be incompatible with the available empirical evidence: if

$$H_0[P(y)] \cap H[P(y)] = \varnothing,$$

one should conclude that $P(y)$ does not lie in the set $H_0[P(y)]$.

A common objective of empirical research is to infer a parameter of $P(y)$. Let $\tau : \Gamma_Y \longrightarrow T$ be a function mapping a probability distribution on Y into a space T, and consider the problem of inference on the parameter $\tau[P(y)]$: its identification region is

$$H\{\tau[P(y)]\} \doteq \{\tau(\eta) \ \ , \eta \in H[P(y)]\} \tag{3.1.4}$$

if only empirical evidence is available, and is

$$H_1\{\tau[P(y)]\} \doteq \{\tau(\eta) \ \ , \eta \in H_1[P(y)]\} \tag{3.1.5}$$

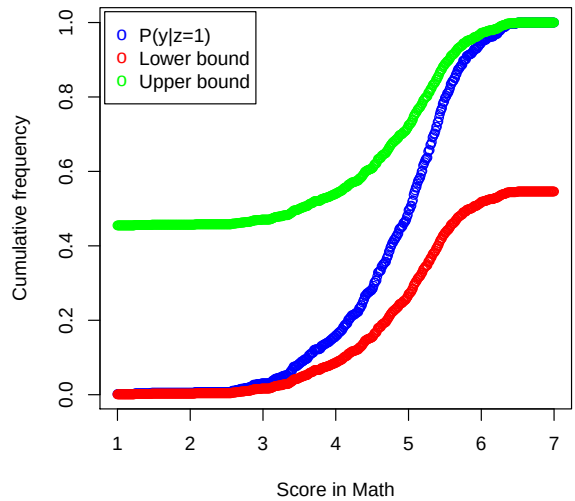
given distributional assumptions.

3.2 Partial identification of means

Let $\mathbb{R} = [-\infty, \infty]$. Let G be the space of measurable functions that map Y into $\mathbb{R}$ and that attain their lower and upper bounds g_0 and g_1, where

$$g_0 = \inf_{y \in Y} g(y), \qquad g_1 = \sup_{y \in Y} g(y).$$

Figure 3.2: Identification region of the probability distribution $P(y)$



Thus, $g \in G$ if there exists a $y_{0g}, y_{1g} \in Y$ such that $g(y_{0g}) = g_0$ and $g(y_{1g}) = g_1$. The problem is to partially identify $E[g(y)]$ using the empirical evidence only. The Law of Total Probability gives

$$E[g(y)] = E[g(y) \mid z = 1]P(z = 1) + E[g(y) \mid z = 0]P(z = 0). \tag{3.2.6}$$

The sampling process reveals $E[g(y) \mid z = 1]$ and $P(z)$. However, it is uninformative regarding $E[g(y) \mid z = 0]$, which can take any value in the interval $[g_0, g_1]$.

Proposition 3.1 *Let $g \in G$. Given the empirical evidence alone, the identification region for $E[g(y)]$ is the closed interval*

$$H\{E[g(y)]\} \doteq [E[g(y) \mid z = 1]P(z = 1) + g_0\, P(z = 0),\ E[g(y) \mid z = 1]P(z = 1) + g_1\, P(z = 0)].$$

Note that $H\{E[g(y)]\} \subset [g_0, g_1]$. The width of the region is $(g_1 - g_0)\, P(z = 0)$. Thus, the severity of the identification problem varies directly with the probability $P(= 0)$ if missing data. If $g_0 = -\infty$, the identification region is

$$[-\infty,\ E[g(y) \mid z = 1]P(z = 1) + g_1\, P(z = 0)].$$

Similarly, if $g_1 = +\infty$, the identification region is

$$[E[g(y) \mid z = 1]P(z = 1) + g_0\, P(z = 0),\ +\infty].$$

Ejemplo 3.2.1 Continuing with Example 3.1.1, we have $E(y \mid z = 1) = 4.8$. The support of the random variable y is 1 and 7, so $g_0 = 1$ and $g_1 = 7$. Therefore,

$$3.1 \leqslant E(y) \leqslant 5.8.$$

■

Let $B \subset Y$ be a measurable set. Let $g_B(y) \doteq 1\!\!1_B(y)$. Then

$$E[g_B(y)] = P(y \in B), \quad E[g_B(y) \mid z = 1] = P(y \in B \mid z = 1).$$

It follows that

$$H[P(y \in B)] \doteq [P(y \in B \mid z = 1)P(z = 1),\, P(y \in B \mid z = 1)P(z = 1) + P(z = 0)].$$

The width of this interval is $P(z = 0)$ *whatever the set B may be*. This is due to the fact that $P(y)$ is unidentified. The location of the interval does vary with B. In particular, if $B' \subset B$, the interval $H[P(y \in B)]$ shifts the interval $H[P(y \in B')]$ rightward.

3.3 Parameters that respect stochastic dominance

Let us consider the identification region discussed in Example 3.1.1, Figure 3.2. The fact that

$$P(y \leqslant t, z = 1) \leqslant P(y \leqslant t) \quad \forall\, t \in [1, 7],$$

means that the distribution $P(y \leqslant t, z = 1)$ stochastically dominates $P(y \leqslant t)$. In this case, the mean of the dominating distribution (namely, 3.1; see Example 3.2.1) is smaller or equal than the mean of the dominated distribution (namely, 5.8). It is said that the mean is a parameter that *respects the stochastic dominance*. This motivates the analysis of partial identifiability of parameters respecting stochastic dominance, which will be discussed in this section.

Let us introduce the following definitions:

Definition 3.1 Let $\Gamma_{\mathbb{R}}$ be the space of probability distributions on the extended real line $\mathbb{R}$. Distribution $F \in \Gamma_{\mathbb{R}}$ stochastically dominates the distribution $F' \in \Gamma_{\mathbb{R}}$ is

$$F(t) \leqslant F'(t) \quad \forall\, t \in \mathbb{R}.$$

Definition 3.2 An extended real-valued function $D : \Gamma_{\mathbb{R}} \longrightarrow \mathbb{R}$ respects stochastic dominance (is a D-parameter) if $D(F) \geqslant D(F')$ whenever F stochastically dominates F'.

The following proposition is in the line of the previous comment:

Proposition 3.2 *Let $D(\cdot)$ respect stochastic dominance. Let $g \in G$ and let $R_g \doteq \{g(y) : y \in Y\}$ be the range set of g. Let Γ_g be the space of probability distributions on R_g. Let $\gamma_{0g}, \gamma_{1g} \in \Gamma_g$ be the degenerate distributions that place all the mass on $g_0$and g_1, respectively. Given the empirical evidence alone, the smallest and largest points in the identification region for $D\{P[g(y)]\}$ are*

$$D\{P[g(y) \mid z = 1]P(z = 1) + \gamma_{0g}P(z = 0)\}, \quad D\{P[g(y) \mid z = 1]P(z = 1) + \gamma_{1g}P(z = 0)\}.$$

Proof: The identification region for the distribution $P[g(y)]$ is

$$H\{P[g(y)]\} \doteq \{P[g(y) \mid z = 1]P(z = 1) + \gamma\, P(z = 0),\ \ \gamma \in \Gamma_g\}.$$

Consider the distribution

$$P[g(y) \mid z = 1]P(z = 1) + \gamma_{0g}\, P(z = 0)$$

which supposes that all missing data take a value y_{0g} that minimizes g. This distribution belongs to $H\{P[g(y)]\}$ and is stochastically dominated by all other members of $H\{P[g(y)]\}$ because

$$\inf_{y \in Y} g(y) \leqslant g(y) \leqslant t \Longrightarrow P[g(y) \leqslant t \mid z = 0] \leqslant P\left[\inf_{y \in Y} g(y) \leqslant t \mid z = 0\right].$$

Similarly,

$$P[g(y) \mid z = 1]P(z = 1) + \gamma_{1g}\, P(z = 0) \in H\{P[g(y)]\}$$

and stochastically dominates all other members of $H\{P[g(y)]\}$. The result follows by the definition of D-parameter.

■

A relevant corollary of the previous proposition is related to the α-quantile, with $\alpha \in (0, 1)$, which respects stochastic dominance. The α-quantile of $P[g(y)]$ is

$$Q_\alpha[g(y)] \doteq \inf\{t : P[g(y) \leqslant t] \geqslant \alpha\}.$$

According to Proposition 3.2, the smallest feasible value of $Q_\alpha[g(y)]$ is the α-quantile of the distribution

$$P[g(y) \mid z = 1]P(z = 1) + \gamma_{0g}P(z = 0)$$

and the largest feasible value is the α-quantile of

$$P[g(y) \mid z = 1]P(z = 1) + \gamma_{1g}P(z = 0).$$

Examination of this quantiles led to the following corollary:

Corollary 3.1 *Let $\alpha \in (0, 1)$. Define $r(\alpha$ and $s(\alpha)$ as follows:*

$$r(\alpha) \doteq \begin{cases} \left[1 - \frac{(1-\alpha)}{P(z=1)}\right]\text{-quantile of } P[g(y) \mid z = 1], & \text{if } P(z = 1) \geqslant 1 - \alpha; \\ g_0, & \text{otherwise}. \end{cases}$$

$$s(\alpha) \doteq \begin{cases} \frac{\alpha}{P(z=1)}\text{-quantile of } P[g(y) \mid z = 1], & \text{if } P(z = 1) \geqslant \alpha; \\ g_1, & \text{otherwise}. \end{cases}$$

The smallest and largest points in the identification region o for $Q_\alpha[g(y)]$ are $r(\alpha)$ and $s(\alpha)$.

It can be noticed that the identification region for $Q_\alpha[g(y)]$ shifts to the right as α increases.

The use of quantiles and their partial identifiability are relevant for analyzing the so-called *score equating problem* in educational measurement. Score equating is essential for any testing program that continually produces new editions of a test and for which the expectation is that scores from these editions have the same meaning over time. Different editions may be built to a common blueprint and be designed to measure the same constructs (mathematics ability, for instance), but they almost invariably differ somewhat in their psychometric properties (for instance, difficulties of the items). If one edition is more difficult than another, examinees would be expected to receive lower scores on the harder form. Score equating seeks to eliminate the effects on scores of these unintended differences in test form difficulty. Score equating is necessary to be fair to examinees and to provide score users with scores that mean the same thing across different editions or forms of the test.

Following Lord (1950) and Angoff (1984), "two scores, one on Form X and the other on Form Y [. . .] may be considered equivalent if their corresponding percentile ranks in any given group are equal" (Angoff, 1984, p.86). The equating transformation accordingly establishes equated scores as those that have the same percentile rank in a given group of test takers. More precisely, let $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ be the random variables representing test scores from tests forms X and Y, respectively; and let F_X and F_Y be their respective distribution functions. The X- and Y-scores are comparable if for each $x \in \mathcal{X}$, there exists a unique $y \in \mathcal{Y}$ such that

$$F_X(x) = P(X \leqslant x) = p = P(Y \leqslant y) = F_Y(y),\ p \in (0,1),$$

namely an equality of quantiles. In Chapter 9 we analyze this problem, making explicit the underlying identification problem.

3.4 Combining Multiple Sampling Processes

The previous considerations implicitly assume that the available empirical evidence is generated by *one* random sampling process. It is relevant to consider the case in which the empirical evidence is generated by multiple sampling sampling processes. Each sampling process draws statistical units at random from population J, and each yields some observable outcomes. Outcomes that are observable under some sampling processes may be missing under others. The objective is to combine the data generated by the sampling processes to identify $P(y)$. The possibility of combining data from multiple sampling processes arise often in surveys: one survey of a population of interest may attempt to interview respondents face-to-face, another by mail or e-mail, etc. Each interview mode yield its own pattern of nonresponse.

In order to identify $P(y)$, let M denote the set of sampling processes. For each $j \in J$ and $m \in M$, let $z_{jm} = 1$ if y_j is observable under sampling process m, and $z_{jm} = 0$ otherwise. It should be noticed that this means that all the statistical units can be interviewed by each survey. For each $m \in M$, the Law of Total Probability gives

$$P(y) = P(y \mid z_m = 1)P(z_m = 1) + P(y \mid z_m = 0)P(z_m = 0) \tag{3.4.7}$$

and therefore

$$P(y) \in \{P(y \mid z_m = 1)P(z_m = 1) + \gamma_m P(z_m = 0),\ \gamma_m \in \Gamma\}. \tag{3.4.8}$$

Thus, the following proposition follows:

Proposition 3.3 *The identification region for $P(y)$ is*

$$H_M[P(y)] \doteq \bigcap_{m \in M} \{P(y \mid z_m = 1)P(z_m = 1) + \gamma_m P(z_m = 0),\ \gamma_m \in \Gamma\}. \tag{3.4.9}$$

The previous proposition raises the question about the size and shape of the identification region. The following corollary provides a useful characterization when M is a finite set: part (a) shows that a distribution is a feasible value for $P(y)$ if and only if the probability that it places on each measurable subset of Y is no less than an explicit lower bound. This characterizations simplifies when Y is countable: in this case, it is only necessary to evaluate the feasible probability on each atom of Y; this is summarized in part (b). Finally, part (c) provides a necessary and sufficient condition for the existence of a unique feasible distribution.

Corollary 3.2 *Let $H_M[P(y)]$ be given by (3.4.9), with M a finite set. Let $\eta \in \Gamma_Y$. For each measurable set $B \subset Y$, define*

$$\pi_M(B) \doteq \max_{m \in M} P(y \in B \mid_m = 1)P(z_m = 1).$$

(a) Then $\eta \in H_M[P(y)]$ if and only if $\eta(B) \geqslant \pi_M(B)$ for all $B \subset Y$.

(b) Let Y be countable. Then $\eta \in H_M[P(y)]$ if and only if $\eta(y) \geqslant \pi_M(y)$ for all $y \in Y$.

(c) Let Y be countable. Let

$$S_M = \sum_{y \in Y} \pi_M(y).$$

The region $H_M[P(y)]$ contains multiple distributions if $S_M < 1$ and a unique distribution if $S_M = 1$. If $S_M = 1$, the unique feasible distribution is

$$\eta_M(y) \equiv \pi_M(y) \quad \forall\, y \in Y.$$

Proof:

(a) Let $\eta \in H_M[P(y)]$. For each $m \in M$, there exists a distribution $\gamma_m \in \Gamma_Y$ such that

$$\eta = P(y \mid z_m = 1)P(z_m = 1) + \gamma_m P(z_m = 0).$$

It follows that

$$\eta(B) \geqslant P(y' in B \mid z_m = 1)P(z_m = 1) \qquad \forall\, B \subset Y.$$

Therefore,

$$\eta(B) \geqslant \pi_M(B) \quad B \subset Y.$$

Conversely, let

$$\gamma_m = \frac{\eta - P(y \mid z_m = 1)P(z_m = 1)}{P(z_m = 0)}.$$

γ_m is a probability because $\eta(B) \geqslant P(y \in B \mid z_m = 1)$ implies that $\gamma_m \geqslant 0$. On the other hand, $\gamma_m(B) \leqslant 1$ for all $B \subset Y$ because

$$\begin{aligned} \eta(B) - P(y \in B, z_m = 1) &\leqslant P(z_m = 0) \\ &= 1 - P(z_m = 1) \\ &= 1 - P(y \in B, z_m = 1) - P(y \in B^c, z_m = 1), \end{aligned}$$

which is equivalent to

$$\eta(B) \leqslant 1 - P(y \in B^c, z_m = 1) \Longleftrightarrow P(y \in B^c, z_m = 1) \leqslant 1 - \eta(B) = \eta(B^c),$$

which is true by hypothesis.

From the definition of γ_m it follows that

$$\eta = P(y \mid z_m = 1)P(z_m = 1) + \gamma_m P(z_m = 0),$$

so $\eta \in H_M[P(y)]$.

(b) Part (a) shows directly that $\eta \in H_M[P(y)]$, then $\eta(y) \geqslant pi_M(y)$ for all $y \in Y$.

Let $\eta(y) \geqslant \pi_M(y)$ fir all $y \in Y$. Then for all $B \subset Y$, with Y countable,

$$\eta(B) = \sum_{y \in B} \eta(y) \geqslant \sum_{y \in B} \pi_M(y) \geqslant \pi_M(B),$$

where the final inequality holds because $\pi_M(\cdot)$ is sub-additive (it is a maximum). Therefore $\eta \in H_M[P(y)]$ by part (a).

(c) If $S_M < 1$, the empirical evidence leaves indeterminate the allocation of $(1 - S_M)$ of probability mass among the atoms of Y. Hence $H_M[P(y)]$ contains multiple distributions.

If $S_M = 1$, then η_M is a probability measure. Par (b) shows that $\eta_M \in H[P(y)]$. Let η be a measure with

$$\eta(y) \geqslant \pi_M(y) \quad y \in Y,$$

and $\eta(y) > \pi_M(y)$ for some $y \in Y$. Then $\eta(Y) > 1$, son η is not a probability measure. Hence η_M is the only element of $H_M[P(y)]$.

Note that $\eta_M \neq \pi_M$, which is sub-additive and hence not a probability measure. That is, $\eta_M(y) = \pi_M(y)$ for $y \in Y$ nut $\eta_M(B) \geqslant \pi_M(B)$ for $B \subset Y$.

□

This corollary provides conditions to get point identification: if $S_M = 1$ then $P(y)$ is point identified.

3.5 Example: The National Longitudinal Survey for Young

The National Longitudinal Survey for Young (NLSY) surveyed peoples born between January 1, 1957 and December 31, 1964 who resided in the United States in 1979. From 1979 on, the NLSY has periodically sought to interview a random sample of this population and supplemented samples of some sub-populations. In this example, the random sample is used.

Table 3.1: Employment status of NLSY respondents

Emplyment status	**Number of respondets**
Employed ($y = 2$)	4332
Unemployed ($y = 1$)	297
Out of labor force ($y = 0$)	927
Ever-interviewed nonrespondents	555
Never-interviewed nonrespondents	701
Total	6812

The outcome y indicates an individual's employment status at the time of the 1991 interview. In the 1979 base year, the NLSY sought to interview a random sample of 6812 individuals and succeeded in obtaining interviews from 6111 if the sample members. Data on employment status in 1991 are available for 5556 of the 6111 individuals interviewed in the base year. The remaining 555 are correspondents, some because they declined to be interviewed in 1991 and some because they did not answer the employment-status question in the 1991 interviews. Table 3.1 summarizes the available information.

Note that the 5556 respondents interviewed in the base year are those such that $y = 2$ or $y = 1$ or $y = 0$.

The empirical nonresponse rate is

$$P(z = 0) = \frac{1256}{6812} = 0.184;$$

here 1256 is equal to the ever-interviewed nonrespondents (555) and the never-interviewed nonrespondents (701). Researcher computing nonresponse rates to questions in the later year of longitudinal surveys often condition on the event that a sample member was interviewed in the base year. This event, denoted as BASE, is always observed. Then the *ever-interviewed* nonrespondents rate for employment status in 1991 is

$$P(z = 0 \mid \text{BASE}) = \frac{555}{6111} = 0.091,$$

where 6111=4332+297+927+555. Therefore

$$P(z = 1 \mid \text{BASE}) = \frac{5556}{6111} = 0.909.$$

The empirical probability of employment rate for the 5556 individuals who responded to the 1991 employment-status question is

$$P(y = 2 \mid z = 1) = \frac{4332}{5556} = 0.780.$$

The probability of employment among nonrespondents can take any value in the interval $[0, 1]$. Hence the identification region for the population empirical employment probability $P(y = 2)$ is given by

$$P(y = 2) \in [\,(0.780)\,(0.816),\ (0.780)\,(0.816) + (0.184)\,] = [\,0.636,\ 0.820\,]\,.$$

The identification region for the ever-interviewed empirical employment probability $P(y = 2 \mid \text{BASE})$ is given by

$$P(y = 2 \mid \text{BASE}) \in [\,(0.780)\,(0.909),\ (0.780)\,(0.909) + (0.091)\,] = [\,0.709,\ 0.800\,]\,.$$

3.6 Instrumental variables

It is relevant to make explicit the standard way to solve the identification problem in the sense of point identifying the distribution $P(y)$. As a matter of fact, distributional assumptions may enable one to shrinks identification regions obtained using empirical evidence alone. This shows how strong could be the distributional assumptions in such a way that the conclusions highly depend on them rather than on the empirical evidence alone. A tension between credibility of such additional assumptions and stronger conclusions becomes explicit, which is stated through the *Law of Decreasing Credibility* (Manski, 2003):

> Inference using the empirical evidence alone sacrifices strength of conclusions in order to maximize credibility. Inference invoking point-identifying distributional assumptions sacrifices credibility in order to achieve strength conclusions.

Between these two poles, there is a vast middle ground of possible modes of inference asserting assumptions that may shrink the identification region $H[P(y)]$ but not reduce it to a point.

3.6.1 Preliminaries

Each person $j \in J$ is characterized by a covariate $v \in V$. Moreover, for each $j \in J$, y_j is observable if $z = 1$ and it is unobservable if $z = 0$. We will assume that all the realizations of v are observable. The basic idea is that the observability of v provides an instrument that may help to identify $P(y)$. The sampling process reveals the distributions

$$P(z), \quad P(y, v \mid z = 1), \quad P(v \mid z = 0).$$

It is uninformative about

$$\{P(y \mid v = \text{v}, z = 0),\ \text{v} \in V\}.$$

3.6.2 Distributional assumptions

The following distributional assumptions are typically used in empirical research:

1. Outcomes missing-at-random (MAR): $y \perp\!\!\!\perp z \mid v$, that is,

$$P(y \mid v) = P(y \mid v, z = 0) = P(y \mid v, z = 1).$$

2. Statistical independence of outcomes and instrument (SI): $y \perp\!\!\!\perp v$, that is,

$$P(y \mid v) = P(y).$$

3. Mean missing-at-random (MMAR): $g(y) \perp z \mid v$, where $\cdot \perp \cdot \mid \cdot$ denotes *conditional orthogonality* (for definition and properties, see Florens & Mouchart, 1985); that is,

$$E[g(y) \mid v] = E[g(y) \mid v, z = 1] = E[g(y) \mid v, z = 1].$$

4. Mean independence of outcomes and instrument (MI): $g(y) \perp\!\!\!\perp v$, that is,

$$E[g(y) \mid v] = E[g(y)].$$

5. Mean missing monotonically (MMA):

$$E[g(y) \mid v, z = 1] \geqslant E[g(y) \mid v] \geqslant E[g(y) \mid v, z = 0].$$

6. Mean monotonicity of outcome and instrument (MM): Let V be an ordered set, then

$$E[g(y) \mid v = \mathrm{v}] \geqslant E[g(y) \mid v = \mathrm{v}'] \quad \forall(\mathrm{v}, \mathrm{v}') \in V \times V \text{ such that } \mathrm{v} \geqslant \mathrm{v}'.$$

Researchers contemplating the use of instrumental variables should pay due attention to the credibility of these assumptions. Credibility depends not on the covariate per-se, but on the assumption that the distribution $P(y, z, v)$ is assumed to satisfy.

3.6.3 Consequences of the distributional assumptions

In order to simplify the discussion, we assume $P(v = \mathrm{v}, z = 1) > 0$ for all $\mathrm{v} \in V$. The following propositions summarize the main conclusions of the distributional assumptions:

Proposition 3.4 *Suppose MAR holds. Then $P(y)$ is point-identified with*

$$P(y) = \sum_{v \in V} P(y \mid v = v, z = 1) P(v = v).$$

Furthermore, MAR is non empirically refutable.

Proof: Applying successively the Law of Total Probability and the MAR assumption, we have

$$\begin{aligned} P(y) &= \sum_{\mathrm{v} \in V} P(Y \mid v = \mathrm{v}) P(v = \mathrm{v}) \\ &= \sum_{\mathrm{v} \in V} P(Y \mid v = \mathrm{v}, z = 1) P(v = \mathrm{v}). \end{aligned}$$

But $P(Y \mid v = \mathrm{v}, z = 1)$ and $P(v = \mathrm{v})$ are point-identified, and therefore the conclusion follows.

MAR is not empirically refutable because the empirical evidence reveals nothing about $P(y \mid v, z = 0)$.

□

Proposition 3.5

(a) Suppose SI holds. Then the identification region for $P(y)$ is

$$H_{SI}[P(y)] = \bigcap_{v \in V} \{P(y \mid v = v, z = 1)(z = 1 \mid v = v) + \gamma_v \, P(z = 0 \mid v = v), \ \gamma_v \in \Gamma_Y\}.$$

(b) Let the set $H_{SI}[P(y)]$ empty. Then assumption SI does not hold.

Proof:
(a): The identification region for $P(y \mid v = \mathrm{v})$ is

$$H[P(y \mid v = \mathrm{v})] = \{P(y \mid v = \mathrm{v}, z = 1)P(z = 1 \mid v = \mathrm{v}) + \gamma_{\mathrm{v}} \, P(z = 0 \mid v = \mathrm{v}), \ \gamma_{\mathrm{v}} \in \Gamma_Y\}.$$

The identification region for the set of distributions $\{P(y \mid v = \mathrm{v}), \ \mathrm{v} \in V\}$ is

$$\bigtimes_{\mathrm{v} \in V} H[P(y \mid v = \mathrm{v})].$$

SI states that $\{P(y \mid v = \mathrm{v}), \ \mathrm{v} \in V\}$ coincide, all being equal to $P(y)$. Hence $P(y)$ must lie in

$$\bigcap_{\mathrm{v} \in V} H[P(y \mid v = \mathrm{v})].$$

(b): If SI holds, then $H_{\mathrm{SI}}[P(y)]$ is necessarily non-empty. Therefore, statement (b) implies that SI is refutable.

□

Proposition 3.6 *Suppose MMAR holds. Then $E[g(y)]$ is point-identified with*

$$E[g(y)] = \sum_{v \in V} E[g(y) \mid v = v, z = 1]P(v = v).$$

MMAR is not empirically refutable.

The proof is similar to Proposition 3.4.

Proposition 3.7

(a) Suppose MI holds. Then the closed interval

$$H_{MI}[E(g(y))] \doteq \left[\max_{v \in V} E[g(y)z + g_0(1 - z) \mid v = v], \min_{v \in V} E[g(y)z + g_1(1 - z) \mid v = v \right]$$

is the identification region for $E[g(y)]$.

(b) If $H_{MI} = \emptyset$ then MI does not hold. Therefore, MI is empirically refutable.

Proof: Note that

$$E[g(y) \mid z = 1]P(Z = 1) + g_0 P(z = 0) = E[g(y)z + g_0(1 - z)];$$

it is enough to iterate in z. The remaining of the proof is similar to the proof of Proposition 3.5.

□

Proposition 3.8 *Suppose MMM holds. The identification region for $E[g(y)]$ is the closed intervl*

$$H_{MMM}[E(g(y))] \doteq \left[E[g(y) \mid Z = 1]P(z = 1) + g_0 P(z = 0), \sum_{v \in V} E[g(y) \mid v = v, z = 1]P(v = v) \right].$$

MMM is non-refutable.

Proof: Let $\mathrm{v} \in V$. Under MMM the identification region for $E[g(y) \mid v = \mathrm{v}, z = 0]$ is the closed interval

$$H_{\mathrm{MMM}}[E(g(y) \mid v = \mathrm{v}, z = 0)] \doteq [\, g_0, E[g(y) \mid v = \mathrm{v}, z = 1] \,].$$

The identification region for

$$\{E[g(y) \mid v = \mathrm{v}, z = 0] : \mathrm{v} \in V\}$$

is the $|V|$-dimensional rectangle

$$\bigtimes_{\mathrm{v} \in V} H_{\mathrm{MMM}}\{E[g(y) \mid v = \mathrm{v}, z = 0]\}.$$

The Law of Total Probability gives

$$E[g(y)] = \sum_{\mathrm{v} \in V} E[g(y) \mid v = \mathrm{v}, z = 1]\, P(v = \mathrm{v}, z = 1) + E[g(y) \mid v = \mathrm{v}, z = 0]P(v = \mathrm{v}, z = 0).$$

Combining this decomposition with the previous regions, the conclusion follows.

□

3.6.4 Estimation with nonresponses weights

Construction of nonresponse weights presumes the existence of an instrumental variable. The standard use of such weights to infer a population mean $E[g(y)]$ yields a consistent estimate if MMAR holds but not otherwise. Suppose that a random sample of size N has been drawn from a population J. Let $\mathcal{N}(1)$ denote the sample of members for whom $z = 1$ and let N_1 be the cardinality of $\mathcal{N}(1)$. Let

$s : V \longrightarrow [0, \infty]$ be a weighting function. Consider estimation of $E[g(y)]$ by the weighted sample average

$$\theta_N \doteq \frac{1}{N_1} \sum_{i \in \mathcal{N}(1)} s(\mathrm{v}_i) g(y_i).$$

By the Strong Law of Large Numbers,

$$\lim_{N \to \infty} \theta_N = E[s(v)g(y) \mid z = 1].$$

The standard weights provided by survey organizations have the form

$$s(\mathrm{v}) = \frac{P(v = \mathrm{v})}{P(v = \mathrm{v} \mid z = 1)} \quad \mathrm{v} \in V.$$

Then

$$\begin{aligned}
E[s(v)g(y) \mid z = 1] &= \sum_{\mathrm{v} \in V} E[s(v)g(y) \mid v = \mathrm{v}, z = 1] P[v = \mathrm{v} \mid z = 1] \\
&= \sum_{\mathrm{v} \in V} E[g(y) \mid v = \mathrm{v}, z = 1] P[v = \mathrm{v}] \\
&= \sum_{\mathrm{v} \in V} E[g(y) \mid v = \mathrm{v}, z = 1] P[v = \mathrm{v} \mid z = 1] P(z = 1) + \\
&\quad \sum_{\mathrm{v} \in V} E[g(y) \mid v = \mathrm{v}, z = 1] P[v = \mathrm{v} \mid z = 0] P(z = 0) \\
&= E[g(y) \mid z = 1] P(z = 1) + \\
&\quad \sum_{\mathrm{v} \in V} E[g(y) \mid v = \mathrm{v}, z = 1] P[v = \mathrm{v} \mid z = 0] P(z = 0).
\end{aligned}$$

The right side of this equation equals $E[g(y)]$ if MMAR holds but it generically differs from $E[g(y)]$ otherwise.

3.7 Concluding remark

In this chapter we discussed the problem of missing outcomes as an identification problem. We provided two types of results: the first one is based on the empirical evidence alone and this leads us to obtain identification regions for $E[g(y)]$. The width of the identification regions essentially depends on $P(z = 1)$. The second type of results are based on distributional assumptions which in general allow us to point identify $E[g(y)]$. Both types of results show the trade-off underlying the Law of Decreasing Credibility. In empirical research, this type of results compels us to make explicit the assumptions we are ready to belief in order to solve identification problems.

Chapter 4

Conditional Prediction with Missing Data

In empirical research, we focus our attention on the conditional probability $P(y \mid x)$, where y is an outcome of interest and x a vector of covariates. Nevertheless, the presence of missing data induces identification problems for this distribution. Following Manski (2005b), we study this problem, considering three missing-data patterns: missing outcomes, jointly missing outcomes and covariates, and missing covariates.

More specifically, let X be the space of covariates. Each member of the population is characterized by $(y, x) \in Y \times X$ with distribution $P(y, x)$. The objective is to learn on $P(y \mid x = \mathrm{x})$ for $\mathrm{x} \in X$ when the realizations of (y, x) may be observable in whole, in part, or not at all. Two binary random variables (z_y, z_x) indicate observability. A realization of y is observable if $z_y = 1$ but not if $z_y = 0$. A realization of x is observable if $z_x = 1$ but not if $z_x = 0$. The sampling process reveals distributions $P(z_x, z_y)$, $P(y, x \mid z_y = 1, z_x = 1)$, $P(y \mid z_y = 1, z_x = 0)$ and $P(x \mid z_y = 0, z_x = 1)$. The problem is to use this empirical evidence to identify $P(y \mid x = \mathrm{x})$, $\mathrm{x} \in X$. In what follows, it is assumed that X is finite and that $P(x = \mathrm{x}, z_x = 1) > 0$ for all $\mathrm{x} \in X$.

4.1 Missing outcomes

The results obtained in Chapter 3 apply immediately to $P(y \mid x = \mathrm{x})$ if realizations of x are always observable. Thus,

$$H[P(y \mid x = \mathrm{x})] = \{P(y \mid x = \mathrm{x})P(z_y = 1 \mid x = \mathrm{x}) + \gamma\, P(z_y = 1 \mid x = \mathrm{x}),\ \gamma \in \Gamma_Y\}. \tag{4.1.1}$$

4.2 Jointly missing outcome and covariates

Jointly missing outcomes and covariates is a regular occurrence in surveys: realizations of (y, x) may be missing in their entirety when sample members refuse to be interviewed or cannot be contacted by survey administrators. Joint missingness also occurs when outcomes are missing and the objective is to learn a distribution of the form $P(y \mid y \in B)$, where $B \subset Y$.

In a first approach, the identification of $P(y \mid x = \mathrm{x})$ when (y, z) realizations are jointly missing may be studied as an instance of the problem of missing outcomes: it is enough to consider (y, x) as the outcome rather than y alone. Let $\Gamma_{Y \times X}$ denote the space of all probability distributions on $Y \times X$. Let

$$z_{yx} = \begin{cases} 1, & \text{if } z_y = z_x = 1; \\ 0, & \text{otherwise.} \end{cases}$$

Then the identification region for the joint distribution $P(y, x)$ is

$$H[P(y, x)] \doteq \{P(y, x \mid z_{yx} = 1)P(z_{yx} = 1) + \gamma\, P(z_{yx} = 0),\ \gamma \in \Gamma_{Y \times X}\}.$$

However, this identification region does not provide a useful information in terms of the conditional probability $P(y \mid x)$. The following proposition improves the previous identification region:

Proposition 4.1 *Let* $P(z_y = z_x = 1) + P(z_y = z_x = 0) = 1$. *Then*

$$H[P(y \mid x = x)] \doteq \{P(y \mid x = x, z_{yx} = 1)\, r(x) + \gamma\, [1 - r(x)],\ \gamma \in \Gamma_Y\}, \tag{4.2.2}$$

where

$$r(x) = \frac{P(x = x \mid z_{yx} = 1)\, P(z_{yx} = 1)}{P(x = x \mid z_{yx} = 1)P(z_{yx} = 1) + P(z_{yx} = 0)}. \tag{4.2.3}$$

Proof: By the Law of Total Probability, we have

$$\begin{aligned} &P(y \mid x = \mathrm{x}) = \\ &\quad P(y \mid x = \mathrm{x}, z_{yx} = 1)P(z_{yx} = 1 \mid x = \mathrm{x}) + P(y \mid x = \mathrm{x}, z_{yx} = 0)P(z_{yx} = 0 \mid x = \mathrm{x}). \end{aligned} \tag{4.2.4}$$

We need to carefully consider the conditional probability $P(z_{yx} = 0 \mid x = \mathrm{x})$. In fact, by the Bayes-Laplace Theorem, we have

$$P(z_{yx} = i \mid x = \mathrm{x}) = \frac{P(x = \mathrm{x} \mid z_{yx} = i)P(z_{yx} = i)}{P(x = \mathrm{x} \mid z_{yx} = 1)P(z_{yx} = 1) + P(x = \mathrm{x} \mid z_{yx} = 0)P(z_{yx} = 0)}$$

for $i = 0$ or 1. The sampling process identifies $P(z_{yx})$, $P(x = \mathrm{x} \mid z_{yx} = 1)$ and $P(y \mid x = \mathrm{x}, z_{yx} = 1)$. It is uninformative about $P(x = \mathrm{x} \mid z_{yx} = 0)$ and $P(y \mid x = \mathrm{x}, z_{yx} = 0)$. Therefore, the identification region for $P(y \mid x = \mathrm{x})$ is

$$\begin{aligned} &H[P(y \mid x = \mathrm{x})] = \\ &\quad \bigcup_{p \in [0,1]} \Bigg\{ P(y \mid x = \mathrm{x}, z_{yx} = 1) \frac{P(x = \mathrm{x} \mid z_{yx} = 1)P(z_{yx} = 1)}{P(x = \mathrm{x} \mid z_{yx} = 1)P(z_{yx} = 1) + pP(z_{yx} = 0)} \\ &\quad + \ \gamma\, \frac{pP(z_{yx} = 0)}{P(x = \mathrm{x} \mid z_{yx} = 1)P(z_{yx} = 1) + pP(z_{yx} = 0)},\ \gamma \in \Gamma_Y \Bigg\}. \end{aligned} \tag{4.2.5}$$

For each $p \in [0, 1]$, the distributions in brackets are mixtures of $P(y \mid x = \mathrm{x}, z_{yx} = 1)$ and arbitrary distributions of Y. The set of mixtures enlarges as p increase from 0 to 1. Hence, it suffices to set $p = 1$.

□

Note that the identification regions 4.1.1 and 4.2.2 are similar. $r(\mathrm{x})$ is the smallest value of $P(z_{yx} = 1 \mid x = \mathrm{x})$ and is obtained by conjecturing that all the missing covariate realizations have the value x. Thus, joint missingness of (y, x) exacerbates the identification problem produced by missingness of y alone.

The degree to which joint missingness exacerbates the identification problem depends on the prevalence of the value x among the observable realizations of x. In particular, $r(\mathrm{x}) = P(z_{yx} = 1)$ when $P(x = \mathrm{x} \mid z_{yx} = 1) = 1$ and decreases to zero as $P(x = \mathrm{x} \mid z_{yx} = 1)$ falls to zero. Thus, region 4.2.2 is uninformative if the observable covariates distribution $P(x \mid z_{yx} = 1)$ places zero mass on value x.

If we replace $r(\mathrm{x})$ by $P(z = 1 \mid x = \mathrm{x})$ and z_{yx} by z, Proposition 3.2 is rewritten as follows:

Proposition 4.2 *Let D respect stochastic dominance. Let $g \in G$. Let $P(z_y = z_x = 1) + P(z_y = z_x = 0) = 1$. Then the smallest and largest points in the identification region for $D\{P[g(y) \mid x = x]\}$ are*

$$D\{P[g(y) \mid z_{yx} = 1, x = x]r(x) + \gamma_{0g}[1 - r(x)]\}$$

and

$$D\{P[g(y) \mid z_{yx} = 1, x = x]r(x) + \gamma_{1g}[1 - r(x)]\}.$$

Similarly, Proposition 3.3 has the following generalization to settings in which data are available from multiple sampling process, with only outcomes missing in some sampling process and (y, x) jointly missing in the others.

Proposition 4.3 *Let there be a set M of sampling processes for which $P(z_x = 1) = 1$ and set M' of sampling processes fir which $P(z_y = z_x = 1) + P(z_y = z_x = 0) = 1$. Then*

$$\begin{aligned} H_{(M,M')}[P(y \mid x = x)] \quad = \quad & \bigcap_{m \in M} [P(y \mid x = x, z_{my} = 1)P(z_{my} = 1 \mid x = x) + \gamma_m P(z_{my} = 0 \mid x = x],\ \gamma_m \in \Gamma_Y] \\ & \bigcap_{m \in M'} [P(y \mid x = x, z_{my} = 1)r_m(x) + \gamma_m[1 - r(x)],\ \gamma_m \in \Gamma_Y] \end{aligned}$$

is the identification region for $P(y \mid x = x)$.

Here, data are available from multiple sampling processes, with only outcomes missing in some sampling process and (y, x) jointly missing in the others.

4.3 Missing covariates

We assume now that realizations of the outcome y are always observed but realizations of the covariate x may be missing. The following proposition gives the identification region for $P(y \mid x = \mathrm{x})$ using the empirical evidence alone.

Proposition 4.4 *Let $P(z_y = 1) = 1$. Then*

$$H[P(y \mid x = x)] = \bigcup_{p \in [0,1]} \left\{ P(y \mid x = x, z_x = 1) \frac{P(x = x \mid z_x = 1)P(z_x = 1)}{P(x = x \mid z_x = 1)P(z_x = 1) + p\,P(z_x = 0)} + \right. \tag{4.3.6}$$

$$\left. + \eta \frac{p\,P(z_x = 0)}{P(x = x \mid z_x = 1)P(z_x = 1) + p\,P(z_x = 0)}, \quad \eta \in \Gamma_Y(p) \right\},$$

where

$$\Gamma_Y(p) \doteq \Gamma_Y \cap \left\{ \frac{P(y \mid z_x = 1) - \gamma(1-p)}{p}, \ \gamma \in \Gamma_Y \right\}.$$

Proof: By the Law of Total Probability and the Bayes-Laplace Theorem, we have

$$P(y \mid x = \mathrm{x}) = P(y \mid x = \mathrm{x}, z_x = 1) \frac{P(x = \mathrm{x} \mid z_x = 1)P(z_x = 1)}{P(x = \mathrm{x} \mid z_x = 1)P(z_x = 1) + P(x = \mathrm{x} \mid z_x = 0)P(z_x = 0)} +$$

$$P(y \mid x = \mathrm{x}, z_x = 0) \frac{P(x = \mathrm{x} \mid z_x = 0)P(z_x = 0)}{P(x = \mathrm{x} \mid z_x = 1)P(z_x = 1) + P(x = \mathrm{x} \mid z_x = 0)P(z_x = 0)}.$$

The sampling process reveals $P(z_x)$, $P(x = \mathrm{x} \mid z_x = 1)$ and $P(y \mid x = \mathrm{x}, z_x = 1)$ but not $P(x = \mathrm{x} \mid z_x = 0)$ and $P(y \mid x = \mathrm{x}, z_x = 0)$. It does also reveal $P(y \mid z_x = 0)$, which is related to $P(x = \mathrm{x} \mid z_x = 0)$ and $P(y \mid x = \mathrm{x}, z_x = 0)$ because by the Law of Total Probability

$$P(y \mid z_x = 0) = P(y \mid x = \mathrm{x}, z_x = 0)P(x = \mathrm{x} \mid z_x = 0) + P(y \mid x \neq \mathrm{x}, z_x = 0)P(x \neq \mathrm{x} \mid z_x \tag{4.3.7}$$

Suppose that $P(x = \mathrm{x} \mid z_x = 0) = p$. Then a plausible value of $P(y \mid x = \mathrm{x}, z_x = 0)$ is given by

$$\frac{P(y \mid z_x = 0) - \gamma(1-p)}{p}$$

for $\gamma \in \Gamma_Y$. Therefore, the set of values of $P(y \mid x = \mathrm{x}, z_x = 0)$ that are consistent with (4.3.7) is $\Gamma_Y(p)$. Now p ranges over the interval $[0, 1]$, which implies the identification region (4.3.6).

□

A specific aspect of the missing covariates problem is given by equality (4.3.7). Missing covariates pose a less severe observational problem than do jointly missing outcomes and covariates. As a matter of fact, comparison of (4.3.6) in Proposition 4.4 with (4.2.5) in Proposition 4.1 makes this precise: whereas γ could range over the space Γ_Y of all distributions in (4.3.6), it can only range over the restricted space $\Gamma_Y(p)$, $p \in [0, 1]$, in (4.3.6.

4.4 General Missing-Data Pattern

In this section, we consider a sampling process with a general pattern of missing data in which some realizations of (y, x) may be completely observed, others observed in part, and others not observed at

all. The structure of the identifiability of $P(y \mid x = \mathrm{x})$ is displayed by the Law of Total Probability and the Bayes-Laplace Theorem:

$$P(y \mid x = \mathrm{x}) = \tag{4.4.8}$$

$$\sum_{j=0}^{1} \sum_{k=0}^{1} P(y \mid x = \mathrm{x}, z_x = j, z_y = k) \frac{P(x = \mathrm{x} \mid z_x = j, z_y = k) P(z_x = j, z_y = k)}{\sum_{l=0}^{1} \sum_{m=0}^{1} P(x = \mathrm{x} \mid z_x = l, z_y = m) P(z_x = l, z_y = m)}.$$

The sampling process identifies

- $P(z_x, z_y)$,
- $P(x = \mathrm{x} \mid z_x = 1, z_y)$,
- $P(y \mid x = \mathrm{x}, z_x = 1, z_y = 1)$.

It does not identify

- $P(x = \mathrm{x} \mid z_x = 0, z_y)$,
- $P(y \mid x = \mathrm{x}, z_x = 0, z_y = 1)$,
- $P(y \mid x = \mathrm{x}, z_x, z_y = 0)$.

However, the sampling process identifies $P(y \mid z_x = 0, z_y = 1)$, which is related to $P(x = \mathrm{x} \mid z_x = 0, z_y = 1)$ and $P(y \mid x = \mathrm{x}, z - x = 0, z_y = 1)$ by the Law of Total Probability

$$\begin{aligned} P(y \mid z_x = 0, z_y = 1) &= P(y \mid x = \mathrm{x}, z_x = 0, z_y = 1) P(x = \mathrm{x} \mid z_x = 0, z_y = 1) \\ &\quad + P(y \mid x \neq \mathrm{x}, z_x = 0, z_y = 1) P(x \neq \mathrm{x} \mid z_x = 0, z_y = 1). \end{aligned} \tag{4.4.9}$$

Thus, using the empirical evidence alone, the identification region for $P(y \mid x = \mathrm{x})$ is displayed in the following proposition.

Proposition 4.5 *Let $P_{jk} \doteq P(z_x = j, z_y = k)$ for $j, k \in \{0, 1\}$. Then*

$$
\begin{aligned}
H[P(y \mid x = x)] \quad = \quad & \\
& \left\{ P(y \mid x = x, z_x = 1, z_y = 1) \frac{P(x = x \mid z_x = 1, z_y = 1) P_{11}}{\sum_{k=0}^{1} P(x = x \mid z_x = 1, z_y = 1) P_{1k} + p_0 P_{00} + p_1 P_{01}} \right. \\
& + \eta_{10} \frac{P(x = x \mid z_x = 1, z_y = 0) P_{10}}{\sum_{k=0}^{1} P(x = x \mid z_x = 1, z_y = 1) P_{1k} + p_0 P_{00} + p_1 P_{01}} \\
& + \eta_{00} \frac{p_0 P_{00}}{\sum_{k=0}^{1} P(x = x \mid z_x = 1, z_y = 1) P_{1k} + p_0 P_{00} + p_1 P_{01}} \\
& + \eta_{01} \frac{p_1 P_{01}}{\sum_{k=0}^{1} P(x = x \mid z_x = 1, z_y = 1) P_{1k} + p_0 P_{00} + p_1 P_{01}}, \\
& (\eta_{10}, \eta_{00}, \eta_{01}) \in \Gamma_Y \times \Gamma_Y \times \Gamma_Y(p_1) \; : \; (p - 0, p_1) \in [0, 1]^2 \Big\}
\end{aligned}
$$

where

$$
\Gamma_Y(p_1) \doteq \Gamma_Y \cap \left\{ \frac{P(y \mid z_x = 0, z_y = 1) - \gamma(1 - p_1)}{p_1}, \; \gamma \in \Gamma_Y \right\}.
$$

4.5 Examples

4.5.1 Motivation

Official statistics are based on survey data[1]. Such statistics are typically reported as *point estimates*, without accompanying measures of error or indicators of reliability. Sometimes, official technical reports acknowledge that official statistics are subject to error but they provide limited guidance on the magnitude of sampling errors and typically do not attempt to quantify nonsampling errors. As Manski (2016) mentions, prominent examples include the reporting of employment and income statistics by the Bureau of Labor Statistics (BLS) and the Census Bureau:

- On the first Friday of each month, the BLS issues The Employment Situation, a monthly news release reporting official employment statistics for the previous month. The unemployment-rate

[1]This section is based on Manski (2016).

statistic is based on data on households sampled in the Current Population Survey (CPS). The one on nonfarm employment is based on data collected from employer establishments sampled in the Current Employment Statistics survey (CES). The employment statistics are reported as point estimates, without measures of potential error. A Technical Note issued with the news release contains a section on Reliability of the estimates that acknowledges the possible presence of errors, beginning with the statement "Statistics based on the household and establishment surveys are subject to both sampling and nonsampling error". The section describes the conventional use of standard errors and confidence intervals to measure sampling error, providing some numerical illustrations. The Technical Note then turns to nonsampling errors, stating that they "can occur for many reasons, including the failure to sample a segment of the population, inability to obtain information for all respondents in the sample, inability or unwillingness of respondents to provide correct information on a timely basis, mistakes made by respondents, and errors made in the collection or processing of the data".

- Each year the Census Bureau reports statistics on the income distribution based on data collected in the Annual Social and Economic (ASEC) Supplement to the CPS. The Census release provides point estimates, acknowledged but did not quantify sampling error, and did not mention nonsampling error. The Census Bureau's P60 Current Population Report provides numerous statistics characterizing the income distribution and measures sampling error by providing 90% confidence intervals for various estimates. However, the report does not measure nonsampling errors. A supplementary technical document describes some sources of nonsampling error, but it does not quantify them. Reporting official statistics as point estimates manifests a common tendency of policy analysts to project incredible certitude, encouraging the public to believe that errors are small and inconsequential (Manski, 2013c). In the absence of agency guidance, persons who understand that official statistics are subject to error must fend for themselves and conjecture the error magnitudes. Thus, users of official statistics may misinterpret the information that the statistics provide.

One of the nonsampling errors is due to survey nonresponse. Statistical agencies use traditional assumptions –namely that nonresponse is random conditional on specified observed covariates– to form point estimates with nonresponse. These assumptions are implemented as weights for unit nonresponse and imputations for item nonresponse. In particular, the Census Bureau applies hot-deck imputations to the CPS and other surveys, describing the hot deck this way:

> This method assigns a missing value from a record with similar characteristics, which is the hot deck. Hot decks are defined by variables such as age, race, and sex. Other characteristics used in hot decks vary depending on the nature of the unanswered question. For instance, most labor force questions use age, race, sex, and occasionally another correlated labor force item such as full- or part-time status (U.S. Census Bureau, 2006, p.9).

Table 4.1: Annual median household income

Year	Number of interviewed households	Fraction with missing data	Interval estimates	Point estimate with imputations
2001	78265	0.461	[32000, 102000]	44200
2002	78310	0.466	[31225, 104612]	44231
2003	77149	0.464	[32040, 106000]	45100
2004	76447	0.463	[33000, 106847]	46324
2005	75939	0.440	[35600, 101000]	48078
2006	75477	0.423	[36462, 99999]	50002
2007	75872	0.428	[38000, 101747]	52000
2008	76185	0.417	[39157, 100100]	52004
2009	76260	0.416	[37481, 99999]	51157
2010	75188	0.414	[36909, 100000]	51016
2011	74838	0.408	[38000, 100000]	52000

4.5.2 Some findings

Official statistics describing income in the United States are based on data collected in the CPS. The basic CPS sampling unit is a household, which may include one or more families or unrelated individuals who reside together. Data on annual income of household members in the preceding calendar year are collected in the ASEC Supplement, which is administered in the period February through April. Data on the current employment status of civilian adult household members are collected in the monthly administration of the CPS. Applying the findings of this chapter, Manski (2016) uses ASEC data collected in 2002-2012 to form interval estimates of median household. Table 4.1 shows the interval estimates: in the absence of assumptions, the lower (upper) bound on median household income would be the value that the median would take if all households with missing data were to have zero (infinite) income. It can be appreciated how unreliable are the point estimates with imputation due to the nonresponses rates. In any case, such estimates are a plausible value for the median household income because they belong to the identification region.

Chapter 5

Dissecting Chilean Surveys: The Missing Outcomes Case

Broadly speaking[1], public surveys are applied either to get a better gauge of citizens' political opinions (Berinsky, 2017) or to collect information that is useful for policy makers. These surveys are perceived as reliable tools as it is argued that they are applied to "representative samples". If this were the case, the analysis of the strength of a survey would be reduced to indicating how a sample design ensures access to a "representative sample". However, it is necessary to emphasize that the expression "representative sample" is *not* a statistical concept because it is logically contradictory. As a matter fact, a survey is applied to know the behavior of a population in relation to an outcome of interest. Doing so means that we have no idea about this outcome: how then can we ensure the representativeness of the survey? On the other hand, if we know this outcome, why do we need to conduct a survey?

A question then arises: how can we assess a survey? This chapter intends to answer this question in a specific but quite typical case, namely when some surveyed individuals do not answer a specific question. Our approach is based on the tension between the following two questions: what *can* be learned from the data provided by a survey?; and, what do we *want* to learn from those data? The difference between these two question relies on the statistical concept of *identifiablity*.

As a matter of fact, a statistical model is a family of probability distributions indexed by a parameter and defined on a sample space; that is,

$$\mathcal{E} = \{(M, \mathcal{M}),\ P^{\theta} : \theta \in \Theta\}$$

where $(M, \mathcal{M})$ is a measurable space, called *sample space*; P^{θ} is a probability distribution defined on $(M, \mathcal{M})$ and indexed by the parameter θ; and Θ is the set of plausible values of θ, called *parameter space*. It should be remarked that this structure is not only valid for the so-called parametric models, but also for semi-parametric and non-parametric models. Thus, in particular, a probability distribution can

[1]This chapter reproduces the paper entitled *Dissecting Chilean Surveys: The Missing Outcomes Case* by E. San Martín and E. Alarcón-Bustamante, recently submitted to the *Chilean Journal of Statistics*. This journal was chosen by intellectual coherence.

also be a parameter; for details, see San Martín, González, and Tuerlinckx (2015).

From a modelling point of view, a set of data is fully represented by a probability distribution that generates them. Consequently, a parameter of this distribution represents a specific characteristic of the set of data under analysis; see Fisher (1922). Technically speaking, these correspond to the *identified parameter*. However, if we attribute a characteristic to a set of data that cannot be represented by a parameter (that is, it is not a functional of the probability distribution underlying the data), then we face an identification problem. Technically speaking, these correspond to a *parameter of interest*. Thus, the identified parameters summarize what can be learned from the data, whereas the parameters of interest represent what we want to learn from the data. When an injective relationship is established between them, then the identification problem is solved. For details and references, see Koopmans and Reiersøl (1950), San Martín (2018) and San Martín and González (2022).

In this chapter we will use this distinction to assess both the strengths and weaknesses of three Chilean surveys: two of political opinion (CADEM survey and Araucanía citizen consultation), and one related to the income distribution of employees (CASEN). We will analyze the identification problem raised by missing outcome. To do that, we will use Manski (2007a)'s technique of partial identification, which allows us to evaluate how strong are the ignorability conditions typically used to impute missing data. Based on this discussion, we will emphasize the way in which these survey should report their results.

This chapter is organized as follows. In Section 5.1 the political opinion survey CADEM is analyzed. Section 5.2 focus its attention on the National Socioeconomic Characterisation Survey CASEN. Finally Section 5.3 analyze a recent citizen consultation applied in the Araucanía region in Chile. In each of these sections, we provide the corresponding methodological information of each survey and also the political and/or economical context in which the survey is used. The chapter ends by a general discussion.

5.1 CADEM survey

5.1.1 General objective and methodological information

According to the information provided on its website, the CADEM survey is one of the many services offered by the market research company *CADEM research & estrategia*. Specifically, it is related to the service called *Plaza Pública*, which describes itself as "the first and only polling platform that measures public opinion on a weekly basis to provide data and analysis on a wide range of topics of interest"[2]. This particular aspect is related to one of the general objectives of this marketing company: "We want to connect people with decision makers, through data and not from intuition, providing strategies and action plans to achieve the expected results based on a deep knowledge of the new consumer/citizen"[3].

CADEM survey delivers "reliable, timely and contingent information on the political, economic and social debate in Chile on a weekly basis". The study published by CADEM "contemplates a probabilistic survey of 700 weekly cases (with a monthly consolidation that goes from 2800 surveys to 3500 depending on whether the month has 4 or 5 weeks), applied 100% through cell phones, using CADEM's own

[2]Retrieved from https://cadem.cl/sobre-cadem/ on December 30, 2021.
[3]Retrieved from https://cadem.cl/plaza-publica/ on December 30, 2021.

database that contains more than 18 million cell phones considering both prepaid and postpaid numbers, all obtained through Random Digit Dialing and consolidated during the last four years". Its target group is, therefore, all individuals living in the national territory, Chileans and immigrants, men and women over 18 years old, inhabitants of the 15 regions of the country. This led to perform a previous stratification of the total population based on the population projections made by the National Institute of Statistics (NIS) of the Chilean Government for the year 2017 at the national level. Table 5.1 presents the estimated population aged 18 and over for each region of the country as of 2017 and the number of surveys proposed for each region in order to comply with the national proportionality. In addition to the distribution by region, the previous stratification considers, only as a control, the combination of sex and age variables; for more details, see CADEM (2018).

Table 5.1: NIS population projections for 2017 and number of surveyed

Region	Population over 18 years old	Theoretical sample
XV	182.301	9
I	252.814	13
II	471.980	24
III	234.933	12
IV	595.594	30
V	1.430.182	72
VI	706.014	35
VII	804.214	40
VIII	1.634.325	82
IX	756.349	38
XIV	313.112	16
X	636.432	32
XI	80.797	4
XII	126.772	6
RM	5.713.842	287
Total	13.939.661	700

It is important to emphasize that this general information is not published week by week, except for the total number of people selected and the total number of people who agreed to answer the survey.

5.1.2 How are the missing responses treated?

Taking into account that the survey is conducted by telephone, the main issue is the non-response rate. CADEM is not only aware of this problem, but distinguishes three cases of non-response: cases of no contact, namely no one answers the call either because the phone is busy or out of service; cases of a non-eligible person, namely a person who answers the call, but does not satisfy the requirements of the

target group; and a person who is correctly selected but refuses to answer the survey. The impact of the non-response rate is assessed in the following terms:

> Estimating the magnitude of non-response is critical because of the direct relationship it may have with self-selection biases in public opinion polls. The calculation of the non-response rate is also used as a measure of validation of the results. Under the assumption that those who rejects to answer the survey are equal to those who answers it, the magnitude of the non-response rate does not offer major disadvantages, but when there is evidence that the two groups are not equivalent, the non-response can introduce serious distortions in the results (CADEM, 2018).

CADEM accordingly reports the rate of non-response. Three types of results are reported by the survey: those that make explicit the number of cases surveyed, which is approximately equal to 700; those that use a subset of these cases; and trends over time, using previous survey results. However, the impact of the non-response rate on both the results of the survey and their report are not discussed. For an example, see the survey published on the fourth week of December 2021 (CADEM, 2022).

5.1.3 Dissecting the CADEM survey

The objective of this section is to answer the following questions: What can be learned from the data collected by the CADEM survey? How reliable is the CADEM survey? In order to be consistent with a certain degree of reliability, how should its results be communicated?

Example

Let us consider the collected results during the fifth week of December 2021 (CADEM, 2022). As mentioned above, each study contains a methodological sheet, which indicates that the sampling is a probability sample with random selection of individuals and previously stratified by region; that the sample consists of 705 cases, which required making 6,401 telephone calls, so the response rate is equal to 11%. Let us focus our attention on the first question of the study:

> Do you have a very positive, positive, negative or very negative image of Gabriel Boric?

The results are the following: 63% have a very positive or positive (denoted by a) image of Gabriel Boric; 27% have a negative or very negative (denoted by b) image; and 10% do not know or non-response (denoted by c).

What we can learn from the data

Let M be the sample space whose components are the numbers of cellular phones. On this space we define the vector of random variables $(E, R, S, C, G) : M \longrightarrow \{0, 1\}^4 \times \{1, \ldots, 15\}$, where for each $m \in M$

- $E(m) = 1$ if the person associated with cell phone m is eligible, and $E(m) = 0$ if not.
- $R(m) = 1$ if the cell phone m answers the call, and $R(m) = 0$ if not.
- $S(m) = 1$ if the person associated with cell phone m is selected, and $S(m) = 0$ if not.
- $C(m) = 1$ if the person associated with cell phone m answers the survey, and $C(m) = 0$ if not.
- $R(m) = r$ with $r \in \{1, \ldots, 15\}$ if the person associated to cell phone m belongs to region r.

From these definitions, it follows that

$$\{m \in M : S(m) = 1\} \subset \{m \in M : E(m) = 1\} \cap \{m \in M : R(m) = 1\}; \tag{5.1.1}$$

$$\{m \in M : S(m) = 1\} = \{m \in M : C(m) = 0\} \cup \{m \in M : C(m) = 1\}. \tag{5.1.2}$$

Let Y be the outcome of interest, taking values in the set $\{a, b, c\}$. The data inform about the conditional distribution of Y given $(E = 1, R = 1, S = 1, C = 1)$; that is,

$$P(Y = a \mid E = 1, R = 1, S = 1, C = 1) = 0.63; \quad P(Y = b \mid E = 1, R = 1, S = 1, C = 1) = 0.27$$

$$P(Y = c \mid E = 1, R = 1, S = 1, C = 1) = 0.10; \quad P(C = 1 \mid E = 1, R = 1, S = 1) = 0.11.$$

Both $P(Y = y \mid E = 1, R = 1, S = 1, C = 1)$ for $y \in \{a, b, c\}$, and $P(C = c \mid E = 1, R = 1, S = 1)$ for $c \in \{0, 1\}$ correspond to the identified parameter, and therefore they represent all that can be learned from the data.

What we want to learn from the data

The results of the CADEM survey can be interpreted conditionally to different sub-populations.

First level of analysis

The first level corresponds to what we can learn from the data and it is captured by the identified parameter $P(Y = y \mid E = 1, R = 1, S = 1, C = 1)$ for $y \in \{a, b, c\}$.

Second level of analysis

A second level corresponds to focus the attention on the surveyed persons, namely $\{m \in M : S(m) = 1\}$, which by (5.1.1) is equivalent to $\{m \in M : E(m) = 1, S(m) = 1, R(m) = 1\}$. In this case, it is not

longer possible to identified $P(Y = y \mid E = 1, S = 1, R = 1)$. As a matter of fact, by the Law of Total Probability (?, ?),

$$\begin{aligned} P(Y = y \mid E = 1, S = 1, R = 1) &= P(Y = y \mid S = 1) \quad \text{by (5.1.1)} \\ &= P(Y = y \mid S = 1, C = 1)P(C = 1 \mid S = 1) + P(Y = y \mid S = 1, C = 0)P(C = 0 \mid S = 1) \end{aligned} \tag{5.1.3}$$

for each $y \in \{a, b, c\}$. In this decomposition, $P(Y = y \mid S = 1, C = 1)$ and $P(C = 1 \mid S = 1)$ are identified, whereas $P(Y = y \mid S = 1, C = 0)$ is not identified because it depends of those persons who refuse to answer the survey. Taking into account that such a probability takes values between 0 and 1, we can provide an interval of all plausible values for $P(Y = y \mid E = 1, S = 1, R = 1)$ which are compatible with the observed information: for each $y \in \{a, b, c\}$,

$$\begin{aligned} &P(Y = y \mid S = 1, C = 1)P(C = 1 \mid S = 1) \leqslant \\ &\quad \leqslant P(Y = y \mid S = 1) \leqslant \\ &\quad \leqslant P(Y = y \mid S = 1, C = 1)P(C = 1 \mid S = 1) + P(C = 0 \mid S = 1). \end{aligned} \tag{5.1.4}$$

Following ? (?), this interval corresponds to the region where $P(Y = y \mid S = 1)$ is partially identified. This interval deserves the following comments:

1. Considering the example of Section 5.1.3, we have that $P(C = 1 \mid S = 1) = 0.11$ and $P(Y = a \mid S = 1) = 0.63$. Therefore

   $$0.0693 \leqslant P(Y = a \mid S = 1) \leqslant 0.9593.$$

   Thus, the survey report should be phrased in the following terms: at least 6.93% of the surveyed people have a positive or very positive image of Gabriel Boric, and at most 95.93% of the surveyed people have such positive or very positive image.

2. This interval provides information about the uncertainty inherent to the non-response rate. In fact, the width of (5.1.11) is equal to $P(C = 0 \mid S = 1)$, which in this example is equal to 89%.

3. Different scenarios should be considered when reporting $P(Y = a \mid S = 1)$, $P(Y = b \mid S = 1)$ and $P(Y = c \mid S = 1)$ because these three probabilities belongs to the 2-dimensional simplex $S_3 = \{(p_1, p_2, p_3) \in [0, 1]^3 : p_1 + p_2 + p_3 = 1\}$. Thus, for instance, it can be said that 95.93% of surveyed people have a positive or very positive image of Gabriel Boric and, consequently, a 4.07% have a poor or very poor image or Gabriel Boric, or do not known or non-response, that is,

   $$\begin{aligned} &1 - [P(Y = 1, C = 1 \mid S = 1) + P(C = 0 \mid S = 1)] = \\ &\quad = P(C = 1 \mid S = 1) + P(C = 0 \mid S = 1) - P(Y = a, C = 1 \mid S = 1) - P(C = 0 \mid S = 1) \\ &\quad = P(C = 1 \mid S = 1) - P(Y = a, C = 1 \mid S = 1) \\ &\quad = P(Y \neq a, C = 1 \mid S = 1) \\ &\quad = P(Y \in \{b, c\}, C = 1 \mid S = 1) \\ &\quad = P(Y = b, C = 1 \mid S = 1) + P(Y = c, C = 1 \mid S = 1), \end{aligned}$$

which is the lower bound of $P(Y \in \{b, c\} \mid S = 1)$. In the example, $P(Y = b, C = 1 \mid S = 1) = 0.0297$ and $P(Y = c, C = 1 \mid S = 1) = 0.011$.

Once the partial identification of $P(Y = y \mid S = 1)$ is established, it is possible to qualify CADEM's claims about non-responses. As it was mentioned in Section 5.1.2, CADEM considers that, "under the assumption that those who rejects to answer the survey are equal to those who answers it, the magnitude of the non-response rate does not offer major disadvantages, but when there is evidence that the two groups are not equivalent, the non-response can introduce serious distortions in the results". If we consider the decomposition (5.1.3), the assumption advanced by CADEM corresponds to the following equality:

$$P(Y = y \mid S = 1, C = 1) = P(Y = y \mid S = 1, C = 0) \qquad \text{for all } y \in \{a, b, c\},$$

which, by definition of conditional independence, is equivalent to

$$Y \perp\!\!\!\perp C \mid \{S = 1\}; \tag{5.1.5}$$

here $V \perp\!\!\!\perp W \mid Z$ corresponds to the conditional independence between V and W given Z; for details and properties on conditional independence, see Florens, Mouchart, and Rolin (1990, Chapter 2). This condition, typically known as *missing at random* (?, ?; R. J. Little & Rubin, 2019), is not empirically refutable because it depends on the component $P(Y = y \mid S = 1, C = 0)$ which in turn is not based on actual observations. Consequently, it is *impossible* to find out evidence establishing that "the two groups are not equivalent".

Correctly stated, condition (5.1.5) is an identification restriction (San Martín & González, 2022) under which $P(Y = y \mid S = 1)$ is point identified in the sense that

$$P(Y = y \mid S = 1) = P(Y = y \mid S = 1, C = 1) \qquad \text{for all } y \in \{a, b, c\}.$$

In other words, under assumption (5.1.5), the uncertainty induced by the non-response decreases from an interval of width $P(C = 0 \mid S = 1)$ to the singleton $\{P(Y = y \mid S = 1, C = 1)\}$. Thus, *what we want to learn from the data* coincides with *what we can learn from the data*. In passing, let us mention that condition (5.1.5) should be viewed as a characterization of *absence of (self-)biased* and, consequently, the identification problem induced by the non-response is exactly the same as the identification problem induced by self-selection.

Third level of analysis

A third level of analysis corresponds to focus the attention on the eligible persons, namely $\{m \in E(m) = 1\}$. In this case, the parameter of interest is given by $P(Y = y \mid E = 1)$ for $y \in \{a, b, c\}$. Let us analyze its identifiability using only the information available at the CADEM survey as published.

Using the Law of Total Probability, we have

$$\begin{aligned} &P(Y = y \mid E = 1) = \\ &\quad = P(Y = y \mid E = 1, R = 1)P(R = 1 \mid E = 1) + P(Y = y \mid E = 1, R = 0)P(R = 0 \mid E = 1) \end{aligned} \tag{5.1.6}$$

for $y \in \{a, b, c\}$. In this decomposition, $\gamma \doteq P(Y = y \mid E = 1, R = 0)$ is not identified because it is impossible to know whether a person associated with a cell phone that does not answer a call is eligible or not. On the other hand, $P(Y = y \mid E = 1, R = 1)$ can be decomposed as follows:

$$\begin{aligned} P(Y = y \mid E = 1, R = 1) &= P(Y = y \mid E = 1, R = 1, S = 1)P(S = 1 \mid R = 1, E = 1) + \\ &\quad P(Y = 1 \mid E = 1, R = 1, S = 0)P(S = 0 \mid R = 1, S = 1) \\ &= P(Y = y \mid S = 1)P(S = 1 \mid R = 1, E = 1) + \\ &\quad P(Y = 1 \mid E = 1, R = 1, S = 0)P(S = 0 \mid R = 1, E = 1) \quad \text{by (5.1.1).} \end{aligned}$$

Note that

$$\{m \in M : E(m) = 1, R(m) = 1, S(m) = 0\} = \varnothing$$

because there are no eligible persons associated with a cell phone that answered the call who are not selected. Consequently,

$$P(S = 0 \mid R = 1, E = 1) = \frac{P(S = 0, R = 1, E = 1)}{P(R = 1, E = 1)} = 0.$$

Moreover, $P(Y = 1 \mid E = 1, R = 1, S = 0)$ is a probability conditional on an event of probability 0 and, therefore, takes an arbitrary value in $[0, 1]$ (see Remark 5.1). It follows that

$$P(Y = y \mid E = 1, R = 1) = P(Y = y \mid S = 1)P(S = 1 \mid R = 1, E = 1). \tag{5.1.7}$$

Thus, for each $y \in \{a, b, c\}$,

$$\begin{aligned} P(Y = y \mid E = 1) &= P(Y = y \mid S = 1)P(S = 1 \mid R = 1, E = 1)P(R = 1 \mid E = 1) + \\ &\quad \gamma\, P(R = 0 \mid E = 1) \\ &= P(Y = 1 \mid S = 1)P(S = 1 \mid E = 1) + \gamma\, P(R = 0 \mid E = 1) \end{aligned}$$

for all $\gamma \in [0, 1]$. In this decomposition, $P(Y = 1 \mid S = 1)$ is partially identified by the interval (5.1.11); by (5.1.1), $P(S = 1 \mid E = 1)$ corresponds to the ratio

$$\frac{\sharp\{\text{selected persons}\}}{\sharp\{\text{eligible persons}\}},$$

which is identified; and $P(R = 0 \mid E = 1)$ corresponds to the proportion of eligible persons who did not respond to the telephone call. Taking into account that a person can be classified as eligible once he/she has answered the telephone call (see Section 5.1.1), then it is impossible to identify this parameter. Nevertheless, (5.1.1) implies that

$$\{m \in M : R(m) = 0\} \subset \{m \in M : S(m) = 0\}$$

and, therefore,

$$P(R = 0 \mid E = 1) \leqslant P(S = 0 \mid E = 1) = 1 - P(S = 1 \mid E = 1) = \frac{\sharp\{\text{non-selected persons}\}}{\sharp\{\text{eligible persons}\}},$$

which is identified.

Thus, $P(Y = y \mid E = 1)$ is partially identified, where the lower bound of the identification region is given by

$$P(Y = 1 \mid S = 1, C = 1)P(C = 1 \mid S = 1)P(S = 1 \mid E = 1),$$

which by (5.1.1) reduces to $P(Y = y, S = 1, C = 1 \mid E = 1)$; and its upper bound is given by

$$[\, P(Y = 1 \mid S = 1, C = 1)P(C = 1 \mid S = 1) + P(C = 0 \mid S = 1)\,]P(S = 1 \mid E = 1) + P(S = 0 \mid E = 1),$$

which by (5.1.1) reduces to

$$P(Y = y, S = 1, C = 1 \mid E = 1) + P(C = 0, S = 1 \mid E = 1) + P(S = 0 \mid E = 1).$$

Using the data of the Example,

$$P(S = 1 \mid E = 1) \sim \frac{6.401}{14 \times 10^6},$$

and

$$0.0003198 \leqslant P(Y = a \mid E = 1) \leqslant 0.9999814.$$

Clearly, this interval is non-informative.

Remark 5.1 Let $(M, \mathcal{M}, P)$ be a finite probability space. Let $\mathcal{C} = (C_1, \ldots, C_n\} \subset \mathcal{M}$ be a partition of M such that $P(C_1) = 0$ and $P(C_j) > 0$ for $j = 2, \ldots, n$. Finally, let $A \in \mathcal{M}$. In this case, the conditional probability $P(A \mid \mathcal{C})$ is a random variable defined as

$$P(A \mid \mathcal{C}) = \sum_{j=1}^{n} P(A \mid C_j) 1\!\!1_{C_j},$$

where $1\!\!1_{C_j}$ is the indicator function of the event C_j (?, ?, §6); here the numbers $P(A \mid C_j)$ are computed using the following rule

$$P(A \mid C_j) = \begin{cases} \dfrac{P(A \cap C_j)}{P(C_j)} & \text{if } P(C_j) > 0; \\ \eta \in [0, 1], & \text{if } P(C_j) = 0 \end{cases} \tag{5.1.8}$$

with η arbitrary. This rule is a correct rule (that is, it avoids paradoxes) because it satisfies the following equality:

$$P(A) = E[P(A \mid \mathcal{C})],$$

which ensures the existence of the conditional probability. As a matter of fact, under rule (5.1.8), this equality reduces to the Law of Total Probability –in the general case, it corresponds to the Radon-Nikodym theorem. Moreover, the number $P(A \mid C_a)$ can be arbitrarily chosen because the random variable $P(A \mid \mathcal{C})$ does not change since $P(C_1) = 0$. For more details, see Rao (2005, Chapter 2). ■

Fourth level of analysis

The non-informativity of the above identification region is primarily due to the fact that $P(S = 1 \mid E = 1)$ is extremely small, so $P(S = 0 \mid E = 1)$ is extremely large. This undesired effect could be counteracted by taking into account the information provided by the CADEM survey regarding how persons are selected: "Probabilistic sampling with random selection of individuals and previously stratified by region" (CADEM, 2022).

By the CADEM sampling design, the reasoning should be done conditionally on $\{m \in M : E(m) = 1, R(m) = 1\}$: it is impossible to know whether a person is eligible if he/she has not answered the phone call. Thus, the statement "random selection of individuals and previously stratified by region" corresponds to the following condition:

$$P(Y = y \mid E = 1, R = 1, G, S = 1) = P(Y = y \mid E = 1, R = 1, G, S = 0),$$

which, by definition of conditional independence, is equivalent to

$$Y \perp\!\!\!\perp S \mid \{E = 1, R = 1\}, G. \tag{5.1.9}$$

By the Law of Total Probability, this condition implies that

$$\begin{aligned} P(Y = y \mid E = 1, R = 1, G) &= P(Y = y \mid S = 1, E = 1, R = 1, G) \\ &= P(Y = y \mid S = 1, G) \quad \text{by (5.1.1).} \end{aligned} \tag{5.1.10}$$

Thus, in order to identify $P(Y = y \mid E = 1, R = 1)$, we marginalize with respect to G, namely

$$\begin{aligned} P(Y = y \mid E = 1, R = 1) &= \sum_{g=1}^{15} P(Y = y \mid E = 1, R = 1, G = g) P(G = g \mid E = 1, R = 1) \\ &= \sum_{g=1}^{15} P(Y = y \mid S = 1, G = g) P(G = g \mid E = 1, R = 1) \quad \text{by (5.1.10).} \end{aligned}$$

In this decomposition, the conditional probability $P(G = g \mid E = 1, R = 1)$ is in principle identified, although the current information provided by CADEM does not allow to identify it. Moreover, the conditional probability $P(Y = y \mid S = 1, G = g)$ has the same identification problem that was discussed in the second level of analysis and, therefore, it is partially identified: for each $y \in \{a, b, c\}$ and $g \in \{1, \ldots, 15\}$,

$$\begin{aligned} &P(Y = y \mid S = 1, C = 1, G = g) P(C = 1 \mid S = 1, G = g) \leqslant \\ &\quad \leqslant P(Y = y \mid S = 1, G = g) \leqslant \\ &\quad \leqslant P(Y = y \mid S = 1, C = 1, G = g) P(C = 1 \mid S = 1, G = g) + P(C = 0 \mid S = 1, G = g). \end{aligned} \tag{5.1.11}$$

Therefore, the random selection of each individual in each stratum is far from helping to identify $P(Y = y \mid E = 1, R = 1)$. Furthermore, it does not help to identify $P(Y = y \mid E = 1)$ either, since $P(Y = y \mid E = 1, R = 0)$ is still unidentified.

5.1.4 Discussion

CADEM research & estrategia offers services that "connect people with decision makers, through data and not from intuition". With regard to the CADEM survey, once it has been dissected, we find that the motto is not entirely transparent. Indeed, what makes this survey a disservice is that the way in which the results are reported hides the reliability of the results. However, the reliability of the results depends on the parameter of interest to be reported.

The first level of analysis corresponds to a description of the collected data. For the sake of transparency, CADEM must not only remember for each question of the survey the total number of people who answered it, but also indicate, together with the percentages of preference for each option, the absolute frequencies. This will warn the readers and especially the press that the results reflect the opinion of a *very small number of people*. The second level of analysis makes explicit the uncertainty induced by the non-response. CADEM should be made explicit such uncertainty by reporting both the lower and the upper bound of the identification region of $P(Y = y \mid S = 1)$. In the example, the impact of the non-response rate can be appreciated, which prevents the reader from a false illusion of certainty. It should be emphasized that condition (5.1.5) is a plausible way to treat the non-responses. A transparent treatment of non-response should shows the impact of such a condition on the conclusions of the study. As we have seen in the example, the conclusion depends much more on (5.1.5) than on the data itself. The third level of analysis focuses on the eligible population. Once again, for the sake of transparency, it is necessary to report both the lower and the upper bound of the identification region. The example we have used shows how uninformative the survey results are. This information is more than relevant, showing the intrinsic limits of this type of public opinion instruments.

5.2 CASEN survey

The National Socioeconomic Characterisation Survey (CASEN, for their initials in Spanish) is a Chilean household survey that has been applied since 1987. It is used to assess the impact of social programs on the living conditions of the population[4]. According to the Technical data sheet, the target population is *the population residing in private households throughout the national territory*. The units of analysis are families and individuals living in a household. A suitable respondent is the head of household or, alternatively, a man or woman over 18 years old.

The sampling process of the CASEN survey consists of two steps. First, blocks are chosen that correspond to sets of households; second, a household is chosen in which individuals are surveyed. Due to the pandemic by COVID19, the last version of the survey, called *2020 CASEN survey in pandemic*, was carried out in two steps: first, from the households selected in the previously mentioned sampling process, a face-to-face pre-contact was applied to obtain a contact telephone number. Second, the survey was administered by telephone.

In the 2020 CASEN in pandemic survey, 97.848 households were pre-contacted. Of these, only 86.189 households provided at least a telephone number to be contacted. Of these, 62.540 households had

[4]Retrieved from http://casenpandemia2020.cl/ on December 30, 2021

individuals who answered the survey, which amounted to 185.437 individuals[5]. It should be remarked that the available CASEN data set contains information on these individuals[6].

5.2.1 Treatment of missing outcomes in the CASEN survey

One of the objectives of the CASEN survey is to obtain an overview of the income distribution in Chile, and in particular to have an overview of poverty in the country in terms of income. However, the selected individuals did not answer one of its questions, specifically the income question. CASEN considers appropriate to impute these missing data, so that researchers and policy makers can use a database without missing data. The chosen imputation procedure is called *Conditional Mean Imputation*. The rationale of this technique can be summarized as follows: first, observed covariates are used to define classes. Second, individuals who did not report their income and individuals who reported it are classified in the same class if they share the characteristics of that class. For example, those people from city A, with an age range 30-35 years old who do not report the income, are classified in the same class as those people from the same city in the same age range that report the income. Third, it is computed the mean of the observed incomes conditionally on a class: the missing incomes are imputed through this mean (R. J. Little & Rubin, 2019).

More precisely, let Y be an outcome of interest, and let $\mathbf{X}$ be a set of fully observed covariates which are used to define the classes. Let Z be a binary random variable such that $Z = 1$ if the outcome is observed, and $Z = 0$ if not. The conditional mean of both respondents and non-respondents in the same class are given by $E(Y \mid \mathbf{X} = \mathbf{x}, Z = 1)$ and $E(Y \mid \mathbf{X} = \mathbf{x}, Z = 0)$, respectively. The Conditional Mean Imputation assumes that, for each $\mathbf{x}$,

$$E(Y \mid \mathbf{X} = \mathbf{x}, Z = 0) = E(Y \mid \mathbf{X} = \mathbf{x}, Z = 1). \tag{5.2.1}$$

This assumption is also typically known as *Mean Missing at Random* (Manski, 2007a), *Weak Ignorability* (Imbens, 2000; Hirano & Imbens, 2004), and is equivalent to the conditional orthogonality between Y and Z given $\mathbf{X}$. For details on conditional orthogonality, see Florens and Mouchart (1982).

5.2.2 Dissecting the CASEN survey

Example

Let us focus our attention on the incomes of the salaried employees. According to the technical report *Measuring income and poverty in Chile, 2020 Casen Survey in Pandemic*[7], 45.642 individuals were considered in this category. These individuals were exposed to the following question:

> The last month, what was your net income at your main job?

[5]For details, see Nota técnica N7: Desempeño del Trabajo de Campo, Casen en Pandemia en sección Notas Técnicas 2020: http://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen-en-pandemia-2020

[6]The data base can be downloaded from http://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen-en-pandemia-2020.

[7]Retrieved from http://observatorio.ministeriodesarrollosocial.gob.cl on January 11, 2022

The non-response rate was approximately 11.4% (40.418 valid responses); only 5.062 responses were imputed; the remaining responses (namely, 162) were kept as missing. The following covariates were used to define the classes to impute the missing incomes: X_1=Geographic location, X_2=range age, X_3=sex, X_4=educational level, X_5=category of the occupation, X_6=class of activity of the company where the individual works, and X_6=type of occupation into the company[8].

Table 5.2: Quantiles of the income distribution for both original and imputed incomes

Percentage	Quantile of the original data	Quantile of the imputed data
5%	150.000	160.000
10%	230.000	242.000
25%	320.000	320.000
50%	400.000	420.000
75%	750.000	750.000
90%	1.300.000	1.300.000
95%	1.800.000	1.800.000
99%	3.500.000	3.500.000

If we consider the original data (i.e., the people who reported their income), the average income is equal to 653.891,6 Chilean pesos, while the average income considering the imputed data also was equal to 653.327 Chilean pesos. The quantiles of the income distributions for both data sets are given in Table 5.2. Considering the original data, it can be seen that the 5% of the surveyed individuals have an income at most equal to 150.000 Chilean pesos, while the 10% of the salaried surveyed people have an income at most equal to 230.000 Chilean pesos. When the imputed incomes are considered, these values change.

Remark 5.2 Let Y be a real random variable. The quantile function is defined as

$$q_X(\alpha) = inf\{t \in \mathbb{R} : P(Y \leqslant t) \geqslant \alpha\} \qquad \text{for } \alpha \in [0, 1].$$

This corresponds to the generalized inverse of the cumulative distribution function of Y; see Embrechts and Hofert (2013). The quantiles reported in Table 5.2, as in other part of this paper, were calculated using this definition (for a code, see San Martín and Alarcón-Bustamante (2022)), which respects the nature of the data (the income is a discrete random variable), and not using the Hyndman and Fan (1996)'s recommendations which is used, for instance, in R Core Team (2020). ■

Table 5.2 shows the impact of the imputation procedure on the quantiles of the income distribution. How relevant is this impact on a global view of income distribution and poverty? Could we say that it is negligible? These questions can be answered by addressing the following one: what can we learn about the income by using the empirical evidence only? The remaining of this section is devoted to answer this question.

[8]For details on the imputation procedure, see the technical report: *Measuring income and poverty in Chile, Casen Survey in Pandemic 2020*.

What can we learn from the data?

It was previously mentioned that the CASEN data set contains information of 185.437 individuals, that is, those individuals who answered the survey in the application step. For this reason, we will consider the sample space M as the set of these individuals. Let us define the following vector random variables $(C, S, Z, Y) : M \to \{0,1\}^3 \times R^+ \cup \{0\}$: for each $m \in M$:

- $C(m) = 1$ if the individual m answers the survey at the application step, and $C(m) = 0$ if not.
- $S(m) = 1$ if the individual is classified as a salaried employee in the application step, and $S(m) = 0$ if not.
- $Z(m) = 1$ if the individual m reports the income, and $Z(m) = 0$ if not.
- Let $Y(m)$ be the income of individual m.

From these definitions it follows that

$$\begin{aligned} &\text{(i)} \quad \{m \in M : S(m) = 1\} \subset \{m \in M : C(m) = 1\}; \\ &\text{(ii)} \quad \{m \in M : Z(m) = 1\} \subset \{m \in M : S(m) = 1\} \cap \{m \in M : C(m) = 1\}. \end{aligned} \tag{5.2.2}$$

From the CASEN survey, the information summarized in Table 5.3 is available. This shows that the following conditonal probabilities are identified:

$$P(S = 1 \mid C = 1) = 0.246; \quad P(Z = 1 \mid S = 1, C = 1) = 0.885544.$$

Furthermore, the conditional distribution of the income $P(Y \leqslant y \mid Z = 1, C = 1, S = 1)$ is identified, which is depicted in Figure 5.1. In particular, the average income $E(Y \mid Z = 1, C = 1, S = 1)$ is identified, and it is equal to 653.6 Chilean pesos.

Table 5.3: Total of individuals by random variable – 2020 CASEN survey

Event	Cardinality
$\{m \in M : C(m) = 1\}$	185.437
$\{m \in M : S(m) = 1\}$	45.642
$\{m \in M : Z(m) = 1\}$	40.418

What we want to learn from the data

Analogous to the analysis of the CADEM survey, the results of the CASEN survey can be interpreted conditionally to different sub-populations. This is the content of this section.

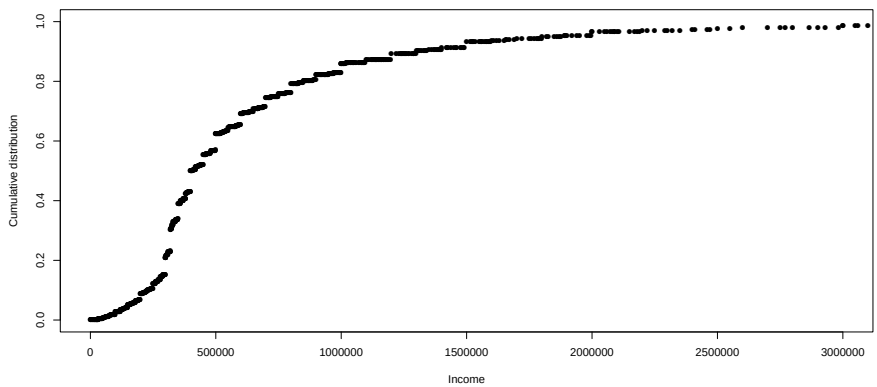


Figure 5.1: The observed income distribution $P(Y \leqslant y \mid Z = 1, C = 1, S = 1)$

First level of analysis

The first level corresponds to what we can learn from the data. This level is accordingly captured by the identified parameters above described. Regarding the distribution of the reported incomes, Figure 5.1 shows that the slope of the curve rapidly increases for lower incomes. As a matter of fact, until 75% of the salaried employees, there are non-dramatic changes in the income, so there is a low variability. In contrast, in the 25% of employees with highest incomes this slope increase slowly, which means that there is a great variability among the incomes.

Second level of analysis: Surveyed salaried employees

The second level of analysis is focused on the parameter of interest $P(Y \leqslant y \mid C = 1, S = 1)$, that is, the income distribution of the salaried employees who answered the survey. The objective of this section is to make explicit the impact of the non-response rate on the income distribution, the average income and the corresponding quantiles. By doing so, it will be appreciated how strong is the Conditional Mean Imputation implemented by the CASEN survey.

Income distribution:

Let us start by the income distribution. Using the Law of Total Probability, we have that

$$\begin{aligned} P(Y \leqslant C = 1, S = 1) \quad = \quad & P(Y \leqslant y \mid C = 1, S = 1, Z = 1) P(Z = 1 \mid C = 1, S = 1) + \\ & P(Y \leqslant y \mid C = 1, S = 1, Z = 0) P(Z = 0 \mid C = 1, S = 1). \end{aligned} \tag{5.2.3}$$

In this decomposition, both $P(Y \leqslant y \mid C = 1, S = 1, Z = 1)$ and $P(Z = z \mid C = 1, S = 1)$, $z \in \{0, 1\}$, are identified, whereas $P(Y \leqslant y \mid C = 1, S = 1, Z = 0)$ is not identified because it depends

on the employees who did not report their income. Instead of using an ignorability condition (as the Conditional Mean Imputation), the relevant question is what can be learned about $P(Y \leqslant | \, C = 1, S = 1)$ without introducing additional assumptions. Taking into account that $P(Y \leqslant y \mid C = 1, S = 1, Z = 0) \in [0, 1]$, it is possible to bound $P(Y \leqslant y \mid C = 1, S = 1)$ as follows:

$$\begin{aligned} & P(Y \leqslant y \mid C = 1, S = 1, Z = 1) P(Z = 1 \mid C = 1, S = 1) \leqslant \\ & \quad \leqslant P(Y \leqslant | \, C = 1, S = 1) \leqslant \\ & \quad \leqslant P(Y \leqslant y \mid C = 1, S = 1, Z = 1) P(Z = 1 \mid C = 1, S = 1) + P(Z = 0 \mid C = 1, S = 1), \end{aligned} \tag{5.2.4}$$

where $P(Z = 1 \mid C = 1, S = 1) = 0.866$. This identification region, depicted in Figure 5.2, includes an infinite number of income distributions that are compatible with the observations. Moreover, it reflects the uncertainty induced by the non-response rate: in fact, the width of this interval is equal to the non-response rate, namely $P(Z = 0 \mid C = 1, S = 1)$.

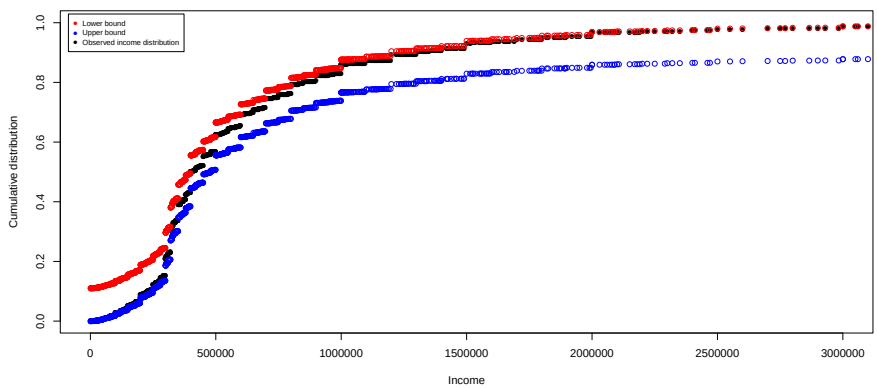


Figure 5.2: Identification region for $P(Y \leqslant y \mid C = 1, S = 1)$

Average income:

At the second level, the average income corresponds to the conditional expectation $E(Y \mid C = 1, S = 1)$, which is decomposed as

$$\begin{aligned} E(Y \mid C = 1, S = 1) \quad = \quad & E(Y \mid C = 1, S = 1, Z = 1) P(Z = 1 \mid C = 1, S = 1) + \\ & E(Y \mid C = 1, S = 1, Z = 0) P(Z = 0 \mid C = 1, S = 1). \end{aligned}$$

In this decomposition, $E(Y \mid C = 1, S = 1, Z = 1)$ and $P(Z = z \mid C = 1, S = 1)$, for $z \in \{0, 1\}$, are identified, whereas $E(Y \mid C = 1, S = 1, Z = 0)$ is not identified because it depends on the employees who did not report their income. However, this last conditional expectation could be partially identified provided the support of Y is bounded. Although theoretically the support of Y is bounded, in practice

the lower bound is known, whereas the upper bound is finite but *unknown*: how large is it? 5.000.000 Chilean pesos? 25.000.000 Chilean pesos? There is no way to answer this question and, therefore, there is no way to provide a partial identification region for $E(Y \mid C = 1, S = 1)$. For additional discussion on partial identifiability of a conditional expectation, see Alarcón-Bustamante, San Martín, and González (2020).

Quantiles of $P(Y \leqslant y \mid C = 1, S = 1)$:

Although the first moment of the income distribution $P(Y \leqslant y \mid C = 1, S = 1)$ is not even partially identified, it is possible to learn from the respective quantiles, and to appreciate the impact of the non-response rate on them. The quantiles of the income distribution $P(Y \leqslant y \mid C = 1, S = 1)$ are given by

$$q_{Y|C=1,S=1}(\alpha) = \inf\{t \in \mathbb{R} : P(Y \leqslant t \mid S = 1, C = 1) \geqslant \alpha\} \quad \text{for } \alpha \in [0, 1].$$

This quantile function is non identified because it is defined in terms of a non identified probability distribution, namely $P(Y \leqslant t \mid S = 1, C = 1)$. However, using the bounds in (5.2.4), it is possible to partially identified the quantile function $q_{Y|C=1,S=1}$ by using the quantiles of the income distribution $P(Y \leqslant y \mid S = 1, C = 1, Z = 1)$: for $\alpha \in (0, 1)$,

$$\begin{aligned} & q_{Y|C=1,S=1,Z=1}\left(\frac{\alpha - P(Z = 0 \mid C = 1, S = 1)}{P(Z = 1 \mid C = 1, S = 1)}\right) \leqslant \\ & \leqslant q_{Y|C=1,S=1}(\alpha) \leqslant \\ & \leqslant q_{Y|C=1,S=1,Z=1}\left(\frac{\alpha}{P(Z = 1 \mid C = 1, S = 1)}\right). \end{aligned} \tag{5.2.5}$$

For a proof, details and reference, see San Martín and González (2022, Section 4).

The identification region (5.2.5) shows the impact of the non-response rate on the quantile function of $P(Y \leqslant y \mid C = 1, S = 1)$ in the sense that one of the bounds of the quantile function is *non-informative* for some values of α. As a matter of fact,

- If $\alpha \leqslant P(Z = 0 \mid C = 1, S = 1)$, then the lower bound in (5.2.5) is equal to the minimum of the support of the conditional distribution $P(Y \leqslant y \mid C = 1, S = 1, Z = 1)$ and, therefore, it is non-informative.
- If $\alpha \geqslant P(Z = 1 \mid C = 1, S = 1)$, then the upper bound in (5.2.5) is equal to the maximum of the support of the conditional distribution $P(Y \leqslant y \mid C = 1, S = 1, Z = 1)$ and, therefore, it is non-informative.

Therefore, the quantile function of $P(Y \leqslant y \mid C = 1, S = 1)$ is informative (that is, provides values in the interior of the support of $P(Y \leqslant y \mid S = 1, C = 1, Z = 1)$) in the following two cases:

1. If $P(Z = 0 \mid S = 1, C = 1) < P(Z = 1 \mid S = 1, C = 1)$ or, equivalently, the non-response rate among the employees individuals is smaller than 50%, then the quantile function $q_{Y|C=1,S=1}$

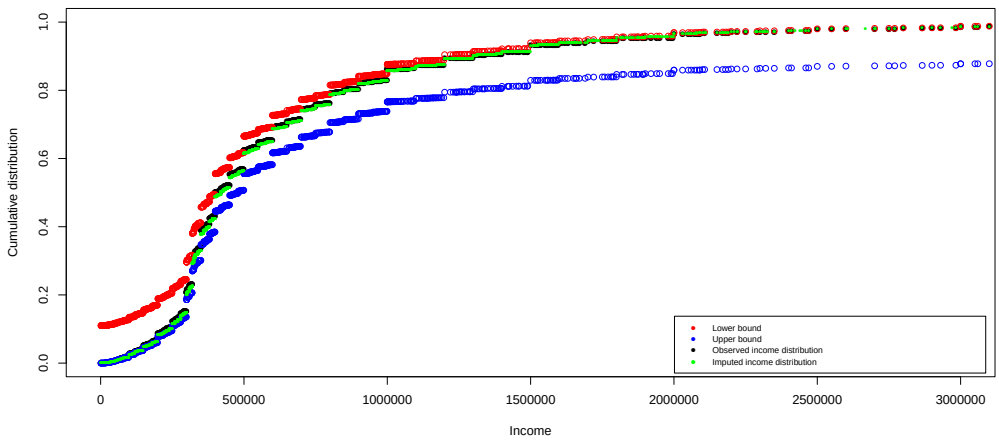


Figure 5.3: Identification region for $P(Y \leqslant y \mid C = 1, S = 1)$ and CASEN income distribution

is informative for all

$$\alpha \in [\, P(Z = 0 \mid S = 1, C = 1),\ P(Z = 1 \mid S = 1, C = 1)\,].$$

2. If $P(Z = 0 \mid S = 1, C = 1) > P(Z = 1 \mid S = 1, C = 1)$ or, equivalently, the non-response rate among the employees individuals is greater than 50%, then the quantile function $q_{Y|C=1,S=1}$ is informative for all

$$\alpha \in [0, P(Z = 1 \mid C = 1, S = 1)\,] \cup [\, P(Z = 0 \mid C = 1, S = 1),\ 1].$$

Let us illustrate this result with the data of the Example. In this case, $P(Z = 0 \mid C = 1, S = 1) = 0.114456$; the corresponding identification regions of the quantile $q_{Y|C=1,S=1}(\alpha)$ for some values of α are summarized in Table 5.4. We also summarize the quantiles of the income distribution with imputations, thereafter called *CASEN income distribution* and denoted as $\widetilde{q}_{Y|C=1,S=1}(\alpha)$. It should be noted that the CASEN income distribution almost overlapped with the distribution of observed incomes. Furthermore, the CASEN income distribution is in the interior of the identification region (5.2.4), as theoretically expected; see Figure 5.3.

Table 5.4 deserves the following comments:

1. For α smaller than the non-response rate, the income of employees can be much lower than the income that can be deduced from the CASEN income distribution. In other words, the non-response rate has such an impact that it is not possible to know how poor the "poorest of the income of employees" are.

2. For α greater that the response rate, the income of employees can be much higher than the income that can be deduced from the CASEN income distribution. In other words, the response rate has

such an impact that it is not possible to know how rich the "richer of the income of employees" are.

3. It can be remarked that for (some) $\alpha_1 \leqslant \alpha_2$, the identification region of $q_{Y|C=1,S=1}(\alpha_1)$ at least intersects the identification region of $q_{Y|C=1,S=1}(\alpha_2)$. This clearly increases the uncertainty of the conclusions that can be drawn using the partially identified income distribution and which, on the other hand, is rendered invisible when using the CASEN income distribution.

Table 5.4: Quantiles of both the partial identified income distribution and the CASEN income distribution

	$q_{Y\|C=1,S=1}(\alpha)$		
α	LB	UB	$\widetilde{q}_{Y\|C=1,S=1}(\alpha)$
0.05	1.200	170.000	160.000
0.10	1.200	250.000	242.000
$P(Z=0 \mid C=1, S=1)$	1.200	265.000	250.000
0.25	300.000	320.000	320.000
0.50	400.000	480.000	420.000
0.75	700.000	1.000.000	750.000
0.80	800.000	1.300.000	865.172
$P(Z=1 \mid C=1, S=1)$	1.100.000	25.000.000	1.200.000
0.90	1.200.000	25.000.000	1.300.000
0.95	1.800.000	25.000.000	1.800.000
0.99	3.500.000	25.000.000	3.500.000
1.00	25.000.000	25.000.000	25.000.000

The previous conclusions allow us to understand the meaning of ignorability conditions, such as the Conditional Mean Imputation technique or, more generally, Missing at Random conditions. These conditions come from the identification restriction

$$Y \perp\!\!\!\perp Z \mid C = 1, S = 1, \mathbf{X}$$

which, by definition of conditional independence, is equivalent to

$$\begin{aligned} P(Y \leqslant y \mid C = 1, S = 1, \mathbf{X}) &= P(Y \leqslant y \mid C = 1, S = 1, Z = 1, \mathbf{X}) \\ &= P(Y \leqslant y \mid C = 1, S = 1, Z = 0, \mathbf{X}). \end{aligned}$$

These equalities means that the missing observations do not provide more relevant information about the output Y, being the only "statistical job" to carefully estimate $P(Y \leqslant y \mid C = 1, S = 1, Z = 1, \mathbf{X})$ –which is the standard procedure.

5.2.3 Discussion

One of the objectives of the CASEN survey is to obtain an overview of the income distribution of employees and, in particular, to have a look at the incomes of the lowest paid employees as well as those of the highest paid. For this purpose, the self-reported income of survey respondents who fall into the category of salaried employees is used. However, individuals who are exposed to the survey are not required

to report their income. As a consequence, the survey includes a non-response rate which, for the 2020 CASEN survey in pandemic, is equal to 11.4456%. Before providing an overview of the distribution of incomes, CASEN applies statistical techniques designed to impute missing income, specifically the Conditional Mean Imputation technique.

Our dissection of the CASEN survey aims to make explicit the policy meaning of this imputation technique. To this end, a partial identification analysis was developed to show the impact of the non-response rate on both the mean of the distribution of the income distribution of employees and on the respective quantiles. One of the main conclusions is that "the poor may be poorer" than what can be asserted from the CASEN income distribution, and that "the rich may be richer" than what can be stated from it.

With this conclusion in mind, it is possible to assess the sense of the imputation technique used by CASEN: the Conditional Mean Imputation technique corresponds to an assumption of *income homogeneity*. As a matter of fact, it is assumed that, among individuals with characteristics $\mathbf{X} = \mathbf{x}$ who did not report their income, there is no relevant income information that was not accessed: all the effectively relevant information has already been observed in those who did report their income. Consequently, the income of an employee who did not report it should be related to the average income of all employees sharing the same characteristics $\mathbf{X} = \mathbf{x}$. The partial identification analysis show how heterogenous may be the indome distributions of employees. Policy decisions should be aware on this uncertainties.

5.3 The Araucanía citizen consultation

5.3.1 Historical and economical context

The capital of the Araucanía region, Temuco, is located 620 kilometers to south of Santiago, the capital of Chile. The Araucanía Region is known for being the original area of the Mapuche People (in the 16th century called "Araucanos"), possibly the only indigenous people with whom the Crown of Spain made a Capitulation of Peace, known as the *Paces de Quilín*, made on January 5 and 6, 1614. This treaty established the Bío Bío River as the border, south of which "the Mapuches lived independently for two hundred and forty years, until 1881" (Bengoa, 2007). In 1881, "Manuel Recabarren, Minister of the Interior [at the time], led Chilean troops to the south and, together with General Gregorio Urrutia, advanced hundreds of kilometers along the border and militarily occupied the area" (Bengoa, 2016). This completed the occupation of Araucanía by the Chilean government.

The Araucanía Region, in addition to the Bío Bío, Los Ríos and Maule regions, develop the country's forestry industry: "the forestry sector represents 1.9% of the domestic GDP, reaching in 2017 USD 5,196 million (3,373 billion of Chilean pesos). Bío Bío region represents 60.0% of the forestry GDP, followed by La Araucanía region with 10.5%, and Los Ríos, and Maule regions with 10.1% each. Regarding the participation of the three forestry subsectors included in the sectorial GDP, the paper, and pulp industry, as well as products derived from paper represents 44.3%, forestry participates with 29.4%, and the wood products, and wood industry represent 26.3%" (Instituto Forestal, 2021).

Many of the conflicts in the area are due to the presence of forestry companies, whose worldview on

nature and its resources is not entirely shared by the Mapuche people's worldview. In addition, part of the forestry exploitation takes place on what were once Mapuche lands, which has triggered a series of territorial claims (Andrade, 2019).

5.3.2 Recent political context

On October 12, 2021, the President of the Chilean Republic declared a *state of emergency* for the provinces of Bío Bío and Arauco, in the Biobío region, and in the provinces of Cautín and Malleco, in the Araucanía Region, for a 15 days period (Diario Oficial de la República de Chile, 2021). According to the Chilean Constitution, this is one of its prerogatives, and it may declare such state of emergency for no more than 15 days. Once a state of emergency is declared, the respective zones will be under the immediate dependence of the Chief of National Defense appointed by the President of the Republic, who will assume the direction and supervision of his jurisdiction with the powers and obligations established by law (Constitución de la República de Chile, 2005, At.42). By declaring a state of emergency, the President of the Republic may restrict the freedom of locomotion and assembly (Constitución de la República de Chile, 2005, At.43).

Among the reasons that led to this decision, the Diario Oficial de la República de Chile (2021) mentions the following ones:

1. An increase of violence acts linked to drug trafficking, terrorism and organized crime, committed by armed groups that have not only made attempts on the lives of members of the Law Enforcement and Security Forces, but have also attacked people and destroyed facilities and machinery used in industrial, agricultural and commercial activities.

2. Since 2018, there has been an increase in crimes and offenses against persons and against property; against public order, including attacks against authority, attacks and threats against prosecutors of the Public Prosecutor's Office and the Judiciary.

3. There has been a 116% increase in reported incidents related to crimes contemplated in Law No. 17.798 on Arms Control, including the seizure of weapons and ammunition.

4. The number, magnitude and seriousness of the crimes and facts indicated, committed in the provinces of the regions of Biobío and Araucanía, imply a serious alteration of public order – understood as the "situation that allows the peaceful exercise of rights and the fulfillment of obligations, ensuring peaceful coexistence"– in the terms established in Article 42 of the Constitution of the Chilean Republic, which allows the enactment of the state of emergency constitutional exception with respect to such areas of the national territory, provided for in said article.

As it was mentioned above, the state of emergency may not be extended for more than fifteen days, notwithstanding that the President of the Republic may extend it for the same period. However, for successive extensions, the President shall always require the consent of the National Congress, specifically the Senate (Constitución de la República de Chile, 2005, Art.42). Until January 2022, the National

Congress has approved the extension of the state of emergency for 6 consecutive times[9]. It should be mentioned that the official account of the recent conflicts in La Araucanía does not relate these conflicts to the territorial claims of the Mapuche people.

5.3.3 Organization of the Araucanía consultation and results

In order to know the opinion of the citizens of the 32 communes of La Araucanía regarding the renewal of the state of emergency in the region, the Regional Intendancy and the Association of Municipalities of La Araucanía organized a citizen consultation, which took place on November 5, 6 and 7, 2021. The consultation was carried out electronically, and all persons over 18 years old registered in the electoral registry in any of the 32 municipalities may participated from a computer, cell phone or another device connected to the internet[10].

The citizen consultation was limited to the following question:

> Do you agree with Congress extending the state of emergency in the Araucanía Region?

The results of the consutlation are summarized in Table 5.5.

Table 5.5: Results of the Araucanía consultation

Option	Votes	% wrt the consultation	% wrt electoral roll
Yes	118.258	81.56	13.34
No	26.655	18.38	3.01
Blank vote	54	0.04	0.01
Null votes	27	0.02	0.00
Total	144.994	100	16.36

5.3.4 How these results were used?

Sections 5.3.1 and 5.3.2 attempt to illustrate the complexity of the political situation in the Araucanía region. This complex context may explain why successive extensions of the state of emergency have been subject to lively debate. In fact, those extensions did not achieve unanimity in the Senate: they were approved not more than 2 or 3 votes in favor. Let us mention the Senate session of November 24, 2021, where the extension of the state of emergency was approved by 16 votes in favor, 14 against, and one abstention. Among the reasons that were mentioned for approving the extension, the Araucanía consultation was explicitly mentioned as an important factor. This was stated by Senator Francisco Chahuán, from the right coalition Chile Vamos, who affirmed that "the state of exception has generated

[9]For details, see https://www.senado.cl/senado/site/cache/search/pags/search164185188127928.html. Retrieved on January 10, 2021.

[10]Retrieved from https://www.consultaaraucania.cl/ on January 10, 2022.

greater tranquility. Attacks against property and arson crimes have decreased. We must listen actively and in La Araucanía there was a citizen consultation that supported this measure"[11]. These expressions are in line with the assessment made by the Governor of La Araucanía, Luciano Rivas, independent, near to the Chile Vamos coalition, at the end of the consultation: "With great respect, but also with great strength, we ask politicians, especially all the deputies and senators of Chile, that our voice be heard, do not turn a deaf ear"[12].

As mentioned by Governor Rivas[13], the Araucanía citizen consultation was one of the first, if not the first, non-binding consultations to be held in Chile. This, added to the complex political situation in the Araucanía region, could explain the interest that this consultation aroused, especially in the relationship that its results have with recent elections, namely the 2020 national referendum on the possibility of a new constitution and the 2021 governor elections. One of these studies is the one conducted by Cayul, Durán, and Jaimovich (2021), which was initially published in the electronic journal CIPER[14]. This study analyzes the representativeness at the minicipality level of the Araucanía Consultation on three axes: Mapuche population, rurality, and population that voted for the *non-approval* option in the 2020 national referendum. According to the authors, "these axes are fundamental to establish whether or not there is a bias in the results, since it analyzes the cultural, socioeconomic and political dimension". To achieve this objective, the authors analyze, on the one hand, the participation in the second round of the election of Regional Governors in Araucanía with the percentage of Mapuche population, the percentage of rural population and the percentage of non-approval in the 2020 referendum; and, on the other hand, the participation in the citizen consultation in Araucanía with the same percentages already mentioned. The choice of the regional governors is due to the fact that in that election "a similar universe of approximately 125,000 people participated".

We are able to reproduce the third analysis by considering the data summarized at Table 5.6. Figures 5.4 and 5.5 reproduce their analysis. Cayul et al. (2021) conclude that "those municipalities with a higher percentage of votes for the non-approval to a new Constitution also had a higher participation in both the citizen consultation and in the second round of governors' elections, but the effect is significantly lower in the latter. That is, there would be a political bias of those who participate in the consultation". The final conclusion of the study is the following:

> We observe then that, when comparing two elections with a similar participation rate, the people who participate in them are very different. While participation in the consultation was higher in urban, non-Mapuche municipalities that voted for non-approval, these same biases are not observed in the second round of governors election.
>
> electors, then, are not representative at the municipal level, and this suggests that the consultation is not necessarily representative of the population of Araucanía. This implies that

[11] Retrieved from https://www.senado.cl/estado-de-excepcion-constitucional on January 10, 2021.

[12] Retrieved from https://assets.eldesconcierto.cl/2021/11/Copia-de-Copia-de-Discurso-Consulta-Araucani%CC%81a.pdf on January 11, 2022.

[13] See his speech of November 7, 2021 in https://assets.eldesconcierto.cl/2021/11/Copia-de-Copia-de-Discurso-Consulta-Araucani%CC%81a.pdf.

[14] At https://www.ciperchile.cl/2021/11/10/consulta-ciudadana-en-la-araucania/

the interpretation of the results should be done with caution, and without extrapolating conclusions for the entire region, especially given the relevance that has been sought to give to the consultation.

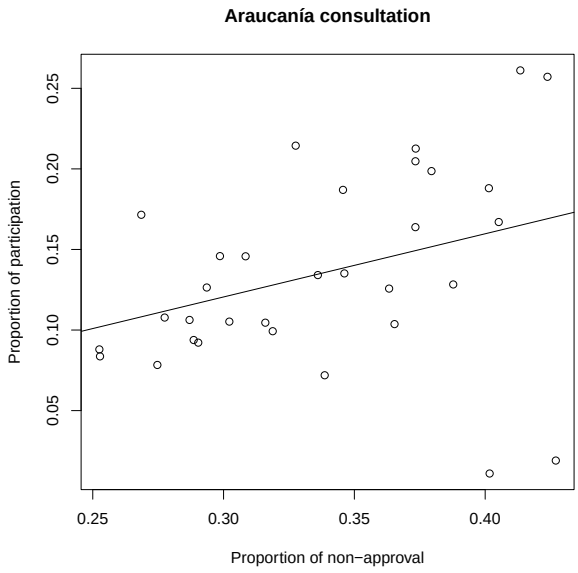


Figure 5.4: Relationship between 2020 referendum and 2021 Araucanía consultation participation

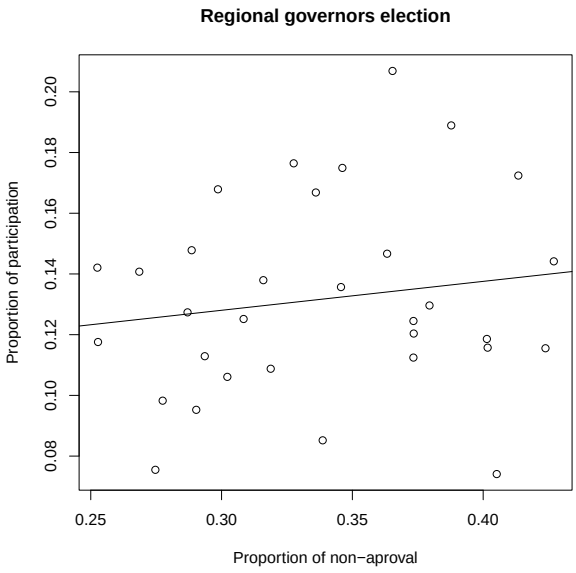


Figure 5.5: Relationship between 2021 governor election and 2021 Araucanía consultation participation

Table 5.6: 2020-2022 elections in the Araucanía Region by municipality

		2021 electoral roll	**2020 referendum for a new constitution**				**2021 regional governors election**				**2022 Araucanía consultation**
c	Municipality	Total	Approve	Non-approve	Null votes	Blank votes	Tuma	Rivas	Null Votes	Blank votes	Total of votes
1	Angol	46.976	12.017	7.349	97	57	2.050	3.924	85	30	9.331
2	Carahue	24.663	4.978	2.220	39	25	1.552	1.481	45	9	3.595
3	Cholchol	11.149	2.951	1.197	51	19	815	804	24	5	1.046
4	Collipulli	22.172	5.159	3.075	65	41	1.120	1.500	32	17	4.714
5	Cunco	20.105	4.090	1.914	47	31	1.041	1.108	24	14	1.996
6	Curacautín	21.097	3.491	2.378	33	20	599	911	39	14	3.524
7	Curarrehue	8.521	1.740	891	12	12	337	371	11	7	613
8	Ercilla	8.036	1.521	1.020	26	26	433	508	8	4	1.511
9	Freire	22.892	5.198	2.401	51	20	1.578	1.535	31	14	2.394
10	Galvaino	12.092	2.472	995	46	24	877	629	24	10	1.285
11	Gorbea	15.651	3.290	2.084	25	20	1.170	1.743	34	10	2.008
12	Lautaro	34.520	8.037	4.788	68	22	1.615	2.186	66	15	5.656
13	Loncoche	23.234	6.125	2.072	27	13	1.411	1.280	32	9	1.944
14	Lonquimay	12.115	1.963	1.318	42	26	852	523	11	16	133
15	Los Sauces	7.464	1.409	1.050	24	12	346	712	12	6	142
16	Lumaco	9.187	1.711	1.206	31	31	478	1.080	18	8	2.399
17	Melipeuco	7.723	1.542	584	15	18	335	2.36	10	2	605
18	Nueva Imperia	30.850	8.765	3.218	83	34	2.407	1.849	64	22	5.292
19	Padres las Casas	59.516	16.892	7.020	142	67	3.063	3.522	107	28	7.523
20	Perquenco	6.814	1.778	757	11	7	721	400	14	9	994
21	Pitrufquén	23.226	5.566	2.947	44	10	1.529	2.494	28	12	3.139
22	Pucón	29.594	8.176	3.140	36	23	1.054	1.776	59	19	3.189
23	Purén	12.240	2.657	1.583	30	19	648	824	38	14	2.506
24	Renaico	9.806	2.640	1.080	17	8	430	477	20	7	904
25	Saavedra	13.239	2.960	1.000	34	27	831	983	42	25	1.165
26	Temuco	238.028	78.332	38.159	360	169	15.375	25.684	740	200	51.039
27	Teodoro Scmidt	14.052	3.026	1.742	28	23	1.304	1.559	32	12	1.457
28	Toltén	10.554	2.126	1.213	30	29	526	1.001	15	6	1.327
29	Traiguén	18.595	4.144	2.189	34	35	1.012	1.445	48	18	3.477
30	Victoria	32.607	7.060	5.193	76	44	1.418	2.264	60	25	8.385
31	Vilcún	24.718	5.971	3.022	70	56	1.614	2.455	41	14	3.315
32	Villarrica	56.170	15.087	6.534	89	40	2.301	3.537	95	27	5.911

5.3.5 Dissecting the use of Araucanía citizen consultation

Statement of the problem

The previous analysis consists in comparing two or more elections that share a common electoral roll. Now, for each election, there are specific probability distributions that are identified, namely (i) the distributions of participation and non-participation, and (ii) the distribution of preferences conditionally on the electors participating in the election. More precisely, let M be the sample space composed of the electors, and define the following random variables on M:

- $V_1(m) = 1$ if the elector m participated at the 2020 referendum, and $V_1(m) =$ if not.
- $V_2(m) = 1$ if the elector m participated at the 2021 governor election, and $V_2(m) =$ if not.
- $V_3(m) = 1$ if the elector m participated at the 2021 citizen consultation, and $V_3(m) =$ if not.
- Let Y_1 be the preference at the 2020 referendum, namely $\mathcal{Y}_1 = \{\text{approve}, \text{non-approve}, \text{blank vote}, \text{null vote}\}$.
- Let Y_2 be the preference at the 2021 governor election, namely $\mathcal{Y}_2 = \{\text{Tuma}, \text{Rivas}, \text{blank vote}, \text{null vote}\}$.
- Let Y_3 be the preference at the 2021 citizen consultation, namely $\mathcal{Y}_3 = \{\text{yes}, \text{no}, \text{blank vote}, \text{null vote}\}$.
- Let C be the municipality in which each elector is registered. C takes 32 different values because there are 32 municipalities; see Table 5.6.

If we consider each election separately, then the identified parameters are the following:

$$P(Y_i = y_i \mid V_i = 1, C = c) \quad (y_i, c) \in \mathcal{Y}_i \times \{1, \ldots, 32\}; \qquad P(V_i = 1 \mid C = c) \quad c \in \{1, \ldots, 32\}$$

for $i = 1, 2, 3$.

If these elections are jointly used, the first thing to verify is if the set of electors is the same, that is, if the following equality is present:

$$\begin{aligned}\{m \in M : V_1(m) = 1\} \cup \{m \in M : V_1(m) = 0\} &= \\ = \{m \in M : V_2(m) = 1\} \cup \{m \in M : V_2(m) = 0\} \\ = \{m \in M : V_3(m) = 1\} \cup \{m \in M : V_3(m) = 0\}.\end{aligned}$$

Certainly, the Chilean Electoral Service (SERVEL for the initials in Spanish) has access to this information and it can verify such equality. In what follows, we will assume that this equality is fulfilled.

The analysis described in Section 5.3.4 consists in comparing

$$\{P(Y_1 = \text{non-aproval} \mid V_1 = 1, C = c) \,:\, c \in \{1, \ldots, 32\}\}$$

with

$$\{P(V_3 = 1 \mid C = c) \,:\, c \in \{1, \ldots, 32\}\}.$$

However, for each $c \in \{1, \ldots, 32\}$, $\{m \in M : V_1(m) = 1, C(m) = v\}$ is not necessarily equal to $\{m \in M : V_3(m) = 1, C(m) = c\}$.

A fair comparison needs to use the same electors, which in turn lead to consider

$$\{P(Y_1 = \text{non-aproval} \mid V_1 = 1, V_3 = 1, C = c) \,:\, c \in \{1, \ldots, 32\}\}$$

and

$$\{P(V_3 = 1 \mid V_1 = 1, C = c) \,:\, c \in \{1, \ldots, 32\}\}.$$

This is due to the fact that the political behavior of those who participate in both elections is not necessarily the same as the political behavior of those who participate in one, or the other, or both. It is, therefore, necessary to identify $P(Y_1 = \text{non-aproval} \mid V_1 = 1, V_3 = 1, C = c)$ and $P(V_3 = 1 \mid V_1 = 1, C = c)$ for each c.

Partial identification of $P(V_1 = v_1, V_3 = v_3 \mid C = c)$

Both $P(Y_1 = \text{non-aproval} \mid V_1 = 1, V_3 = 1, C = c)$ and $P(V_3 = 1 \mid V_1 = 1, C = c)$ require the identifiability of $P(V_1 = v_1, V_3 = v_3 \mid C = c)$ for $(v_1, v_3) \in \{0, 1\}^2$. Taking into account that $P(V_1 = v_1 \mid C = c)$ and $P(V_3 = v_3 \mid C = c)$ are identified, the way to relate them to the joint distribution $P(V_1 = v_1, V_3 = v_3 \mid C = c)$ is through the Fréchet inequality (Fréchet, 1960a, 1960b), namely for each $c \in \{1, \ldots, 32\}$

$$\begin{aligned} &\max\{1, P(V_1 = v_1 \mid C = c) + P(V_3 = v_3 \mid C = c) - 1\} \leqslant \\ &\quad \leqslant P(V_1 = v_1, V_3 = v_3 \mid C = c) \leqslant \\ &\quad \leqslant \min\{P(V_1 = v_1 \mid C = c), P(V_3 = v_3 \mid C = c)\} \quad \forall\, (v_1, v_3) \in \{0, 1\}^2 \end{aligned} \tag{5.3.1}$$

It should be emphasized that these bounds are the best ones; see the constructive proof in Fréchet (1960a). Thus, for $(v_1, v_3) = (1, 1)$, it follows that

$$\begin{aligned} &\max\{0, P(V_1 = 1 \mid C = c) - P(V_3 = 0 \mid C = c)\} \leqslant \\ &\quad \leqslant P(V_1 = 1, V_3 = 1 \mid C = c) \leqslant \min\{P(V_1 = 1 \mid C = c), P(V_3 = 1 \mid C = c)\}. \end{aligned} \tag{5.3.2}$$

For municipality c the lower bound is informative (that is, greater than 0) if $P(V_1 = 1 \mid C = c) > P(V_3 = 0 \mid C = c)$, that is, if the rate of participation at the 2020 referendum is greater than the rate of non-participation at the 2021 citizen consultation; or, equivalently, if the rate of non-participation at the 2020 referendum is smaller than the rate of participation at the 2021 citizen participation. Table 5.7 summarizes the results, where LB_{13} is the lower bound of (5.3.2) and UB_{13} is the corresponding upper bound. It can be seen that, for each municipality, the lower bound is always 0, which means that a plausible assumption is that none of those who participated in one election participated in the other. Another plausible assumption is that

$$P(V_1 = 1, V_3 = 1 \mid C = c) = P(V_3 = 1 \mid C = c), \tag{5.3.3}$$

that is, the rate of joint participation is equal to the rate of participation at the 2021 citizen consultation. In this case, $P(V_1 = 0, V_3 = 1 \mid C = c) = 0$, that is, no elector did not participate at the 2020 referendum and participated at the 2021 citizen consultation. Certainly this conclusion may seem implausible, which in turn would imply that (5.3.3) is implausible as an assumption.

Table 5.7: Bounds of joint participation ratios

c	**Municipality**	$P(V_1 = 1 \mid C = c)$	$P(V_2 = 1 \mid C = c)$	$P(V_3 = 1 \mid C = c)$	LB_{13}	UB_{13}	LB_{23}	UB_{23}
1	Angol	0,42	0,13	0,20	0,00	0,20	0,00	0,13
2	Carahue	0,29	0,13	0,15	0,00	0,15	0,00	0,13
3	Cholchol	0,38	0,15	0,09	0,00	0,09	0,00	0,09
4	Collipulli	0,38	0,12	0,21	0,00	0,21	0,00	0,12
5	Cunco	0,30	0,11	0,10	0,00	0,10	0,00	0,10
6	Curacautín	0,28	0,07	0,17	0,00	0,17	0,00	0,07
7	Curarrehue	0,31	0,09	0,07	0,00	0,07	0,00	0,07
8	Ercilla	0,32	0,12	0,19	0,00	0,19	0,00	0,12
9	Freire	0,34	0,14	0,10	0,00	0,10	0,00	0,10
10	Galvaino	0,29	0,13	0,11	0,00	0,11	0,00	0,11
11	Gorbea	0,35	0,19	0,13	0,00	0,13	0,00	0,13
12	Lautaro	0,37	0,11	0,16	0,00	0,16	0,00	0,11
13	Loncoche	0,35	0,12	0,08	0,00	0,08	0,00	0,08
14	Lonquimay	0,28	0,12	0,01	0,00	0,01	0,00	0,01
15	Los Sauces	0,33	0,14	0,02	0,00	0,02	0,00	0,02
16	Lumaco	0,32	0,17	0,26	0,00	0,26	0,00	0,17
17	Melipeuco	0,28	0,08	0,08	0,00	0,08	0,00	0,08
18	Nueva Imperia	0,39	0,14	0,17	0,00	0,17	0,00	0,14
19	Padres las Casas	0,41	0,11	0,13	0,00	0,13	0,00	0,11
20	Perquenco	0,37	0,17	0,15	0,00	0,15	0,00	0,15
21	Pitrufquén	0,37	0,17	0,14	0,00	0,14	0,00	0,14
22	Pucón	0,38	0,10	0,11	0,00	0,11	0,00	0,10
23	Purén	0,35	0,12	0,20	0,00	0,20	0,00	0,12
24	Renaico	0,38	0,10	0,09	0,00	0,09	0,00	0,09
25	Saavedra	0,30	0,14	0,09	0,00	0,09	0,00	0,09
26	Temuco	0,49	0,18	0,21	0,00	0,21	0,00	0,18
27	Teodoro Scmidt	0,34	0,21	0,10	0,00	0,10	0,00	0,10
28	Toltén	0,32	0,15	0,13	0,00	0,13	0,00	0,13
29	Traiguén	0,34	0,14	0,19	0,00	0,19	0,00	0,14
30	Victoria	0,38	0,12	0,26	0,00	0,26	0,00	0,12
31	Vilcún	0,37	0,17	0,13	0,00	0,13	0,00	0,13
32	Villarrica	0,39	0,11	0,11	0,00	0,11	0,00	0,11

Partial identification of $P(V_3 = 1 \mid V_1 = 1, C = c)$

From (5.3.2) it can be deduced the identification region for $P(V_3 = 1 \mid V_1 = 1, C = c)$, namely

$$\max\left\{0, \frac{P(V_1 = 1 \mid C = c) - P(V_3 = 0 \mid C = c)}{P(V_1 = 1 \mid C = c)}\right\} \leqslant \tag{5.3.4}$$
$$\leqslant P(V_3 = 1 \mid V_1 = 1, C = c) \leqslant \min\left\{1, \frac{P(V_3 = 1 \mid C = c)}{P(V_1 = 1 \mid C = c)}\right\}.$$

For each municipality c, the lower bound is informative if $P(V_1 = 1 \mid C = c) > P(V_3 = 0 \mid C = c)$, whereas the upper bound is informative (that is, smaller than 1) if $P(V_3 = 1 \mid C = c) < P(V_1 = 1 \mid C = c)$, that is, if the rate of participation at the citizen consultation is smaller than the rate of participation at the 2020 referendum. Table 5.8 shows the corresponding lower and upper bound. It can be seen that the lower bound is uninformative, whereas the upper bound is informative: it corresponds to the ratio of participation at the citizen consultation given that electors participated at the 2020 referendum.

Table 5.8: Partial identification of $P(V_3 = 1 \mid V_1 = 1, C = c)$ and $P(Y_1 = \text{non-approve} \mid V_1 = 1, V_3 = 1, C = c)$

		$P(V_3 = 1 \mid V_1 = 1, C = c)$		$P(Y_1 = \text{non-approve} \mid V_1 = 1, V_3 = 1, C = c)$	
c	**Municipality**	LB	UB	LB	UB
1	Angol	0,00	0,48	0,00	0,72
2	Carahue	0,00	0,50	0,00	0,61
3	Cholchol	0,00	0,25	0,05	0,38
4	Collipulli	0,00	0,57	0,00	0,85
5	Cunco	0,00	0,33	0,00	0,47
6	Curacautín	0,00	0,60	0,00	0,99
7	Curarrehue	0,00	0,23	0,14	0,44
8	Ercilla	0,00	0,58	0,00	0,94
9	Freire	0,00	0,31	0,00	0,46
10	Galvaino	0,00	0,36	0,00	0,44
11	Gorbea	0,00	0,37	0,02	0,61
12	Lautaro	0,00	0,44	0,00	0,66
13	Loncoche	0,00	0,24	0,02	0,33
14	Lonquimay	0,00	0,04	0,37	0,41
15	Los Sauces	0,00	0,06	0,39	0,45
16	Lumaco	0,00	0,81	0,00	1,00
17	Melipeuco	0,00	0,28	0,00	0,38
18	Nueva Imperia	0,00	0,44	0,00	0,47
19	Padres las Casas	0,00	0,31	0,00	0,42
20	Perquenco	0,00	0,39	0,00	0,49
21	Pitrufquén	0,00	0,37	0,00	0,54
22	Pucón	0,00	0,28	0,00	0,38
23	Purén	0,00	0,58	0,00	0,89
24	Renaico	0,00	0,24	0,06	0,38
25	Saavedra	0,00	0,29	0,00	0,35
26	Temuco	0,00	0,44	0,00	0,58
27	Teodoro Scmidt	0,00	0,30	0,08	0,52
28	Toltén	0,00	0,39	0,00	0,59
29	Traiguén	0,00	0,54	0,00	0,75
30	Victoria	0,00	0,68	0,00	1,00
31	Vilcún	0,00	0,36	0,00	0,52
32	Villarrica	0,00	0,27	0,04	0,41

Partial identification of $P(V_2 = 1, V_3 = 1 \mid C = c)$

Following the arguments developed in Section 5.3.5, it follows that

$$\begin{aligned} &\max\{0, P(V_2 = 1 \mid C = c) - P(V_3 = 0 \mid C = c)\} \leqslant \\ &\quad \leqslant P(V_2 = 1, V_3 = 1 \mid C = c) \leqslant \min\{P(V_2 = 1 \mid C = c), P(V_3 = v_3 \mid C = c)\}. \end{aligned} \tag{5.3.5}$$

Table 5.7 shows the corresponding lower and upper bounds. Lower bounds are always uninformative because, for each municipality, the rate of participation at the 2021 governor election is smaller than the rate of non-participation at the 2021 citizen consultation. This means that, although the overall participation rates in both elections are very similar (16% for the citizen consultation, 14% for the governor election), a plausible assumption is that there are no electors who participated in both elections. On the other hand, some times the upper bound is equal to $P(V_2 = 1 \mid C = c)$, sometimes to $P(V_3 = 1 \mid C = c)$. In the first case, namely when it is assumed that $P(V_2 = 1, V_3 = 1 \mid C = c) = P(V_2 = 1 \mid C = c)$, then there are no electors who participated in the 2021 governors election and did not participate in the 2021 citizen consultation. In the second case, namely $P(V_2 = 1, V_3 = 1 \mid C = c) = P(V_3 = 1 \mid C = c)$, then there are no electors who did not participate in the 2021 governors election and who participated in the 2021 citizen consultation. Again, it can be stated that these assumptions may not seem entirely plausible, which in turn shows that it is possible to have turnout rates in both elections lower than the upper bound. By passing, this jeopardizes the argument according to which the governor election and the citizen consultation can be compared because their rate of participation are similar.

Partial identification of $P(Y_1 = y \mid V_1 = 1, V_2 = 1, C = c)$

For each municipality c, the conditional probability $P(Y_1 = y \mid V_1 = 1, V_2 = 1, C = c)$ can not vary arbitrarily because it is related to the identified conditional probability $P(Y_1 = y \mid V_1 = 1, C = c)$ through the following decomposition:

$$\begin{aligned} P(Y_1 = y \mid V_1 = 1, C = c) &= P(Y_1 = y \mid V_1 = 1, V_3 = 1, C = c)P(V_3 = 1 \mid V_1 = 1, C = c) + \\ &\quad P(Y_1 = y \mid V_1 = 1, V_3 = 0, C = c)P(V_3 = 0 \mid V_1 = 1, C = c). \end{aligned}$$

In this decomposition, $\gamma_c \doteq P(Y_1 = y \mid V_1 = 1, V_3 = 0, C = c)$ is non identified, whereas $P(V_3 = 1 \mid V_1 = 1, C = c)$ and $P(V_3 = 0 \mid V_1 = 1, C = c)$ are partially identified by (5.3.5).

Let $C \in \{1, \ldots, 32\}$ and $p_c \doteq P(V_3 = 0 \mid V_1 = 1, C = c)$ be fixed. It follows that $P(Y_1 = y \mid V_1 = 1, V_3 = 1, C = c)$ belongs to the set

$$\left\{ \frac{P(Y_1 = y \mid V_1 = 1, C = c) - \gamma_c p_c}{1 - p_c} \; : \; \gamma_c \in [0, 1] \right\}$$

which reduces to the interval

$$A_{p_c} \doteq \left[\frac{P(Y_1 = y \mid V_1 = 1, C = c) - p_c}{1 - p_c}, \; \frac{P(Y_1 = y \mid V_1 = 1, C = c)}{1 - p_c} \right].$$

Now, if $p_{1,c} < p_{2,c}$, then

$$A_{p_{1,c}} \subset A_{p_{2,c}}.$$

Therefore, for each $c \in \{1, \ldots, 32\}$

$$\begin{aligned} P(Y_1 = y \mid V_1 = 1, V_3 = 1, C = c) &\in \bigcup_{p_c \in [l_c, u_c]} A_{p_c} \\ &= A_{u_c}, \end{aligned}$$

where $[l_c,\, u_c]$ is given by (5.3.4). It follows that, for each $c \in \{1, \ldots, 32\}$, $P(Y_1 = y \mid V_1 = 1, V_3 = 1, C = c) \in$ belongs to an identification region where the lower bound is given by

$$\max\left\{0, \frac{P(Y_1 = y \mid V_1 = 1, C = c) - \min\left\{1, \frac{P(V_3=1|C=c)}{P(V_1=1|C=c)}\right\}}{1 - \min\left\{1, \frac{P(V_3=1|C=c)}{P(V_1=1|C=c)}\right\}}\right\},$$

and the upper bound is given by

$$\min\left\{1, \frac{P(Y_1 = y \mid V_1 = 1, C = c)}{1 - \min\left\{1, \frac{P(V_3=1|C=c)}{P(V_1=1|C=c)}\right\}}\right\}.$$

Table 5.8 shows the corresponding lower and upper bound of $P(V_3 = 1 \mid V_1 = 1, C = c)$ and $P(Y_1 = \text{non-approve} \mid V_1 = 1, V_3 = 1, C = c)$. It can be observed the uncertainty induced by the joint participation in both elections. In particular, four municipalities have an extreme uncertainty because the width of their identification regions is at least equal to 0.9: Curacautín, Ercilla, Lumaco and Victoria. In these municipalities, the proportion of non-approval conditionally on the participation at both the 2020 referendum and the 2021 consultation is any value. Moreover, the conditional probability to participate at the citizen referendum given that electors participated at the 2020 referendum is, respectively, 0.6 0.58, 0.81 and 0.68. On the other hand, two municipalities, Lonquimay and Los Sauces, have the smaller uncertainty and, consequently, the rate of approval conditionally on the joint participation is less uncertainty: between 0.37 and 0.41 for Lonquimay; and between 0.39 and 0.45 for Los Sauces. Nevertheless, the rate of participation at the 2021 consultation given participation at the 2020 referendum are quite small: 0.04 and 0.06, respectively.

5.3.6 Discussion

The partial identification analysis shows the impact of the uncertainty due to the joint participation in both elections, namely 2020 referendum and 2021 consultation, on the proportion of electors who chose the *non-approve* option at the 2020 referendum. This impact can be diminished if, for each municipality, the joint distribution $P(V_1 = v_1, V_3 = v_3 \mid C = c)$ with (v_1, v_3) were known. This seems to be feasible for the Chilean Electoral Service, without having to transgress elector identity protection. If this were the case, then $P(V_3 = 1 \mid V_1 = 1, C = c)$ would be identified. However, this fact does not ensure that $P(Y_1 = y \mid V_1, = 1, V_3 = 1, C = c)$ is point identified because $P(Y_1 = y \mid V_1 = 1, V_3 = 0, C = c)$ is not identified given the secrecy of the vote. Consequently, following the arguments developed in Section 5.3.5, $P(Y_1 = y \mid V_1, = 1, V_3 = 1, C = c)$ belongs to an identification interval with a lower bound given by

$$\max\left\{0, \frac{P(Y_1 = y \mid V_1 = 1, C = c) - P(V_3 = 1 \mid V_1 = 1, C = c)}{P(V_3 = 0 \mid V_1 = 1, C = c)}\right\}$$

and an upper one given by

$$\min\left\{1, \frac{P(Y_1 = y \mid V_1 = 1, C = c)}{P(V_3 = 0 \mid V_1 = 1, C = c)}\right\}.$$

It can be deduced that this interval is informative (that is, strictly included in $[0, 1]$) if

$$P(V_3 = 1 \mid V_1 = 1, C = c) < P(V_3 = 0 \mid V_1 = 1, C = c),$$

which is a surprising result.

It could be argued that, under "mild conditions", it is possible to ignore joint participation, and thus argue for the reliability of studies such as the one reported in Section 5.3.4. The partial identification analysis developed in Section 5.3.5 shows that there are two possible assumptions that could be made: the first one would be to assume that $P(Y_1 = \text{non-approve} \mid V_1 = 1, V_3 = 0) = 0$, that is, that no elector who participated in the 2020 referendum and did not participate in the 2021 citizen consultation chose the option *non-approve*. It should be mentioned that this assumption is quite strong and hard to believe. A second assumption would be $Y_1 \perp\!\!\!\perp V_3 \mid \{V_1 = 1\}, C$, which is equivalent to the following two equivalent conditions:

$$\begin{aligned} &P(Y_1 = y \mid V_1 = 1, V_3 = 1, C = c) = P(Y_1 = y \mid V_1 = 1, C = c); \\ &P(V_3 = v_3 \mid Y = y, V_1 = 1, C = c) = P(V_3 = v_3 \mid V_1 = 1, C = c) \quad v_3 \in \{0, 1\}. \end{aligned}$$

The last condition means that, once an elector of a specific municipality participated at the 2020 referendum, the participation at the 2021 consultation does not depend on the preference at the 2020 referendum: again hard to believe.

Finally, it should be emphasized that the previous analysis of partial identification is applicable to critically assess the comparisons over time of political surveys as they would be correctly done if made conditionally on joint participation.

5.4 Concluding remarks

This paper illustrates a possible service that Applied Statistics can render to society. Both political opinion polls and socioeconomic surveys have different uses. The former try to monitor political opinions in relation to issues of citizen interest; the latter are especially used by policy makers who want to assess policies or to have a global view of aspects such as poverty or income. In a democratic society these uses are relevant because they can, for example, change political opinions or question the impact of certain public policies. This is precisely the reason for assessing the strengths and weaknesses of the surveys.

From a statistical point of view, we focus our assessment of the surveys on the correct way of communicating their results, so that the uncertainty induced by non-responses is made explicit. The results can be reported at different levels depending on the population of interest to which the results are to be generalized. The advantage of this strategy is that it makes explicit how this uncertainty could be reduced, and which is impossible to reduce unless very strong assumptions are introduced, which make the conclusions not credible –that is, Law of Decreasing Credibility (?, ?). For instance, in the Araucanía citizen consultation, the uncertainty of the option at the 2020 referendum conditionally on the joint participation at both the 2020 referendum and 2021 consultation can decreased whether the Chilean Electoral Service provides information on such joint participation. However, the uncertainty can not decrease to a point value because of the secrecy of the vote.

At the methodological level, the assessment or dissection of the Chilean surveys was performed making a distinction between identified parameters and parameters of interest: what we can learn from the data is represented by the identified parameters, while what we want to learn from the data is represented by the parameters of interest. In (almost) all empirical research there is a gap between both types of parameters; it is quite relevant to highlight the difference and to study their possible relationships –which is equivalent to solving an identification problem. The way ? (?)expresses himself is illuminating and perhaps summarizes the perspective developed in this paper:

> Anyone who has tried to make sense to real data will, sooner or later, have come across the problem of nonidentifiability. Broadly speaking this means that their first explanation of the data is not the only one. The existence of alternative explanations becomes important when decisions have to be made and particularly so when different explanations suggest completely different courses of action.

The identification regions we established for each of the Chilean survey contain such different explanations.

Chapter 6

How to report the uncertainty due to missing data?

6.1 Introduction

In Section 5.1.3 we discuss some aspects related to the way in which surveys should be reported when they are analyzed under a partial identification perspective. Let us come back to the example discussed there. We defined a random variable y which can take three possible values:

- $y(m) = a$ if the participant m have a very positive or positive image of the Chilean President.
- $y(m) = b$ if the participant m have a negative or very negative image of the Chilean President.
- $y(m) = c$ if the participant m does not known;

here *participant* refers to a cell phone m associated to a person who answers the call, or not. According to the available data, we have

$$P(y = a \mid s = 1, c = 1) = 0.63, \qquad P(y = b \mid s = 1, c = 1) = 0.27, \qquad P(y = c \mid s = 1, c = 1) = 0.1,$$

where $s(m) = 1$ if the person associated with cell phone m is selected, and $s(m) = 0$ if not; and $c(m) = 1$ if the person associated with cell phone m answers the survey, and $c(m)00$ if not. Moreover,

$$P(c = 1 \mid s = 1) = 0.11.$$

The partial identification intervals are the following:

Probability of interest	Upper bound	Lower bound
$P(y = a \mid s = 1)$	$P(y = a \mid s = 1, c = 1)P(c = 1 \mid s = 1)$	$P(y = a \mid s = 1, c = 1)P(c = 1 \mid s = 1)+$ $P(c = 0 \mid s = 1)$
$P(y = b \mid s = 1)$	$P(y = b \mid s = 1, c = 1)P(c = 1 \mid s = 1)$	$P(y = b \mid s = 1, c = 1)P(c = 1 \mid s = 1)+$ $P(c = 0 \mid s = 1)$
$P(y = c \mid s = 1)$	$P(y = c \mid s = 1, c = 1)P(c = 1 \mid s = 1)$	$P(y = c \mid s = 1, c = 1)P(c = 1 \mid s = 1)+$ $P(c = 0 \mid s = 1)$

Using the data at hand, these partial identification intervals become

Probability of interest	Upper bound	Lower bound
$P(y = a \mid s = 1)$	0.0693	0.9593
$P(y = b \mid s = 1)$	0.0297	0.9197
$P(y = c \mid s = 1)$	0.011	0.901

If we report for each possible value of y, then we have the following sentences:

- At least 6.93% of contacted people have a very positive or positive image of the Chilean president, and at most 95.93%.
- At least 2.97% of contacted people have a negative or very negative image of the Chilean president, and at most 91.97%.
- At least 1.1% of contacted people don't know about the image of the Chilean president and at most 90.1%.

However, the challenge is to report *jointly* in such a way that the uncertainty be perceived by people. For instance, if we say that *at mots 95.93% of contacted people have a very positive or positive image of the Chilean president*, then at least 4.07% of contacted people have either a negative or very negative image, or don't know. Note that

$$4.07\% = 2.97\% + 1.1\%,$$

namely the addition of the lower bounds corresponding to $P(y = b)$ ad $P(y = c)$. Consequently, the report need to be complemented in the following way:

> ... and at least 2.97% have a negative or very negative image, and at least 1.1% don't know about the image of the Chilean president.

But the previous case is just one of two other cases: the one in which we take the upper bound of $P(y = b)$; and the other in which we take the upper bound of $P(y = c)$. So we will have the following report:

- At most 95.93% of contacted people have a very positive or positive image of the Chilean president, and at least 2.97% have a negative or very negative image, and at least 1.1% don't know about the image of the Chilean president;

  or

- At most 91.97% of contacted people have a negative or very negative image of the Chilean president, and at most 6.93% have a very positive or positive image, and 1.1% don't know;

  or

- At most 90.1% don't know about the image of the Chilean president, and at least 6.93% have a very positive or positive image, and at least 2,97% have a negative or very negative image.

The idea of this report is to make explicit the uncertainty due to the missing data. However, it is hard to understand.

6.2 Proposal of reporting

The idea of the above report is to make explicit the uncertainty due to missing data: it is the uncertainty of the induction that is made taking into account the observed data and the population of interest on which the induction is performed. In this section we will suggest an alternative way of reporting the results. To do so, we note that

$$P(y = t \mid s = 1, c = 1)P(c = 1 \mid s = 1) = P(y = t, c = 1 \mid s = 1) \qquad t \in \{a, b, c\}.$$

The probability on the right-hand side of the above equality corresponds to the proportion of respondents who chose level t for the random variable y and answered the survey. It should be emphasized that these joint probabilities represent all what we observe. More precisely, $P(c = 0 \mid s = 1)$ corresponds to the proportion of selected person who refuses to answer the survey and, therefore

$$P(c = 1 \mid s = 1) = 1 - P(c = 0 \mid s = 1)$$

is the proportion of persons who provides information. This proportion can be decomposed with respecto to the random variable y, that is,

$$\begin{aligned} P(c = 1 \mid s = 1) &= P(y = a, c = 1 \mid s = 1) + P(y = b, c = 1 \mid s = 1) + \\ &\quad P(y = c, c = 1 \mid s = 1). \end{aligned}$$

Each of the summands corresponds to the proportion of respondents who choose a level of y and agreed to answer the survey. Furthermore, taking into acocunt that

$$\begin{aligned} 1 &= P(c = 1 \mid s = 1) + P(c = 0 \mid s = 1) \\ &= P(y = a, c = 1 \mid s = 1) + P(y = b, c = 1 \mid s = 1) + \\ &\quad P(y = c, c = 1 \mid s = 1) + P(c = 0 \mid s = 1), \end{aligned}$$

this decomposition reflects not only all the available information with respect to the sample space (population of interest), but also reveals the uncertainty induced by the missing data through $P(c = 0 \mid s = 1)$. Note also that

$$P(y = t, c = 1 \mid s = 1)$$

corresponds to the lower bound of $P(y = t \mid s = 1)$ for each $t \in \{a, s, c\}$. This decomposition can be visualized as it is shown at Figure 6.1. This example shows how great is the uncertainty of the induction made from the survey data: it can be seen at a glance, and therefore the report should consist of this type of graphics.

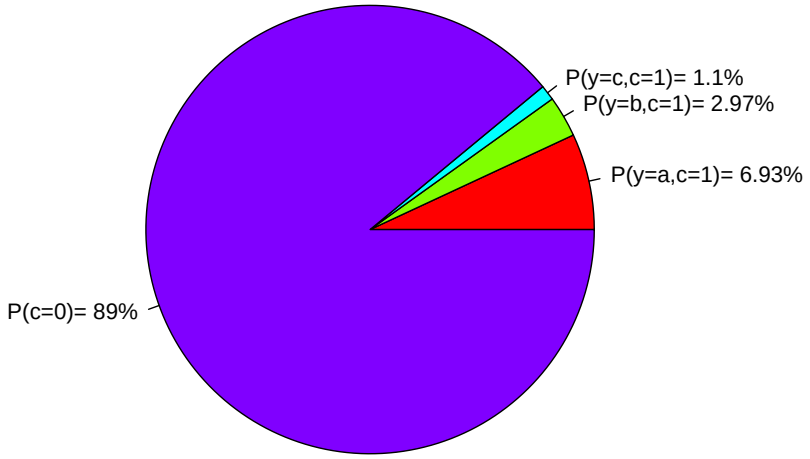


Figure 6.1: Making explicit the uncertainty through the observed information

Chapter 7

Contaminated outcomes

7.1 Motivation

Social researchers[1] typically use the term *data errors* to describe situation in which the available data imperfectly measure variables of interest. Data errors produce identification problems, some of which we will study in this chapter.

Data errors can be conceptualized through a mixture model, which views the available data as realizations of a probability mixture of the variable of interest and another random variable. Let each member j of population J have a pair of outcomes $(\mathrm{y}_j^*, \mathrm{e}_j)$ in the space $Y \times Y$. Let the random variable $(y^*, e) : J \longrightarrow Y \times Y$ have distribution $P(y^*, e)$. Let y^* be the outcome of interest. Let a sampling process draw persons at random from J. The mixture model views the available data as realizations of the probability mixture

$$y \doteq y^* z + e(1 - z), \tag{7.1.1}$$

where z is an unobservable binary random variable indicating whether e or y^* is observed. Thus, y^* is observed if $z = 1$ and e is observed if $z = 0$. Realizations of y with $z = 0$ are said to be data errors, those with $z = 1$ are said to be *error-free*, and y itself is said to be a *contaminated measure of* y^*.

Researchers often assume that the *error probability* $P(z = 0)$ is known, or at least that it can be bounded non-trivially from above. Now, let $p \doteq P(z = 0)$. The identification problem is displayed by the Law of Total Probability:

$$P(y) = P(y \mid z = 1)(1 - p) + P(y \mid z = 0)p, \tag{7.1.2}$$

and

$$P(y^*) = P(y \mid z = 1)(1 - p) + P(y^* \mid z = 0)p. \tag{7.1.3}$$

The sampling process reveals only the distribution $P(y)$ on the left side of (7.1.2). Empirical knowledge of $P(y)$ per se is uninformative about $P(y \mid z = 1)$ and hence about $P(y^*)$. However, informative

[1] This chapter is based on J. Horowitz and Manski (1995).

identification regions emerge if knowledge of $P(y)$ is combined with a non-trivial upper bound on p. This type of hypothesis can be justified in some applications; see the example discussed in Section 7.3.

We will study the identification of two outcomes distributions: the distribution of error-free realizations

$$P(y \mid z = 1) = P(y^* \mid z = 1); \tag{7.1.4}$$

and the marginal distribution $P(y^*)$ of the outcome of interest, These two distributions generally are different, but the are identical if the occurrence of data errors is statistically independent of the outcome of interest.

7.2 Partial identification of $P(y^* \mid z = 1)$ and $P(y^*)$

The main results are summarized in the following proposition:

Proposition 7.1

(a) Let p known, with $p < 1$. Then the identification regions for $P(y \mid z = 1)$ and $P(y^*)$ are

$$H_p[P(y \mid z = 1)] \doteq \Gamma_Y \cap \left\{ \frac{P(y) - p\gamma}{1 - p},\ \gamma \in \Gamma_Y \right\}, \tag{7.2.5}$$

and

$$H_p[P(y^*)] \doteq \{(1 - p)\eta + p\gamma,\ (\eta, \gamma) \in H_p[P(y \mid z = 1)] \times \Gamma_y\}. \tag{7.2.6}$$

(b) Let $\delta > 0$ and $p + \delta < 1$. Then

$$H_p[P(y \mid z = 1)] \subset H_{p+\delta}[P(y \mid z = 1)], \quad H_p[P(y^*)] \subset H_{p+\delta}[P(y^*)].$$

(c) For given $\lambda < 1$, let is be known that $p \leqslant \lambda$. Then the identification regions for $P(y \mid z = 1)$ and $P(y^*)$ are $H_\lambda[P(y \mid z = 1)]$ and $H_\lambda[P(y^*)]$.

Proof:

(a) By (7.1.2), the joint identification region for $[P(y \mid z = 1), P(y \mid z = 0)]$ is

$$H_p[P(y \mid z = 1), P(y \mid z = 0)] \doteq \{(\eta, \gamma) \in \Gamma_Y \times \Gamma_Y \ :\ P(y) = (1 - p)\eta + p\gamma\}.$$

The identification region (7.2.5) follows immediately. The identification region (7.2.6) follows from (7.1.3) because the sampling process is uninformative about $P(y^* \mid z = 0)$.

(b) To show that $H_p[P(y \mid z = 1)] \subset H_{p+\delta}[P(y \mid z = 1)]$, consider $(\eta, \gamma) \in H_p[P(y \mid z = 1), P(y \mid z = 0)]$. Let the error probability increases from p to $p + \delta$. Let

$$\gamma_\delta \doteq \frac{\gamma p + \eta\delta}{p + \delta}.$$

Then γ_δ is a probability distribution because it belongs to $[0,1]$ and $(\eta, \gamma_\delta) \in \Gamma_Y \times \Gamma_Y$ solves the equation

$$P(y) = \eta(1-p-\delta) + \gamma_\delta(p+\delta).$$

Hence $(\eta, \gamma_\delta) \in H_{p+\gamma}[P(y \mid z=1), P(y \mid z=0)]$.

The inclusion $H_p[P(y^*)] \subset H_{p+\delta}[P(y^*)]$ follows from the result above and (7.2.6).

(c) The identification region for $P(y \mid z=1)$ is

$$\bigcup_{p \leqslant \lambda} H_p[P(y \mid z=1)].$$

Part (b) showed that

$$\bigcup_{p \leqslant \lambda} H_p[P(y \mid z=1)] = H_\lambda[P(y \mid z=1)].$$

Similarly for obtaining the identification region $H_\lambda[P(y^*)]$.

□

Observe that, whatever the error probability p may be, the distribution $P(y)$ of observed outcomes necessarily belongs to $H_p[P(y \mid z=1)]$. Hence the hypothesis

$$P(y^*) = P(y \mid z=1) = P(y)$$

is not refutable.

7.3 Contamination through imputation

Organizations conducting major surveys often impute for missing data and report statistics that are mix of real and imputed data. Consider a researcher who sees the reported statistics but who does not see the raw survey data and does not know the imputation rule used when data are missing. To this researcher, imputation are data errors that may be analyzed using previous findings.

As it was mentioned in Section 4.5.1, data on the household income distribution in the U.S.A. are collected annually in the Current Population Survey (CPS). Summary statistics are published by the U.S. Bureau of the Census in Series P-60 of its Current Population Reports. Two sampling problems identified by the Bureau of the Census are *interview nonresponse*, wherein some households in the CPS sampling frame are not interviewed, and *item nonresponse*, wherein some of those interviewed do not provide complete income responses. The Bureau of Census uses available information to impute missing income data. The Bureau mixes actual and imputed data to produce the household income statistics reported in its Series P-60 publications.

Using the notation introduced above, y^* corresponds to the income a household selected for interview in the CPS would report if it were to complete the survey, e is the income the Bureau would impute to the household if the household were not to complete the survey, and $z=1$ if a CPS household actually completes the survey. $P(y \mid z=1)$ is the distribution of income reported by those CPS households

who complete the survey, $P(y^*)$ is the distribution of income that would be reported if all households in the CPS sampling frame were to complete the survey, and $P(y)$ is the distribution of household income found in the Series P-60 publications. The error probability p is the probability that a CPS household does not complete the survey.

Consider the year 1989. U. S. Bureau of the Census states that in the March 1990 CPS, which provides data on income during 1989, approximately 4.5% of the 60.000 households in the sampling frame were not interviewed and that incomplete income data were obtained from approximately 8% of the persons in interviewed households. The Bureau's publication does not report how the latter group are spread across households but we can be sure that no more than $(0.08) \times (0.95) = 0.076$ of the households have item nonresponse, so

$$\lambda = 0.076 + 0.045 = 0.121$$

provides an upper bound on p.

Now consider $P(y)$. U. S. Bureau of the Census provides findings for each of 21 income intervals (in thousands of dollars):

$P[0,5)$	$=$	0.053	$P[35,40)$	$=$	0.066	$P[70,75)$	$=$	0.018
$P[5,10)$	$=$	0.103	$P[40,45)$	$=$	0.060	$P[75,80)$	$=$	0.015
$P[10,15)$	$=$	0.097	$P[45,50)$	$=$	0.048	$P[80,85)$	$=$	0.013
$P[15,20)$	$=$	0.092	$P[50,55)$	$=$	0.043	$P[85,90)$	$=$	0.009
$P[20,25)$	$=$	0.087	$P[55,60)$	$=$	0.032	$P[90,95)$	$=$	0.008
$P[25,30)$	$=$	0.083	$P[60,65)$	$=$	0.028	$P[95,100)$	$=$	0.006
$P[30,35)$	$=$	0.076	$P[65,70)$	$=$	0.023	$P[100,\infty)$	$=$	0.039

Let us suppose the auxiliary assumption on $P(y)$ that income is distributed uniformly within each interval expect the last. We may obtain bounds for $P(y \mid z = 1)$ and $P(y^*)$.

Consider the probability that household income is below \$30.000. We have

$$P[0,30) = 0.515.$$

Hence the bound on $P(y \leqslant 30 \mid z = 1)$ is $[0.448,\ 0.586]$ because

$$\frac{0.515 - \lambda\gamma}{1 - \lambda}$$

for $\gamma = 0$ takes the value $0.515/0.879 = 0.586$; and for $\gamma = 1$ takes the value of

$$\frac{0.515 - 0.121}{0.879} = 0.448.$$

Similarly, the bound on $P(y^* \leqslant 30)$ is $[0.394,\ 0.636]$.

These bounds illustrate what one can learn about the distribution of income using the statistics reported in Series P-60 publications.

Chapter 8

A Qualitative Analysis of Peru's 2021 Presidential Elections: A Partial Identification Strategy

8.1 Introduction

Last June[1], the Peruvian population went to the polls to cast their vote to elect the republic's next president. However, the election has been under the shadow of questioning due to allegations of fraud. Some of the allegations of fraud point out that a candidate cannot obtain zero votes in a polling place. Besides, it is claimed that an unusual number of votes favoring the candidate of the Perú Libre party occurred. This situation has been observed in the frequency distribution of the votes cast. Other accusations are related to the frequency with which extreme observations concerning the number of votes occur, *i.e.*, polling places with all voters supporting a single candidate. However, neither of these arguments have been proven from a technical point of view.

The main objective of this study is to propose, through a strategy of partial identification and parameterization of fraud, an analytical tool that allows us to put ourselves in different electoral scenarios. For this, on the one hand, we make explicit the implications of having different levels of participation and, therefore, what we can affirm regarding the electoral results. On the other hand, we develop a model that incorporates different forms and amounts of electoral fraud in different contexts to generate concrete hypotheses regarding the impact that the existence of electoral fraud may have.

8.2 Motivation

On May 18th, 2003, voters in Belgium went to the polls. Voting was done on a computer in many regions, but the system had a backup. Late that night, as the votes were being tabulated, one of the

[1]This chapter reproduces a 2021-draft paper by Pablo Vargas, Ernesto San Martín and Mauricio Castro.

election officials detected a problem with the results from Schaerbeek, a municipality in central Brussels. Maria Vindevoghel, a little-known candidate with her party, received more votes than possible. The officials started a votes recount, obtaining that the vote totals for every candidate were precisely the same as before, except for Maria that was less than the original by 4,096. Everything about the hardware seemed to be in perfect working order. Computers worked using binary strings of zeros and ones, each corresponding to a power of two. A string of bits representing the number of votes Maria received was somewhere inside the computer tabulating all the votes. It started the day all zeros, and then as each vote for her came in, it would increment by one. Physically, this is done by turning on a transistor for one and turning it off for zero. What is remarkable about the number 4096 is that it is exactly a power of two (two to the power of twelve). That is the 13th bit so, for Maria Vindevoghel to receive an extra 4096 votes, the 13th bit had to flip from a zero to a one.

An investigation concluded that a single event upset (SEU) gave the candidate 4,096 extra votes. SEU is a change of state caused by one single ionizing particle striking a sensitive node (*e.g.*, memory "bit") in the microelectronic device, a type of soft error. The error is soft because the device has not been damaged, but the user could erase it and rewrite it with no problems. However, what happens if we do not have a backup, or if it is a person with political biases who does the counting of the votes. The fraud takes on a panoply of forms that we will deepen later. The question is whether we can determine if a minor distortion can affect the elections' outcome and if this constitutes a "soft error" that can be corrected with no problems.

8.3 Background on electoral fraud

The study of electoral fraud is not something new. There are numerous contributions in the literature devoted to the statistical modeling of the behavior of an election to infer whether or not the results obtained show signs of fraud. In other words, these *electoral forensics* approaches aim to detect manipulations of ballots and the vote count. Leemann and Bochsler (2014) argue that a quickly growing literature has been developed on two types of tools of vote forensics. One part of the literature discusses whether the analysis of single digits of the reported electoral results at the ward level can reveal that these numbers are based on the actual count of the votes, or whether they have been altered, relying on Benford's law on the frequency distribution of digits in numbers. The other part of the literature investigates the plausibility of electoral results from wards and is based on detailed information.

Lehoucq (2003) proposes that there is a fine line between fraud and political pressure. To take a particularly glaring example, landlords are not guilty of electoral fraud when all retainers vote for their party, regardless of whether peasants have been duped into voting in favor of the landlord's party or wish to avoid the reprisals associated with voting against their employer. Moreover, these activities are not fraudulent unless a law has been broken, even if they are morally reprehensible. A vital advantage of this legalistic conception of fraud is that it allows assessing the location of the boundaries between acceptable and unacceptable political activity. The research on electoral fraud tells a consistent story. First, fraud takes on a panoply of forms. It ranges from procedural violations of electoral law (that may or may not intend to distort results) to the outright use of violence to intimidate voters and poll watchers.

Second, ballot-rigging does not appear to be decisive most of the time. The colorful history of vote fabrication probably exaggerates its role in determining election outcomes. Fraud, nevertheless, undermines political stability because, in close races, it can be decisive. Third, political competition shapes the rhythm and nature of electoral fraud. Studying electoral fraud is not easy because it is supposed to remain clandestine. However, analyzing ballot rigging is far from impossible.

López-Pintor (2010) refers to electoral fraud as deceptive or negligent interference with the electoral process that intends to prevent the outcome from reflecting the will of the people. There are two main categories of electoral fraud: outcome-determinative fraud, where the fraud affects the outcome of the election such that the winners and losers are different from what they would have been had the fraud not been committed; and nonoutcome-determinative fraud, when the outcomes are not affected (i.e. the winners and losers would be the same even absent the commission of fraud). Both types of electoral fraud entail criminal behavior that should be punished according to the law. However, outcome-determinative fraud has more serious political implications in that it allows a party or candidate to take over public positions contrary to the popular will.

For example, Hausmann and Rigobon (2011) have studied Venezuela's Recall Referendum in 2004, defining electoral fraud as the difference between the elector's intent and the official vote tally. Since the voters' intent cannot be observed directly, the authors considered independent variables correlated with the voter's intent but not with the fraud. These variables were the number of signatures in favor of holding a recall referendum and the proportion of *yes* responses in the exit polls. In this context, the authors consider an instrumental variable approach where the intention of voters is used to model the number of signatures in favor of holding a recall referendum and the proportion of *yes* responses.

In this thesis, we will talk about election forensics, which is the field devoted to using statistical methods to determine whether the results of an election accurately reflect the intentions of the electors. Election forensics techniques apply statistical methods to low-level aggregates of votes such as polling stations or ballot box counts of eligible voters and votes recorded for parties (Ferrari, McAlister, & Mebane Jr, 2018). It is important to consider whether any of the "frauds" estimated represent genuine frauds in the sense of maleficent acts arising from regular political activity such as strategic voting. In some cases, we can draw on supplemental information to reach judgments, but in other cases, not (Mebane Jr, 2016).

Another situation where the statistical analysis of the election intends to detect systematic fraud is described by Klimek, Yegorov, Hanel, and Thurner (2012). In that paper, the authors developed a parametric model for quantifying systematic irregularities in an election process. This model includes two parameters representing the incremental and extreme fraud situations. The first one refers to the situation where votes (ballots) are taken away from both the nonvoters and the opposition, and they are added to the winning party's votes (ballots), while the latter one means that almost all votes (ballots) from the nonvoters and the opposition are added to the winning party's votes (ballots). In words of Ferrari et al. (2018) two kinds of election fraud refer to how many of the opposition and nonvoters votes are shifted: with "incremental fraud," moderate proportions of the votes are shifted; with "extreme fraud," almost all of the votes are shifted. The model relies on the Gaussian assumption for the number of votes and turnouts, and the conclusions are reported for different levels of data aggregation (territorial units).

8.3.1 Measuring electoral fraud

Not all authors agree with the same assumptions or methods to determine whether or not there was electoral fraud and the magnitude. For example, Klimek et al. (2012) claim that it is not enough to discuss the approximate normality of participation, vote, or logarithmic distributions of the voting rate to decide whether the election results may be corrupt. These methods can lead to ambiguous signals because the results depend to a great extent on the level of aggregation of the electoral data. Therefore, we can agree that the levels of aggregation are a critical dimension to consider when analyzing the behavior of electoral fraud. Ideally, the results and conclusions reached are robust and consistent with these different levels of aggregation.

Other works focus more on a point estimation of electoral fraud than on determining whether a voting process was fraudulent or not. For example, Christensen and Schultz (2014) state that election fraud in established democracies typically involves only a tiny percentage of votes in a closely contested race. Such a small number of fraudulent ballots can easily be concealed in a sea of otherwise legitimate ballots. Those who steal elections rely on this fact to hide their activities. For example, they know that adding or changing 10% or 20% of the ballots in a few precincts will not stand out as anomalous, making it nearly impossible to differentiate legitimately cast votes from fraudulent votes.

Additionally, a recent study of Gonzalez (2021) examines the impact of cell phone access on election fraud during the 2009 Afghan presidential election using a spatial regression discontinuity design. Polling centers just inside coverage report a drop in the share of fraudulent votes of 4 percentage points, while the likelihood of a fraudulent station decreases by eight percentage points. Their results provide considerable evidence that cell phone coverage lowers electoral fraud. It is important to stress that the estimated effects exhibit a considerable degree of spatial heterogeneity: average impacts in the south and east are economically and statistically significant while non-distinguishable from zero in other parts of the country. These results indicate that there may be patterns or spatial characteristics related to elections that strongly mimic the observed pattern of fraud.

8.3.2 Other perspectives

Some studies analyze electoral fraud from other perspectives somewhat different from those previously exposed. For example, the work of Ichino and Schündeln (2012) extends the empirical scholarship on electoral fraud to the study of misconduct at the pre-election stage. It presents findings from a randomized field experiment on the effect of domestic observers on the extent of irregularities in voter registration in Ghana in 2008. This research design and analysis explicitly consider the spillovers that may result from the organization of political parties across multiple electoral areas and the capacity of party agents to transport supporters from one electoral area to another.

On the other hand, Rundlett and Svolik (2016) wonders why do incumbents who could arguably win a clean election engage in fraud. This question is motivated because, according to those authors, most fraud is ultimately executed by a large number of local operatives with an incentive to conceal their actions. Each agent understands that the difference between her promotion and prosecution rests on whether her involvement in fraud will result in the incumbent's eventual victory or defeat. In turn, the

agent's perception of the incumbent's popularity and the ensuing burden on their collective complicity plays a crucial coordinating role. The aggregate result is too much fraud for incumbents who need it the least and too little fraud for those who need it the most. The seeming invincibility of some incumbents and the surprising fragility of others are thus two sides of the same political logic.

Finally, different papers that we have mentioned and others that we will present later analyze the role of observers. For example, A. T. Little et al. (2012) states that the good behavior induced by a convincing election result gives an incentive to commit fraud independent of winning the election by the official tally. The solid incentive for invite election monitoring to reduce (but not eliminate) expectations about how much fraud will occur are consistent with incumbents voluntarily inviting monitors and still cheating in front of them.

8.4 Data

On April 11, 2021, Peru held its general elections where the next president, two vice presidents, 130 congressmen, and 5 Andean parliamentarians were elected for the 2021-2026 government period. In Peru, there are 27 electoral districts: 24 departments, Metropolitan Lima, the Constitutional Province of Callao, and residents abroad. The electoral roll consisted of 24.290.921 voters in the national territory and 997.033 residents abroad, 12.745.409 are women, equivalent to 49,60% and 12.542.545 are men, equivalent to 50,40%. Metropolitan Lima concentrates 29,89% of voters. The total electoral roll included 25.287.954 qualified voters who exercised their right to vote in 12.846 voting places enabled for this process.

In Peru, the president and vice-presidents of the Republic are elected by direct suffrage and a single electoral district. A second electoral round is held if no presidential candidate reaches more than 50% of the validly cast votes. For the first electoral round, 18 presidential candidacies were presented. 17.713.716 of voters participated, corresponding to 70,048% of the electoral roll, and institutions such as the Organization of American States (OAS) and the European Union (EU) participated as international electoral observers. In the results, there were 14.400.630 valid votes (81,296% of votes cast) where the first two candidates were those of Pedro Castillo, Peru Libre National Political Party, with 2.724.752 (18,921% of valid votes) and Keiko Fujimori, Fuerza Popular, with 1.930.762 (13,407% of valid votes). Therefore, these two candidates had to participate in the second round for Presidential Election 2021.

On June 6, 2021, the second round for Presidential Election 2021 was held, using the same electoral roll of the first round. In contrast to the first round, electoral participation in the second round increased by 4,52 percentage points, reaching 74,568% of the electoral roll. In the results, there were 17.628.497 valid votes (93,486% of votes cast) where Pedro Castillo prevailed with 8.836.380 (50,126% of valid votes) over Keiko Fujimori with 8.792.117 (49,874% of valid votes). In other words, there were only 44.263 votes of difference, corresponding to 0,175% of the electoral roll and 0,251% of the validly cast votes.

We collect data from the National Office of Electoral Processes (ONPE). In particular, we use disaggregated data at the *mesa* level, which is the most disaggregated level that exists, for the first (Gobierno de Datos, 2021a) and second presidential round (Gobierno de Datos, 2021b). The data already includes

the modifications, nullity requests, and corrections made by the pertinent institutions.

8.4.1 Share of valid votes

The valid votes consider all the votes that went to any candidate, that is, all the votes that are not null or blank. Figure 8.4.1 shows that the support for the Peru Libre candidate and the Fuerza Popular candidate is territorially heterogeneous, with clear majority support for Pedro Castillo in southern Peru and general in rural areas, sometimes called "Perú profundo" referring to marginalized areas in terms of political and economic power and access to some essential services. At the same time, the majority of support for Keiko Fujimori can be observed in more urban areas, particularly in large cities such as Lima that make up a third of the electorate.

8.4.2 Voter turnout

As we mentioned in Section 8.3, an important consideration to analyze the integrity of elections is that the number of votes does not increase or decrease with some irregular pattern. Some previous studies about electoral fraud analyze the relationship between participation electoral process and the votes for any of the candidates at the most disaggregated level possible. In this context, in Figure 8.2, we can see a *mesa* level graph that shows the percentage of electoral participation on the x-axis and the y-axis the percentage of votes for the respective party. From this figure, we conclude that we do not observe any anomaly. An irregularity would be, for example, *mesas* that have greater electoral participation show a greater preference for a candidate, or that in *mesas* where the participation is close to 100% all the votes go in favor of a single candidate. Those situations would make one suspect that votes favor some candidates or remove votes from the opponent.

8.5 Participation and voting in first and second round

Voter turnout is not necessarily random. We could not even claim that it is unrelated to fraud. For example, more observers can make it more difficult for elections to be stolen by reducing the rate of fraud and increasing the likelihood that fraud will be caught (Hyde, 2007). However, Beaulieu and Hyde (2009) argue that international benefits for democratic elections give electoral autocrats the incentive to invite international observers and manipulate elections to minimize international criticism. This increase in "strategic manipulation" has led to changed incentives for opposition political parties, which have the most to lose from a manipulated but an internationally certified election. Consequently, international monitors increase boycott probability, which means greater electoral abstention. On the other hand, Birch (2010) show that perceptions of electoral integrity are indeed positively associated with propensity to vote.

Let C denote a vector that summarizes the information related to both geographical characteristics of the country and electoral organization, for example, a level of aggregation such as regions, electoral

districts, polling stations, or *mesas*. For $i = 1, 2$, we define the following binary random variables:

$$V_i = \begin{cases} 1, & \text{if the voter participated in the } i\text{-th round of the election;} \\ 0, & \text{if not.} \end{cases}$$

It should be remarked that both V_1 and V_2 are defined on the same finite sample space M, namely the set of voters. This means that the electoral register is the same for both rounds, as in Peru for the first and second rounds.

8.5.1 Partial identification of the joint distribution of participation

It is impossible to identify the joint distribution of (V_1, V_2) conditionally on C with the public and available data since we do not know if a voter has voted on the first and second rounds or only in one of them. On the other hand, the available data only allows us to identify the corresponding marginal distributions of V_1 and V_2, conditionally on C. However, using the Frechét inequality (Fréchet, 1960a), it is possible to partially identify the joint distribution of (V_1, V_2) conditionally on C, namely

$$\begin{aligned} \max\{0, P(V_1 = v_1 \mid C) + P(V_2 = v_2 \mid C) - 1\} &\leqslant P(V_1 = v_1, V_2 = v_2 \mid C) \\ &\leqslant \min\{P(V_1 = v_1 \mid C), P(V_2 = v_2 \mid C)\}, \quad (v_1, v_2) \in \{0, 1\}^2. \end{aligned} \tag{8.5.1}$$

Let us introduce the following convention: we will say that *a partial identification interval is non-informative* when it is equal to $[0, 1]$. Similarly, we will say that *the upper bound of the interval is non-informative* when it is equal to 1; and *the lower bound of the interval is non-informative when it is equal to 0*. Therefore, the following comments seem to be relevant:

1. The upper bound of the partial identification interval (8.5.1) is always informative. This is due to the fact that the proportion of participation in each round is always less than 100%.

2. The lower bound of the partial identification interval (8.5.1) is informative in the following cases:

   (a) For $(v_1, v_2) = (1, 1)$, if $P(V_1 = 1 \mid C) > P(V_2 = 0 \mid C)$, that is, when the participation rate in the first round is higher than the non-participation rate in the second round.

   (b) For $(v_1, v_2) = (1, 0)$, if $P(V_1 = 1 \mid C) > P(V_2 = 1 \mid C)$, that is, when the participation rate in the first round is higher than the participation rate in the second round.

   (c) For $(v_1, v_2) = (0, 1)$, if $P(V_1 = 0 \mid C) > P(V_2 = 0 \mid C)$, that is, when the non-participation rate in the first round is higher than the non-participation rate in the second round.

   (d) For $(v_1, v_2) = (0, 0)$, if $P(V_1 = 0 \mid C) > P(V_2 = 1 \mid C)$, that is, when the non-participation rate in the first round is higher than the participation rate in the second round.

8.5.2 Partial identification of the conditional probabilities of participation

Using (8.5.1), we will develop a partial identification analysis for the conditional probabilities $P(V_2 = v_2 \mid V_1 = v_1, C)$ and $P(V_1 = v_1 \mid V_2 = v_2, C)$, for $(v_1, v_2) \in \{0,1\}^2$. We focus on those that will be useful later.

- For $(v_1, v_2) = (1, 1)$,

$$\begin{aligned} P(V_2 = 1 \mid V_1 = 1, C) &= \frac{P(V_1 = 1, V_2 = 1 \mid C)}{P(V_1 = 1 \mid C)} \\ &\leqslant \min\left\{1, \frac{P(V_2 = 1 \mid C)}{P(V_1 = 1 \mid C)}\right\} \\ &\geqslant \max\left\{0, \frac{P(V_2 = 1 \mid C) + P(V_1 = 1 \mid C) - 1}{P(V_1 = 1 \mid C)}\right\}. \end{aligned}$$

  This interval is informative if $P(V_1 = 0 \mid C) < P(V_2 = 1 \mid C) < P(V_1 = 1 \mid C)$.

- For $(v_1, v_2) = (1, 1)$,

$$\begin{aligned} P(V_1 = 1 \mid V_2 = 1, C) &= \frac{P(V_1 = 1, V_2 = 1 \mid C)}{P(V_2 = 1 \mid C)} \\ &\leqslant \min\left\{1, \frac{P(V_1 = 1 \mid C)}{P(V_2 = 1 \mid C)}\right\} \\ &\geqslant \max\left\{0, \frac{P(V_2 = 1 \mid C) + P(V_1 = 1 \mid C) - 1}{P(V_2 = 1 \mid C)}\right\}. \end{aligned}$$

  This interval is informative if $P(V_2 = 0 \mid C) < P(V_1 = 1 \mid C) < P(V_2 = 1 \mid C)$.

- For $(v_1, v_2) = (1, 0)$,

$$\begin{aligned} P(V_2 = 0 \mid V_1 = 1, C) &= \frac{P(V_1 = 1, V_2 = 0 \mid C)}{P(V_1 = 1 \mid C)} \\ &\leqslant \min\left\{1, \frac{P(V_2 = 0 \mid C)}{P(V_1 = 1 \mid C)}\right\} \\ &\geqslant \max\left\{0, \frac{P(V_2 = 0 \mid C) + P(V_1 = 1 \mid C) - 1}{P(V_1 = 1 \mid C)}\right\}. \end{aligned}$$

  This interval is informative if $P(V_1 = 0 \mid C) < P(V_2 = 0 \mid C) < P(V_1 = 1 \mid C)$.

- For $(v_1, v_2) = (0, 1)$,

$$\begin{aligned} P(V_1 = 0 \mid V_2 = 1, C) &= \frac{P(V_1 = 0, V_2 = 1 \mid C)}{P(V_2 = 1 \mid C)} \\ &\leqslant \min\left\{1, \frac{P(V_1 = 0 \mid C)}{P(V_2 = 1 \mid C)}\right\} \\ &\geqslant \max\left\{0, \frac{P(V_2 = 1 \mid C) + P(V_1 = 0 \mid C) - 1}{P(V_1 = 1 \mid C)}\right\}. \end{aligned}$$

  This interval is informative if $P(V_2 = 0 \mid C) < P(V_1 = 0 \mid C) < P(V_2 = 1 \mid C)$.

From these intervals, the following analysis can be made:

1. If $P(V_2 = 1 \mid C) < P(V_1 = 1 \mid C)$, that is, if the participation rate in the second round is lower than the participation rate in the first round, then the upper bound for $P(V_2 = 1 \mid V_1 = 1, C)$ is informative (different from 1) and the upper bound for $P(V_1 = 1 \mid V_2 = 1, C)$ is non-informative (is exactly one). On the other hand, the informativeness criterion of the lower bound for $P(V_2 = 1 \mid V_1 = 1, C)$ and $P(V_1 = 1 \mid V_2 = 1, C)$ is obtained in a similar way, namely, it is informative if $P(V_1 = 0 \mid C) < P(V_2 = 1 \mid C)$ or, equivalently, $P(V_1 = 1 \mid C) > P(V_2 = 0 \mid C)$. Thus, the minimum values for $P(V_2 = 1 \mid V_1 = 1, C)$ and $P(V_1 = 1 \mid V_2 = 1, C)$ are different from zero.

2. If $P(V_2 = 1 \mid C) > P(V_1 = 1 \mid C)$, that is, if the participation rate in the second round is greater than the participation rate in the first round, then the upper bound for $P(V_2 = 1 \mid V_1 = 1, C)$ is non-informative and the upper bound for $P(V_1 = 1 \mid V_2 = 1, C)$ is informative. The lower bound for $P(V_2 = 1 \mid V_1 = 1, C)$ and $P(V_1 = 1 \mid V_2 = 1, C)$ is informative if $P(V_1 = 0 \mid C) < P(V_2 = 1 \mid C)$ or, equivalently, $P(V_1 = 1 \mid C) > P(V_2 = 0 \mid C)$.

3. If $P(V_1 = 0 \mid C) < P(V_2 = 0 \mid C)$, that is, if the non-participation rate in the first round is lower than the non-participation rate in the second round, then the lower bound for $P(V_2 = 0 \mid V_1 = 1, C)$ is informative and the lower bound for $P(V_1 = 0 \mid V_2 = 1, C)$ is non-informative. On the other hand, the upper bound of $P(V_2 = 0 \mid V_1 = 1, C)$ and $P(V_1 = 0 \mid V_2 = 1, C)$ is informative if $P(V_2 = 0 \mid C) < P(V_1 = 0 \mid C)$ or, equivalently, $P(V_2 = 1 \mid C) > P(V_1 = 1 \mid C)$.

4. If $P(V_1 = 0 \mid C) > P(V_2 = 0 \mid C)$, that is, if the non-participation rate in the first round is greater than the non-participation rate in the second round, then the lower bound for $P(V_2 = 0 \mid V_1 = 1, C)$ is non-informative and the lower bound for $P(V_1 = 0 \mid V_2 = 1, C)$ is informative. The upper bound for $P(V_2 = 0 \mid V_1 = 1, C)$ and $P(V_1 = 0 \mid V_2 = 1, C)$ is informative if $P(V_2 = 0 \mid C) < P(V_1 = 0 \mid C)$ or, equivalently, $P(V_2 = 1 \mid C) > P(V_1 = 1 \mid C)$.

8.5.3 Partial identification of the distributions for each round

Let X be the vector of possible choices in the first round, including blank and null votes. Similarly, let Y be the vector of possible choices in the second round, including blank and invalid votes. Using the Law

of Total Probability, we have that

$$P(X = x \mid C = c) = P(X = x \mid C = c, V_1 = 1)P(V_1 = 1 \mid C = c) + P(X = x \mid C = c, V_1 = 0)P(V_1 = 0 \mid C = c).$$

The sampling process identifies $P(X = x \mid C = c, V_1 = 1)$ for $(x, c) \in \mathcal{X} \times \mathcal{C}$, where $\mathcal{X}$ and $\mathcal{C}$ are well defined collection of subsets, and $P(V_1 = v_1 \mid C = c)$ for $c \in \mathcal{C}$. However, $P(X = x \mid C = c, V_1 = 0)$ can not be identified because we do not know the preferences of the people who do not vote. Taking into account that $P(X = x \mid C = c, V_1 = 0) \in [0, 1]$, the plausible values of $P(X = x \mid C = c)$ that are coherent with the observed data are the following:

$$\begin{aligned} P(X = x \mid C = c, V_1 = 1)P(V_1 = 1 \mid C = c) &\leqslant P(X = x \mid C = c) \\ &\leqslant P(X = x \mid C = c, V_1 = 1)P(V_1 = 1 \mid C = c) + P(V_1 = 0 \mid C = c), \quad (x, c) \in \mathcal{X} \times \mathcal{C} \end{aligned} \tag{8.5.2}$$

Note that the width of each interval is equal to $P(V_1 = 0 \mid C = c)$, that is, the proportion of voters who, conditional on $C = c$, did not vote. Consequently, the higher this proportion, the higher the bounds for $P(X = x \mid C = c)$. Similarly, for the second round, we have

$$\begin{aligned} P(Y = y \mid C = c, V_1 = 1)P(V_1 = 1 \mid C = c) &\leqslant P(Y = y \mid C = c) \\ &\leqslant P(Y = y \mid C = c, V_1 = 1)P(V_1 = 1 \mid C = c) + P(V_1 = 0 \mid C = c), \quad (y, c) \in \mathcal{Y} \times \mathcal{C} \end{aligned} \tag{8.5.3}$$

Note that, from in (8.5.2) and (8.5.3), the length of the interval for $P(X = x \mid C = c)$ and $P(Y = y \mid C = c)$ is the proportion of citizens who did not participate in the presidential election.

8.5.4 Empirical analysis of participation and voting using partial identification

Based on the empirical participation percentages that we observe in each round, we can establish intervals for the probability of voting in the first and second rounds and the probability of voting in the second round conditional on a vote in the first, as we discussed in Subsection 8.5.2. This probability is of particular interest since the winner of the elections is defined in the second round.

Table 8.1 shows the analysis of the participation in each round (in each department), including the probability intervals, described previously. Electoral participation at this level of aggregation shows that the participation in the second round was more significant than the first one. If we analyze PUNO (a department in the south of Peru), we can observe that even with high participation in both elections (over 80%), the proportion of people who vote in both rounds is at least 64% and at most 81.91%.

Now, we focus on the probability of voting in the second round, conditional on voting in the first one. In the last columns, Table 8.1, we find lower and upper bounds. Note that, in all the Departments, the upper bound is 100%. However, it is important to remark that it is possible to find cases where this does not occur at a more disaggregated level.

Turning the attention to the lower bound, we found that PUNO presents more than 80% participation in both rounds. Consequently, the proportion of people who vote in the second round conditional on voting in the first one is at least one 78.13%. On the other hand, AMAZONAS presents a lower bound of 38.98%, with participation in both rounds about 60%.

	% of Participation		Voting in the First and Second Round		Voting in Second Round Conditional to Voting in First Round	
Departments	**First Round**	**Second Round**	**Lower**	**Upper**	**Lower**	**Upper**
AMAZONAS	60,11%	63,32%	23,43%	60,11%	38,98%	100,00%
ANCASH	69,26%	73,26%	42,52%	69,26%	61,39%	100,00%
APURIMAC	69,39%	71,96%	41,35%	69,39%	59,59%	100,00%
AREQUIPA	78,78%	79,69%	58,47%	78,78%	74,22%	100,00%
AYACUCHO	68,63%	71,92%	40,55%	68,63%	59,08%	100,00%
CAJAMARCA	62,60%	69,30%	31,90%	62,60%	50,96%	100,00%
CALLAO	75,23%	79,53%	54,76%	75,23%	72,79%	100,00%
CUSCO	73,52%	75,85%	49,37%	73,52%	67,16%	100,00%
HUANCAVELICA	67,55%	68,99%	36,54%	67,55%	54,09%	100,00%
HUANUCO	68,28%	69,10%	37,38%	68,28%	54,75%	100,00%
ICA	75,97%	79,87%	55,85%	75,97%	73,51%	100,00%
JUNIN	71,93%	74,37%	46,30%	71,93%	64,37%	100,00%
LA LIBERTAD	68,93%	74,36%	43,29%	68,93%	62,80%	100,00%
LAMBAYEQUE	71,40%	76,32%	47,72%	71,40%	66,83%	100,00%
LIMA	74,57%	79,66%	54,23%	74,57%	72,72%	100,00%
LORETO	60,93%	61,92%	22,85%	60,93%	37,50%	100,00%
MADRE DE DIOS	71,07%	73,66%	44,73%	71,07%	62,94%	100,00%
MOQUEGUA	77,20%	78,37%	55,57%	77,20%	71,99%	100,00%
PASCO	63,65%	67,23%	30,88%	63,65%	48,51%	100,00%
PIURA	66,75%	75,34%	42,09%	66,75%	63,06%	100,00%
PUNO	81,91%	82,09%	64,00%	81,91%	78,13%	100,00%
SAN MARTIN	69,17%	73,07%	42,23%	69,17%	61,06%	100,00%
TACNA	77,80%	79,35%	57,14%	77,80%	73,45%	100,00%
TUMBES	74,56%	78,32%	52,88%	74,56%	70,92%	100,00%
UCAYALI	66,28%	69,20%	35,47%	66,28%	53,52%	100,00%

Table 8.1: Analysis of participation in each round

Using the results provided in Subsection 8.5.3, we can provide lower and upper bounds for the probability of voting for a particular candidate in a specific department. Table 8.2 shows the votes in favor of Fuerza Popular in each department. For example, in the second round in TACNA, 25.97% of the citizens who voted supported Fuerza Popular (the participation here was 79.35%). If all the citizens who did not vote would vote by another option (Perú Libre, null or blank), then the proportion of support for Fuerza Popular would have been 20.61%. On the other hand, if all the citizens who did not vote would vote for Fuerza Popular, the support would be 41.26%.

Finally, Figure 8.3 shows the bounds described above by department and district. If we analyze the bounds at the district level, which is the lowest level of granularity in terms of territory, we find an essential heterogeneity in the electoral support of the Fuerza Popular party. The upper bounds are low at the south districts of Peru, principally in PUNO, CUZCO, and AYACUCHO. On the other hand, higher upper bounds can be found in LORETO districts.

8.6 Electoral fraud

In this section, we propose a simple probability argument for analyzing the fraud, focusing on the second round results. For simplicity, we assume that fraud occurs when votes are added (or removed) from one of the candidates. Note that further analysis can be performed assuming that votes were changed from one candidate to another. Our goal is to determine at what level of fraud the results change.

	First Round		P(X = x | C = c)		Second Round		P(Y = y | C = c)	
Fuerza Popular	**% votes**	**% Participation**	**Lower**	**Upper**	**% votes**	**% Participation**	**Lower**	**Upper**
AMAZONAS	9,68%	60,11%	5,82%	45,71%	31,18%	63,32%	19,74%	56,43%
ANCASH	10,98%	69,26%	7,60%	38,34%	38,14%	73,26%	27,94%	54,68%
APURIMAC	4,96%	69,39%	3,44%	34,06%	17,44%	71,96%	12,55%	40,59%
AREQUIPA	4,46%	78,78%	3,51%	24,73%	33,16%	79,69%	26,42%	46,73%
AYACUCHO	5,46%	68,63%	3,75%	35,12%	16,47%	71,92%	11,85%	39,93%
CAJAMARCA	7,96%	62,60%	4,98%	42,38%	26,87%	69,30%	18,62%	49,32%
CALLAO	12,85%	75,23%	9,67%	34,44%	62,66%	79,53%	49,83%	70,30%
CUSCO	3,60%	73,52%	2,65%	29,12%	15,85%	75,85%	12,03%	36,17%
HUANCAVELICA	4,17%	67,55%	2,82%	35,27%	14,40%	68,99%	9,93%	40,95%
HUANUCO	8,20%	68,28%	5,60%	37,32%	30,08%	69,10%	20,79%	51,69%
ICA	12,55%	75,97%	9,53%	33,56%	49,52%	79,87%	39,55%	59,68%
JUNIN	11,33%	71,93%	8,15%	36,22%	39,05%	74,37%	29,04%	54,68%
LA LIBERTAD	13,38%	68,93%	9,22%	40,30%	55,36%	74,36%	41,16%	66,80%
LAMBAYEQUE	17,37%	71,40%	12,40%	41,00%	54,04%	76,32%	41,24%	64,93%
LIMA	12,15%	74,57%	9,06%	34,49%	60,55%	79,66%	48,23%	68,58%
LORETO	12,27%	60,93%	7,48%	46,55%	48,05%	61,92%	29,75%	67,83%
MADRE DE DIOS	8,79%	71,07%	6,25%	35,17%	27,23%	73,66%	20,06%	46,40%
MOQUEGUA	4,03%	77,20%	3,11%	25,92%	25,44%	78,37%	19,94%	41,56%
PASCO	9,87%	63,65%	6,28%	42,64%	32,55%	67,23%	21,89%	54,66%
PIURA	18,66%	66,75%	12,46%	45,70%	55,56%	75,34%	41,86%	66,52%
PUNO	2,32%	81,91%	1,90%	19,99%	10,27%	82,09%	8,43%	26,34%
SAN MARTIN	10,61%	69,17%	7,34%	38,17%	40,61%	73,07%	29,68%	56,61%
TACNA	4,25%	77,80%	3,31%	25,51%	25,97%	79,35%	20,61%	41,26%
TUMBES	29,10%	74,56%	21,70%	47,14%	60,93%	78,32%	47,72%	69,40%
UCAYALI	15,68%	66,28%	10,39%	44,11%	48,27%	69,20%	33,40%	64,21%

Table 8.2: Analysis of voting in each round

8.6.1 Electoral fraud as a contaminated data problem

Suppose two candidates run in the second ballot of a presidential election. Voters may cast their vote for one of the candidates, none of them (blank vote), or they may invalidate their vote (null vote). The winner is the candidate who obtains an absolute majority of the valid votes.

Our framework to analyse the situation is probabilistic. As a matter of fact, the result of a candidate is the proportion of favorable votes with respect to the number of *valid* votes; this proportion is what statistician call a *relative frequency* of observations. The probability space, Ω_n, contains the identification number of all n voters with a valid vote. The function y from Ω_n into $\{0, 1\}$ is a random variable taking value 0 if the voter supports the first candidate and value "1" if she supports the other candidate. In this setting, candidate "0" wins if and only if

$$P_n(y = 0) > P_n(y = 1)$$

where P_n denotes the probability of an event. In this notation the index n reminds that the probability is computed on discrete events (given that Ω_n is discrete). The probability P_n is therefore a statistical frequency of votes. Note that this framework satisfies all axioms of probability (?, ?) even though Ω_n

contains the labels of the whole population. Sometimes, in statistics, the probability function P_n is called the *empirical measure*.

A fraud is defined as an observed value $y(i)$ that is not equal to the actual vote of the voter i. **we do not consider ballot stuffing!** This situation may occur from a variety of reasons (see e.g. XXX for a discussion) but all with one and only one consequence, being the mismatch between an observed vote and an actual vote. Let us define a binary random variable, denoted z, indicated if a vote is subject to fraud or not. The random variable z takes value 1 if there is no mismatch, while the value 0 in the case of fraud. The *observed* vote is, therefore, decomposed as follows:

$$y = y^* z + e^*(1 - z)$$

where y^* denotes the value of the *actual* vote and e^* denotes the value of the fraudulent vote. Obviously when $z = 1$ the observed vote equals the actual vote. In case where $z = 0$ the observed vote is e^* whose value is complementary to y^*.

The above model for the observed vote y is a *contaminated* model because the pure value of y^* is subject to an additive error; for details, see Manski (2007a).

8.6.2 Assessing electoral fraud

Fraudulent votes imply that the proportion of observed votes differ from the proportion of the actual vote. When the difference is too big, it might imply a different ranking of candidates. This is the situation we address in our study. An electoral fraud has an impact on the ranking if either

$$P_n(y = 1) > P_n(y = 0) \quad \text{and} \quad P_n(y^* = 1) < P_n(y^* = 0)$$

or

$$P_n(y = 1) < P_n(y = 0) \quad \text{and} \quad P_n(y^* = 1) > P_n(y^* = 0) \ .$$

These definitions are not operational because we cannot compute $P_n(y^* = t)$, with $t \in \{0, 1\}$, from data. However, we can raise the following question : what is the minimal fraud rate such that fraud has an impact on the ranking? We now provide a formal answer to this question.

Denote the fraud rate by $p = P_n(z = 1)$. From the law of total probability we may write

$$P_n(y = 1) = P_n(y = 1|z = 0)(1 - p) + P_n(y = 1|z = 1)p$$

According to our model for y, when there is no fraud we can replace $P_n(y = 1|z = 0)$ by $P_n(y^* = 1|z = 0)$. Similarly when there is fraud then $P_n(y = 1|z = 1)$ equals $P_n(y^* = 0|z = 1) = 1 - P_n(y^* = 1|z = 1)$. Finally, we get

$$P_n(y = 1) = P_n(y^* = 1|z = 0)(1 - p) + \{1 - P_n(y^* = 1|z = 1)\}p \ . \tag{8.6.4}$$

Moreover, the law of total probability also implies

$$P_n(y^* = 1) = P_n(y^* = 1|z = 0)(1 - p) + P_n(y^* = 1|z = 1)p \ .$$

In this last equation, we substitute equation (8.6.4) to get

$$P_n(y^* = 1) = P_n(y = 1) - \{1 - 2P_n(y^* = 1|z = 1)\}p \ .$$

On the right hand side, the probability $P_n(y = 1)$ is computable from data, but the conditional probability $P_n(y^* = 1|z = 1)$ is unknown. The last represents the proportion of voters who effectively voted for candidate "1", but their vote was changed in the final counting. Except if an expert has a prior about this proportion, we are agnostic with respect to this quantity. The only information we have is that this probability is between 0 and 1, which lead to the following bounds for the effective proportion of voters:

$$P_n(y = 1) - p \leqslant P_n(y^* = 1) \leqslant P_n(y = 1) + p \ . \tag{8.6.5}$$

By the same reasoning, we get a similar bounds for the result of candidate "0":

$$P_n(y = 0) - p \leqslant P_n(y^* = 0) \leqslant P_n(y = 0) + p \ . \tag{8.6.6}$$

Now that we framed the two probabilities of interested in bounds containing observable quantities on y, we can answer the above question. Suppose, say, that we observe that candidate "1" wins, *i.e.* $P_n(y = 0) < P_n(y = 1)$. Then a sufficient condition to claim that this ranking is not impacted by fraud is that the upper bound of the inequality (8.6.6) is lower than the lower bound of inequality (8.6.5), that is,

$$P_n(y = 0) + p < P_n(y = 1) - p$$

or, equivalently

$$p < \frac{1}{2}\{P_n(y = 1) - P_n(y = 0)\} \ .$$

Applied to the situation of Peru in 2021, where Pedro Casillo defeated Keiko Fujimori by 44,263 votes, our result indicates that the confidence about this ranking is conditioned by a fraud rate that is not larger than 0.0013, that is, a bit more that one in a thousand.

8.7 More than two candidates

In an election with more that two candidates, the question that can be raised is the following: what is the minimal fraud rate that preserves the rank between the candidates? The previous result for 2 candidates can be generalized, although we need an argument that does not rely on the binarity of the choice.

The observed vote is now a random variable y taking values in $\{1, 2, \ldots, J\}$ if we consider J candidates. Note that they may be a subset of candidates among a larger number of candidates, as we shall illustrate in the peruvian election below. Again we base on argument on the model with contamination, namely

$$y = y^* z + e^*(1 - z)$$

where y^* is the value of the actual vote among the J candidates, and e^* is the value of the reported vote. Note that now e^* can take any over $J - 1$ values different from y^*.

For any candidate j in $\{1, \ldots, J\}$, the observed vote is, using the law of total probabilities, decomposed as

$$P_n(y = j) = P_n(y = j|z = 1)(1 - p) + P_n(y = j|z = 0)p,$$

where the probability of fraud is $p = P_n(z = 1)$. On the right hand side, we have in principle no information about $P_n(y = j|z = 0)$; except the fact that it is between 0 and 1. Consequently, we can write

$$\frac{P_n(y = j) - p}{1 - p} \leqslant P_n(y = j|z = 1) \leqslant \frac{P_n(y = 1)}{1 - p}.$$

Thus, we have partly identified the conditional probability to observe a correctly reported vote for candidate "j". This partial identification is useful because it appears in the following decomposition of the actual vote:

$$P_n(y^* = j) = P_n(y = j|z = 1)(1 - p) + P_n(y^* = j|z = 0)p$$

and, therefore,

$$P_n(y = j) - p + P_n(y^* = j|z = 0)p \leqslant P_n(y^* = j) \leqslant P_n(y = 1) + P_n(y^* = j|z = 0)p\ .$$

In this last expression, we also bound the unknown conditional probability $P_n(y^* = j|z = 0)$ between 0 and 1, and finally get the partial identification of the actual vote, namely

$$P_n(y = j) - p \leqslant P_n(y^* = j) \leqslant P_n(y = j) + p \tag{8.7.7}$$

for every candidate j in $\{1, 2, \ldots, J\}$.

We are now in position to answer the initial question. Suppose we label the candidates such that we observe the following voting order : $P_n(y = 0) < P_n(y = 1) < \ldots < P_n(y = J)$. Then a sufficient condition to claim that this ranking is not impacted by fraud is that partial identification intervals found in equation (8.7.7) do not overlap. This implies

$$P_n(y = j) + p \leqslant P_n(y = j + 1) - p \qquad j = 1, \ldots, J - 1$$

or

$$p \leqslant \frac{P_n(y = j + 1) - P_n(y = j)}{2} \qquad j = 1, \ldots, J - 1\ .$$

In other words, the condition on the fraud rate to preserve the ranking is

$$p < \min\left\{\frac{P_n(y = j + 1) - P_n(y = j)}{2}\ :\ j = 1, \ldots, J - 1\right\}\ . \tag{8.7.8}$$

To illustrate the result, we focus now on the first round of the presidential election in Peru in 2021. For this round 18 candidates run for the office. We report in Table 8.3 the results obtained by the first six candidates, who all obtain more than one million observed votes. Therefore in this example $J = 6$. The last column of the Table shows the difference of probabilies appearing in the inequality (8.7.8). In this election there is tight outcome between candidate ranked 3 (R. López Aliaga) and 4 (H. de Soto). If

Table 8.3: First round of the presidential election in Peru on April 11th, 2021. The first $J = 6$ candidates are showed. The total of valid votes is 14,400,630. Source: `https://en.wikipedia.org/wiki/2021_Peruvian_general_election`

Candidate	**Party**	**Votes**	%	$P_n(y = j + 1) - P_n(j)$
Pedro Castillo	Free Peru	2,724,752	18.92	0.0551
Keiko Fujimori	Popular Force	1,930,762	13.41	0.0166
Rafael López Aliaga	Popular Renewal	1,692,279	11.75	0.0013
Hernando de Soto	Go on Country...	1,674,201	11.63	0.0255
Yonhy Lescano	Popular Action	1,306,288	9.07	0.0121
Verónika Mendoza	Together for Peru	1,132,577	7.86	—

we consider those first six candidates, we can say that our confidence on this ranking is conditional on a fraud rate that should be lower that 0.0006.

The use of our result must depend on the political analysis of interest. In a two-rounds election, the interest might be to focus on the three first candidates only, given that the second run retains the two first. In that case $J = 3$ and the level of confidence requires a fraud rate lower than 0.0082 (that is 0.0166/2). Considering all the 18 candidates in a two round election is probably useless since there are very small differences at the bottom of the table, and those little difference will drive the minimum value in inequality (8.7.8).

8.8 Discussion

In this study we find firstly that electoral participation entails significant uncertainty when it is low and that therefore we must be careful when it comes to understanding and analyzing electoral support when we are under this scenario, secondly than when constructing a model of electoral fraud with parameters that incorporate this fraud both in its quantity and in its direction we can understand how the results of an election vary against it, thirdly under this same fraud model we can formulate a series of hypotheses that account for of the impact that the cancellation of polling stations and their subsequent reinsertion can have, correcting fraudulent votes instead of canceling them, and in fourth, we could observe throughout this study that in a vote as narrow as that of Peru, electoral fraud, and in general, the analysis of electoral results, distortions in the electoral process and their respective complaints has n to be taken seriously and with the weight that corresponds because they compromise electoral integrity.

The practical contribution of this study lies in the simulation of the allegations of electoral fraud, especially in one of the most widespread such as the alleged impossibility that the defeated candidate has received 0 votes in some polls, which we show that even if it were true, It did not change who was proclaimed president of Peru. On the other hand, regarding the theoretical contribution, this study proposes a different perspective regarding the proposed methodologies and we believe that the parameterization of electoral fraud, although simple as in the models we observe, can provide us with a valuable tool to study the impact Regarding the electoral results that the different forms of fraud exposed may have, however,

we propose in what remains of this section possible extensions of this model that may be of particular interest to different sectors of society.

The participation and fraud tools could interact, but for this it would be quite attractive to assume certain assumptions from the political sciences for the behavior of the voters, for example, that the one who voted in the first round for an ideologically right-wing candidate votes in the second round by the candidate who is on the ballot that is closest to that ideology, or even that assumption could be relaxed and simply assume that whoever voted for one of the candidates who went to the second round and voted in both rounds, vote in the second round for that same candidate. Where, as we saw in sectors with high participation in both rounds, the probability of voting in the second given that they voted in the first is also high even if we partially achieve that identification.

However, the techniques presented here are analytical tools that can be extensible and flexible, where we could include in the model's parameterization different types of fraud, such as differentiating between changing the vote of one candidate to another, or changing the vote of one candidate. candidate to null, or change a vote from null to a candidate. In any case, in this work we show a first step for the analysis of the electoral process where we warn the sensitivity of the results to electoral participation, as well as we propose a model that allows proposing multiple hypotheses regarding the size and direction of the electoral fraud to conclude if they are hypotheses or accusations regarding fraud would have modified who is proclaimed the winner in the elections.

For cases in which the analysis focuses on suspicions of fraud that are originated by the characteristics of the polling place, a C could be defined that comes from the tradition of point estimation or econometrics where the vector includes characteristics as if the Voting place belongs to rural areas, if it has telephone coverage, public investment in the sector, poverty rates, etc.

This model can also be extended to the analysis of the organizations and institutions in charge of the electoral process, where they can observe the impact of certain decisions, irregularities and nullity of the acts within the same day of the elections as with subsequent requests in around this process.

A future development of this model may be to model the value of p through intervals as in the Horowitz and Manski study J. L. Horowitz and Manski (1995), where although they do it to find "Contaminated and Corrupted Data" in a database of households, we in this study could do the same, considering the "not interviewed" as those who did not participate in the electoral process while on the electoral roll, and from some assumptions from the political sciences it could be possible to determine these bound of p. In addition, the aforementioned work is strongly recommended if you want to deepen regarding the partial identification and parameterization of errors from its algebraic and statistical development.

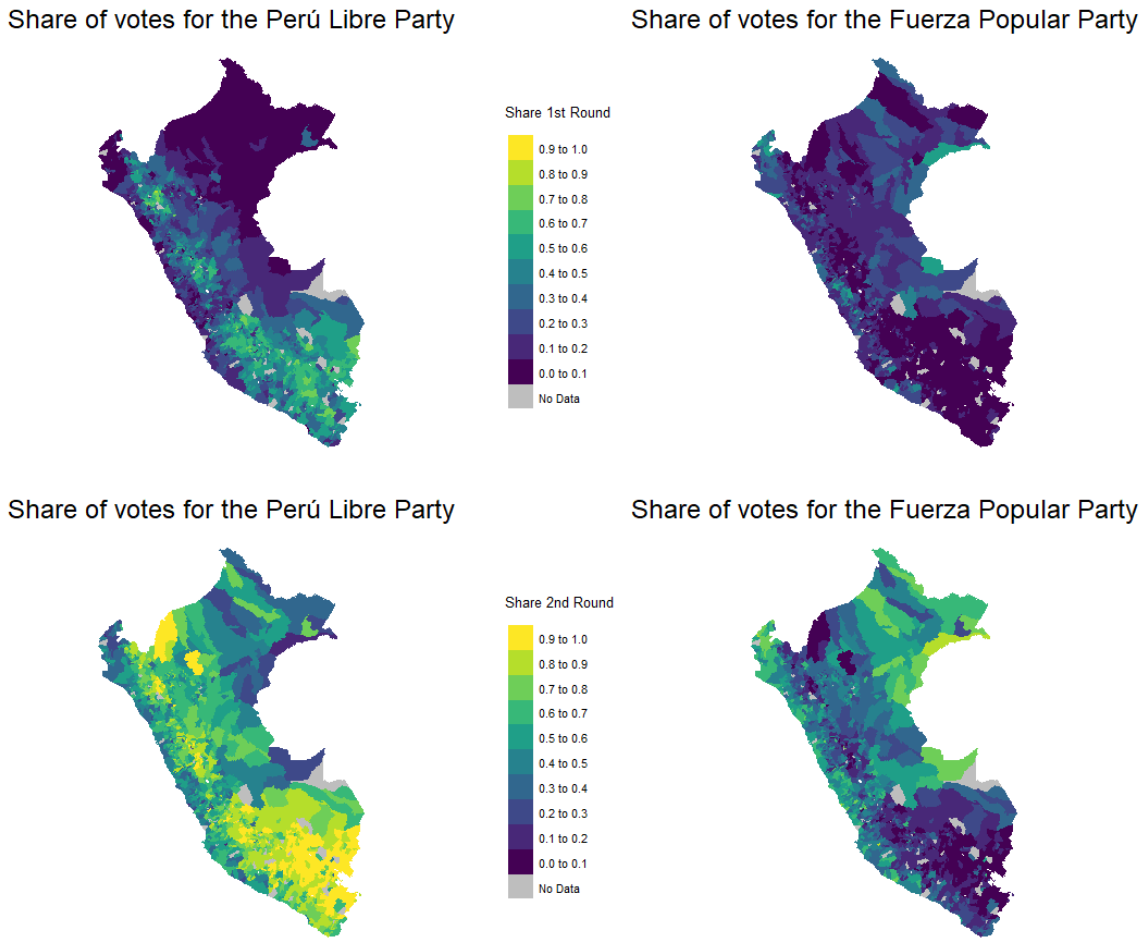


Figure 8.1: Share of valid votes in first and second round

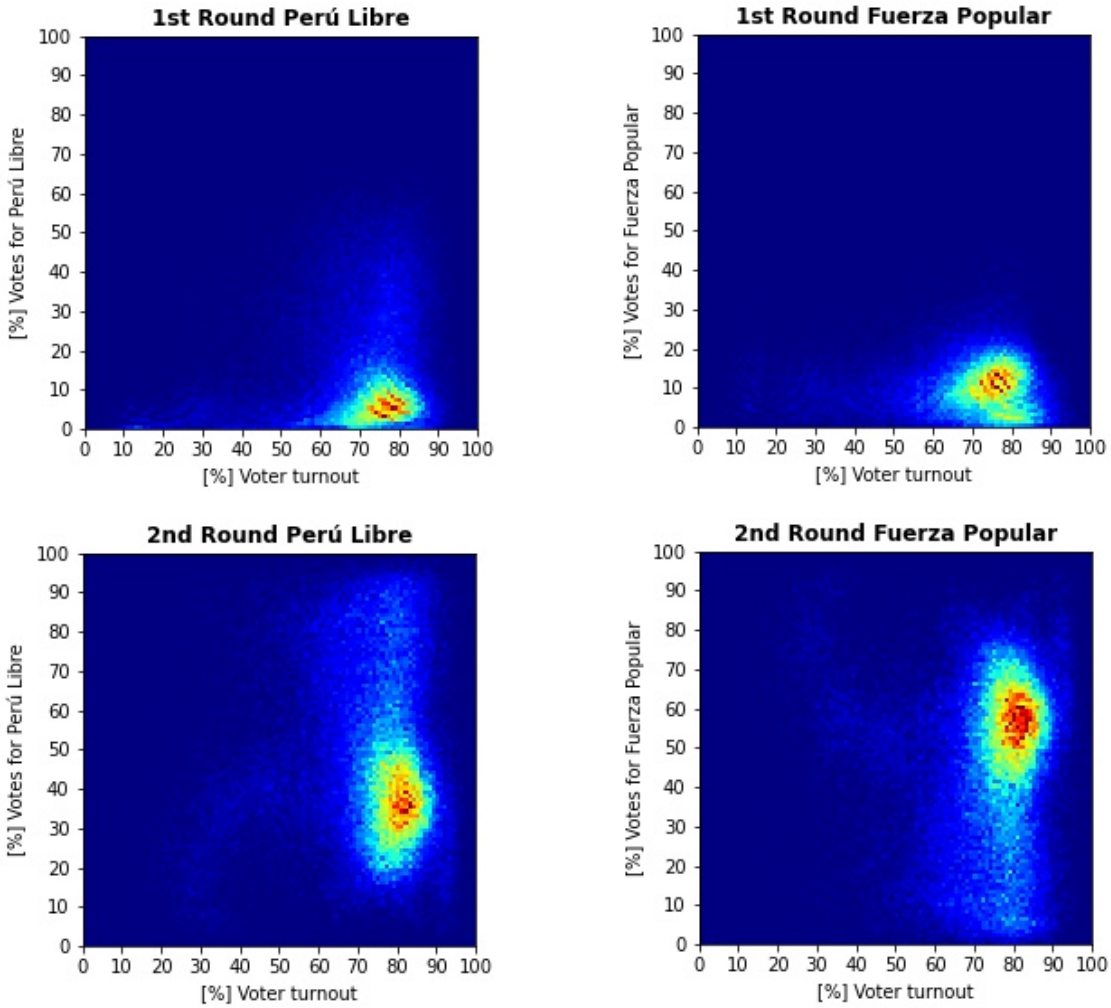


Figure 8.2: Election fingerprints

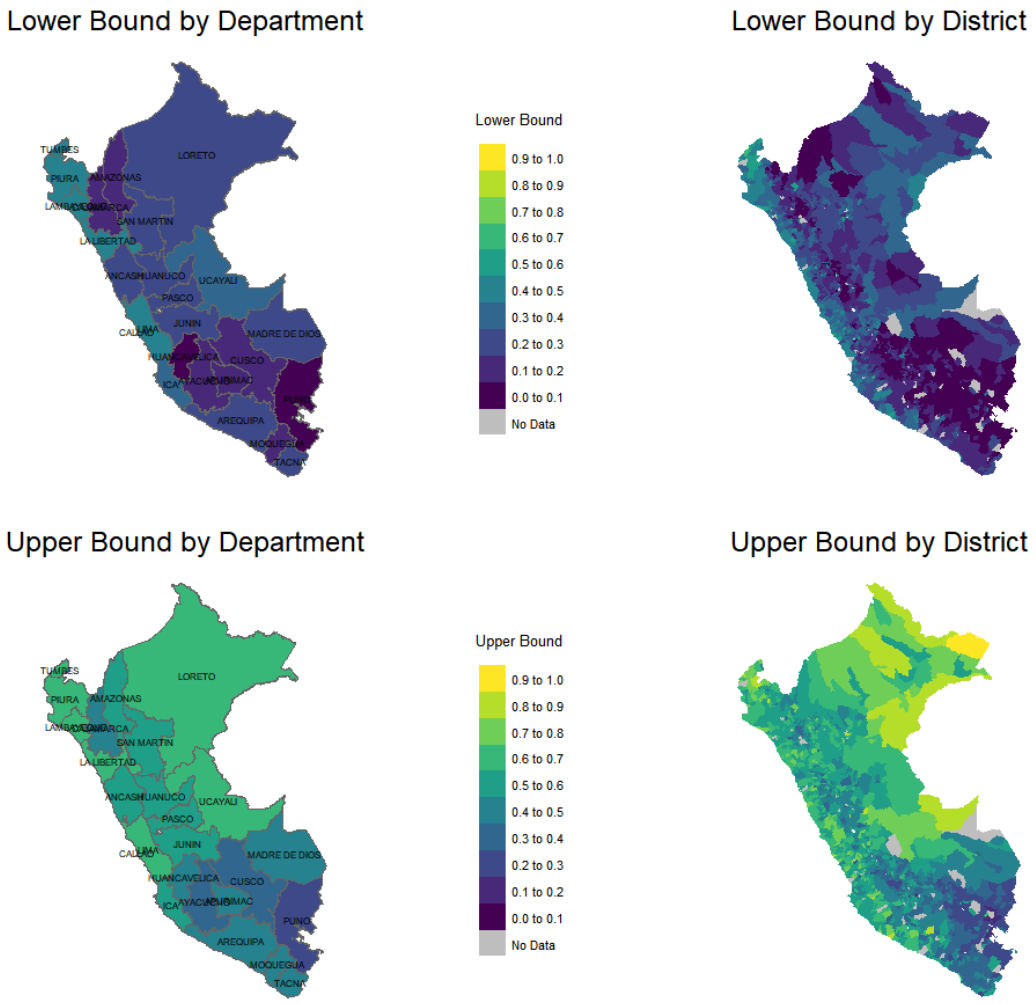


Figure 8.3: Bounds for the Fuerza Popular party

Chapter 9

A Critical View on the NEAT Equating Design: Statistical Modelling and Identifiability Problems

9.1 Introduction

Test equating[1] is conducted to adjust the score scales of different test forms in order to compensate for differences in relative difficulty and thus make scores equivalent and comparable (Lord, 1950; Angoff, 1984; Kolen & Brennan, 2014). The *equating problem* consists in mapping scores defined on one scale into their equivalents on the other scale. Such mapping is achieved using what is called an equating transformation function (for details, see González & Wiberg, 2017, Chapter 1). Although different types of score linkages are distinguished in the literature (?, ?), with equating being a particular case of score linking, the same linking function can be used in each case (Kolen, 2007) leading to scores that are interpreted as either interchangeable (equating) or comparable (linking).

Because score differences can also arise due to ability differences of test takers, comparable groups of examinees must be used when collecting score data to estimate the equating function. Different strategies for collecting score data have been proposed in the literature, leading to what are called equating designs; see, e.g., von Davier, Holland, and Thayer (2004), Kolen and Brennan (2014) and González and Wiberg (2017). These designs differ in that either common persons or common items are used to perform the score transformation, and in the conditions and assumptions made to collect the score data. In this chapter we will focus the attention on the nonequivalent groups with anchor test design (NEAT).

The NEAT design (a.k.a the common item nonequivalent group design, CINEG) is widely used in test equating. Under this design, two groups of test takers are administered different test forms that are intended to measure a common variable, with each test form containing a subset of common items.

[1]This chapter reproduces the chapter entitled *A Critical View on the NEAT Equating Design: Statistical Modelling and Identifiability Problems* by E. San Martín and J. González, which was accepted for publication in the *Journal of Educational and Behavioral Statistics*.

Because test takers from different groups respond to only one test form, *missing* score data emerge by design rendering some of the score distributions unavailable. The partially observed score data formally leads to an identifiability problem, which has not been recognized as such in the equating literature, and has been considered from different perspectives, all of them making different assumptions in order to estimate the unidentified score distributions; see, *e.g.*, Sinharay and Holland (2010), Miyazaki, Hoshino, Mayekawa, and Shigemasu (2009) and Holland, Sinharay, von Davier, and Han (2008).

In this chapter, we specify the statistical model underlying the NEAT design using a fully probabilistic approach. By so doing, we not only show that the notion of synthetic population is meaningless, but also offer an alternative formal and useful conceptualization of it. One of the consequences of this conceptualization is that the population weights used in the definition of a synthetic population cannot be arbitrarily chosen. Moreover, using a partial identification approach (Manski, 2007a; Tamer, 2010a), we make explicit the devastating results implied by the identification problem that are hidden when the missing at random condition is assumed.

The chapter is organized as follows: in Section 9.2 current definitions and assumptions considered when conducting equating under the NEAT design are revised. Next, we criticize aspects of the current view on NEAT equating, make explicit the statement of the problem to be discussed, and explain the strategy used in the chapter. In Section 9.3 the statistical model underlying the NEAT design is defined and the identifiability problem is described. In Section 9.4 we describe two approaches to tackle the identifiability problem and study the impact of the lack of identifiability in the actual equating using the theory of quantiles. An empirical illustration of the main findings is also presented at the end of this section. In the context of score linking, in Section 9.5 we discuss alternative identification restrictions leading to improve the previous results. The chapter finalizes in Section 9.6 summarizing the main points and discussing on how severe the identifiability problem can be in the context of test equating.

9.2 Modelling problems underlying the NEAT design

In order to show what motivates our critique of the NEAT design, we first review the basic definitions and strategies given in the literature for estimating the corresponding equating transformation.

9.2.1 Equating under the NEAT design: a summary of the literature

Let X and Y be random variables representing the scores on tests forms X and Y, respectively. The equating problem is addressed by using an *equating function* $\varphi : \mathcal{X} \longrightarrow \mathcal{Y}$, where $\mathcal{X}$ and $\mathcal{Y}$ are the sets of all possible scores on test forms X and Y (the score scales), so that for each raw score x we can compute the corresponding $y = \varphi(x)$ (Braun & Holland, 1982; González & Wiberg, 2017). Following Lord (1950) and Angoff (1984), a function φ can be defined taking into account that "two scores, one on Form X and the other on Form Y [. . .] may be considered equivalent if their corresponding percentile ranks in any given group are equal" (Angoff, 1984, p.86). More specifically, the function $\varphi : \mathcal{X} \longrightarrow \mathcal{Y}$ is defined as follows: for each $x \in \mathcal{X}$, we associate $y \doteq \varphi(x) \in \mathcal{Y}$ such that

$$F_Y[\varphi(x)] = F_X(x); \tag{9.2.1}$$

the symbol $\doteq$ means "defined as". Although equation (9.2.1) can always be solved using generalized inverses (Embrechts & Hofert, 2013), meaningful equating results are obtained only when scores are continuous (van der Linden, 2019). This is why all practical applications of test equating make use of what is called a continuization step; for details, see Braun and Holland (1982), Kolen and Brennan (2014) and González and Wiberg (2017).

Consider now a third random variable, A, which represent the scores obtained on an anchor test form A. Two definitions of the NEAT design appearing in the standard literature on equating are the following:

> In the Non-Equivalent groups with Anchor Test (NEAT) Design there are two populations, $\mathcal{P}$ and $\mathcal{Q}$, of test-takers and a sample of examinees from each. The sample from $\mathcal{P}$ takes test X, the sample from $\mathcal{Q}$ takes test Y, and both samples take a set of common items, the anchor test A (von Davier et al., 2004, p.32).

> For [the NEAT] design, two groups of examinees from different populations are each administered different test forms that have a set of items in common (Kolen & Brennan, 2014, p.103).

According to Braun and Holland (1982), the function φ should be defined with respect to a single population. This is a challenging task when performing equating under the NEAT design as, in fact, two populations, $\mathcal{P}$ and $\mathcal{Q}$, are involved. Consequently, $\mathcal{P}$ and $\mathcal{Q}$ "must be combined to obtain a single population for defining [the] equating function" (Kolen & Brennan, 2014, p.103), which leads to introduce a *synthetic population*, also called *target population*, denoted by $\mathcal{T}$. This is done by "conceptualizing a larger population that has $\mathcal{P}$ and $\mathcal{Q}$ as two mutually and exhaustive strata" (Braun & Holland, 1982, p.21), a "combined group, some of whom would be taking Form X, other taking Form Y, and all taking Form A, the anchor test" (?, ?, p.296), "a mixture of both $\mathcal{P}$ and $\mathcal{Q}$" (von Davier et al., 2004, p.34). Each of these statements is assumed to be represented by the following expression:

$$\mathcal{T} = \omega\,\mathcal{P} + (1-\omega)\,\mathcal{Q}, \tag{9.2.2}$$

where the populations $\mathcal{P}$ and $\mathcal{Q}$ are weighted by $\omega \in [0,1]$ and $1-\omega$, respectively. However, $\mathcal{P}$ and $\mathcal{Q}$ are just labels denoting the two populations and thus the expression (9.2.2) would represent a "weighted average of labels" which, of course, does not make any sense. A consequence of using this expression is that the target score distributions are arbitrarily chosen, as it will be seen in Section 3.

The equating transformation is then typically computed after defining the score distributions of X, Y and A on the synthetic population $\mathcal{T}$. Thus, for instance, for a specific and arbitrary choice of ω, the distribution $F_{X\mathcal{T}}(x)$, which has also been denoted in the literature as $P(X \leqslant x \mid \mathcal{T})$ (von Davier et al., 2004, p.7) (Sinharay & Holland, 2010), is obtained by first computing the distribution of raw scores X for each strata $\mathcal{P}$ and $\mathcal{Q}$, and then forming weighted averages of these distributions using ω and $1-\omega$, namely

$$F_{X\mathcal{T}}(x) \doteq \omega\,F_{X\mathcal{P}}(x) + (1-\omega)\,F_{X\mathcal{Q}}(x). \tag{9.2.3}$$

The score distributions $F_{X\mathcal{P}}(x)$ and $F_{X\mathcal{Q}}(x)$ have been denoted in the literature as $P(X \leqslant x \mid \mathcal{P})$ and $P(X \leqslant x \mid \mathcal{Q})$, respectively, with similar notation used for $F_{Y\mathcal{T}}(y)$ and $F_{A\mathcal{T}}(a)$. From the data collected under the NEAT design, the distributions $F_{X\mathcal{P}}(x)$, $F_{Y\mathcal{Q}}(y)$, $F_{A\mathcal{P}}(a)$ and $F_{A\mathcal{Q}}(a)$ may be

computed. Nevertheless, the distributions $F_{X\mathcal{Q}}(x)$ and $F_{Y\mathcal{P}}(y)$ are not directly estimable. They are typically obtained by conditioning on the anchor score A and using the following assumptions:

$$\text{(i) } F_{X\mathcal{P}|A}(x \mid a) = F_{X\mathcal{Q}|A}(x \mid a); \qquad \text{(ii) } F_{Y\mathcal{P}|A}(y \mid a) = F_{Y\mathcal{Q}|A}(y \mid a), \tag{9.2.4}$$

where $F_{X\mathcal{P}|A}$ is the distribution function of X conditionally on A in $\mathcal{P}$; and the other distributions have a similar meaning. Under these assumptions, the target score distributions are given by

$$\begin{aligned} &\text{(i)} \quad F_{X\mathcal{T}}(x) = \omega\, F_{X\mathcal{P}}(x) + (1-\omega) \sum_{a\in\mathcal{A}} F_{X\mathcal{P}|A}(x \mid a)\, P(A = a \mid \mathcal{Q}); \\ &\text{(ii)} \quad F_{Y\mathcal{T}}(y) = \omega \sum_{a\in\mathcal{A}} F_{Y\mathcal{Q}|A}(y \mid a)\, P(A = a \mid \mathcal{P}) + (1-\omega)\, F_{Y\mathcal{Q}}(y), \end{aligned} \tag{9.2.5}$$

where $\mathcal{A}$ is the set of possible anchor-test scores. These score distributions are then used to compute $\varphi(x)$ for a given $x \in \mathcal{X}$ using the equality $\varphi(x) = F_{Y\mathcal{T}}^{-1}[F_{X\mathcal{T}}(x)]$. This method is called frequency estimation equating (FEE, Angoff, 1984). Other methods used for equating under the NEAT design such as the chained equipercentile equating (CEE) and IRT-Observed Score equating (IRT-OSE), together with the assumptions needed in each case to obtain the distributions that are not directly estimable are described in Sinharay and Holland (2010).

9.2.2 What we question: A statement of the problem

From a modelling point of view, equating under the NEAT design depends on both the synthetic population (9.2.2) and the assumptions in (9.2.4), which are typically known as the *missing-at-random assumptions* (Liou & Cheng, 1995; Miyazaki et al., 2009; Sinharay & Holland, 2010). Bearing in mind that the equating function is defined in probabilistic terms, the following questions arise:

1. Taking into account that $\mathcal{P}$ and $\mathcal{Q}$ are not random variables but mere labels, under which conditions are $P(X \leqslant x \mid \mathcal{P})$ and $P(X \leqslant x \mid \mathcal{Q})$ properly understood as conditional probability functions?

2. If such conditions exist, would it be possible to define the concept of a synthetic population in probabilistic terms?

3. Is it possible to provide alternative assumptions to the ones given in (9.2.4) to solve the problem of missing score data underlying the NEAT design? If so, what would be the impact of such alternative assumptions on the NEAT design in specific contexts?

The answers to the previous questions depend on the answer to a more fundamental one: how can "an understanding of the way in which the data" collected under the NEAT design "are supposed to, or did in fact, originate" be obtained (Fisher, 1973, p.8)? This question can be answered once the parameters of interest that "describe [the observations] exhaustively in respect of all qualities under discussion" (?, ?, p.311) are made explicit. Nevertheless, the data generating process is not necessarily characterized by the parameters of interest. When this is the case, we face an identifiability problem. This is the problem we will unveil in the NEAT design.

9.3 Statistical specification of the NEAT design and the inherent identification problem

One way to answer these questions is to follow González and Wiberg (2017)'s perspective, which consists in specifying the NEAT design as a *statistical model*. This means to make explicit the following three components:

1. The *sample space* $(M, \mathcal{M})$, where M contains the elementary events and $\mathcal{M}$ is a class containing the events of interest and their combinations through unions of sets as well as complement of a set –this is why $\mathcal{M}$ is a σ-field of subsets of M. The pair $(M, \mathcal{M})$ is technically called a measurable space.
2. The *sampling probabilities* P^γ, indexed by a parameter γ, where each P^γ is defined on $(M, \mathcal{M})$.
3. The *parameter space* Γ, which represents the set of all plausible values of γ;

for details, see San Martín et al. (2015) and references therein.

What are the practical advantages of this formalism? First, the population of interest is made explicit by listing it through the elements of the sample space. Second, features of the population under study are fully characterized by the parameters indexing the sampling probabilities. Third, this formalism allows us to evaluate whether certain characteristics relevant to researchers can indeed be represented as (a function of) parameters: when this is not the case, then we face an identification problem (?, ?).

The specification of the statistical model underlying the NEAT design will be performed in a sequential way, which leads to decompose the joint probability distribution defined on all the pertinent random variables into specific conditional sub-models. The sequential specification makes explicit the way in which the data are (supposed to be) generated (Wunsch, Mouchart, & Russo, 2014). This sequential specification is critically based on the Law of Total Probability in the sense that it ensures the *existence* of the conditional distribution of a new random variable given the previously introduced variables. This corresponds to the formal definition of a conditional probability function as introduced by ? (?, §6). By doing so, it will be made explicit which are the parameters of the statistical model and the lack of identifiability of the parameters of interest.

9.3.1 The sample space in the NEAT design

In order to construct both the random variables and the probability distributions underlying the NEAT design, it is necessary to make explicit the sample space M. In the case of the NEAT design, M corresponds to the union of the index sets labeling the two mutually exclusive groups of examinees who were *exposed* to the test forms X and Y, namely

$$M = \mathcal{P} \cup \mathcal{Q}, \quad \text{where} \quad \mathcal{P} = \{i_1, \ldots, i_{n_\mathcal{P}}\}, \quad \mathcal{Q} = \{j_1, \ldots, j_{n_\mathcal{Q}}\}, \quad \mathcal{P} \cap \mathcal{Q} = \varnothing,$$

$n_\mathcal{P}$ and $n_\mathcal{Q}$ are the total number of examinees exposed to test forms X and Y, respectively. Thus, M corresponds to the population of interest underlying the NEAT design. Since it is a finite set, the class $\mathcal{M}$ of events of interest corresponds to the class of subsets of M.

M	X	Y	A	Z
i_1	1	-	2	1
i_2	4	-	1	1
i_3	3	-	2	1
j_1	-	1	3	0
j_2	-	5	3	0
j_3	-	5	1	0
j_4	-	2	0	0
j_5	-	0	2	0

Table 9.1: Toy data set example

9.3.2 The random variable Z

All the random variables underlying the NEAT design should be defined from M to a set of possible values. In addition to the variables representing the test scores, namely X, Y, A, we construct a random variable representing the groups underlying the NEAT design. To do so, remind that, by definition, given two measurable spaces $(M, \mathcal{M})$ and $(N, \mathcal{N})$, a function $f : M \longrightarrow N$ is a random variable if $f^{-1}(B) \in \mathcal{M}$ for all $B \in \mathcal{N}$ (?, ?, ?). In the finite discrete case, as it is in the NEAT design, this formal definition reduces to the following characterization: a mapping of the set M into a set of values is a random variable if and only if the mapping induces a partition of M. For instance, for the case when $\mathcal{P} = \{i_1, i_2, i_3\}$ and $\mathcal{Q} = \{j_1, j_2, j_3, j_4, j_5\}$, consider the toy data set shown in Table 9.1. If $A : M \longrightarrow \mathcal{A} \doteq \{0, 1, 2, 3\}$ is a random variable representing the scores obtained in an anchor test of three items, then $A^{-1}\{0\} = \{j_4\}, A^{-1}\{1\} = \{i_2, j_3\}, A^{-1}\{2\} = \{i_1, i_3, j_5\}$ and $A^{-1}\{3\} = \{j_1, j_2\}$. These sets belong to $\mathcal{M}$ and constitute a partition of M.

This characterization is relevant for at least two reasons: first, different examinees with equal values on a random variable will be treated as equivalent and put in the same equivalence class (i.e., the members of the partition) of examinees. Second, a random variable is the class of events in $\mathcal{M}$ (in the finite case, subsets of M) that is induced by the possible values the random variable can take (?, ?), which means that all information provided by a random variable is captured in the sample space $(M, \mathcal{M})$.

The sample space M has been defined through the partition $\{\mathcal{P}, \mathcal{Q}\}$, which implies that a binary random variable Z can naturally be defined as

$$Z = \begin{cases} 1, & \text{if the examenee belongs to group } \mathcal{P}; \\ 0, & \text{if the examenee belongs to group } \mathcal{Q}. \end{cases} \tag{9.3.1}$$

That is, the preimage $Z^{-1}\{1\}$ corresponds to the event $\mathcal{P}$, and the preimage $Z^{-1}\{0\}$, to the event $\mathcal{Q}$: this is the semantic meaning of Z.

A natural first approach to construct a probability function of Z would be thinking in terms of experimental units (the members of the sample space M) and a random mechanism that assigns numbers to the experimental units, as for instance scores or measurements (Lord & Novick, 1968, Sections 1.3 and

2.2). Under this perspective, the groups $\mathcal{P}$ and $\mathcal{Q}$ could be viewed as "realizations" or "occurrences" of such a mechanism. However, using R. J. Little and Rubin (1994)'s terminology, the groups $\mathcal{P}$ and $\mathcal{Q}$ in the NEAT design are considered as *selected samples* and, consequently, subject to selection bias. Thus, this approach is not useful to reveal some random assignment mechanism that makes sense of Z in the NEAT design.

Nevertheless, it is possible to propose an alternative approach that suits better the characteristics of the NEAT design. To do so, we resort to Carnap (1962, §12)'s distinction between what a researcher says about a term in his formulation (semantics) and what he actually does with that term in the corpus of his formulation (syntax): the sense of a certain term is revealed from its use in formal arguments. Taking into account that, from an axiomatic perspective, "random" and "realization" or "occurrence" are mere terms, not probabilistic concepts (see ? (?, p.2) and Itô (1984, pp.1,3), respectively), an axiomatically well defined probability function of Z can be defined as follows: when $M = \{m_1, \ldots, m_k\}$ is a finite set, it is enough to take an arbitrary set of non-negative numbers $\{p_1, \ldots, p_k\}$ with their sum equal to 1 such that $P\{m_i\} = p_i$ for $i = 1, \ldots, k$ (?, ?, p.3). Consequently, a possible probability function of Z can accordingly be constructed by considering the relative sizes of the groups $\mathcal{P}$ and $\mathcal{Q}$ (Braun & Holland, 1982), namely

$$P(Z = 1) = \frac{n_{\mathcal{P}}}{n_{\mathcal{P}} + n_{\mathcal{Q}}}, \qquad P(Z = 0) = \frac{n_{\mathcal{Q}}}{n_{\mathcal{P}} + n_{\mathcal{Q}}}. \tag{9.3.2}$$

Under this approach, both $\mathcal{P}$ and $\mathcal{Q}$ are viewed as constituting the population of interest, rather than realizations from some sampling mechanism (see, Manski, 2013b, p.126). This is due to the fact that we focus our attention on the identification problem underlying the NEAT design, which will be the same regardless of the type of sampling process it is reasonable to assume; see also Manski and Nagin (1998, specially p.109). It is in fact possible to specify a different probability function for Z, but its identifiability should be justified, as it is the case for the one defined in (9.3.2). Furthermore, in the identification analysis that will follow, the role played by the random variable Z is only through the *probability function that is defined on it*.

As for its semantic meaning, the partition of M induced by Z can be interpreted as a "sample by specification" (Lord & Novick, 1968, Section 2.5), thus avoiding the idea of a possible random assignment mechanism. In any case, we define a "random assignment mechanism" in terms of "absence of bias", which in turn is rigorously described in terms of conditional probabilities; for details, see Appendix 9.7. This definition is actually motivated by a semantic relationship between both notions (Stephenson & Imrie, 1998; Odgaard-Jensen et al., 2011; Bhide, Shah, & Acharya, 2018). To the best of our knowledge, no formal proof exists deriving "absence of bias" from "random assignment" precisely because this last notion is not a probabilistic concept. As it will be seen in what follows, the identification problem arises from the selection bias inherent to the NEAT design, which formally means that $P(X \leqslant x \mid A, Z = 1) \neq P(X \leqslant x \mid A, Z = 0)$ and $P(Y \leqslant y \mid A, Z = 1) \neq P(Y \leqslant y \mid A, Z = 0)$. Before discussing this aspect, let us follow through the sequential specification of the NEAT design.

9.3.3 The anchor random variable A

Once Z has been specified, we introduce a random variable A defined from M into the set $\mathcal{A}$ of possible scores in the anchor test. Following the sequential specification strategy, the existence of the conditional probability of A given Z is ensured by the following representation: for all $a \in \mathcal{A}$,

$$\begin{aligned} P(A = a) &= P(A = a \mid Z = 1)P(Z = 1) + P(A = a \mid Z = 0)P(Z = 0). \\ &= E\left[P(A = a \mid Z = 1)\mathbb{1}_{\{Z=1\}} + P(A = a \mid Z = 0)\mathbb{1}_{\{Z=0\}}\right] \\ &= E[P(A = a \mid Z)]. \end{aligned} \tag{9.3.3}$$

It should be remarked that the existence of the conditional probability function $P(A = a \mid Z)$ depends on Z through the partition on M induced by it. Because both numbers $P(A = a \mid Z = 1)$ and $P(A = a \mid Z = 0)$ can be computed from the data generated by the sampling process, we conclude that, for each $a \in \mathcal{A}$, $P(A = a \mid Z)$ are identified parameters and, consequently, the joint distribution $P(A = a, Z = z)$ for $(a, z) \in \mathcal{A} \times \{0, 1\}$ is identified. Note that these statements do not depend on specific *values* of the probabilities $P(Z = 1)$ and $P(Z = 0)$, but only on the fact that Z is well defined and can be endowed with a probability distribution. Finally, it can be verified that A and Z are not independent by construction. For instance, for $a \in \mathcal{A}$

$$P(Z = 1, A = a) \doteq P[Z^{-1}\{1\} \cap A^{-1}\{a\}] = P\{m \in \mathcal{P} : A(m) = a\} \neq P(Z = 1)\, P(A = a).$$

9.3.4 The random variables X and Y

Once (A, Z) has been specified, we introduce two random variables X and Y defined from M into $\mathcal{X}$ and $\mathcal{Y}$, respectively. As discussed previously, the equating function depends on two probability distributions, namely $F_X(x) = P(X \leqslant x)$ for all $x \in \mathcal{X}$ and $F_Y(y) = P(Y \leqslant y)$ for all $y \in \mathcal{Y}$: these are actually the parameters of interest. However, these parameters are not identified because it is *not possible to construct the respective conditional distributions* $P(X \leqslant x \mid A, Z)$ and $P(Y \leqslant y \mid A, Z)$. As a matter of fact, the marginal distribution $P(X \leqslant x)$ can only be obtained once $P(X \leqslant x \mid A)$ is obtained, which in turn is defined through the following representation: for all $x \in \mathcal{X}$,

$$\begin{aligned} P(X \leqslant x \mid A) &= P(X \leqslant x \mid A, Z = 1)P(Z = 1 \mid A) + P(X \leqslant x \mid A, Z = 0)P(Z = 0 \mid A) \\ &= E\left[P(X \leqslant x \mid A, Z = 1)\mathbb{1}_{\{Z=1\}} + P(X \leqslant x \mid A, Z = 0)\mathbb{1}_{\{Z=0\}} \mid A\right] \\ &= E[P(X \leqslant x \mid A, Z) \mid A]. \end{aligned} \tag{9.3.4}$$

In this representation, the function $P(X \leqslant x \mid A, Z = 0)$ is not identified because it is impossible to determine from the sampling process the event $X^{-1}\{x\} \cap Z^{-1}\{0\} \subset M$, that is, it is not observed which examinees exposed to test form Y would have obtained a score equal to x in test form X. This makes explicit the underlying selection bias problem according to which $P(X \leqslant x \mid A, Z = 1) \neq P(X \leqslant x \mid A, Z = 0)$.

The non-identifiability of $P(X \leqslant x)$ immediately follows: for all $x \in \mathcal{X}$,

$$\begin{aligned} F_X(x) \doteq P(X \leqslant x) &= P(X \leqslant x \mid Z=1)P(Z=1) + P(X \leqslant x \mid Z=0)P(Z=0) \\ &= E[P(X \leqslant x \mid Z)] \\ &= E[E[P(X \leqslant x \mid A, Z)] \mid A] \\ &\overset{\text{by (9.3.4)}}{=} E[P(X \leqslant x \mid A)]. \end{aligned} \tag{9.3.5}$$

By similar arguments, in the following representation, for $y \in \mathcal{Y}$,

$$\begin{aligned} P(Y \leqslant y \mid A) &= P(Y \leqslant y \mid A, Z=1)P(Z=1 \mid A) + P(Y \leqslant y \mid A, Z=0)P(Z=0 \mid A) \\ &= E\left[P(Y \leqslant y \mid A, Z=1)\mathbb{1}_{\{Z=1\}} + P(Y \leqslant y \mid A, Z=0)\mathbb{1}_{\{Z=0\}} \mid A\right] \\ &= E[P(Y \leqslant y \mid A, Z) \mid A], \end{aligned} \tag{9.3.6}$$

the function $P(Y \leqslant y \mid A, Z=1)$ is not identified because it is impossible to determine from the sampling process the event $Y^{-1}\{y\} \cap Z^{-1}\{1\} \subset M$, that is, it is not observed which examinees exposed to test form X would have obtained a score equal to y in test form Y. This makes explicit the underlying selection bias problem according to which $P(Y \leqslant y \mid A, Z=1) \neq P(Y \leqslant y \mid A, Z=0)$. Therefore, $P(Y \leqslant y)$ is not identified because, for all $y \in \mathcal{Y}$,

$$\begin{aligned} F_Y(y) \doteq P(Y \leqslant y) &= P(Y \leqslant y \mid Z=1)P(Z=1) + P(Y \leqslant y \mid Z=0)P(Z=0) \\ &= E[P(Y \leqslant y \mid Z)] \\ &= E[E[P(Y \leqslant y \mid A, Z)] \mid A] \\ &\overset{\text{by (9.3.6)}}{=} E[P(Y \leqslant y \mid A)]. \end{aligned} \tag{9.3.7}$$

Summarizing, the statistical model in the NEAT design can be written as $\{(M, \mathcal{M}) : (F_X, F_Y) \in \mathcal{F}(M, \mathcal{M})\}$, where $\mathcal{F}(M, \mathcal{M})$ denotes the space of probability distributions defined on $(M, \mathcal{M})$. The type of data that can be analyzed under the NEAT design are characterized by the sequential specification developed above.

9.3.5 Consequences of the model specification

The previous specification deserves several comments. First, one might ask what is the role of the anchor test in the previous identification analysis? More precisely, why it is necessary to use the conditional distribution $P(X \leqslant x \mid A, Z)$ (see (9.3.4)) to establish the non-identifiability of $P(X \leqslant x)$, instead of establishing it using $P(X \leqslant x \mid Z)$ directly (see (9.3.5), last line)? In fact, the non-identifiability of both $P(X \leqslant x)$ and $P(Y \leqslant y)$ follows after marginalizing Z. However, we will show that the anchor test A is useful for introducing identification restrictions leading to either point-identify or partially identify them. Furthermore, the sampling process characterizing the NEAT design identifies

$$P(A=a, Z=z), \qquad P(X \leqslant x \mid A, Z=1), \qquad P(Y \leqslant y \mid A, Z=0)$$

for all $(z, a, x, y) \in \{0, 1\} \times \mathcal{A} \times \mathcal{X} \times \mathcal{Y}$. These distributions cannot be obtained from

$$P(A = a, Z = z), \quad P(X \leqslant x \mid Z = 1), \quad P(Y \leqslant y \mid Z = 0)$$

for all $(z, a, x, y) \in \{0, 1\} \times \mathcal{A} \times \mathcal{X} \times \mathcal{Y}$: it is necessary to make explicit the dependency between X and A for $\{Z = 1\}$, and between Y and A for $\{Z = 0\}$.

Second, an advantage of this fully probabilistic specification of the NEAT design is to show that the term *synthetic population* is useless. As a matter of fact, the population of interest underlying the NEAT design is given by $M = \mathcal{P} \cup \mathcal{Q}$: once this is made explicit, all the necessary random variables can explicitly be defined. Moreover, the score distributions $P(X \leqslant x)$ and $P(Y \leqslant y)$ are decomposed in a *unique* way through the representations (9.3.5) and (9.3.7). In these decompositions, $P(Z = 1)$ is an identified parameter of the statistical model that, once it is fixed (for instance, through (9.3.2)), *can not be modified arbitrarily* as suggested by Braun and Holland (1982), Brennan and Kolen (1987), and Kolen and Brennan (2014, Section 4.5.2), among many others. Therefore, the "target score distribution" used in the equating literature does not always coincide with $P(X \leqslant x)$ as decomposed in (9.3.5), except when $\omega = P(Z = 1)$. In other words, the existence of the target score distribution is *not* well established.

9.4 Approaches to tackle the Identification Problem

In this section we discuss two different approaches to tackle the identification problem underlying the NEAT design. The first one is based on a missing-at-random condition, widely used in the equating literature. In order to grasp how strong such a condition is, we consider a second approach to the identification problem based on the theory of partial identification. The theoretical results are illustrated with a well known data set appearing in the literature (Kolen & Brennan, 2014).

9.4.1 Strong ignorability condition

In the equating literature the lack of identifiability of the marginal distributions F_X and F_Y has been considered as a missing data problem; see, among many others, Liou and Cheng (1995), ? (?), Bolsinova and Maris (2016), and Sinharay and Holland (2010). As we have seen in Section 9.2.1, the problem is typically solved assuming that the anchor scores are informative enough such that $F_{X|Z=1,A}(x) = F_{X|Z=0,A}(x)$ for all $x \in \mathcal{X}$, and $F_{Y|Z=1,A} = F_{Y|Z=0,A}$ for all $y \in \mathcal{Y}$, which corresponds to

$$\text{(i)} \quad X \perp\!\!\!\perp Z \mid A; \quad \text{(ii)} \quad Y \perp\!\!\!\perp Z \mid A; \tag{9.4.1}$$

see Braun and Holland (1982), Kolen and Brennan (2014), von Davier et al. (2004) and González and Wiberg (2017). Here $U \perp\!\!\!\perp V \mid W$ denotes the conditional independence of U and V given W; for details, see Florens et al. (1990, Chapter 2). Note that condition (9.4.1) is equivalent to the assertion of absence of bias. For details, see Appendix A.

In the econometric literature, this condition is known as the *switching condition* (Maddala, 1983), whereas in the causal inference literature it is known as the *strong ignorability condition* (Rosenbaum

& Rubin, 1983). Condition (9.4.1) is not empirically refutable (Manski, 2007a), but only justified in a specific application. Such justification could lead to answer affirmatively to the following question: Are we ready to believe that, conditionally on A, the score distribution of the examinees taking test form X would be the same as if they were exposed to test form Y?

What is important to emphasize is that condition (9.4.1) is *an identification restriction* allowing to identify $F_{X|A}$ and $F_{Y|A}$ and, by extension, F_X and F_Y. As a matter of fact, under condition (9.4.1), decompositions (9.3.4) and (9.3.6) imply that $F_{X|A}(x) = F_{X|Z=1,A}(x)$ for all $x \in \mathcal{X}$, and that $F_{Y|A}(y) = F_{Y|Z=0,A}(y)$ for all $y \in \mathcal{Y}$; and therefore

$$\begin{aligned} &\text{(i)} \quad F_X(x) = \sum_{a \in \mathcal{A}} P(X \leqslant x \mid Z = 1, A = a)\, P(A = a) \quad \forall\, x \in \mathcal{X}; \\ &\text{(ii)} \quad F_Y(y) = \sum_{a \in \mathcal{A}} P(Y \leqslant y \mid Z = 0, A = a) P(A = a) \quad \forall\, y \in \mathcal{Y}. \end{aligned} \tag{9.4.2}$$

The equating function can thus be obtained using (9.4.2), which is equivalent to (9.2.5) provided that $\omega = P(Z = 1)$; for a proof, see Appendix 9.8. Thus, the strong ignorability condition (9.4.1) is simply an identification restriction that leads to being able to identify $F_{X|A}$ and $F_{Y|A}$, which in turn implies the identifiability of the parameters of interest F_X and F_Y. Now, the question is: how strong is the condition (9.4.1)? The remainder of this chapter is focused in answering this question.

9.4.2 Partial identification analysis

A natural starting point is to see what the data alone reveal about $F_{X|A}$ and $F_{Y|A}$. This can be done by means of a partial identification strategy, widely used in empirical research; see, *e.g.*, Pepper (2000), Blundell, Gosling, Ichimura, and Meghir (2007), Gundersen and Kreider (2009), Molinari (2010).

In comparison with the traditional concept of identifiability that gives a binary status for the parameters of interest in a statistical model, namely either identified or not, partial identification is an approach that recognizes that identification is not an all-or-nothing concept and that models that do not identify parameters of interest can, and typically do, contain valuable information about these parameters (Tamer, 2010a). Following Manski (2007a), a parameter is partially identified if the sampling process and maintained assumptions reveal that the parameter lies in a set, its "identification region", that is smaller than the logical range of the parameter but larger than a single point. In this sense, the smaller (larger) the identification region is, the larger (smaller) the information we have about the parameter.

Partial identification of the parameters of interest

In order to make explicit what the data alone reveal about $F_{X|A}$, it is enough to use that the unidentified conditional probability distribution $F_{X|Z=0,A}$ lies between 0 and 1. Decomposition (9.3.4) implies, therefore, that the conditional probability $F_{X|A}$ lies in the interval

$$F_{X|Z=1,A}(x)P(Z = 1 \mid A) \leqslant F_{X|A}(x) \leqslant F_{X|Z=1,A}(x)P(Z = 1 \mid A) + P(Z = 0 \mid A) \tag{9.4.3}$$

for all $x \in \mathcal{X}$. Similarly, $F_{Y|Z=1,A}$ lies between 0 and 1 and, therefore, decomposition (9.3.6) implies that the conditional probability $F_{Y|A}$ lies in the interval

$$F_{Y|Z=0,A}(y)P(Z=0 \mid A) \leqslant F_{Y|A}(y) \leqslant F_{Y|Z=1,A}(y)P(Z=0 \mid A) + P(Z=1 \mid A) \tag{9.4.4}$$

for all $y \in \mathcal{Y}$. After marginalizing with respect to A, these inequalities provide the partial identification intervals of the parameters of interest, which is summarized in the following theorem:

Theorem 9.1 *In the NEAT design, the parameters of interest F_X and F_Y are partially identified by the following identification intervals:*

$$\begin{aligned} &(i) \quad F_{X|Z=1}(x)P(Z=1) \leqslant F_X(x) \leqslant F_{X|Z=1}(x)P(Z=1) + P(Z=0), \quad x \in \mathcal{X}; \\ &(ii) \quad F_{Y|Z=0}(y)P(Z=0) \leqslant F_Y(y) \leqslant F_{Y|Z=0}(y)P(Z=0) + P(Z=1), \quad y \in \mathcal{Y}. \end{aligned} \tag{9.4.5}$$

Let us comment on some important consequences of this theorem. First, the partial identification of F_X and F_Y can be directly obtained using decompositions (9.3.5) and (9.3.7). However, the partial identification intervals (9.4.3) and (9.4.4) will be useful in Section 9.5 to obtain a more informative identification region.

Second, the partial identification intervals (9.4.5.i) and (9.4.5.ii) do not depend on any assumption; they accordingly provide all the plausible values of F_X and F_Y coherent with the data and the probabilities identified by them. In particular, the solution given in (9.4.2) is a plausible one (see Figure 9.1). It should be stressed that other solutions of the type (9.2.5), as the ones suggested in the literature where $\omega \neq P(Z=1)$, are *not* plausible solutions because *some of the components are not based on the identified parameters underlying the NEAT design*.

Third, the width of interval (9.4.5.i) is $P(Z=0)$, whereas the width of interval (9.4.5.ii) is $P(Z=1)$. Taking into account that $P(Z=0) + P(Z=1) = 1$, the longer one interval is, the shorter the other. This trade-off –which is valid even if $P(Z=1)$ and $P(Z=0)$ do not correspond to the relative weight sizes of both groups– shows how strong the ignorability condition is: $F_X(x)$ with $x \in \mathcal{X}$ (respect., $F_Y(y)$ with $y \in \mathcal{Y}$) belongs to an identification interval of width $P(Z=0)$ (respect., $P(Z=1)$); which under the ignorability condition collapses to a point, namely $F_{X|Z=1}(x)$ (respect., $F_{Y|Z=0}(y)$). The degree of information that the ignorability condition hides can be precisely quantified: it corresponds to $P(Z=1)$ and $P(Z=0)$, which can not be arbitrarily manipulated once the groups of examinees have responded the tests. Figure 9.1 shows this trade-off graphically.

Fourth, the larger the partial identification intervals, the lesser is the information on the unidentified score distributions used to compute the equating function. It could be argued that smaller identification intervals can be obtained simultaneously for both score distributions by conditioning on the anchor scores A. This is in fact one of the ways in which the strong ignorability condition is motivated, namely that A is so informative that allows us to ignore Z. However, this is not the case. It is enough to compare the width of the conditional partial identification intervals (9.4.3) and (9.4.4) with the intervals (9.4.5.i) and (9.4.5.ii), respectively. For each $a \in \mathcal{A}$, we consider the following cases:

$$\begin{aligned} &\text{(i)} \quad P(Z=1) > P(Z=1 \mid A=a) \iff P(Z=0) < P(Z=0 \mid A=a) \\ &\text{(ii)} \quad P(Z=1) < P(Z=1 \mid A=a) \iff P(Z=0) > P(Z=0 \mid A=a) \end{aligned} \tag{9.4.6}$$

It can be seen that after conditioning on A, the identification interval for F_Y is indeed smaller but the one for F_X turns to be larger in Case (i). Likewise in Case (ii), the identification interval for F_X is smaller but the one for F_Y is larger. Thus, it is not possible to reduce simultaneously the width of the identification intervals by conditioning on A. Note that these conditions are empirically testable. Figure 9.2 shows a graphical representation of these comparisons.

Partial identification of the quantiles

The impact of the lack of identifiability of F_X and F_Y, and the amount of information that can be obtained from a partial identifiability approach on the actual equating, may be studied by analysing the quantile functions involved in the process. In fact, the equating function actually equates the quantiles of the score distributions F_X and F_Y (see Section 9.2.1).

Let $\alpha \in (0, 1)$. Taking as a starting point the identifiability bounds in (9.4.5), we define the following quantiles functions:

$$\begin{aligned} q_X(\alpha) &\doteq \inf\{t : F_X(t) > \alpha\}; \\ r_X(\alpha) &\doteq \inf\{t : F_{X|Z=1}(t)P(Z = 1) + P(Z = 0) > \alpha\}; \\ s_X(\alpha) &\doteq \inf\{t : F_{X|Z=1}(t)P(Z = 1) > \alpha\}. \end{aligned}$$

Note that $r_X(\alpha)$ and $s_X(\alpha)$ are identified, whereas $q_X(\alpha)$ is unidentified. The aim is thus to partially identify $q_X(\alpha)$ using both $r_X(\alpha)$ and $s_X(\alpha)$. To do that, we establish the following relationships using (9.4.5.i): let $t < r_X(\alpha)$, it follows that $F_{X|Z=1}(t)P(Z = 1) + P(Z = 0) < \alpha$, which in turn implies that $F_X(t) < \alpha$ and, therefore, $q_X(\alpha) > t$. It follows that $r_X(\alpha) < q_X(\alpha)$: if not, take $t = q_X(\alpha)$ and conclude that $F_X[q_X(\alpha)] < \alpha$, which is a contradiction with the definition of $q_X(\alpha)$.

On the other hand, let $t \geqslant s_X(\alpha)$, it follows that $F_{X|Z=1}(t)P(Z = 1) > \alpha$, which implies $F_X(t) \geqslant \alpha$, which in turn implies that $q_X(\alpha) \leqslant t$. It follows that $q_X(\alpha) \leqslant s_X(\alpha)$ because $\alpha/P(Z = 1) \geqslant \alpha$.

Similarly, for $\alpha \in [0, 1]$, we define the quantiles:

$$\begin{aligned} q_Y(\alpha) &\doteq \inf\{t : F_Y(t) > \alpha\}; \\ r_Y(\alpha) &\doteq \inf\{t : F_{Y|Z=0}(t)P(Z = 0) + P(Z = 1) > \alpha\}; \\ s_Y(\alpha) &\doteq \inf\{t : F_{Y|Z=0}(t)P(Z = 0) > \alpha\}. \end{aligned}$$

By using (9.4.5.ii), a similar argument leads to conclude that $r_Y(\alpha) \leqslant q_Y(\alpha) \leqslant s_Y(\alpha)$. Summarizing, we obtain the following theorem:

Theorem 9.2 *In the NEAT design, the quantiles of the partially identified probability distributions F_X and F_Y are partially identified by the following intervals:*

$$\textit{(i)} \quad r_X(\alpha) \leqslant q_X(\alpha) \leqslant s_X(\alpha); \quad \textit{(ii)} \quad r_Y(\alpha) \leqslant q_Y(\alpha) \leqslant s_Y(\alpha). \tag{9.4.7}$$

In what follows we give some comments on Theorem 9.2 that highlight relevant aspects that can be learned from what the data is able to identify, and to show how severe is the identification problem

underlying the NEAT design. Although the analyses are valid for any identified specification of $P(Z = z)$, $z \in \{0, 1\}$, throughout the exposition we assume the specification given in (9.3.2).

In order to describe both the lower and upper bounds of the quantiles $q_X(\alpha)$ and $q_Y(\alpha)$ as a function of the relative sizes of groups $\mathcal{P}$ and $\mathcal{Q}$, we introduce additional notation: let $S_{(X|Z=1)}$ and $S_{(Y|Z=0)}$ be the supports of the conditional distributions $F_{X|Z=1}$ and $F_{Y|Z=0}$, respectively. Let $t_m^{X|Z=1} \doteq \min\{t : t \in S_{(X|Z=1)}\}$ and $t_M^{X|Z=1} \doteq \max\{t : t \in S_{(X|Z=1)}\}$. Similarly for $t_m^{Y|Z=0}$ and $t_M^{Y|Z=0}$. The lower and upper bounds of $q_X(\alpha)$ can be expressed as quantiles of the conditional distribution $F_{X|Z=1}$, namely

$$\begin{aligned} r_X(\alpha^*) &= \inf\left\{t : F_{X|Z=1}(t) > \frac{\alpha - P(Z=0)}{P(Z=1)}\right\} = q_{X|Z=1}\left(\frac{\alpha - P(Z=0)}{P(Z=1)}\right), \\ s_X(\alpha^*) &= \inf\left\{t : F_{X|Z=1}(t) > \frac{\alpha}{P(Z=1)}\right\} = q_{X|Z=1}\left(\frac{\alpha}{P(Z=1)}\right), \end{aligned}$$

where $\alpha^* \in (0, 1)$ is the percentage of the observations lying below t and it depends on the relative sizes of groups $\mathcal{P}$ and $\mathcal{Q}$. Similarly, the lower and upper bounds of $q_Y(\alpha)$ can be expressed as quantiles of the conditional distribution $F_{Y|Z=0}$, namely

$$\begin{aligned} r_Y(\alpha^*) &= \inf\left\{t : F_{Y|Z=0}(t) > \frac{\alpha - P(Z=1)}{P(Z=0)}\right\} = q_{Y|Z=0}\left(\frac{\alpha - P(Z=1)}{P(Z=0)}\right), \\ s_Y(\alpha^*) &= \inf\left\{t : F_{Y|Z=0}(t) > \frac{\alpha}{P(Z=0)}\right\} = q_{Y|Z=0}\left(\frac{\alpha}{P(Z=0)}\right). \end{aligned}$$

Because these quantiles depend on the group's relative sizes, three cases can be distinguished, namely $P(Z = 1) < P(Z = 0)$, $P(Z = 1) > P(Z = 0)$ and $P(Z = 1) = P(Z = 0)$. Tables 9.2, 9.3 and 9.4 show the corresponding upper and lower bounds for each of these cases, respectively. The identification regions shown in these tables make explicit how severe is the non-uniqueness of the equated values due to the identification problem inherent to the NEAT design. Figure 9.3 complements the information given in these tables showing plots of the quantile functions in each case. We comment on the uniqueness issue by making reference to Table 9.2, focusing attention on the role of the relative sizes of groups $\mathcal{P}$ and $\mathcal{Q}$:

Interval 1: Let $\alpha \in [0, P(Z = 1)]$. Suppose that $F_{X|Z=1}$ stochastically dominates $F_{Y|Z=0}$, that is, $F_{X|Z=1}(t) \leqslant F_{Y|Z=0}(t)$ for all $t \in \mathcal{W} = \mathcal{X} \cap \mathcal{Y}$. This implies that $S_{(Y|Z=0)} \supseteqq S_{(X|Z=1)}$. Given that the quantile function respects the stochastic dominance (see, e.g., ?, ?), it follows that

$$q_{Y|Z=0}\left(\frac{\alpha}{P(Z=0)}\right) \leqslant q_{X|Z=1}\left(\frac{\alpha}{P(Z=0)}\right). \tag{9.4.8}$$

Because $P(Z = 1) < P(Z = 0)$, it holds that $\alpha/P(Z = 0) < \alpha/P(Z = 1)$ and

$$\begin{aligned} s_Y(\alpha^*) = q_{Y|Z=0}\left(\frac{\alpha}{P(Z=0)}\right) &\leqslant q_{X|Z=1}\left(\frac{\alpha}{P(Z=0)}\right) \\ &\leqslant q_{X|Z=1}\left(\frac{\alpha}{P(Z=1)}\right) = s_X(\alpha^*); \end{aligned}$$

see Figure 9.3(a). If for simplicity we also assume that $t_m^{X|Z=1} = t_m^{Y|Z=0}$, equating a X-score to a Y-score would lead to shrunk values of the X-scores. Notice that in this case, no explicit order between $r_X(\alpha^*)$ and $r_Y(\alpha^*)$ can be established; see Figure 9.3(d).

Remark 9.1 *The stochastic dominance assumption is only meaningful in the context of equating where the test forms to be equated might be assembled as parallel as possible, which leads to score distributions with common supports. Nevertheless, the same argument follows if instead of assuming the stochastic dominance we assume the quantile inequality (9.4.8). By doing so, the conclusions are also valid in a less restricted linking context as can be seen in ? (?, p.427).*

Interval 2: If $\alpha \in (\,P(Z=1), P(Z=0)\,)$, $q_X(\alpha^*)$ is partially identified on $(r_X(\alpha^*), s_X(\alpha^*)) = S_{(X|Z=1)}$ and thus no bounded information is obtained to learn about it. On the other hand, $q_Y(\alpha^*)$ provides information characterized by the respective identification interval. As a consequence, Y-scores on the interval $(r_Y(\alpha^*), s_Y(\alpha^*))$ would be considered equivalent to all possible X-scores; see Figure 9.3(a).

Interval 3: Let $\alpha \in [\,P(Z=0), 1\,]$. Suppose now that $F_{Y|Z=0}$ stochastically dominates $F_{X|Z=1}$, that is, $F_{Y|Z=0}(t) \leqslant F_{X|Z=1}(t)$ for all $t \in \mathcal{W} = \mathcal{X} \cap \mathcal{Y}$, which implies that $S_{(X|Z=1)} \supseteqq S_{(Y|Z=0)}$. It follows that

$$q_{X|Z=1}\left(\frac{\alpha - P(Z=0)}{P(Z=1)}\right) \leqslant q_{Y|Z=0}\left(\frac{\alpha - P(Z=0)}{P(Z=1)}\right).$$

But $P(Z=1) < P(Z=0)$, so $[\,\alpha - P(Z=1)\,]/P(Z=0) > [\,\alpha - P(Z=0)\,]/P(Z=1)$. It follows that

$$\begin{aligned} r_X(\alpha^*) = q_{X|Z=1}\left(\frac{\alpha - P(Z=0)}{P(Z=1)}\right) &\leqslant q_{Y|Z=0}\left(\frac{\alpha - P(Z=0)}{P(Z=1)}\right) \\ &\leqslant q_{Y|Z=0}\left(\frac{\alpha - P(Z=1)}{P(Z=0)}\right) = r_Y(\alpha^*). \end{aligned}$$

If for simplicity we assume that $t_M^{X|Z=1} = t_M^{Y|Z=0}$, equating an X-score to a Y-score would lead to X-scores defined on an expanded sample space. Note that in this case, no explicit order between $s_X(\alpha^*)$ and $s_Y(\alpha^*)$ can be established.

Similar comments apply for Table 9.3; see in particular Figure 9.3(b). Regarding Table 9.4, it can be noted that if $P(Z=1) = P(Z=0)$, the asymmetry observed (i.e., shrunk or expanded sample spaces) disappears but the severity of the identification problem still persists: an X-score is not uniquely transformed to a Y-score; see Figure 9.3(c). In other words, the results show that partial identification implies lack of equatability for examinees with scores inside the identification region.

9.4.3 Empirical illustration

To illustrate Theorems 9.1 and 9.2 and complement the comments given in the previous sections, we use a data set appearing in Kolen and Brennan (2014). The data consists of two 36-item test forms for which 12

Table 9.2: Lower and upper bounds for $q_X(\alpha)$ and $q_Y(\alpha)$ when $P(Z=1) < P(Z=0)$

α	$r_X(\alpha^*)$	$s_X(\alpha^*)$	$r_Y(\alpha^*)$	$s_Y(\alpha^*)$
$[\,0, P(Z=1)\,]$	$t_m^{X\|Z=1}$	$q_{X\|Z=1}\left(\frac{\alpha}{P(Z=1)}\right)$	$t_m^{Y\|Z=0}$	$q_{Y\|Z=0}\left(\frac{\alpha}{P(Z=0)}\right)$
$(\,P(Z=1), P(Z=0)\,)$	$t_m^{X\|Z=1}$	$t_M^{X\|Z=1}$	$q_{Y\|Z=0}\left(\frac{\alpha-P(Z=1)}{P(Z=0)}\right)$	$q_{Y\|Z=0}\left(\frac{\alpha}{P(Z=0)}\right)$
$[\,P(Z=0), 1\,]$	$q_{X\|Z=1}\left(\frac{\alpha-P(Z=0)}{P(Z=1)}\right)$	$t_M^{X\|Z=1}$	$q_{Y\|Z=0}\left(\frac{\alpha-P(Z=1)}{P(Z=0)}\right)$	$t_M^{Y\|Z=0}$

Table 9.3: Lower and upper bounds for $q_X(\alpha)$ and $q_Y(\alpha)$ when $P(Z=1) > P(Z=0)$

α	$r_X(\alpha^*)$	$s_X(\alpha^*)$	$r_Y(\alpha^*)$	$s_Y(\alpha^*)$
$[\,0, P(Z=0)\,]$	$t_m^{X\|Z=1}$	$q_{X\|Z=1}\left(\frac{\alpha}{P(Z=1)}\right)$	$t_m^{Y\|Z=0}$	$q_{Y\|Z=0}\left(\frac{\alpha}{P(Z=0)}\right)$
$(\,P(Z=0), P(Z=1)\,)$	$q_{X\|Z=1}\left(\frac{\alpha-P(Z=0)}{P(Z=1)}\right)$	$q_{X\|Z=1}\left(\frac{\alpha}{P(Z=1)}\right)$	$t_m^{Y\|Z=0}$	$t_M^{Y\|Z=0}$
$[\,P(Z=1), 1\,]$	$q_{X\|Z=1}\left(\frac{\alpha-P(Z=0)}{P(Z=1)}\right)$	$t_M^{X\|Z=1}$	$q_{Y\|Z=0}\left(\frac{\alpha-P(Z=1)}{P(Z=0)}\right)$	$t_M^{Y\|Z=0}$

out of the 36 items are common between both test forms (items 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, and 36). Test Form X was administered to $n_{\mathcal{P}} = 1,655$ examinees whereas the Test Form Y was administered to $n_{\mathcal{Q}} = 1,638$ examinees, so that for this data $P(Z=1) = \frac{n_{\mathcal{P}}}{n_{\mathcal{P}}+n_{\mathcal{Q}}} = \frac{1,655}{1,655+1638} = 0.503$. Subsamples were used to illustrate cases when the relative sizes of both groups differed.

We derived the identification regions for both F_X and F_Y and for the associated quantiles $q_X(\alpha)$ and $q_Y(\alpha)$. Figure 9.1 shows the identifiability bounds for different values of the relative sizes, and the corresponding target score distributions proposed in the literature that are obtained under the traditional practice of identifying using the ignorability condition for the case when $\omega = P(Z=1)$. It can be seen that the target score distributions computed as in (9.2.5) are indeed a plausible solution that lies between the identification bounds. Figure 9.1 also illustrates how the degree of information associated with F_X and F_Y due to the identification problem is quantified by the width of the identification intervals, which corresponds to the relative sizes of groups $\mathcal{P}$ and $\mathcal{Q}$ respectively. The degree of information on F_X is quantified by $P(Z=0)$, while for F_Y is quantified by $P(Z=1)$.

Figure 9.2 illustrates the fact that conditioning on A does not necessarily lead to improve the identification intervals simultaneously for F_X and F_Y (as it was commented after Theorem 9.1).

The identifiability bounds for the quantile functions are shown in Figure 9.3. The Figure corroborates the findings related to the asymmetry of sample spaces derived from Theorem 9.2. For instance, if we consider Figure 9.3(a), we can conclude that transforming Y-scores to X-scores, increases the lack of information inherent to the Y-scale; if we do the converse transformation, the degree of information inherent to the X-scale decreases (see Figure 9.3(b)). This result could be considered as a criterion to choose which direction the equating transformation should be performed. Nevertheless, it also shows

Table 9.4: Lower and upper bounds for $q_X(\alpha)$ and $q_Y(\alpha)$ when $P(Z=1)=P(Z=0)$

α	$r_X(\alpha^*)$	$s_X(\alpha^*)$	$r_Y(\alpha^*)$	$s_Y(\alpha^*)$
$[0, \frac{1}{2})$	$t_m^{X\|Z=1}$	$q_{X\|Z=1}(2\alpha)$	$t_m^{Y\|Z=0}$	$q_{Y\|Z=0}(2\alpha)$
$[\frac{1}{2}, 1]$	$q_{X\|Z=1}(2\alpha-1)$	$t_M^{X\|Z=1}$	$q_{Y\|Z=0}(2\alpha-1)$	$t_M^{Y\|Z=0}$

that the degree of information can *not* be increased arbitrarily because it depends on the relative sizes of populations $\mathcal{P}$ and $\mathcal{Q}$. Moreover, it is palatable that the lack of symmetry is due to the fact that $P(Z=0) \neq P(Z=1)$. When both relative sizes are equal, symmetry seems to be recovered (see Figure 9.3(c)), but the lack of information by design persists: if a score on $\mathcal{X}$ is transformed to the $\mathcal{Y}$ scale using φ, no *unique* equivalent Y-score is obtained.

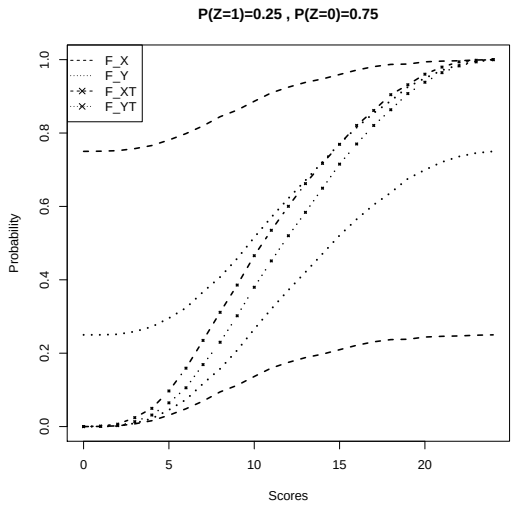
9.5 Improved partial identification intervals

In this section we show how the partial identification interval can be improved in the sense that their widths decrease. This can be done by introducing alternative assumptions. We motivate them in the context of a real data set example.

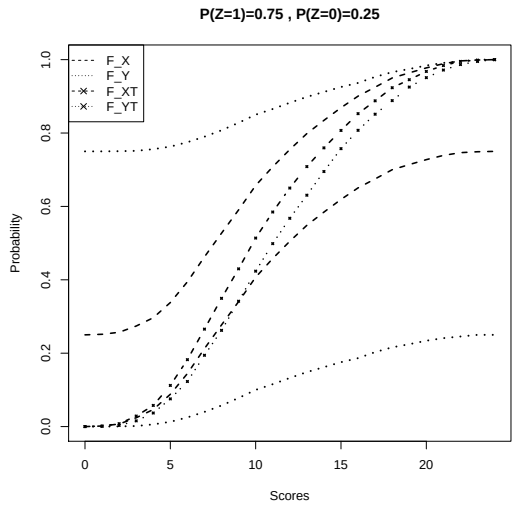
9.5.1 Motivation

The Chilean *Prueba de Selección Universitaria* test (PSU, by their initials in Spanish) is a university entrance test composed of four different sections: Language, Mathematics, History and Sciences; (for details, see ?, ?). For the Sciences section, there are three different test forms, each of them composed of 80 items and sharing 54 out of the 80 items in common. More specifically, an examinee can choose among a Sciences-Physics, Sciences-Biology and Sciences-Chemistry test form, where each form contains 26 subject-specific items and $18 \times 3 = 54$ common items, 18 of each subject. A particular feature of the sciences test is that a unique *score on sciences* should be reported, no matter what of the three forms was chosen or, in other words, no matter to which of the group the examinee belongs to. Thus, to report a score on sciences, a linking procedure under a NEAT design is needed.

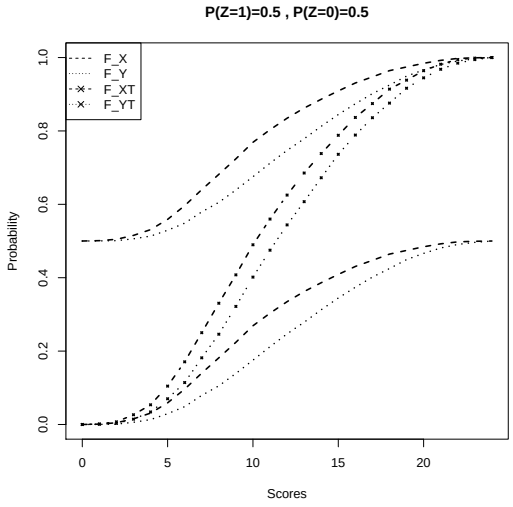
For illustrative purposes, let us focus the attention on the following two groups: group $\mathcal{P}$ corresponds to the examinees who choose the Sciences-Chemistry form; and group $\mathcal{Q}$ corresponds to the examinees who choose the Sciences-Physics form. Following the notation of the main text, Form X corresponds to Chemistry and Form Y, to Physics. Table 9.5 shows descriptive summaries of the anchor scores. Both show that the distribution of anchor scores in groups $\mathcal{P}$ and $\mathcal{Q}$ is similar. Furthermore, for group $\mathcal{P}$, the correlation between the X-scores and the anchor scores is 0.9, whereas that for group $\mathcal{Q}$, the correlation between the Y-scores and the anchor scores is 0.86.



(a)

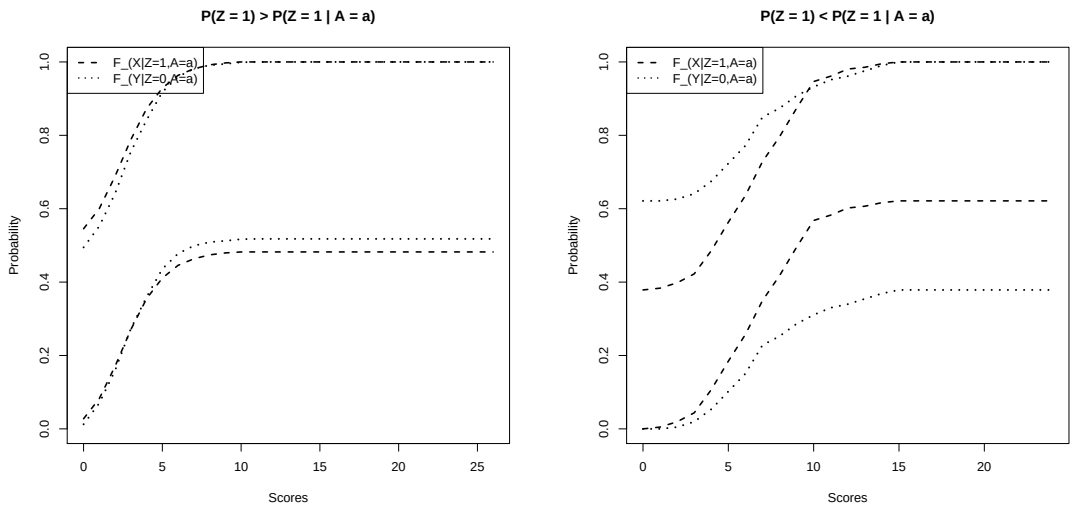


(b)



(c)

Figure 9.1: Identifiability bounds for different relative sizes



(a) (b)

Figure 9.2: Identifiability bounds for conditional score distributions

Table 9.5: Anchor scores for different quantiles of the distributions in $\mathcal{P}$ and $\mathcal{Q}$

Group	Size	$q(0.25)$	$q(0.5)$	$q(0.75)$	Min.	Max.
$\mathcal{P}$	15.048	10	17	28	0	53
$\mathcal{Q}$	13.771	11	18	30	0	53

9.5.2 Partial identification of the parameters of interest

Modelling self-selection

The fact that each examinee can choose which specific form to take raises a problem, namely how to model a process of self-selection in the personal choice of the test form. A plausible assumption to represent a "rational choice" of a test form is to assume that those who choose to take the form X do so because they believe they will score better more likely than if they had chosen form Y; and those who take the Form Y do so because they believe they will score better more likely than if they had chosen Form X. This type of strategy is indeed used in the context of the Sciences PSU test as some students are better trained in a subject than in other. As can be seen, these assumptions are intended to formalize the idea that an examinee chooses one form of testing or another because they want to maximize their score.

Given that the test forms include a set of common items, the assumption that examinees choose form X aiming to score better more likely had they chosen form Y can be represented probabilistically by at

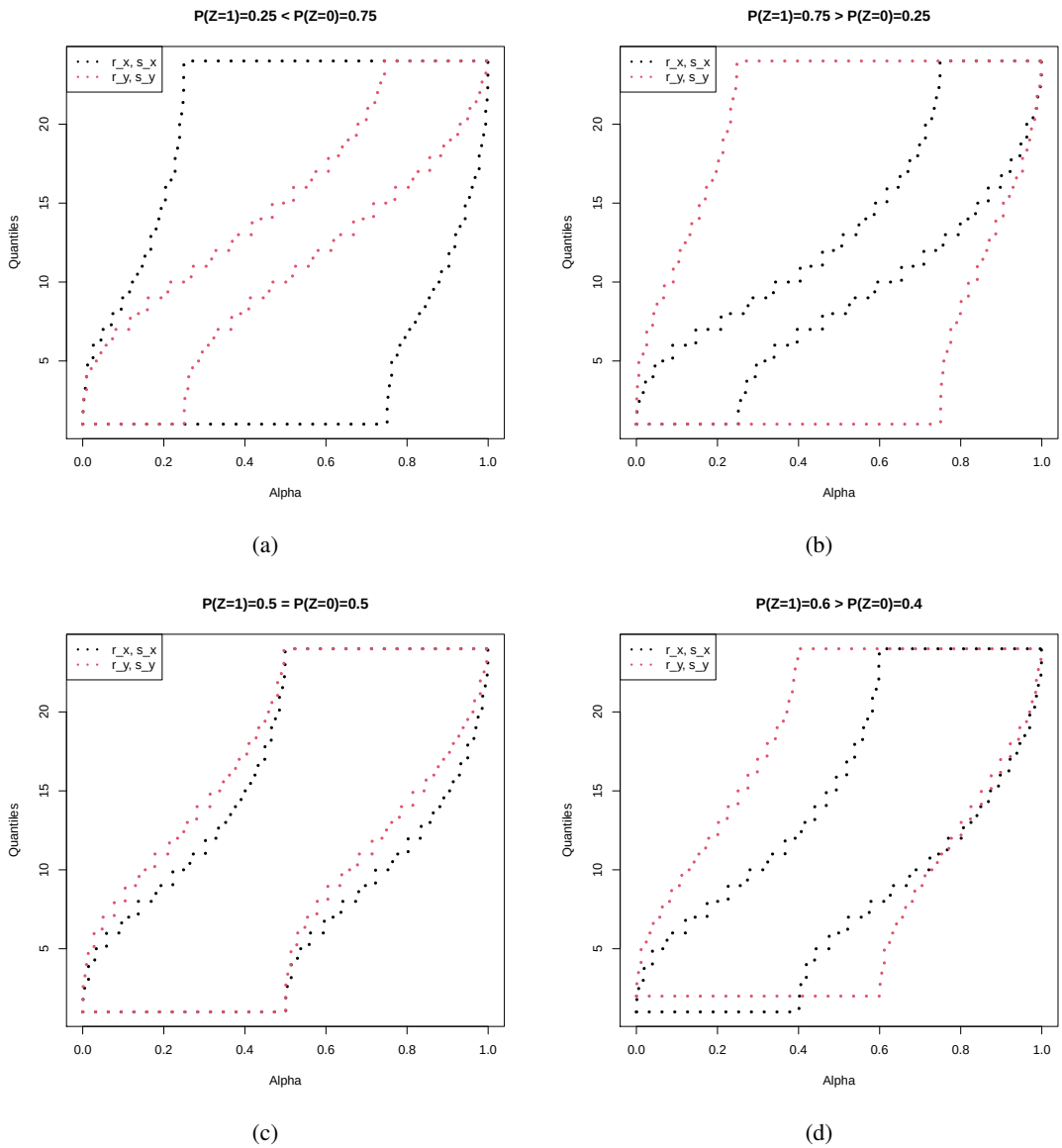


Figure 9.3: Identifiability bounds of quantiles for different relative sizes

least the following three set of conditions:

$$
\begin{aligned}
P(X > t \mid Z = 1, A = a) &> P(Y > t \mid Z = 1, A = a), \quad \forall\, (t, a) \in \mathcal{W} \times \mathcal{A}; && (9.5.1)\\
P(Y > t \mid Z = 0, A = a) &> P(X > t \mid Z = 0, A = a), \quad \forall\, (t, a) \in \mathcal{W} \times \mathcal{A}; && (9.5.2)\\
P(X > t \mid Z = 1, A \in \mathcal{A}_1) &> P(Y > t \mid Z = 1, A \in \mathcal{A}_1), \quad \forall\, t \in \mathcal{W}, \;\; \mathcal{A}_1 \subsetneq \mathcal{A}; && (9.5.3)\\
P(Y > t \mid Z = 0, A \in \mathcal{A}_1) &> P(X > t \mid Z = 0, A \in \mathcal{A}_1), \quad \forall\, t \in \mathcal{W}, \;\; \mathcal{A}_1 \subsetneq \mathcal{A}; && (9.5.4)\\
P(X > t \mid Z = 1) &> P(Y > t \mid Z = 1), \quad \forall\, t \in \mathcal{W}; && (9.5.5)\\
P(Y > t \mid Z = 0) &> P(X > t \mid Z = 0), \quad \forall\, t \in \mathcal{W}. && (9.5.6)
\end{aligned}
$$

Conditions (9.5.1) and (9.5.2) depend on all the A-scores, whereas (9.5.3) and (9.5.4) depend on some specific A-scores values, for instance $\mathcal{A}_1 = \{a \in \mathcal{A} : a \geqslant a_1\}$. Here, we further assume that examinees who obtain an anchor score $a \in \mathcal{A}_1$ are supposed to maximize their specific scores by choosing either the Form X or the Form Y. Consequently, for the remaining examinees obtaining an A-score in $\mathcal{A}_1^c$, the complement of the set $\mathcal{A}$, no self-selection assumptions are considered. Finally, conditions (9.5.5) and (9.5.6) do not depend on the anchor. It should be noted that all these condition, like the strong ignorability (9.4.1), are not empirically testable because they all depend on unidentified parameters. However, they all have an impact on the identification intervals in the sense that the stronger are the conditions, the shorter are the identification intervals obtained, which is in line with the so-called *mean monotonicity assumptions* used in econometrics (see Manski, 2007a).

Impact of the self-selection assumption on the identifiability of F_X and F_Y

The following theorem summarizes the impact of the self-selection assumptions on the identifiability of F_X and F_Y:

Theorem 9.3 *In the NEAT design, under the assumptions (9.5.1) and (9.5.2), the parameters of interest F_X and F_Y are partially identified by the following intervals: for all $t \in \mathcal{W}$,*

$$
\begin{aligned}
F_{X|Z=1}(t)P(Z=1) + F_{Y|Z=0}(t)P(Z=0) &\leqslant F_X(t) \leqslant F_{X|Z=1}(t)P(Z=1) + P(Z=0); && (9.5.7)\\
F_{X|Z=1}(t)P(Z=1) + F_{Y|Z=0}(t)P(Z=0) &\leqslant F_Y(t) \leqslant F_{Y|Z=0}(t)P(Z=0) + P(Z=1). && (9.5.8)
\end{aligned}
$$

Under assumptions (9.5.3) and (9.5.4), the parameters of interest F_X and F_Y are partially identified by the following intervals: for all $t \in \mathcal{W}$

$$
\begin{aligned}
&F_{X|Z=1}(t)P(Z=1) + F_{Y|Z=0,A\in\mathcal{A}_1}(t)P(Z=0, A\in\mathcal{A}_1) \leqslant\\
&\qquad F_X(t) \leqslant F_{X|Z=1}(t)P(Z=1) + P(Z=0); && (9.5.9)\\
&F_{X|Z=1,A\in\mathcal{A}_1}(t)P(Z=1, A\in\mathcal{A}_1) + F_{Y|Z=0}(t)P(Z=0) \leqslant\\
&\qquad F_Y(t) \leqslant F_{Y|Z=0}(t)P(Z=0) + P(Z=1). && (9.5.10)
\end{aligned}
$$

Finally, under assumptions (9.5.5) and (9.5.6), the parameters of interest F_X and F_Y are partially identified by the intervals (9.5.7) and (9.5.8), respectively.

Table 9.6: Widths of identification intervals

Identification interval	Width of the interval
(9.4.5.i)	$P(Z=0)$
(9.4.5.ii)	$P(Z=1)$
(9.5.7)	$P(Y>t, Z=0)$
(9.5.8)	$P(X>t, Z=1)$
(9.5.9)	$P(Z=0, A\in\mathcal{A}_1^c)+P(Y>t, Z=0, A\in\mathcal{A}_1)$
(9.5.10)	$P(Z=1, A\in\mathcal{A}_1^c)+P(X>t, Z=1, A\in\mathcal{A}_1)$

For a proof, see Appendix 9.9.

The upper bounds for the three sets of intervals are the same as the ones derived in Theorem 9.1. Furthermore, intervals (9.5.7) and (9.5.8) have a common lower bound despite that these identification intervals follow from two different assumptions, namely (9.5.1) and (9.5.2), or (9.5.5) and (9.5.6). In order to see how the identification intervals in Theorem 9.3 improve those in Theorem 9.1, Table 9.6 summarizes the corresponding width of the identification intervals. The identification interval (9.5.7) improves the interval (9.5.9), which in turn improves the interval (9.4.5.i) because

$$\begin{aligned} P(Y>t, Z=0) &= P(Y>t, Z=0, A\in\mathcal{A}_1)+P(Y>t, Z=0, A\in\mathcal{A}_1^c) \\ &\leqslant P(Y>t, Z=0, A\in\mathcal{A}_1)+P(Z=0, A\in\mathcal{A}_1^c) \\ &\leqslant P(Z=0, A\in\mathcal{A}_1)+P(Z=0, A\in\mathcal{A}_1^c) \\ &= P(Z=0). \end{aligned}$$

Similarly, the identification interval (9.5.8) improves the interval (9.5.10), which in turn improves the interval (9.4.5.ii). Note finally that the width of the intervals (9.5.9), (9.5.10), (9.5.7) and (9.5.8) depend on t and, consequently, their width is not constant.

These results show that the stronger an assumption is, the smaller the intervals, thus alleviating the severity of the identification problem; or, using Manski (2007a)'s jargon, this is an example of the *Law of Decreasing Credibility*: "The credibility of inference decreases with the strength of the assumptions maintained" (p.1).

Illustration

We come back to our motivating example to illustrate the results presented in this section. The identification intervals for F_X and F_Y are shown graphically in Figure 9.4. The results when no particular assumptions are made are shown in Figure 9.4(a) and are used here as a reference for comparison. The improvement of the identification intervals is graphically illustrated in Figure 9.4(b), for $\mathcal{A}_1=\{a\in\mathcal{A}: a\geqslant 30\}$, and 9.4(c), when no conditioning on A is made. It can be seen that compared to Figure 9.4(a), both identification intervals are narrower. The differences in the width of the intervals and the dependency on t is

also reflected in these figures. Note that the width of the partial identification intervals decreases as the scores increases which seems to be relevant in the context of a selection university process.

Summarizing, under assumptions (9.5.3) and (9.5.4); and (9.5.5) and (9.5.6), the partial identification intervals of the parameters of interest are better than the corresponding intervals derived in Sections 9.4.2 and 9.4.2. The example shows that if self-selection assumptions are introduced we learn something additional from the evidence (examinees' scores).

9.6 Discussion

The objective of statistical modelling is to specify the probability distribution that generates the *observables*. This modelling process corresponds to a combination of evidence (the observables, the data) with the researcher's ideas on the explanation or formation of the phenomenon studied, which in turn are assumptions about unobserved quantities. "Knowledge is the set of conclusions that one draws by combining evidence with assumptions about unobserved quantities" (Manski, 2013b). The content of this chapter is precisely in this line and tries to make explicit the *knowledge* we obtain by combining the scores provided by examinees under the NEAT design (the evidence provided by the phenomenon under analysis) with two type of assumptions used to tackle the identification problem inherent to the NEAT design: strong ignorability (discussed in Section 9.4.1) that leads to identify the score distributions; and partial identifiability (discussed in Sections 9.4.2 and 9.4.2), where no assumption regarding the unobserved quantities are needed because the aim is to learn what the data alone (i.e., without assumptions) can inform us.

The modelling process consisted in specifying the statistical model (developed in Section 9.3), that is, to specify (i) a set of probability distributions generating the observations, (ii) making explicit their parameters, and (iii) pointing out the parameters of interest. Following ? (?), the parameters of the sampling probabilities are characteristics of the observations under study. However, the researcher focuses the attention on additional characteristic of interest (represented by parameters of interest) and thus we want to know if such characteristics can be expressed as functional of the sampling probabilities: when this is not possible, we face an identification problem.

When conducting equating under the NEAT design, the parameters of interest are F_X and F_Y: our modelling strategy allows us to make explicit their meaning as well as their lack of identifiability. We suppose the researcher knows that those distributions are necessary to define the equating function (see Section 9.2.1), and that it is possible to *explicitly show* why those parameters can not be derived from the statistical model, which is done through the decompositions (9.3.5) and (9.3.7): the lack of identifiability of F_X and F_Y is due to the lack of identifiability of $F_{X|Z=0}$ and $F_{Y|Z=1}$ –which in turn follows from the lack of identifiability of $F_{X|A,Z=0}$ and $F_{Y|A,Z=1}$.

Decompositions (9.3.5) and (9.3.7) are a key step that surprisingly have not been used in the equating literature, even though it has been recognized that the lack of identifiability of $F_{X|A,Z=0}$ and $F_{Y|A,Z=1}$ can be thought as a missing data problem; see Holland et al. (2008) and Sinharay and Holland (2010). The missing data problem and, in more general terms, the selection problem becomes palatable after using the Law of Total Probability: it allows us to correctly relate F_X and F_Y with the identified condi-

tional probability distributions. Consequences of this key step are twofold: on the one hand, the notion of a target *synthetic population* is meaningless; on the other hand, the target distributions are arbitrary and therefore difficult to interpret with respect to the statistical model underlying the NEAT design.

What is the knowledge we gain when we combine the evidence (examinees' scores) with assumptions on the unobserved quantities? If we are ready to believe the strong ignorability condition, then $F_{X|A}$ and $F_{Y|A}$ become identified and, in fact, are equal to $F_{X|A,Z=1}$ and $F_{Y|A,Z=0}$. From these distributions, we obtain F_X and F_Y, and the equating function can be computed. Note that in this case $P(Z = 1)$ and $P(Z = 0)$ do not play any role.

If no assumption regarding the unobserved quantities is made, we have shown the severity of the lack of identifability of F_X and F_Y. The partial identification intervals show what we learn from the evidence in absence of assumptions regarding the unidentified parameters $F_{X|Z=0}$ and $F_{Y|Z=1}$. In particular, $P(Z = 1)$ and $P(Z = 0)$ quantify the *lack of information by design* that underlies F_X and F_Y: it involves a trade-off in the sense that the bigger $P(Z = 1)$ (respect., $P(Z = 0)$) is, the less we know about F_Y (respect., F_X).

The lack of information on the unidentified parameters can somehow be reduced by obtaining improved (shorter) identification intervals as those discussed in Section 9.5. The researcher's assumptions reflect what they think about the performance of the examinees taking the test forms. When these assumptions are combined with the observations, it is possible to assess their impact in the sense that it will become explicit to what extent the inference depends largely on such assumptions and not just on the observations. A concrete example of this modelling strategy is developed in Section 9.5.2.

In this chapter we have used the equipercentile equating transformation (see, Eq. 9.2.1) that is more appealing for the frequency estimation equating method (Angoff, 1984), also called postratification (von Davier et al., 2004). Other equating transformations based on IRT are also not exempted of the identifiability problem. For instance, in Observed-Score Equating, different constraints must be imposed to solve the identifiability problem (Sinharay & Holland, 2010; ?, ?). Also, in Local Equating (?, ?), which is based on the assumption that X, Y and A all measure the same ability, θ, the parameters of interest are the conditional distributions $F_{X|\theta}(x)$ and $F_{Y|\theta}(y)$. These parameters are identified when the items in the two forms and the anchor are jointly calibrated with appropriate identification restriction under the same response model; or when all parameters are linked afterwards through the anchor in the case of separate calibration of the two forms.

Scientific knowledge is obtained when evidence is combined with assumptions about unobserved quantities. Such assumptions become relevant when an identification problem is present. A constructive modelling process like the one developed in this article is relevant because it makes explicit to what extent scientific knowledge is highly dependent on those assumptions. Psychometrics needs to travel these avenues in order to be honest (Pielke Jr, 2007). "Scientific honesty demands that the specification of a model be based on prior knowledge of the phenomenon studied and possibly on criteria of simplicity, but not on the desire for identifiability of characteristics that the researcher happens to be interested in" (Koopmans & Reiersøl, 1950, p.169f).

Appendix

9.7 A definition of random assignment in terms of bias

A way of conceiving the "realization of score data in test equating" is motivated by the assignment of tests forms to "randomly selected" students. A clear example in test equating is the assignment of test forms to randomly selected test takers under the equivalent groups design, a scheme that can be associated with what is called a Randomized Controlled Trial. But, what is actually the meaning of random selection/random assignment/random allocation, when terms such as "random" and "realization" are mere denominations? In what follows we define "random assignment" in terms of "absence of bias", which in turn is formally described in terms of conditional probabilities.

Stephenson and Imrie (1998) point out that a randomized control trial (RCT) is the best way of measuring the efficacy of intervention "because of its ability to minimise bias and avoid false conclusions. Random assignment of individuals to different treatment groups is the best way of achieving a balance between groups for the known and unknown factors that influence outcome" (p.611). Odgaard-Jensen et al. (2011) are even more explicit in affirming that bias is reduced thanks to random assignment:

> Randomised trials use the play of chance to assign participants to comparison groups. The unpredictability of the process, if not subverted, should prevent systematic differences between comparison groups (selection bias). Differences due to chance will still occur and these are minimised by randomising a sufficiently large number of people.

Bhide et al. (2018) contrasts a RCT with an observational study: the evidence based on RCTs is the highest one because "evidence based on observational data is prone to bias". *Bias* is understood as "the systematic tendency of any factors associated with the design, conduct, analysis, evaluation and interpretation of the results of a study to make the estimate of the effect of a treatment or intervention deviate from its true value". The presence of bias in an observational study is described in the following terms: "If two or more groups are being compared in an observational study, there are often systematic differences between the groups, so much so that the outcome of the groups may be different because of these differences rather than actual exposure or intervention" (p.381). The way to overcome this serious drawback is through a random assignment:

> The only way to eliminate these differences is to allocate each individual to one or the other intervention at random. Therefore, the probability of any individual receiving one intervention or the other is decided solely by chance (p.381).

All these statements provide us with an intuition, but they fail to be formal in probabilistic terms because *random* is not a probabilistic concept, just a designation, as we mentioned in Section 9.3.2. However, they provide us with a term that can indeed be defined in probabilistic terms: *bias*. We propose a *structural definition* of bias, that is, one that makes explicit the concept of bias and, therefore, is applicable to many concrete situations.

This definition arise from the Law of Total Probability because it allows us to make explicit the counterfactual aspect underlying bias selection. More specifically, let M be the sample space representing the

population of interest. Let V be a random variable defined on M representing the outcome of interest (for instance, the consequence of a treatment), and C a random vector defined on M describing characteristics of the statistical units. The target is to learn about $P(V \leqslant v \mid C)$ from $P(V \leqslant v \mid C, Z = 1)$, where Z denotes the random assignment variable, namely

$$Z = \begin{cases} 1, & \text{if a statistical unit in } M \text{ is selected;} \\ 0, & \text{if a statistical unit in } M \text{ is not selected.} \end{cases}$$

Since Z induces a partition on M, it is natural to decompose $P(V \leqslant v \mid C)$ through the Law of Total Probability, namely

$$P(V \leqslant v \mid C) = P(V \leqslant v \mid C, Z = 1)P(Z = 1 \mid C) + P(V \leqslant v \mid C, Z = 0)P(Z = 0 \mid C);$$

here $P(V \leqslant v \mid C, Z = 0)$ corresponds to the conditional probability of the outcome that the statistical units would have experienced if they had been assigned to the treatment. Consequently, to learn about $P(V \leqslant v \mid C)$, it is possible to ignore this last conditional probability only if

$$P(V \leqslant v \mid C, Z = 1) = P(V \leqslant v \mid C, Z = 0), \tag{9.7.1}$$

or equivalently, if $V \perp\!\!\!\perp Z \mid C$, which means *absence of bias*.

Additionally, if the statistical units in $Z^{-1}\{1\}$ were chosen without taking into account their characteristics captured by C, then

$$Z \perp\!\!\!\perp C. \tag{9.7.2}$$

A RCT can thus accordingly be defined through the structural conditions (13.1.1) and (13.1.2). Furthermore, *bias* should be defined as the logical negation of (13.1.1). Let us remark that we call these conditions *structural* because we emphasize the "form" or "anatomy" of the concept, which need to be revealed for specific problems. Moreover, up to the best of our knowledge, there not exist a formal proof deriving conditions (13.1.1) and (13.1.2) from "random assignment mechanism": if the proof existed, these conditions would be a consequence; if not, they would be a formal definition.

9.8 Proof of the equivalence between (9.2.5) and (9.4.2)

If $\omega = P(Z=1)$, then

$$\begin{aligned}
F_X(x) &= F_{X|Z=1}(x)P(Z=1) + F_{X|Z=0}(x)P(Z=0) \\
&= F_{X|Z=1}(x)P(Z=1) + \sum_{a\in\mathcal{A}} P(X \leqslant x \mid A=a, Z=0)P(A=a \mid Z=0)P(Z=0), \quad \text{namely (2.5)} \\
&= F_{X|Z=1}(x)P(Z=1) + \sum_{a\in\mathcal{A}} P(X \leqslant x \mid A=a, Z=1)P(A=a \mid Z=0)P(Z=0) \quad \text{by (4.1)} \\
&= \sum_{a\in\mathcal{A}} P(X \leqslant x \mid Z=1, A=a)P(A=a \mid Z=1)P(Z=1) + \\
&\quad \sum_{a\in\mathcal{A}} P(X \leqslant x \mid Z=1, A=a))P(A=a \mid Z=0)P(Z=0) \\
&= \sum_{a\in\mathcal{A}} P(X \leqslant x \mid Z=1, A=a)P(A=a, Z=1) + \sum_{a\in\mathcal{A}} P(X \leqslant x \mid Z=1, A=a)P(A=a, Z=0) \\
&= \sum_{a\in\mathcal{A}} P(X \leqslant x \mid Z=1, A=a)P(A=a),
\end{aligned}$$

which is precisely (4.2.i). Similar arguments can be used to obtain (4.2.ii).

9.9 Proof of Theorem 9.3

Proof of 9.5.7: Using (9.5.2) to see that $F_{Y|A,Z=0} < F_{X|A,Z=0} \leqslant 1$, the result follows directly from decomposition (9.3.5) after marginalizing with respect to A. The proof of (9.5.8) is obtained similarly using (9.5.1) and (9.3.7).

Proof of 9.5.9: Note first that

$$\begin{aligned}
&\text{(i)} \quad F_X(t) = F_{X|A\in\mathcal{A}_1}(t)P(A\in\mathcal{A}_1) + F_{X|A\in\mathcal{A}_1^c}(t)P(A\in\mathcal{A}_1^c); \\
&\text{(ii)} \quad F_Y(t) = F_{Y|A\in\mathcal{A}_1}(t)P(A\in\mathcal{A}_1) + F_{Y|A\in\mathcal{A}_1^c}(t)P(A\in\mathcal{A}_1^c),
\end{aligned} \tag{9.9.1}$$

where

$$\begin{aligned}
&\text{(i)} \quad F_{X|A\in\mathcal{A}_1}(t) = F_{X|A\in\mathcal{A}_1,Z=1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + F_{X|A\in\mathcal{A}_1,Z=0}(t)P(Z=0 \mid A\in\mathcal{A}_1); \\
&\text{(ii)} \quad F_{X|A\in\mathcal{A}_1^c}(t) = F_{X|A\in\mathcal{A}_1^c,Z=1}(t)P(Z=1 \mid A\in\mathcal{A}_1^c) + F_{X|A\in\mathcal{A}_1^c,Z=0}(t)P(Z=0 \mid A\in\mathcal{A}_1^c); \\
&\text{(iii)} \quad F_{Y|A\in\mathcal{A}_1}(t) = F_{Y|A\in\mathcal{A}_1,Z=1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + F_{Y|A\in\mathcal{A}_1,Z=0}(t)P(Z=0 \mid A\in\mathcal{A}_1); \\
&\text{(iv)} \quad F_{Y|A\in\mathcal{A}_1^c}(t) = F_{Y|A\in\mathcal{A}_1^c,Z=1}(t)P(Z=1 \mid A\in\mathcal{A}_1^c) + F_{Y|A\in\mathcal{A}_1^c,Z=0}(t)P(Z=0 \mid A\in\mathcal{A}_1^c),
\end{aligned} \tag{9.9.2}$$

for all $t \in \mathcal{W}$. In (9.9.2.i), $F_{X|A\in\mathcal{A}_1,Z=0}$ is unidentified. Using (9.5.3), the corresponding lower bound of $F_{X|A\in\mathcal{A}_1}$ is given by

$$F_{X|Z=1,A\in\mathcal{A}_1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + F_{Y|Z=0,A\in\mathcal{A}_1}(t)P(Z=0 \mid A\in\mathcal{A}_1). \tag{9.9.3}$$

Furthermore, $F_{X|A\in\mathcal{A}_1,Z=0} \leqslant 1$ and, therefore, the corresponding upper bound of $F_{X|A\in\mathcal{A}_1}$ is given by

$$F_{X|Z=1,A\in\mathcal{A}_1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + P(Z=0 \mid A\in\mathcal{A}_1). \tag{9.9.4}$$

Consider now (9.9.2.ii). Taking into account that no self-selection condition is assumed for those examinees scoring $a \in \mathcal{A}_1^c$, it follows that the lower bound of $F_{X|A\in\mathcal{A}_1^c}$ is given by

$$F_{X|Z=1,A\in\mathcal{A}_1^c}(t)P(Z=1 \mid A\in\mathcal{A}_1^c); \tag{9.9.5}$$

and the corresponding upper bound by

$$F_{X|Z=1,A\in\mathcal{A}_1^c}(t)P(Z=1 \mid A\in\mathcal{A}_1^c) + P(Z=0 \mid A\in\mathcal{A}_1^c). \tag{9.9.6}$$

Combining (9.9.3) and (9.9.2.i), and (9.9.5) and (9.9.2.ii), the lower identification bound for F_X in (9.9.1.) is obtained as

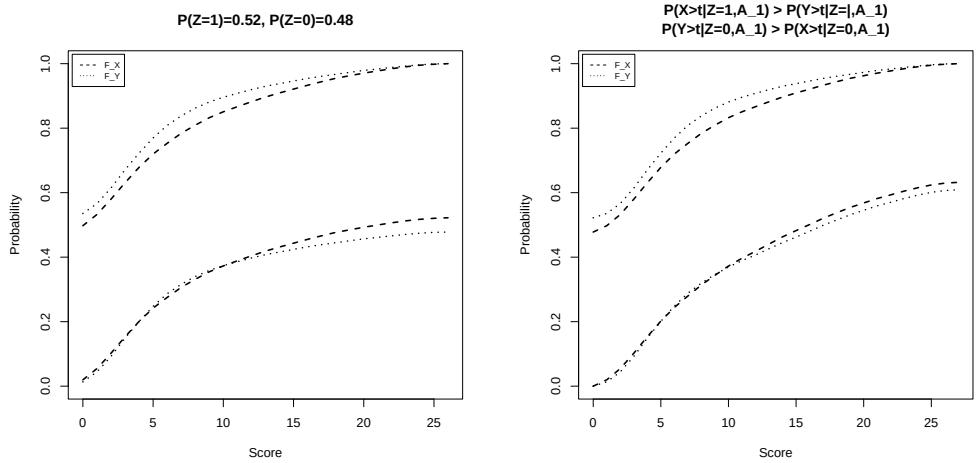
$$\begin{aligned} F_X(t) &\geqslant \left\{F_{X|Z=1,A\in\mathcal{A}_1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + F_{Y|Z=0,A\in\mathcal{A}_1}(t)P(Z=0 \mid A\in\mathcal{A}_1)\right\} P(A\in\mathcal{A}_1) + \\ &\quad F_{X|Z=1,A\in\mathcal{A}_1^c}(t)P(Z=1 \mid A\in\mathcal{A}_1^c)\, P(A\in\mathcal{A}_1^c) \\ &= P(X\leqslant t, Z=1, A\in\mathcal{A}_1) + P(Y\leqslant t, Z=0, A\in\mathcal{A}_1) + P(X\leqslant t, Z=1, A\in\mathcal{A}_1^c) \\ &= F_{X|Z=1}(t)P(Z=1) + F_{Y|Z=0,A\in\mathcal{A}_1}(t)P(Z=0, A\in\mathcal{A}_1). \end{aligned}$$

Similarly, combining (9.9.4) and (9.9.2.i), and (9.9.6) and (9.9.2.ii), the upper bound for F_X in (9.9.1.) is obtained as

$$\begin{aligned} F_X(t) &\leqslant \left\{F_{X|Z=1,A\in\mathcal{A}_1}(t)P(Z=1 \mid A\in\mathcal{A}_1) + P(Z=0 \mid A\in\mathcal{A}_1)\right\} P(A\in\mathcal{A}_1) + \\ &\quad \left\{F_{X|Z=1,A\in\mathcal{A}_1^c}(t)P(Z=1 \mid A\in\mathcal{A}_1^c) + P(Z=0 \mid A\in\mathcal{A}_1^c)\right\} P(A\in\mathcal{A}_1^c) \\ &= P(X\leqslant t, Z=1, A\in\mathcal{A}_1) + P(Z=0, A\in\mathcal{A}_1) + P(X\leqslant t, Z=1, A\in\mathcal{A}_1^c) + \\ &\quad P(Z=0, A\in\mathcal{A}_1^c) \\ &= F_{X|Z=1}(t)P(Z=1) + P(Z=0). \end{aligned}$$

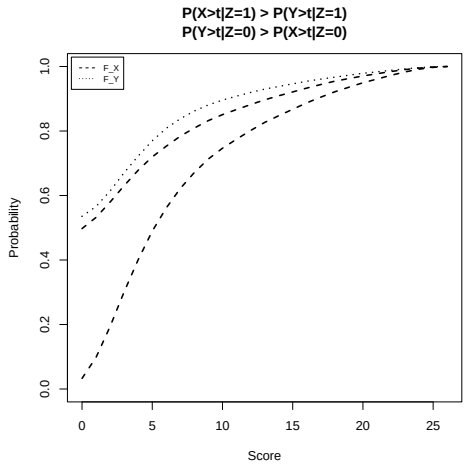
The partial identification intervals for F_Y are obtained using similar arguments.

■



(a)

(b)



(c)

Figure 9.4: Identifiability bounds for the score distributions of the PSU Sciences test: 9.4(a) under no-assumptions; 9.4(b) under a self-selection process conditioning on $\mathcal{A}_1 = \{a \in \mathcal{A} : a \geqslant 30\}$; 9.4(c) under a self-selection process without conditioning on A (for the lower bounds, the point and segmented curves are superimposed in (c))

Chapter 10

Brokering the Evaluation of Public Policies

10.1 Introduction

Most of the time[1], three agents come together in a public policy decision process: policymakers, politicians and scientists. Policymakers perform *decisions* because policy "is a decision, a commitment to particular course of action" (Pielke Jr, 2007), "it is the sum total of government action, from signals of intent to the final outcomes" (Cairney, 2016). Politicians do the necessary "bargaining, negotiation, and compromise in pursuit of desired ends related to *political values*" (Pielke Jr, 2007, the italics are added). Thus, as it is widely known, the political values shape the meaning and scope of the action. Whatever the political values are, public policies intend to improve the welfare of society through interventions. The meaning of society's well-being depends on those values, (implicitly) assuming a causal relationship between interventions and welfare.

10.1.1 Public policy implementation and uncertainty

Such a casual relation, which can be considered as a belief (Müller, Garcia-Retamero, Cokely, & Maldonado, 2011), is subject to uncertainty due at least to some of the following issues: a specific intervention could imply outcomes that are inconsistent with the expectations of the policymaker (Pielke Jr, 2007, Chapter 5), leading even to unintended and perverse impacts (Van Woensel, 2020, Chapter 3); a policymaker can understand and try to solve a policy problem in many ways because a single event can take place in a multitude of possible states (Cairney, 2016); and a policymaker takes a decision with a partially known welfare function and/or probabilities of the outcomes (Manski, 2007a, Chapter 11).

In a context of uncertainty, science seems to be useful because it also looks for welfare of society and, consequently, their findings are considered as progress of and for humanity. Just some leading examples: the American Psychological Association is called "to make a positive impact on critical societal issues" and "to promote the advancement, communication, and application of psychological science and

[1]This chapter is based on San Martín and González-Larrondo (2020).

knowledge to benefit society and improve lives"[2]. The mission of the American Educational Research Association is "to promote the use of research to improve education and serve the public good"[3]. Science then plays the role of reducing the uncertainty of the causal intervention-welfare relationship. To do this, scientific research not only produces evidence, but through an inter-subjective agreement defines criteria that allow to graduate the quality of scientific evidence, in addition to developing statistical methodologies relevant to such graduation (for instance, systematic revision of randomized clinical trials or meta-analysis; causal inference for observational data; and so on). This is the so-called *Evidence-Based Practice* (EBP) framework, from which Evidence Based Medicine and Evidence Based Policy are key instances. For references and applications of Evidence Based Medicine, see Cochrane Initiative (www.cochrane.org); and for Evidence Based Policy, see The Campbell Collaboration (campbellcollaboration.org).

10.1.2 The features of the EBP paradigm

The EBP paradigm is highly appreciated by both academicians and politicians because "evidence *matters* for public policymaking" (Parkhurst, 2017, p.4) given that a rigorous use of it could avoid unnecessary harms, and evidence tells us "what works". These advantages are due to the fact that the EBP framework intends to solve the problem of the gap between policy practices and research, improving thus the quality of policies and interventions (Cochrane, 1972; OECD, 2007; European Commission, 2017). Politicians and policymakers perceive quantitative evidence-based research as the more legitimate framework because of its eventual accuracy and trustworthiness (Wiseman, 2010), which leads to consider it as the way to ensure "objectivity" (King, 2010). This perspective stresses that, when relating science and public policy, the emphasis is placed on the quality of the scientific evidence used. The challenge to reduce uncertainty in decision-making is accordingly to recognize and rely on the best available evidence defined according to the inter-subjectivity international agreement mentioned above (Sackett, Straus, Richardson, Rosenberg, & Haynes, 2000; Clearinghouse, 2007; Head, 2010; Burns, Rohrich, & Chung, 2011; Schünemann, 2016; Rousseau & Gunia, 2016; Ryan & Hill, 2016). The motto of Evidence Based Medicine, of Evidence Based Policy and, in general, of Evidence Based Practice is accordingly that *policy decisions should be based on the better available evidence* (Davies & Boruch, 2001).

10.1.3 The consequences of the EBP paradigm

According to the EBP paradigm, the welfare of society is ensured by science (knowledge, research, applications). This fits with the *linear model of science* (V. Bush, 1945; Godin, 2006, 2011; Pielke Jr, 2007, 2012), a paradigm that brings politics into science rather than science into policy. In fact, it argues that science can and should compel political outcomes given that *right science bases better decision-making*. The purpose of the policy action is, consequently, shaped by "right science" in the sense that the right scientific results deserve to become public policies. This attitude corresponds to the politicization of science by scientist, being its final stage to substitute politics by science.

[2]From https:www.apa.orgmonitor201905ceo; retrieved on September 22, 2019.
[3]From https:www.aera.net/About-AERA; retrieved on September 22, 2019.

Following Pielke Jr (2007)'s taxonomy of the role of scientist in policy and politics, the previous attitude could lead to an advocacy scientist role: under the linear model paradigm, science becomes a convenient and necessary means for removing certain options from a debate without explicitly dealing with disputes over values (politics). For examples of this attitude, see Manski (1992)'s analysis of Friedman's perspective on voucher and social mobility; or Pielke Jr (2007)'s analysis of scientists' attitude regarding Lomborg's *The Skeptical Environmentalist*. The conclusions are determined according to a standardized quality of evidence, which is related to what is considered correct science through an uncritical revision of an inter-subjective agreement –in this case, Evidence Based Medicine and Evidence Based Policy standards.

10.1.4 Limitations of the EBP paradigm due to additional uncertainties

Although the decision of public policies seeks to decrease uncertainty through the EBP paradigm, there exists additional uncertainties that limit the power of EBP. These uncertainties emerge when the decision making reasoning is analyzed under a descriptive decision theory rather than a normative theory: instead of assessing the decision making with respect to normative theories as those developed by Wald (1950) and Von Neumann, Morgenstern, and Kuhn (2007), the challenge is to describe how and to what extend the context affects the decision making; for details, see Kahneman and Tversky (1979), Parmigiani and Inoue (2009) and Peterson (2017).

A first uncertainty is due to biases of scientific policy advisors. As a matter of fact, following Van Woensel (2020), scientific policy advisers collect information and on its basis formulate alternative policy options. In this process, biases can cause systematically distorted perceptions of facts and opinions, which lead to incorrectly weight the evidence. Van Woensel (2020) proposes a tool (the *bias-wheel*) for bias-awareness in scientific advising. Six types of biases are necessary to take into account in order to acquire a critical view on the own biases: research biases, cultural and value biases, attention biases, interest biases, availability biases and associative biases; for details, see Van Woensel (2020, Chapter 2). By using the bias-awareness tool, the one's biases toward new information will be reduced, leading to less uncertain evidence-based policy decisions.

A second uncertainty has been made explicit by the Policy Analysis perspective. The distinction between normative and descriptive decision theories allows to realize that policy decisions are not ultimately based on scientific evidence. Several reasons explain this: sometimes scientific evidence does not interest; in other occasions, it is used to manipulate, pushing thus a specific political value; sometimes it is necessary to consider institutional and organizational mechanisms that influence decisions; and so on. The point is that a policy decision making is complex and subject to political ambiguities; see, among many others, Rycroft-Malone and Bucknall (2011), Cairney (2016), Cairney, Oliver, and Wellstead (2016), Cairney (2017), Parkhurst (2017), Newman (2017) and Strassheim (2017).

A third uncertainty comes from deviations of both policymakers and politicians from the normative theory. As a matter of fact, prospect theory (a descriptive decision theory; see Kahneman & Tversky, 1979) has been applied to policy decision making (see, among many others, Dacey, 2003; Levy, 2003). Linde and Vis (2017) has applied two concepts of the prospect theory, namely *probability weighting* and *reflection effect*, to experimentally analyze the decisions made by a sample of Dutch parliamentarians;

for details on these concepts, see Levy (1992) and Kahneman and Tversky (2013). Linde and Vis (2017) builds prospects such as those presented by Allais (1953) and Kahneman and Tversky (1979). In particular, they frame prospects in terms of profits and their reflections in terms of losses; this type of prospects is inspired in the classical Asian disease problem (Kahneman & Tversky, 1984), where a choice between voting in favor of implementing program A or voting against it in which case program B is implemented. The conclusions are that decisions depend on the *framing* of the decision situation (Tversky & Kahneman, 1981), but when participants are "familiar with the type of decision" then decisions are taken according to the normative theory.

10.1.5 Are policies based on scientific evidence or on scientific knowledge?

The previous critical frameworks of the EBP paradigm share a common feature, namely to critically describe how policymaker *relate to* the scientific evidence. Such a relation can be affected either by the scientific advisors biases, or by the complexity and ambiguity of the political context, or by deviations from a normative decision theory. It could be concluded that if such uncertainties were non-existent, the EBP paradigm would work without limitations. This conclusion is reinforced because the three critical frameworks already described do not develop (up to the best of our knowledge) a critique of the way in which *scientific evidence* becomes *scientific knowledge*. It is important to make a distinction between scientific knowledge and scientific evidence: "Knowledge is the set of conclusions that one draws by combining evidence with assumptions about unobserved quantities" (Manski, 2013a). What it is really useful for political decision making is scientific knowledge, and not just evidence. The evidence is not a subject whose light and majesty we bow to (paraphrasing Toland, 1702, one of the first to describe the properties of evidence in the context of reason and its powers), but an object that a subject (the scientist) uses to share (some) knowledge.

What we mean by *assumptions about unobserved quantities*? On the one hand, let us recall a terminological distinction: *assumption* here does not correspond to the concept of *statistical hypothesis* in the sense that such hypothesis can be tested through the hypothesis testing techniques (Oaks, 1986; Lehmann, 2011; Gigerenzer, 2004) and eventually falsifiable (for the potential relationships between hypothesis testing and the popperian concept of falsification, see Oaks, 1986; Lehmann, 1995; Mayo & Spanos, 2006). On the other hand, the concept of *assumptions* is a component of the empirical research, which can schematically be represented as follows (Manski, 2013c):

$$\text{Assumptions} \quad + \quad \text{Data / Empirical evidence} \quad \Longrightarrow \quad \text{Conclusions / Recommendations.}$$

Here *assumptions* correspond to the explanation or formation of the studied phenomenon, which is the spring of the substantive theory (Mouchart & Russo, 2011; Wolpin, 2013; Wunsch et al., 2014; Felline, 2018). Paraphrasing Koopmans (1949), statistical inference unsupported by a substantive theory applies to whatever statistical regularities and, consequently, stable relationships can be discerned in the data. Such purely empirical relationships are likely to be due to the presence and persistence of underlying structural relationships. However, the direction of this deduction cannot be reversed (from the empirical to the structural relationships) expect possibly with the help of a substantive theory which specifies the form of the structural relationships. The possibility of this deduction necessarily implies to solve an

identification problem at the model construction level. As Haavelmo (1997) puts it,

> The one type of problems, seemingly paradoxical, grew out of a rather intricate consequence of successful economic theory. Strangely enough, the fact is that if an economic theory, an economic relation, is a good theory, true to reality, it may not be possible to quantify it by using data from the economic of which that relation is a part. This is the so-called "problem of identification" (p.14).

These considerations are applicable to all social and health sciences, among many others. This is a requirement of a scientific inquiry that focuses on the explanation of the phenomena, and not (only) on the mere prediction with implicit assumptions of structural stability (as is the case with most of the recent data science techniques).

Technically speaking, an observed phenomenon is captured through a set of probability distributions, defining thus the so-called *statistical model* (Fisher, 1922). This modelling approach is due to the fact that observed phenomena are heterogeneous and therefore subject to variability. Now, such probability distributions include parameters, which represent characteristic of the observed population under study. However, most of the time, scientists look deducing facts based on the observed phenomenon: when these facts can not be represented as a functional of the statistical model (that is, when such facts are not represented through a parameter), we face an identification problem. An identification problem can only be resolved through assumptions. There is no single way to solve it: when an assumption is chosen and combined with the evidence, we gain *a* specific scientific knowledge. If we change the assumptions, a *different* scientific knowledge could arise. For details and discussion, see San Martín (2015) and San Martín (2018).

We argue that the evaluation of public policies requires this type of assumptions leading to solve inherent identification problems. In this chapter, we not only show that there is not a unique way to establish such assumptions, but we argue that a scientist should explicitly explore more than one assumption in order to expand policy choices. By so doing, it is possible to make explicit how the scientific evidence becomes scientific knowledge, which in turn is mediated by subjective choices. In this way, we do not intend to look for additional elements leading to improve the EBP paradigm, but rather to consider the relationship science-policy-politics as a problem to be studied, looking for a *prescriptive solution* (Manski, 2019) of the relationship between science, policy and politics, respecting the difference there exists between them.

10.2 How to construct a prescriptive role for science in policy decisions?

Following Pielke Jr (2007), science can play a constructive role in policy if organizations and individuals are willing to expand the range of options available to policymakers "by serving as Honest Broker of Policy Alternative" (p.141). In the context of policy debate the following questions may be useful to ask suppliers of policy analyses that are based on science: *(1) If your recommendation is indeed based on scientific results, what scientific information would be necessary to change your policy position? (2) A range of policies is consistent with particular scientific results. What is the full range of options that you*

see as consistent with the state of science in order to achieve particular desired ends? (Pielke Jr, 2007, pp.141-142).

We propose to broker the evaluation of public policies in the context of observational studies using partial identification. The *basic elements* of the context are the following: (1) The public policy decision problem consists in implementing an innovation or not. (2) The evidence available to take a decision is an observational study, which is summarized by two conditional probabilities, $P(Y = 1 \mid Z = 1, X)$ and $P(Y = 1 \mid Z = 0, X)$, where Y is a policy relevant outcome. Without loosing generality, we assume that it is a binary random variable: when $Y = 1$ we will say that the statistical unit (or, more briefly, individual) experiences the expected beneficial outcome, and if $Y = 0$, that the beneficial outcome is not experienced by the individual. $Z = 1$ means that the individual is under an *innovation*, whereas $Z = 0$ that is under the *status quo*; and X is a set of covariates characterizing subpopulations. (3) The decision should be based on a comparison between $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$, where Y_1 is the outcome as if all the individuals were under the innovation, and similarly for Y_0. The *problem* is that those conditional probabilities are unidentified. This can be viewed by considering the following fundamental decompositions (which follows from the Law of Total Probability):

$$P(Y_1 = 1 \mid X) = P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y_1 = 1 \mid X, Z = 0)P(Z = 0 \mid X); \quad (10.2.1)$$
$$P(Y_0 = 1 \mid X) = P(Y_0 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y_0 = 1 \mid X, Z = 0)P(Z = 0 \mid X). \quad (10.2.2)$$

The evidence alone is enough to identify $P(Z = 1)$ –the proportion of individuals under innovation, and therefore those under the status quo. It is also enough to identify $P(Y_1 = 1 \mid Z = 1, X)$ because it corresponds to the distribution of the outcome that an individual would experience under the innovation when it is actually under the innovation ($Z = 1$); that is, $P(Y_1 = 1 \mid X, Z = 1) = P(Y = 1 \mid X, Z = 1)$. By a similar argument, $P(Y_0 \mid Z = 0, X) = P(Y = 1 \mid X, Z = 0)$ and, therefore, it is identified. However, the evidence is not enough to identify $P(Y_1 = 1 \mid Z = 0, X)$, that is, the distribution of the outcome that an individual would experience under the innovation when it is actually under the status quo. By a similar argument, $P(Y_0 = 1 \mid Z = 1, X)$ is unidentified. This is the so-called Fundamental Problem of Causal Inference (?, ?).

According to the Honest Broker perspective (Pielke Jr, 2007), a public policy evaluation should consider different policy alternatives. These ones intend to make explicit the complexity of the decision making process: once we have empirical evidence in the form of an observational study, it is reasonable to assume that there is more than one alternative to drive the implementation of innovation or the maintenance of the status quo.

10.2.1 The standard solution

Before displaying the complexity of the decision making process by making explicit different alternatives, let us revisit the standard solution to the identification problem which is based on the *exogenous switching condition* (Maddala, 1983) or *strong ignorability condition* (Rosenbaum & Rubin, 1983). It asserts statistical independence of the realized treatments and the outcomes, conditional on a set of covariates. This condition mimics the definition of a randomized experiment (Fisher, 1935; Cochran &

Chambers, 1965; Angrist & Pischke, 2008), but it is not empirically refutable: it can be justified in specific applications only, the justification essentially depending on the choice of specific covariates X. More precisely, it is assumed that

$$(Y_0, Y_1) \perp\!\!\!\perp Z \mid X, \tag{10.2.3}$$

which combined with (10.2.1) and (10.2.2) implies that

$$\begin{aligned} P(Y_1 = 1 \mid X, Z = 1) &= P(Y_1 = 1 \mid X, Z = 0) = P(Y_1 = 1 \mid X); \\ P(Y_0 = 1 \mid X, Z = 1) &= P(Y_0 = 1 \mid X, Z = 0) = P(Y_0 = 1 \mid X). \end{aligned}$$

Condition (10.2.3) deserves two remarks and one question:

1. It is an identification restriction: it restricts the parameter space of $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$ to two points, which precisely are the information provided by the observational study.

2. It does not solve the Fundamental Problem of Causal Inference which states that "we do not observe a same statistical unit under both the innovation and the status quo" (?, ?). Condition (10.2.3) states, for instance, that the distribution of the outcome that an individual would experience under the innovation when it is actually under the status quo is exactly equivalent to the distribution of the outcome that an individual should experience under the innovation when it is actually under the the innovation.

3. Is it possible to provide a policy interpretation of such a condition? Following the logic of the empirical research (see Section 10.1.5, condition (10.2.3)) is combined with evidence in order to provide policy recommendations. However, it is important to stress what kind of policy behavior and/or political value is being affected by such a condition. We will come back to this later.

10.2.2 Optimistic policymaker

Using Manski (2007a)'s approach on partial identification, we propose six solutions leading to solve of the Fundamental Problem of Causal Inference. These solutions are obtained after comparing $P(Y_1 = 1 \mid X, Z = 1)$ with $P(Y_0 = 1 \mid X, Z = 1)$, and $P(Y_1 = 1 \mid X, Z = 0)$ with $P(Y_0 = 1 \mid X, Z = 0)$. A first solution is based on two *policymaker's optimistic perspectives*: the first one is characterized by a *convincing policymaker's behavior*, whereas the second one by an *ambiguous policymaker's perspective*.

Convincing policymaker's optimistic perspective

A convincing policymaker's optimistic (CPO) perspective is defined by the following inequalities:

$$\begin{aligned} &\text{(i)} \quad P(Y_1 = 1 \mid X, Z = 1) > P(Y_0 = 1 \mid X, Z = 1); \\ &\text{(ii)} \quad P(Y_1 = 1 \mid X, Z = 0) \geqslant P(Y_0 = 1 \mid X, Z = 0). \end{aligned} \tag{10.2.4}$$

This perspective is characterized by (i) promoting the idea that if individuals who are actually under the innovation, were under the status quo, their performance would be worse; that is, the status quo harms

individuals under the innovation –this corresponds to condition (10.2.4.i). And (ii) if individuals who are actually under the status quo were under the innovation, their performance would be at least as the performance under the status quo; that is, the innovation does not harm individuals under the status quo -this corresponds to condition (10.2.4.ii).

What the impact of the CPO perspective on the identifiability of both $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$ is? Or, equivalently, what are all the plausible values of $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$ that are compatible with the CPO perspective? These questions can be answered after combining (10.2.4) with the basic decompositions (10.2.1) and (10.2.2). By so doing, we obtain the following partial identification intervals:

$$\begin{array}{ll} \text{(i)} & P(Y = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) \leqslant P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X); \\ \text{(ii)} & P(Y = 1, Z = 0 \mid X) \leqslant P(Y_0 = 1 \mid X) < P(Y = 1 \mid X). \end{array} \tag{10.2.5}$$

For a proof, see Appendix 10.5.1. The intervals (10.2.5) make explicit the consequences of the CPO perspective at both behavioral and expectation levels:

1. The interval (10.2.5.i) shows that a CPO perspective will improve the current situation represented by $P(Y = 1 \mid X)$, that is, the proportion of individuals experiencing the beneficial expected outcome. By rewriting the upper bound as

   $$P(Y = 1, Z = 1 \mid X) + P(Y = 1, Z = 0 \mid X) + P(Y = 0, Z = 0 \mid X),$$

   it can be described how the policymaker's behavior and action are made explicit, and on which groups:

   (a) The positive promotion of the innovation (that is, condition (10.2.4.i)) is a message for individuals under the innovation: the policymaker trusts that such a message will reinforce actions leading to experience the beneficial expected outcome. This corresponds to $P(Y = 1, Z = 1 \mid X)$. This proportion is a granted result in the sense that it does not require the actual intervention of the policymaker.

   (b) The non-negative promotion of the innovation (that is, condition (10.2.4.ii)) is a message for individuals under the status quo experiencing the beneficial expected outcome. This corresponds to $P(Y = 1, Z = 0 \mid X)$. This is also a granted result that does not require the intervention of the policymaker.

   (c) The question is, on which group the implementation of the innovation intends to change the expected outcome? On the proportion of individuals who are under the status quo and does not experience the beneficial expected outcome; that is, $P(Y = 0, Z = 0 \mid X)$.

Summarizing, the implementation of the innovation requires, first, making a distinction between those who do require the implementation and those who don't. Secondly, it requires to trust that those who don't require the implementation will continue to maintain their attitude, which is ensured by the positive message underlying the CPO perspective.

Now, if the policymaker acts against her perspective (that is, she advocates for the status quo rather than the innovation, which can be done by concrete political actions), then the actual proportion of individuals experiencing the beneficial expected outcome (namely, $P(Y = 1 \mid X)$) will decrease, being the worse scenario represented by those individuals under the status quo experiencing the beneficial expected outcome; that is, $P(Y = 1, Z = 0 \mid X)$. That is, a contradictory policymaker's behavior negatively affects those individuals that are under the innovation (that is, $P(Y = 1, Z = 1)$), leaving intact the behavior of the individuals under the status quo (that is, $P(Z = 0 \mid X) = P(Y = 1, Z = 0 \mid X) + P(Y = 0, Z = 0 \mid X)$). This explains why the worse scenario is represented by $P(Y = 1, Z = 0 \mid X)$.

In the standard literature of causal inference, the decision of implementing the innovation or not is based on the Average Treatment Effect (ATE), which is defined by $P(Y_1 = 1 \mid X) - P(Y_0 = 1 \mid X)$: if it is statistically positive, the decision is to implement the innovation; for details, see Angrist and Pischke (2008). The intervals (10.2.5) allow us to obtain a partial identification interval for the ATE, that is,

$$0 < P(Y_1 = 1 \mid X) - P(Y_0 = 1 \mid X) \leqslant P(Y = 1, Z = 1 \mid X) + P(Y = 0, Z = 0 \mid X).$$

Under the CPO perspective, the ATE is strictly positive with an upper bound depending on both the proportion of the individuals under the innovation who experience the beneficial expected outcome and the proportion of the individuals under the status quo who don't experience the beneficial outcome: this is precisely the group on which the policymaker should focus the intervention.

Ambiguous policymaker's optimistic perspective

Although optimistic, the attitude of the policymaker is not necessarily convincing, but it could be *ambiguous*. An ambiguous policymaker's optimistic (APO) perspective is characterized by the following inequalities:

$$\begin{array}{ll} \text{(i)} & P(Y_1 = 1 \mid X, Z = 1) \geqslant P(Y_0 = 1 \mid X, Z = 1); \\ \text{(ii)} & P(Y_1 = 1 \mid X, Z = 0) \geqslant P(Y_0 = 1 \mid X, Z = 0). \end{array} \tag{10.2.6}$$

In other word, the APO perspective consists is believing that the implementation of the innovation is at least as better as the status quo. As a matter of fact, (10.2.6.i) means that the actual proportion of individuals experiencing the expected beneficial outcome under the innovation will be at least equal to the corresponding proportion if those individuals were under the status quo . Similarly, (10.2.6.ii) means that among the individuals that are actually under the status quo, the proportion of them experiencing the expected beneficial outcome as if they were under the innovation will be at least equal to the actual proportion under the status quo.

In order to obtain all the plausible values of $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$ which are compatible with the APO perspective, we combine (10.2.6) with the fundamental decompositions (10.2.1) and (10.2.2). The following partial identification intervals are obtained:

$$\begin{array}{ll} \text{(i)} & P(Y = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) \leqslant P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X); \\ \text{(ii)} & P(Y = 1, Z = 0 \mid X) \leqslant P(Y_0 = 1 \mid X) \leqslant P(Y = 1 \mid X). \end{array} \tag{10.2.7}$$

The proof is similar to the proof implying intervals (10.2.5). The current situation, characterized by $P(Y = 1 \mid X)$, cannot be considered as a *sure result* because it is the point common to both the decision of implementing the innovation and to maintaining the status quo. In other words, the groups characterized by $P(Y = 1, Z = 0)$ and $P(Y = 1, Z = 1)$ could change their behavior due to the APO perspective.

10.2.3 Conservative policymaker

When we evaluate the impact of a public policy, it is also necessary to consider a conservative policymaker. At the political level, a conservative policymaker's perspective may represent the opposition to a Government which in turn is promoting an innovation. Similarly to the optimistic policymaker perspective, we can distinguish two conservative perspectives: convincing policymaker's conservative (CPC) perspective and an ambiguous policymaker's conservative (APC) perspective.

Convincing policymaker's conservative perspective

A CPC perspective is defined by the following inequalities:

$$\begin{array}{ll} \text{(i)} & P(Y_1 = 1 \mid X, Z = 1) < P(Y_0 = 1 \mid X, Z = 1); \\ \text{(ii)} & P(Y_1 = 1 \mid X, Z = 0) \leqslant P(Y_0 = 1 \mid X, Z = 0). \end{array} \tag{10.2.8}$$

In other words, the CPC perspective is characterized by (i) promoting the idea that if individuals who are actually under the innovation, were under the status quo, their performance would be better; that is, the innovation harms individuals under the status quo –this corresponds to condition (10.2.8.i). And (ii) if individuals who are actually under the status quo were under the innovation, their performance would be at most as the performance under the status quo; that is, the status quo does not harm individuals under the innovation -this corresponds to condition (10.2.8.ii).

As in the previous cases, after combining (10.2.8) with the fundamental decompositions (10.2.1) and (10.2.2), we obtain all the plausible values of $P(Y_1 = 1 \mid X)$ and $P(Y_0 = 1 \mid X)$ that are compatible with the CPC perspective. These ones are given by the following partial identification intervals:

$$\begin{array}{ll} \text{(i)} & P(Y = 1, Z = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) \leqslant P(Y = 1 \mid X); \\ \text{(ii)} & P(Y = 1 \mid X) < P(Y_0 = 1 \mid X) \leqslant P(Y = 1 \mid X) + P(Y = 0, Z = 1). \end{array} \tag{10.2.9}$$

The proof is similar to the proof implying interval (10.2.5). Let us comment the consequences of the intervals (10.2.9):

1. The interval (10.2.9.ii) shows that a CPC perspective will improve the current situation represented by $P(Y = 1 \mid X)$, that is, the proportion of individuals experiencing the beneficial expected outcome. By rewriting the upper bound as

   $$P(Y = 1, Z = 1 \mid X) + P(Y = 1, Z = 0 \mid X) + P(Y = 0, Z = 1 \mid X),$$

   it can be described how the policymaker's behavior is made explicit and on which groups:

(a) The positive promotion of the status quo (that is, condition (10.2.8.ii)) is a message for individuals under the status quo: the policymaker trusts that such a message will reinforce actions leading to experience the beneficial expected outcome. This corresponds to $P(Y = 1, Z = 0 \mid X)$. This is a granted result in the sense that the group characterized by $P(Y = 1, Z = 0 \mid X)$ does not require the intervention of the policymaker.

(b) The non-negative promotion of the status quo (that is, condition (10.2.8.i)) is a message for individuals under the innovation who experience the beneficial expected outcome. This corresponds to $P(Y = 1, Z = 1 \mid X)$. This is also a granted result in the sense that the group characterized by $P(Y = 1, Z = 1 \mid X)$ that does not require the intervention of the policymaker.

(c) The question is, on which group the implementation of the status quo intends to change the expected outcome? On the group characterized by the proportion of individuals who are under the innovation and does not experience the beneficial expected outcome; that is, $P(Y = 0, Z = 1 \mid X)$.

Summarizing, the implementation of the status quo requires, on the one hand, to make a distinction between those who do require the implementation and those who don't. On the other hand, it requires to trust that those who don't require the implementation will continue to maintain their attitude, which is ensured by the message underlying the CPC perspective.

Now, if the policymaker acts against her perspective (that is, she advocates for the innovation rather than the status quo, which can be done by actual political actions), then the actual proportion of individuals experiencing the beneficial expected outcome (namely, $P(Y = 1 \mid X)$) will decrease, being the worse scenario represented by those individuals under the innovation experiencing the beneficial expected outcome, namely $P(Y = 1, Z = 1 \mid X)$. This means that the contradictory policymaker's behavior affects negatively those individuals that are under the innovation (that is, $P(Y = 1, Z = 0)$), leaving intact the behavior of the individuals under the innovation experiencing the beneficial outcome (that is, $P(Y = 1, Z = 1 \mid X)$). This explains why the worse scenario is represented by $P(Y = 1, Z = 1 \mid X)$.

As in the previous case, the intervals (10.2.9) allow us to obtain a partial identification interval for the ATE, which is given by

$$-P(Y = 1, Z = 0 \mid X) - P(Y = 0, Z = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) - P(Y_0 = 1 \mid X) < 0.$$

The ATE is not only strictly negative (that is, supporting the status quo), but also its minimum bound is given by both the proportion of the individuals under the status quo who experience the beneficial expected outcome, and the subpopulation of the individuals under the innovation who don't experience the beneficial outcome: this is precisely the group on which the policymaker should focus the intervention.

Ambiguous policymaker's conservative perspective

It is also possible to define an ambiguous policymaker's conservative perspective as follows:

$$\begin{aligned} &\text{(i)} \quad P(Y = 1, Z = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) \leqslant P(Y = 1 \mid X); \\ &\text{(ii)} \quad P(Y = 1 \mid X) \leqslant P(Y_0 = 1 \mid X) \leqslant P(Y = 1 \mid X) + P(Y = 0, Z = 1). \end{aligned} \tag{10.2.10}$$

Comments and discussions are similar to those developed in Section 10.2.2 and therefore are left to the reader.

10.2.4 What if the policy decision harms all the individuals?

As it was discussed above, the scientific evidence is mainly characterized by two conditional probabilities, namely $P(Y = 1 \mid X, Z = 1)$ –the proportion of individuals experiencing the beneficial outcome when they are actually under the innovation; and $P(Y = 1 \mid X, Z = 0)$ –the proportion of individuals experiencing the beneficial outcome when they are actually under the status quo. We argue that a public policy evaluation should consider the possibility that a policy decision could harm both groups.

In order to quantify how severe the damage is, we need to write in probabilistic terms this possibility. Let us call both $P(Y_1 = 1 \mid X, Z = 1)$ and $P(Y_0 = 1 \mid X, = 0)$ *factual probabilities*, and $P(Y_1 = 1 \mid X, Z = 0)$ and $P(Y_0 = 1 \mid X, Z = 1)$ *counterfactual probabilities*. Factual probabilities represent what actually we observe, whereas counterfactual probabilities represent what would occur if an individual under innovation (resp., status quo) were under the status quo (resp., innovation). Let suppose that we are ready to accept that *factual is better than counterfactual*, which in probabilistic terms is written as follows:

$$\begin{array}{ll} \text{(i)} & P(Y_1 = 1 \mid X, Z = 1) > P(Y_0 = 1 \mid X, Z = 1); \\ \text{(ii)} & P(Y_1 = 1 \mid X, Z = 0) < P(Y_0 = 1 \mid X, Z = 0). \end{array} \tag{10.2.11}$$

That is, (i) if individuals who are actually under the innovation, were under the status quo, their performance would be worse and, consequently, the innovation is the ideal scenario for those individuals –this corresponds to condition (10.2.11.i). And (ii) if individuals who are actually under the status quo were under the innovation, their performance would be worse and, consequently, the status quo is the ideal scenario for those individuals -this corresponds to condition (10.2.11.ii).

It can be noticed that if policy decision is to implement the innovation, the individuals under the status quo will be negatively affected; and if the policy decision is to maintain the status quo, the individuals under the innovation will be negatively affected. Thus, whatever the policy decision is, one or the other group is harmed. It is possible to quantify the damage. As a matter of fact, by combining (10.2.11) with the fundamental decompositions (10.2.1) and (10.2.2), we get the following intervals of partial identification:

$$\begin{array}{ll} \text{(i)} & P(Y = 1, Z = 1 \mid X) \leqslant P(Y_1 = 1 \mid X) < P(Y = 1 \mid X); \\ \text{(ii)} & P(Y = 1, Z = 0 \mid X) \leqslant P(Y_0 = 1 \mid X) < P(Y = 1 \mid X). \end{array} \tag{10.2.12}$$

For a proof, see Appendix 10.5.2. It can be concluded that, whatever the policy decision is, the current situation represented by $P(Y = 1 \mid X)$ will be worse. How bad the situation could be? If the decision is to implement the innovation, the proportion of individuals experiencing the beneficial expected outcome could decrease up to $P(Y = 1, Z = 1 \mid X)$, that is, the proportion of individuals under the innovation and experiencing the beneficial outcome. If the decision is to maintain the status quo, the proportion of individuals experiencing the beneficial expected outcome could decrease up to $P(Y = 1, Z = 0 \mid X)$, that is, the proportion of individuals under the status quo and experiencing the beneficial outcome. In other words, the worst scenarios are actually represented by the proportion of individuals experiencing

the beneficial outcome in each group. Thus, in a first approach, taking into account that the greatest welfare of society is pursued, the policy decision will depend on a comparison between the lower bounds of both intervals. Consequently, a treatment assignment rule will run as follows:

1. If $P(Y = 1, Z = 0 \mid X) < P(Y = 1, Z = 1 \mid X)$, then the decision is to implement the innovation.

2. If $P(Y = 1, Z = 1 \mid X) < P(Y = 1, Z = 0 \mid X)$, then the decision is to maintain the status quo.

Let us remark that the decision is unclear if the ATE is used as a criterion because the corresponding interval contains 0:

$$-P(Y = 1, Z = 0 \mid X) < P(Y_1 = 1 \mid X) - P(Y_1 = 0 \mid X) < P(Y = 1, Z = 1 \mid X).$$

It is, therefore, desirable to look for a criterion in order to assess the previous treatment assignment rule and eventually to improve it. We postpone the discussion to Section 10.3.

Remark 10.1 It is possible to characterize the scenario *factual is at least as better as counterfactual*. It is enough to rewrite (10.2.11) as

$$\begin{array}{ll} \text{(i)} & P(Y_1 = 1 \mid X, Z = 1) \geqslant P(Y_0 = 1 \mid X, Z = 1); \\ \text{(ii)} & P(Y_1 = 1 \mid X, Z = 0) \leqslant P(Y_0 = 1 \mid X, Z = 0). \end{array} \tag{10.2.13}$$

We leave to the reader to complete details and discussions about this scenario.

10.2.5 What if the policy decision benefits all individuals?

Finally, let us consider the possibility that whatever the policy decision is, it benefits both groups – the group under the innovation and the group under the status quo. This means to assume that *the counterfactual is better than the factual*, which in probabilistic terms is written as follows:

$$\begin{array}{ll} \text{(i)} & P(Y_1 = 1 \mid X, Z = 1) < P(Y_0 = 1 \mid X, Z = 1); \\ \text{(ii)} & P(Y_1 = 1 \mid X, Z = 0) > P(Y_0 = 1 \mid X, Z = 0). \end{array} \tag{10.2.14}$$

Thus, (i) if individuals who are actually under the innovation, were under the status quo, their performance would be better and, consequently, the status quo is the ideal scenario for those individuals –this corresponds to condition (10.2.14.i). And (ii) if individuals who are actually under the status quo were under the innovation, their performance would be better and, consequently, the innovation is the ideal scenario for those individuals -this corresponds to condition (10.2.14.ii).

In this case, whatever the policy decision, both groups benefit positively. The corresponding social benefit can be quantified. It is enough to combine (10.2.14) with the fundamental decompositions (10.2.1) and (10.2.2). We obtain the following intervals of partial identification:

$$\begin{array}{ll} \text{(i)} & P(Y = 1 \mid X) < P(Y_1 = 1 \mid X) \leqslant P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X); \\ \text{(ii)} & P(Y = 1 \mid X) < P(Y_0 = 1 \mid X) < P(Y = 1 \mid X) + P(Y = 0, Z = 1 \mid X). \end{array} \tag{10.2.15}$$

The proof is similar to that underlying intervals (10.2.12). It can be concluded that, whatever the policy decision is, the current situation represented by $P(Y = 1 \mid X)$ will improve. How good the situation could be? If the decision is to implement the innovation, the proportion of individuals experiencing the beneficial expected outcome could increase up to $P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X)$ and, consequently, the focus of intervention is on the individuals under the status quo who don't experience the beneficial outcome. If the decision is to maintain the status quo, the proportion of individuals experiencing the beneficial expected outcome could increase up to $P(Y = 1 \mid X) + P(Y = 0, Z = 1 \mid X)$ and, consequently, the focus of intervention is on the individuals under the innovation who don't experience the beneficial outcome. Thus, in a first approach, taking into account that the greatest welfare of society is pursued, the policy decision will depend on a comparison between the upper bounds of both intervals. Consequently, a treatment assignment rule will run as follows:

1. If $P(Y = 0, Z = 0 \mid X) > P(Y = 0, Z = 1 \mid X)$, then the decision is to implement the innovation.
2. If $P(Y = 0, Z = 0 \mid X) < P(Y = 0, Z = 1 \mid X)$, then the decision is to maintain the status quo.

As in the previous case, the decision is unclear if the ATE is used as a criterion: its partial identification interval contains 0, namely

$$-P(Y = 0, Z = 1 \mid X) < P(Y_1 = 1 \mid X) - P(Y_1 = 0 \mid X) < P(Y = 0, Z = 0 \mid X).$$

Similarly to the previous case, it is desirable to look for an additional criterion allowing to eventually improve this treatment assignment rule.

10.2.6 Overview of the solutions

Let us summarize the four solutions CPO, CPC, factual better than counterfacual (FBC) and counterfactual better than factual (CBF), discussed in Sections 10.2.2, 10.2.3, 10.2.4 and 10.2.5, respectively. These results are summarized as follows:

	$P(Y_1 = 1 \mid X)$		$P(Y_0 = 1 \mid X)$	
Scenario	**Lower bound**	**Upper bound**	**Lower bound**	Upper bound
CPO	$P(Y = 1 \mid X)$	$P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X)$	$P(Y = 1, Z = 0 \mid X)$	$P(Y = 1 \mid X)$
CPC	$P(Y = 1, Z = 1 \mid X)$	$P(Y = 1 \mid X)$	$P(Y = 1 \mid X)$	$P(Y = 1 \mid X) + P(Y = 0, Z = 1 \mid X)$
FBC	$P(Y = 1, Z = 1 \mid X)$	$P(Y = 1 \mid X)$	$P(Y = 1, Z = 0 \mid X)$	$P(Y = 1 \mid X)$
CBF	$P(Y = 1 \mid X)$	$P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X)$	$P(Y = 1 \mid X)$	$P(Y = 1 \mid X) + P(Y = 0, Z = 1 \mid X)$

Regarding CPO and CPC, it should be confessed that the respective policy recommendations are *verbatim* a narrative repetition of the corresponding perspective. Thus, the policy recommendation under the CPO perspective is to implement the innovation, whereas under the CPC perspective, is to maintain the status quo. Then, what the information provided by the partial identification interval is? It provides the

uncertainty inherent to each perspective: such an uncertainty is estimated using the scientific evidence summarized by both $P(Y = 1 \mid X, Z = 1)$ and $P(Y = 1 \mid X, Z = 0)$.

Similarly, the FBC perspective always harms both individuals under the innovation and individuals under the status quo. The policy recommendation is still a *verbatim* narrative of this perspective. However, these intervals show how severe is the damage in each group. Similar comments can be done for the CBF perspective.

An illustration of these considerations is developed in Chapter 11.

10.2.7 A plausible policy interpretation of strong ignorability?

The four solutions, namely CPO CPC, FBC and CBF perspectives, can be compared between them. Such a comparison leads us to answer one of the questions proposed in Section 10.2.1, namely the eventual policy interpretation of the strong ignorability condition (10.2.3).

Different patterns can be obtained depending on the order we impose between $P(Y = 1 \mid X), P(Y = 0 \mid X), P(Z = 1 \mid X)$ and $P(Z = 0 \mid X)$. It is possible to distinguish 16 different patterns. We focus our attention on two of them: $P(Z = 0 \mid X) < P(Y = 0 \mid X) < P(Y = 1 \mid X) < P(Z = 1 \mid X)$ in Figure 10.1, and $P(Y = 1 \mid X) < P(Z = 0 \mid X) < P(Z = 1 \mid X) < P(Y = 1 \mid X)$ in Figure 2 (for simplicity, in the figures we omit to condition on X).

In general terms, the decision of implementing the innovation or not depends on the pursuit of the social welfare. This means to improve the current situation represented by $P(Y = 1 \mid X)$. Under the ignorability condition we have that

$$\begin{aligned} P(Y_1 = 1 \mid X) &= P(Y_1 = 1 \mid X, Z = 1) = P(Y = 1 \mid X, Z = 1), \\ P(Y_0 = 1 \mid X) &= P(Y_0 = 1 \mid X, Z = 0) = P(Y = 1 \mid X, Z = 0). \end{aligned}$$

Now, for the case depicted in Figure 10.1, it follows that

$$P(Y = 1 \mid X, Z = 0) < P(Y = 1 \mid X) < P(Y = 1 \mid X, Z = 1). \tag{10.2.16}$$

Therefore, the solution obtained with the strong ignorabilty condition is compatible with the CPO perspective. Moreoever, it can be proved that (10.2.16) is equivalent to

$$cov\left(\mathbb{1}_{\{Y=1\}}, \mathbb{1}_{\{Z=1\}} \mid X\right) > 0.$$

That is, the innovation is positively correlated with the expected outcome: this explains why the strong ignorability condition is compatible with the CPO perspective.

However, the situation is different if we consider the case depicted in Figure 10.2. This case implies that

$$P(Y = 1 \mid X, Z = 1) < P(Y = 1 \mid X) < P(Y = 1 \mid X, Z = 0). \tag{10.2.17}$$

Therefore, the solution obtained with the strong ignorabilty condition is compatible with the CPC perspective. Condition (10.2.17) is equivalent to

$$cov\left(\mathbb{1}_{\{Y=1\}}, \mathbb{1}_{\{Z=0\}} \mid X\right) > 0.$$

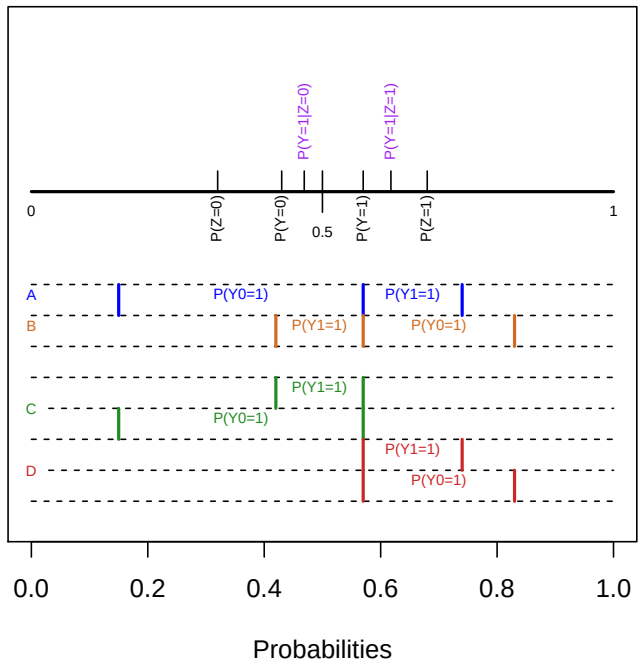


Figure 10.1: Overview of 4 solutions assuming that $P(Z = 0 \mid X) < P(Y = 0 \mid X) < P(Y = 1 \mid X) < P(Z = 1 \mid X)$

That is, the status quo is positively correlated with the expected outcome; this explains why the strong ignorability condition is compatible with the CPC perspective.

Summarizing, both $P(Y = 1 \mid X, Z = 1)$ and $P(Y = 1 \mid X, Z = 0)$ must belong to some partial identification interval and, therefore, the strong ignorability have not an "intrinsic" policy interpretation. The previous cases actually show that the policy interpretation of the solution obtained under strong ignorability depends on the association between outcome and treatment, as well as on the order between $P(Y = 1 \mid X), P(Y = 0 \mid X), P(Z = 1 \mid X)$ and $P(Z = 0 \mid X)$.

10.3 A Decision Problem

Let us come back to the FBC perspective. The policy decision will always harm those under the innovation and those under the status quo. A first solution was based on the comparison of the lower bounds of each partial identification intervals, namely

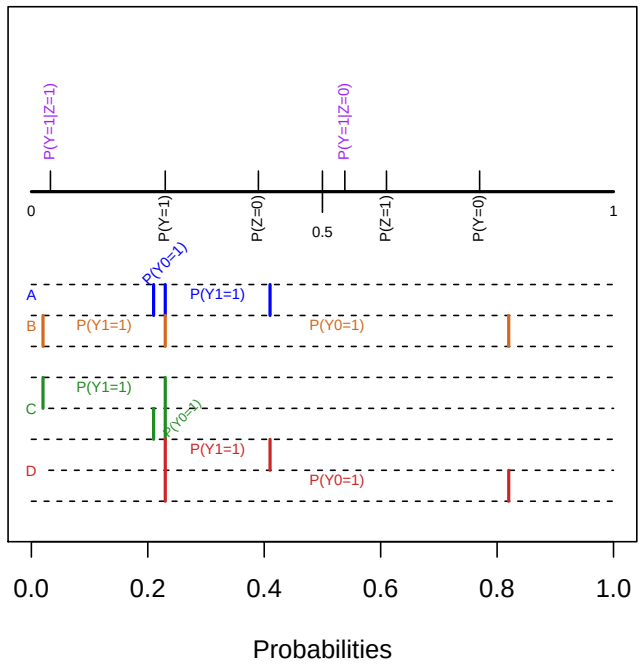


Figure 10.2: Overview of 4 solutions assuming that $P(Y = 1 \mid X) < P(Z = 0 \mid X) < P(Z = 1 \mid X) < P(Y = 1 \mid X)$

1. If $P(Y = 1, Z = 0 \mid X) < P(Y = 1, Z = 1 \mid X)$, then the decision is to implement the innovation.

2. If $P(Y = 1, Z = 1 \mid X) < P(Y = 1, Z = 0 \mid X)$, then the decision is to maintain the status quo.

This solution is extreme in the sense that all individuals, whether under innovation or the status quo, will be harmed by the policy decision. Consequently, we can look for a policy decision based on fractional treatment assignment; that is, deciding to which proportion of individuals to apply the innovation and to which one, the status quo. This problem can be solved by applying the decision theory due to Wald (1950) and Savage (1951).

10.3.1 Basic framework of decision theory

We follow Manski (2005c), Manski (2007a) and Manski (2007b) for a presentation of the decision theory. We consider two treatment, $t = 0$ (status quo) and $t = 1$ (innovation), so $T = \{0, 1\}$. The policymaker's

problem is to choose one of the treatment. Each member j of the population J has a response function $Y_j : T \longrightarrow \mathcal{Y}$ mapping treatments $t \in T$ into outcomes $Y_j(t) \in \mathcal{Y}$. The policymaker, who must assign a treatment to each member of J, observe covariates $X_j \in \mathcal{X}$ for each individual j: the policymaker can differentiate individuals with different values of X, but she cannot distinguish between individuals with the same observed covariates. This is due to the fact that covariates induce a partition (or equivalent class) in the set of individuals. Thus, $X : J \longrightarrow \mathcal{X}$ is the random variable mapping individuals into their covariates. Given that the policymaker observes the covariates of all the individuals, she knows the covariate distribution $P(X = \xi) > 0$ for $\xi \in \mathcal{X}$. The basic steps of the decision process are the following:

A policymaker must choose among a set of feasible actions: A feasible treatment rule assigns either (i) all individuals with the same observed covariates to one treatment, or (ii) randomly allocates such individuals across the treatments. Let Δ denote the space of functions that map $T \times \mathcal{X}$ into the unit interval and that satisfy the adding-up conditions

$$\delta(\cdot,\cdot) \in \Delta \Longrightarrow \delta(0,\xi) + \delta(1,\xi) = 1 \quad \forall\, \xi \in \mathcal{X}.$$

The feasible treatment rules are the elements of Δ: these elements satisfy conditions (i) and (ii) mentioned above.

A policymaker wants to choose a feasible treatment rule that maximizes population mean welfare: The welfare from assignment treatment t to individual j is $u_j(t) \doteq u[Y_j(t), t, X_j]$ (here $\doteq$ denotes "defined as") , where $u : \mathcal{Y} \times T \times \mathcal{X} \longrightarrow \mathbb{R}$. For each treatment rule δ, the mean welfare that would be realized with choice of rule δ is

$$U(\delta, P) \doteq \sum_{\xi \in \mathcal{X}} P(X = \xi)\ \{\, \delta(0,\xi)\, E[u(0) \mid X = \xi] + \delta(1,\xi)\, E[u(1) \mid X = \xi]\,\}\,.$$

This follows from $E[u] = E[E(u \mid X)]$ after interpreting $\delta(t,\xi)$ as a conditional probability. Here $U(\delta, P)$ is the social welfare when treatment rule δ is applied to a population with distribution of treatment response P. The policymaker wants to solve the problem

$$\max_{\delta \in \Delta} U(\delta, P). \tag{10.3.1}$$

The maximum is achieved by a rule that allocates all individuals with covariates ξ to a treatment solving the problem

$$\max_{t \in \{0,1\}} E[u(t) \mid X = \xi].$$

Consequently, the population welfare achieved by an optimal rules is

$$U^*(P) \doteq \sum_{\xi \in \mathcal{X}} P(X = \xi) \left\{ \max_{t \in \{0,1\}} E[u(t) \mid X = \xi] \right\}. \tag{10.3.2}$$

Criteria for planning under uncertainty: A policymaker who knows the treatment-response distribution $P(Y(t) \mid X)$, with $t \in \{0, 1\}$, can choose an optimal rule. However, in the case addressed in this chapter, the policymaker does not know the response distributions, but observes a study population in which treatments have been selected and outcomes realized. Whatever the empirical evidence and maintained assumptions may be, let Γ index the set of feasible *state of nature*. Thus, $\{P_\gamma : \gamma \in \Gamma\}$ is the set of values that the policymaker deems feasible for the distribution of treatment response. Note that this set is related to the partial identification intervals. It was mentioned above that the policymaker wants to choose a feasible treatment rule that maximize the population mean welfare, which essentially reduces to solve problem (10.3.1). However, when the state of nature is unknown, the policymaker faces the ambiguity of ranking treatment rules and therefore the policymaker does not know which rule is optimal. As a matter of fact, let $\delta, \delta' \in \Delta$ be any two feasible treatment rules. Rule δ dominates rule δ' if $U(\delta, P_\gamma) \geqslant U(\delta', P_\gamma)$ for all $\gamma \in \Gamma$ and $U(\delta, P_\gamma) > U(\delta', P_\gamma)$ for some $\gamma \in \Gamma$. It is clear that the policymaker should not choose dominated actions. It is accordingly necessary to reduce the choice to undominated treatment rules, that is, such that $U(\delta, P_\gamma) > U(\delta', P_\gamma)$ for some $\gamma \in \Gamma$ and $U(\delta, P_\gamma) < U(\delta', P_\gamma)$ for other $\gamma \in \Gamma$. It is difficult to determine which treatment rules are undominated. Hence the usual practice is to perform the maximization over the full set Δ of feasible treatment rules. In the case where the state of nature is unknown, this maximization problem requires a criterion to choose an admissible tretement rule:

1. Wald (1950) focused on the *maxmin* criterion, which means that the policymaker would solve

$$\max_{\delta \in \Delta} \min_{\gamma \in \Gamma} U(\delta, P_\gamma).$$

   This criterion has a normative foundation in competitive games. In a competitive game, the policymaker choose a treatment rule from Δ. Then a probability treatment from Γ is chosen by an opponent whose objective is to minimize the realize outcome (Manski, 2007a).

2. Savage (1951) proposes the *minimax regret* criterion, which means that the policymaker would solve

$$\min_{\delta \in \Delta} \max_{\gamma \in \Gamma} \left\{ U^*(P_\gamma) - U(\delta, P_\gamma) \right\}, \tag{10.3.3}$$

   where $U^*(P_\gamma)$ is the optimal population mean welfare that would be achievable if it were known that $P = P_\gamma$ in (10.3.2). This criterion suggests that the policymaker chooses a treatment rule that minimizes the maximum loss of welfare that results from not knowing the probability of the treatment response. As a matter of fact, suppose that the policymaker choose some treatment δ and that the probability of treatment response is some P_γ. The loss in potential welfare resulting from choice of this treatment rule is

$$\max_{\delta' \in \Delta} U(\delta', P_\gamma) - U(\delta, P_\gamma) = U^*(P_\gamma) - U(\delta, P_\gamma).$$

   This loss, called *regret*, is the nonnegative difference between the maximum welfare attainable with P_γ and the welfare attained by choice of δ. The true probability of treatment response is unknown, but we can compute the maximum regret of action δ over Γ. This is

$$\max_{\gamma \in \Gamma} \left\{ U^*(P_\gamma) - U(\delta, P_\gamma) \right\}.$$

A minimax-regret rule choose a treatment rule that minimize the previous function, which means to solve (10.3.3). For details, see Manski (2007a).

10.3.2 Minimax-regret for the FBC perspective

Using the notation of Section 10.2, outcomes Y_0 and Y_1 correspond to $Y(0)$ and $Y(1)$, respectively. These outcomes are viewed as the beneficial expected outcomes, so we define the welfare function $u(t)$ with $t \in \{0,1\}$ as $u(t) \doteq Y(t)$. Let $\delta \doteq \delta(1,\xi)$. Following Manski (2007b)'s arguments, it can be verified that the minimax-regret rule solves the problem

$$\min_{\delta\in[0,1]} \max\left\{(1-\delta)\left[\,(e_{1\xi}p_{1\xi}+u_{1\xi}p_{0\xi})-(e_{0\xi}p_{0\xi}+l_{0\xi}p_{1\xi})\,\right],\delta\left[\,(e_{0\xi}p_{0\xi}+u_{0\xi}p_{1\xi})-(e_{1\xi}p_{1\xi}+l_{1\xi}p_{0\xi})\,\right]\right\}$$

where $e_{0\xi} = P(Y=1 \mid X=\xi, Z=0)$, $e_{1\xi} = P(Y=1 \mid X=\xi, Z=1)$, $p_{0\xi} = P(Z=0 \mid X=\xi)$ and $p_{1\xi} = P(Z=1 \mid X=\xi)$. Moreover, $l_{1\xi}$ and $u_{1\xi}$ are the lower and upper bound of interval (10.2.12.i); and $l_{0\xi}$ and $u_{0\xi}$ are the lower and upper bound of interval (10.2.12.ii). The solution of this problem is given by (see Manski, 2007b, Proposition 1):

1. The innovation is applied to all the population characterized by ξ (that is, $\delta = 1$) if

   $$P(Y=1,Z=0 \mid \xi) < \\ P(Y=1,Z=1 \mid \xi) + \left[P(Y=1,Z=1 \mid \xi)P(Z=0 \mid \xi) - P(Y=1 \mid \xi)P(Z=1 \mid \xi)\right].$$

   In this case, the innovation dominates the status quo.

2. The status quo is applied to all the population characterized by ξ (that is, $\delta = 0$) if

   $$P(Y=1,Z=1 \mid \xi) < \\ P(Y=1,Z=0 \mid \xi) + \left[P(Y=1,Z=0 \mid \xi)P(Z=1 \mid \xi) - P(Y=1 \mid \xi)P(Z=0 \mid \xi)\right].$$

   In this case, the status quo dominates the innovation.

3. The innovation is applied to a δ fraction of the population characterized by ξ and the status quo to a $(1-\delta)$ fraction of the same population, where δ is given by

   $$\delta = \frac{[P(Y=1,Z=1 \mid \xi) - P(Y=1,Z=0 \mid \xi)] + [P(Y=1 \mid \xi)P(Z=0 \mid \xi) - P(Y=1,Z=0 \mid \xi)P(Z=1 \mid \xi)]}{P(Y=1 \mid \xi) - P(Y=1,Z=1 \mid \xi)P(Z=0 \mid \xi) - P(Y=1,Z=0 \mid \xi)P(Z=1 \mid \xi)}$$

   In this case, neither treatment (innovation or status quo) dominates.

According to the first treatment assignment rule, the innovation is applied to all the population characterized by $X = \xi$ if

$$P(Y=1,Z=0 \mid \xi) < P(Y=1,Z=1 \mid \xi). \tag{10.3.4}$$

However, the minimax-regret solution offer a different criterion: the previous inequality needs to be complemented by a second term which in turn is lower than the dependence between Y and Z, namely

$$P(Y=1,Z=1 \mid \xi)P(Z=0 \mid \xi) - P(Y=1 \mid \xi)P(Z=1 \mid \xi) < \\ P(Y=1,Z=1 \mid \xi) - P(Y=1 \mid \xi)P(Z=1 \mid \xi) = cov\left(\mathbb{1}_{\{Y=1\}}, \mathbb{1}_{\{Z=1\}} \mid \xi\right).$$

This means that such a second term could be negative and therefore the criterion to choose $\delta = 1$ is more severe than criterion (10.3.4). Similar comments can be done for the implementation of the status quo to all the population characterized by $X = \xi$.

10.3.3 Ethical issues of fractional treatment rules

It seems relevant to point out the ethical issues raised by fractional treatment rules. Typically, the standard practice in social planning is to consider singleton treatment rules, that is, to assign a unique treatment to all the population. However, as Manski (2007a) argues, "implementation of a fractional minimax-regret rule is appealing because it enables society to diversify a risk that is privately indivisible. An individual cannot diversify; a person receives either [innovation] or [status quo]. Yet society can diversify by having positive fractions of the population receive each treatment" (p. 233); see also Manski (2005a).

Fractional treatment rules ensure, therefore, a diversification of the risk at the society level. This aspect is pertinent when the policy decision will always harm all the population. The singleton treatment rule (based on criterion (10.3.4)) does not diversify the risk in the sense that all individuals under status quo are more harmed than individuals under innovation. One might ask why not "equally" diversify the damage between both groups: the fractional treatment rule is the way to diversify the risk. Let us mention that this type of rules are consistent with the principle *equal treatment of equals* in the *ex ante* sense "that observationally identical people have the same probability of receiving a particular treatment"; but fractional rules "violate the principle in the *ex post* sense that observationally identical persons ultimately receive different treatments" (Manski, 2007a, p. 233-234). This type of considerations is widely discussed in the prospect theory literature, particularly in the context of the *Asian disease-type problems*; for details, see Li and Xie (2006), Jullien (2013), Manski (2013c, Chapter 4) and Dreisbach and Guevara (2017).

10.4 The role of science in policy

Pielke Jr (2007) describes the Honest Broker of Policy Alternative in the following terms:

> The defining characteristic of the honest broker of policy alternatives is an effort to expand (or at least clarify) the scope of choice for decision-making in a way that allows for the decision-maker to reduce choice based on his or her own preferences and values (pp. 2-3).

The Honest Broker perspective is considered as the most relevant and appropriate scientific role in a renewed relationship between science, policy and politics (Turnhout, Stuiver, Klostermann, Harms, &

Leeuwis, 2013; Pedersen, 2014; Van Woensel, 2020). Its difficulties has been also analyzed, proposing strategies to overcome them (e.g., Andereggen, Vischer, & Boutellier, 2012).

10.4.1 The impact of the Honest Broker role at the methodological level

In a different vein, we develop a methodological strategy leading to make explicit how to be an Honest Broker in the context of public policy evaluations. We realize that this is possible when the identification problems and their eventual solution are made explicit. It is important to note that there is a temptation to build models in such a way that the identification problem is hidden or solved without justifying implicit assumptions:

> One might regard problems of identifiability as a necessary part of the specification problem. We would consider such a classification acceptable, *provided the temptation to specify models in such a way as to produce identifiability of relevant characteristics is resisted. Scientific honesty* demands that the specification of a model be based on prior knowledge of the phenomenon studied and possibly on criteria of simplicity, but *not on the desire for identifiability of characteristics in which the researcher happens to be interested* (Koopmans & Reiersøl, 1950, pp.169-170; the italics are added).

In passing, it should be said that most of the methodological statistical contributions seem to be motivated by solving identification problems through strong assumptions (Tamer, 2010b). Moreover, some Bayesian statisticians claim that "unidentifiability causes no real difficulty in the Bayesian approach" (Lindley, 1972), a statement which is conceptually incorrect; for a discussion, see San Martín (2018) and the references therein.

10.4.2 Implications of an Honest Broker strategy

The strategy developed in this chapter shows that an identification problem in the evaluation of public policies can be solved in different ways. Each way is associated with a specific belief of a policymaker which in turn concerns benefits and harms of a policy innovation as well as on the statu quo: when these beliefs are combined with scientific evidence, the uncertainty inherent in that belief arises. We emphasize that it is a matter of *beliefs* because these ones cannot be empirically tested. When the different ways of solving the identification problem were analyzed, it became clear that a political intervention is not reduced to *actions* but also to *speak*. Thus, for example, we saw that in the CPO and CPC perspectives, the policy maker is confident that a *convincing speech* will reinforce the actions of a group of individuals who already experience the expected benefit outcome; and additionally we identify the group in which the *actions* (interventions) should be carried out. In this regard, it is pertinent to refer to Hannah Arendt and her thoughts on political speaking:

> Although I am of the opinion that one must not appear and act in public self-consciously, still I know that in every action the person is expressed as in no other human activity. *Speaking is also a form of action*. That is one venture (pp. 37-38 Arendt, 2013);

for a more detailed digression, see Arendt (1958). This type of speaking could be related to the concept of *performative sentences* introduced by Austin (1962). The name is derived from 'perform', the usual verb with the noun 'action': it indicates that the issuing of the utterance is the performing of an action –it is not normally thought of as just saying something.

The analysis of the different ways in which the identification problem can be solved also allowed us to make explicit that a policy decision can harm a population or benefit it. We argue that a public policy evaluation should consider these possibilities. In doing so, fractional assignment treatment rules appear as a viable alternative, although they face ethical problems, particularly those related to the problem of Asian disease. At the level of policy evaluation, these considerations are relevant because fractional rules are consistent with the principle that *all observationally identical people have the same probability of receiving a particular treatment* in an *ex-ante* sense. As a matter of fact, the design and implementation of policies is performed in the *ex-ante* sense –for instance, progressive implementations of public policies or random choice of groups where an innovation is implemented.

10.4.3 The challenge of the proposed strategy

The strategy of public policy evaluation proposed in this chapter should be considered not as a normative solution, but as a prescriptive solution: to know the implications of decisions after choosing specific beliefs. It may actually be considered as a qualitative analysis tool, which could be useful to a team of policy advisors. In this sense, Lichtenberg's words are a good aphoristic summary of the proposed strategy:

> When we teach men *how* to think, instead of *what* to think, then we also reduce misunderstanding. It is a sort of baptism into the mysteries of humanity (quoted by Lorenzen, 1987, p.3).

One of the main methodological challenges of this strategy is to relate different ways of solving an identification problem with political behaviors and political values. This requires not only transdisciplinary teams, but also (fluid) communication with politicians and policy makers.

This strategy intends to be *a* methodological realization of the Honest Broker of policy alternatives role. It makes explicit that scientific knowledge is mediated by subjective choices, but also shows what (could) mean(s) a rigorous scientific knowledge. This perspective collaborates to the necessity of considering all the sides of a public policy problem, even if those sides are contradictory. This seems to be the way to serve politics and policy:

> The statesman is surrounding, encircled by an army of experts. So that now the question of action lies between the statesman and experts. The statesman has to make the final decision. He can hardly do that realistically, since he can't know everything himself. He must take the advice of experts, indeed of experts who in principle always have to contradict each other. Isn't that so? Every reasonable statesman summons experts with opposing points of view. Because he has to see the matter from all sides. That's true, isn't it? He has to judge between them. And this judgement is a highly mysterious process –in which, then, common sense is made manifest (pp.35-36 Arendt, 2013).

10.5 Proofs

10.5.1 Proof of inequalities (10.2.5)

Using (10.2.4.ii), (10.2.1) implies that

$$P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y_0 = 1 \mid X, Z = 0)P(Z = 0 \mid X) \leqslant P(Y_1 = 1 \mid X).$$

However, $P(Y_1 = 1 \mid X, Z = 1) = P(Y = 1 \mid X, Z = 1)$ and $P(Y_0 = 1 \mid X, Z = 0) = P(Y = 1 \mid X, Z = 0)$. It follows that $P(Y = 1 \mid X) \leqslant P(Y_1 = 1 \mid X)$, namely the lower bound of (10.2.5.i). On the other hand, taking into account that $P(Y_1 = 1 \mid X, Z = 0) \leqslant 1$, (10.2.1) implies that

$$\begin{aligned} P(Y_1 = 1 \mid X) &\leqslant P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Z = 0 \mid X) \\ &= P(Y = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y = 1, Z = 0 \mid X) + P(Y = 0, Z = 0 \mid X) \\ &= P(Y = 1 \mid X) + P(Y = 0, Z = 0 \mid X). \end{aligned}$$

This proves the upper bound of (10.2.5.i).

Similarly, using (10.2.4.i), (10.2.2) implies

$$\begin{aligned} P(Y_0 = 1 \mid X) &< P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y_0 = 1 \mid X, Z = 0)P(Z = 0 \mid X) \\ &= P(Y = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y = 1 \mid X, Z = 0)P(Z = 0 \mid X) \\ &= P(Y = 1 \mid X). \end{aligned}$$

This proves the upper bound of (10.2.5.ii). On the other hand, taking into account that $P(Y_1 = 1 \mid X, Z = 1) \geqslant 0$, (10.2.2) implies that

$$\begin{aligned} P(Y = 1, Z = 0 \mid X) &= P(Y = 1 \mid Z = 0, X)P(Z = 0 \mid X) \\ &= P(Y_0 = 1 \mid Z = 0, X)P(Z = 0 \mid X) \\ &\leqslant P(Y_0 = 1 \mid X); \end{aligned}$$

that is, the lower bound of (10.2.5.ii).

10.5.2 Proof of inequalities (10.2.12)

Using (10.2.11.ii), (10.2.1) implies that

$$\begin{aligned} P(Y_1 = 1 \mid X) &< P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) + P(Y_0 = 1 \mid X, Z = 0)P(Z = 0 \mid X) \\ &= P(Y = 1 \mid X). \end{aligned}$$

On the other hand, $P(Y_1 = 1 \mid X, Z = 0) \in [0, 1]$, and therefore (10.2.1) implies that

$$P(Y = 1, Z = 1 \mid X) = P(Y_1 = 1 \mid X, Z = 1)P(Z = 1 \mid X) \leqslant P(Y_1 = 1 \mid X).$$

Thus, (10.2.12.i) is obtained. Using a similar argument, we obtain (10.2.12.ii).

10.5.3 Proof of (10.2.16)

First, note that $P(Y = 1 \mid X, Z = 1) > P(Y = 1 \mid X)$ is equivalent to $P(Y = 1 \mid X, Z = 1) > P(Y = 1 \mid X, Z = 0)$. Secondly,

$$\begin{aligned} cov\left(\mathbb{1}_{\{Y=1\}}, \mathbb{1}_{\{Z=1\}} \mid X\right) &= E\left(\mathbb{1}_{\{Y=1\}}\mathbb{1}_{\{Z=1\}} \mid X\right) - E\left(\mathbb{1}_{\{Y=1\}} \mid X\right) E\left(\mathbb{1}_{\{Z=1\}} \mid X\right) \\ &= P(Y = 1, Z = 1 \mid X) - P(Y = 1 \mid X)P(Z = 1 \mid X). \end{aligned}$$

In passing, let us mention that this type of equality provides the correct way to define correlations between categorial variables.

Chapter 11

How good/how bad is integrated school leadership for school administration? The Chilean case

This chapter is a direct application of the perspective discussed in Chapter 10.

11.1 The Chilean National System of Quality Assurance of Education

After five years of political discussion, the Chilean government approved in 2011 the Law 20,529 on the *National System of Quality Assurance of Education*, positioning it as the most significant reform of the Chilean educational system. Its main objective is to make a systemic reform to improve and achieve a continuous improvement of students learning. According to the Law, the primary requirement to accomplish this is to increase the educational capabilities of principals and teachers (Law 20.529, Art. 2). To meet its objectives, one of the pillars of the reform of the educational system consisted of the generation of new educational institutions. One of them is the National Agency of Quality of Education; it is in charge of implementing a centralized monitoring system of the performance of schools.

This monitoring system implies to evaluate the achievement of the learning standards as well as to promote standards on the performance of both teachers and principals. The essential part of this process is the School Performance Categorization that leads to classify schools in four performance levels (high, medium, medium-low and insufficient); each category is derived from an indicator that incorporates four dimensions of quality (Law 20.529, Art. 18): the first of these comprises learning standards (67%), which indicate the level of learning achieved by students. The second includes progress measures (3.3%), which indicate an improvement in results over time. The third area is SIMCE scores (3.3%), where the SIMCE is a national standardized test in Mathematics and Language, plus other disciplines when correspond ; for details on the SIMCE test, see Meckes and Carrasco (2010) and Manzi and Preiss (2013). The final dimension is related to Indicators of Personal and Social Development (26.4%). It should be mentioned

that if a school classified as insufficient does not improve its performance level after four years, it risks to be closed (Law 20.529 Art. 11).

11.1.1 Integrated leadership model in the Chilean policy context

According to the Law 20,529, the improvement of students learning requires to increase the educational capabilities of principals and teachers. The Law states that the main function of the principal is to lead the Institutional Educational Project; this implies to manage pedagogical, administrative and financial aspects (Art. 5). From an educational theoretical perspective, this requires to disseminate and apply a series of initiatives to improve the school leadership. The Chilean educational system has been implementing this on different actions across the time as the definition of Indicative Performance Standards for Educational Establishments and their Sustainers (in what follows, EID) (MINEDUC, 2005). Based on these initial advances, in 2015 the Ministry of Education proposed a final version of the standard s, establishing thus the Framework for Good School Leadership (MBD) (MINEDUC, 2014b; MINEDUC, 2015). These standards were defined using the first version of the framework (MINEDUC, 2005), the EID and international literature on school leadership, international empirical research, discussion with civil society, among others type of sources of information.

Currently, the EID and the MBD constitute the general framework considered by the National Agency of Quality of Education to improve the capabilities of the members of a school community. The specific practices determined in the EID are directly related to the school leadership literature as well as to the school leadership dimensions as established by the MBD:

1. Develop and implement a shared vision on the mission and goals of the school.
2. To improve and empowered teachers' and other staff members' abilities and leadership.
3. Coordinating, monitoring, and evaluating curriculum, instruction and assessment.
4. Promote inclusive and collaborative organizational culture, stimulating community relationships among school members.
5. Manage an effective organizational performance ensuring school resources, disseminating results and defining roles of the school members.

Dimensions 1, 3 and 5 are mostly related to collaborative instructional leadership, whereas dimensions 2 and 4 are more related to the transformational leadership model. As a matter of fact, in the collaborative instructional model the principal is oriented to educational development looking for conditions that directly impact on the quality of curriculum and instruction, considering the opinion, perspective and work of teachers. This requires to develop and implement practices leading to coordinate the members of an organization (the school). By so doing, it is expected to mobilize them to the search of a set of common objectives (Robinson, Lloyd, & Rowe, 2008; Hallinger, 2003).

The transformational leadership model provides intellectual direction and aims at innovating within the organization, while empowering and supporting teachers as partners in decision making (Marks & Printy,

2003). In the school context, the effective functions of transformational leadership that has been identified are the following: (a) mission centered (developing a widely shared vision for the school, building consensus about school goals and priorities), (b) performance centered (holding high performance expectations, providing individualized support, supplying intellectual stimulation), and (c) culture centered (modeling organizational values, strengthening productive school culture, building collaborative cultures, and creating structures for participation in school decisions). For details, see Leithwood, Day, Sammons, Harris, and Hopkins (2006); Leithwood and Jantzi (2006); T. Bush and Glover (2003); Waters, Marzano, and McNulty (2003); Barber and Mourshed (2007); Pont, Nusche, and Moorman (2008); Seashore, Leithwood, Wahlstrom, and Anderson (2010); Horn and Marfán (2010); Weinstein and Hernández (2014).

Summarizing, in the Chilean policy context, both collaborative instructional leadership and transformational leadership operate "in tandem" which, according to Marks and Printy (2003)'s terminology, is called *integrated leadership*.

11.1.2 Information and guidance: the inputs provided to schools by the Agency of Quality

The Agency of Quality of Education promotes the school leadership standards by means of a report of results which communicates to each school its performance level obtained according to the School Performance Categorization Methodology (ACE, 2016). The report is organized in such a way that is useful to facilitate and orientate principals' and teachers' actions regarding school improvement. As a matter of fact, the performance category level obtained by a school is explained in terms of the expected achievements attained by the students in each of the 10 indicators used to construct such a category level (for the construction, see Section 11.1). Associated to this information, the Agency of Quality offers three types of supports: (a) online resources that orientate the analysis of the results gathered in the report in such a way that the resulting reflections help to take intra-school decisions related to school improvement, particularly curricular and pedagogical changes; (b) progressive evaluation, a new voluntary and self-applied evaluation that intends to provide information on students progress in Language at second grade of primary school (this is the only grade at which the test is available); (c) learning visits that, thanks to the active participation of the school community, allow to identify practices significant for the development of students as well as to the institutional improvement (Decree 17, 2014).

These supports are aligned with the integrated leadership model. It should be mentioned that this alignment is emphasized by reporting the positive marginal effects of leadership and teacher feedback on the learning results in Language and Mathematics as measured by the SIMCE test. Among the orientations proposed by the Agency of Quality, three are relevant in this respect: (a) teachers are invited to promote a school climate of respect, and provide feedback to students; (b) principals are invited to base curricular and pedagogical decisions on the results obtained by the school; (c) to involve parents/guardians in the design of the mission of the school (Decree 17, 2014).

11.2 The impact of school leadership on intra-school practices

Due to the high stakes consequences of the National System of Quality Assurance of Education, the Chilean Government decided to start in 2015 the implementation of the school monitoring system as a trial period. During this period, the National Agency of Quality of Education applies a national survey to collect information on five dimensions: (a) general conceptions of school communities on issues such as quality of education, the National System of Quality Assurance of Education, and the National Agency of Quality of Education; (b) general knowledge of school communities on the four school performance levels and the associated educational interpretation; (c) perceptions and general evaluation of school communities about the School Classification Methodology; (d) perceptions of school communities about the relationships between the school performance level and support actions provided by the Agency of Quality, which may include the eventual close of a low-performing school; (e) intra-school changes performed by school communities as a reaction to the obtained school performance level (for details, see DESUC-CEPPE, 2015).

The sampling framework of the National Survey corresponds to a stratified-multistage sampling procedure. This procedure was applied to each region of the country (Chile is politically divided in 15 regions). For each region, the official percentages of type of schools is available: public schools (fully financed by the respective county), subsidized schools (financed by both the county and parents/guardians) and private schools (financed by parents/guardians only); for details, see Bellei, Vanni, Valenzuela, and Contreras (2016). At the first stage of the sampling, schools were chosen with a probability proportional to the percentage of type of schools. When a school was chosen, its principal as well as the head of the technical-pedagogical unit were chosen. At the second sampling stage, from each school at most four teachers were randomly chosen. To the final sample, specific weights were applied in such a way that the sampling distribution of schools were similar to the national distribution of schools. The weights were constructed using the number of schools by region and, conditionally to a region, the number of schools by type of school (for details, see Kalton & Flores-Cervantes, 2003). The final sample includes 385 schools, 385 principals, 385 heads of the technical-pedagogical units, and 1218 teachers; at a confidence level of .95, the error of the sample is +/-.2 (for details, see DESUC-CEPPE, 2015).

11.2.1 Relating school leadership and intra-school practices

We focus our attention on two questions from this national survey, question Q34 and Q39:

Q34. Which of the following activities was or will be implemented in your school as a consequence of the school performance level which was informed during the pilot implementation of the school monitoring system?

- **A.** Teachers receive in a writing format the information regarding the obtained school performance level.
- **B.** Meetings with the teachers to explain the obtained performance level.
- **C.** Principals and teacher discuss, analyze or evaluate the results obtained by the school.

Q39. According to the information you manage, which of the following actions was or will be implemented in your school as a consequence of the school performance level which was informed during the pilot implementation of the school monitoring system?

- **A.** Perform pedagogical and curricular changes.

B. Introduce adjustments to the Plan of School Improvement (this plan is agreed between each school and the Ministry of Education).

C. Propose changes to the institutional educative project.

These questions belongs to the fifth dimension mentioned above and deals with the relationship between school leadership and two specific intra-school practices, namely pedagogical and curricular changes. This relationship is an explicit intention of the Quality Agency that promotes that decisions related to school improvement are based on information related to the school performance level (see Section 11.1). More specifically, question Q34 inquires how principals share with teachers the information provided the Agency of Quality. Respondent should choose among three levels at which the information is shared: mere reception of a writing communication, meetings in which principals explain the obtained results, a joint discussion and analysis of the information. The last level represents actions undertaken by principals in order to share with teachers information relevant to take the better joint decisions on school improvement. This is precisely related to both collaborative instructional leadership and transformational leadership –that is, integrated leadership. Question Q39 describes different intra-school changes. We focus our attention to the pedagogical and curricular changes because they are more immediate for students as well as directly related to the school improvement[1]. Question P39 inquires therefore on aspects related to the effects of integrated leadership. Following the substantive framework proposed by Marks and Printy (2003), Q39 can be considered as an effect of Q34 which in turn represents integrated school leadership. As a matter of fact, "when the principal elicits high levels of commitment and professionalism from teachers and works interactively with teachers in a shared instructional leadership capacity, schools have the benefit of integrated leadership; they are organizations that learn and perform at high levels" (Marks & Printy, 2003, p.393). Moreover, question Q39 is precisely what the Agency of Quality of Education expects as an effect of the joint analysis of the information regarding the obtained school performance level. Summarizing, question Q34 is substantively interpreted as the innovation policy (or innovation, to be denoted by Z) that the Agency of Quality of Education searches to implement in the Chilean schools. The question Q39 corresponds to the outcome (to be denoted by Y) which is generated by the treatment policy Z: this causal relationship, illustrated in Figure 11.1, is justified by the theoretical integrated school leadership framework that underlines the action theory of the National System of Quality Assurance of Education.

11.2.2 Summary of the observational evidence

As discussed in the main text, the observational evidence should be summarized in probabilistic terms. The substantive considerations already exposed lead to relate the question Q34 to the innovation Z, and the question Q39 to the outcome Y. In order to operationalize both Y and X, two considerations should be taken into account:

1. The Agency of Quality of Education applied the survey as a means to inquire if intra-school changes at both the pedagogical and the curricular levels are decided on the basis of the information regarding the school performance level.

[1]The changes described in alternatives **B** and **C** of question Q39 have a potential impact at the school level and at the official relationship between the school and the educational authorities.

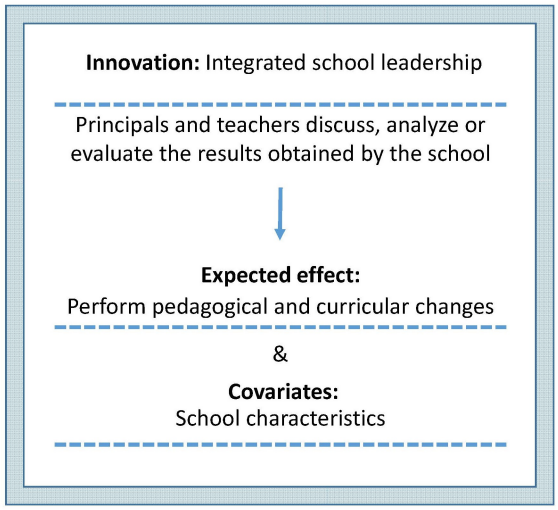


Figure 11.1: Causal relationship between integrated school leadership (innovation) and pedagogical and curricular changes (outcome)

2. According to the theoretical framework already exposed, this type of changes is in practice performed done by teachers as a result of integrated leadership.

Therefore, both Y and Z will be operationalized taken into account the information provided by the teachers only. Given that the survey was applied to at most four teachers by school, three different binary codifications for the outcome can be considered:

- $Y_a = 1$ if at least one teacher of a school reports pedagogical and curricular changes, $Y_a = 0$ otherwise.
- $Y_b = 1$ if 50% of teachers of a school reports pedagogical and curricular changes, $Y_b = 0$ otherwise.
- $Y_c = 0$ if 100% of teachers of a school reports pedagogical and curricular changes, $Y_c = 0$ otherwise.

Similarly, three binary codifications for the innovation assignment variable can be defined:

- $Z_a = 1$ if at least one teacher of a school reports discussion and analysis of the information, $Z_a = 0$ otherwise.
- $Z_b = 1$ if 50% of teachers r of a school reports discussion and analysis of the information, $Z_b = 0$ otherwise.
- $Z_c = 1$ if 100% of teachers r of a school reports discussion and analysis of the information, $Z_c = 0$ otherwise.

Using these variables, Tables 11.1 and 11.2 summarize the information collected in the survey for public schools and subsidized schools. Different combinations of outcomes and innovation assignments can be used to analyze the empirical evidence. In order to show the stability of the observational evidence, we focus our attention on three of them: (Y_b, Z_b), (Y_c, Z_c) and (Y_b, Z_c). Tables 11.3 and 11.4 summarize the estimation of the corresponding conditional probabilities. In the three cases, it is observed that the proportion of schools (public or subsidized) reporting pedagogical and curricula changes under the presence of school integrated is greater than the proportion of schools reporting such changes under the status quo. The only difference between these cases deals with the precision of the estimates of the conditional probabilities: for the case (Y_b, Z_b), the precision is better because the number of observations in each cell is greater than in the other cases. Consequently, the observational evidence suggests to promote that principals and teachers adequate their intra-school practices to the innovation (integrated school leadership). However, from a public policy perspective, the relevant question is the following:

> What would be the percentage of public and subsidized schools performing curricular and pedagogical changes if both principals and teachers were to modify their intra-school practices in line with school integrated leadership?

This question is answered using the strategy developed in the main text.

Table 11.1: Results for Public Schools

	$Z_a = 0$	$Z_a = 1$	$Z_b = 0$	$Z_b = 1$	$Z_c = 0$	$Z_c = 1$
$Y_a = 0$	25	13	30	8	35	3
$Y_a = 1$	5	104	34	74	88	20
$Y_b = 0$	28	42	47	24	65	6
$Y_b = 1$	2	75	18	59	59	17
$Y_c = 0$	30	97	62	65	111	16
$Y_c = 1$	0	20	3	17	12	8

Table 11.2: Results for Subsidized Schools

	$Z_a = 0$	$Z_a = 1$	$Z_b = 0$	$Z_b = 1$	$Z_c = 0$	$Z_c = 1$
$Y_a = 0$	32	17	35	14	48	1
$Y_a = 1$	23	129	48	104	123	29
$Y_b = 0$	44	54	57	41	94	4
$Y_b = 1$	11	92	26	77	77	26
$Y_c = 0$	52	123	77	98	154	22
$Y_c = 1$	5	24	6	20	17	9

Table 11.3: Estimation of the conditional probabilities for Public Schools

Case	$P(Y = 1 \mid Z = 1, \text{Public})$	$P(Y = 1 \mid Z = 0, \text{Public})$	$P(Z = 1 \mid \text{Public})$
(Y_b, Z_b)	0.70	0.28	0.56
(Y_c, Z_c)	0.33	0.10	0.16
(Y_b, Z_c)	0.74	0.48	0.16

Table 11.4: Estimation of the conditional probabilities for Subsidized Schools

Case	$P(Y = 1 \mid Z = 1, \text{Subsidized})$	$P(Y = 1 \mid Z = 0, \text{Subsidized})$	$P(Z = 1 \mid \text{Subsidized})$
(Y_b, Z_b)	0.65	0.31	0.59
(Y_c, Z_c)	0.29	0.10	0.15
(Y_b, Z_c)	0.87	0.45	0.15

11.3 Policy Evaluation

Using the probabilities summarized in the first row of Tables 11.3 and 11.4, we obtain the lower and upper bounds of the partial identification intervals for the CPO, CPC, FBC and CBF perspectives; the results are summarized in Tables 11.5 and 11.6, respectively.

Table 11.5: Impact evaluation for Public Schools

	$P(Y_1 = 1 \mid \text{Public})$		$P(Y_0 = 1 \mid \text{Public})$	
Scenario	**Lower bound**	**Upper bound**	**Lower bound**	**Upper bound**
CPO	0.52	0.84	0.12	0.52
CPC	0.39	0.52	0.52	0.73
FBC	0.39	0.52	0.12	0.52
CBF	0.52	0.84	0.52	0.73

The discussion is focused on public schools; for subsidized schools the considerations below are similar because the estimates are quite similar.

Discussion with respect to the CPO perspective: This perspective is characterized by saying that *the innovation is better than status quo*. Therefore, it is expected that the implementation of the innovation will improve the current situation, namely 52% of the schools experiencing pedagogical and curricular changes. However, the identification region allows to provide information on the following question: How good is the innovation compared with the status quo? According to the results, if the policymaker acts according to her belief, the implementation of the innovation improves the current situation with a maximum of 84%. Nevertheless, the policymaker needs to focus the innovation on $P(Y = 0, Z = 0 \mid \text{Public}) = 31\%$ of the schools, trusting that her message and actions will reinforce the behavior of the schools characterized by $P(Y = 1, Z = 1 \mid \text{Public}) = 39\%$ and $P(Y = 1, Z = 0 \mid \text{Public}) = 12\%$.

Table 11.6: Impact evaluation for Subsidized Schools

	$P(Y_1 = 1 \mid \text{Subsidized})$		$P(Y_0 = 1 \mid \text{Subsidized})$	
Scenario	**Lower bound**	**Upper bound**	**Lower bound**	**Upper bound**
CPO	0.51	0.79	0.13	0.51
CPC	0.38	0.51	0.51	0.72
FBC	0.38	0.51	0.13	0.51
CBF	0.51	0.79	0.51	0.72

Discussion with respect to the CPC perspective: This perspective is characterized by saying that *the status quo is better than innovation*. Therefore, it is expected that the implementation of the innovation will worse the current situation, namely 52% of the schools experience pedagogical and curricular changes. How bad is the innovation compared to the status quo? According to the results, if the policymaker acts contrary to her belief, the implementation of the innovation will worsen the current situation with a minimum of 39%. If she acts according to her belief, the maintenance of the status quo will improve the current situation with a maximum of 73%. Nevertheless, the policymaker needs to focus the status quo on $P(Y = 0, Z = 1 \mid \text{Public}) = 21\%$ of the schools, trusting that her message and actions will reinforce the behavior of the schools characterized by $P(Y = 1, Z = 1 \mid \text{Public}) = 39\%$ and $P(Y = 1, Z = 0 \mid \text{Public}) = 12\%$.

Discussion with respect to the FBC perspective: This perspective is characterized by saying that *the factual situation of schools is better than the counterfactual*. Therefore, if the policymaker decides to implement the innovation or not, in both situations she will take a bad decision. How bad is it? According to the results, if the policymaker takes the wrong decision of implementing the innovation, then the current situation will worsen with a minimum of 39%. However, if the policymaker takes the wrong decision of maintaining the status quo, then the current situation will worsen with a minimum of 15%. Consequently, the maintenance of the status quo is less bad than the implementation of the innovation.

Discussion with respect to the CBF perspective: This perspective is is characterized by saying that *the counterfactual situation of schools is better than the factual*. Therefore, if the policymaker decides to implement the innovation or not, in both situations she will take a good decision. How good is it? According to the results, if the policymaker takes the good decision of implementing the innovation, then the current situation will improve with a maximum of 84%: this means that the policymaker needs to focus the innovation on $P(Y = 0, Z = 0 \mid \text{Public}) = 32\%$ of the schools. If the policymaker takes the good decision of maintenance of the status quo, then the current situation will improve with a maximum of 73%: this means that the policymaker needs to focus the status quo on $P(Y = 0, Z = 1 \mid \text{Public}) = 21\%$ of the schools. Consequently, the implementation of the innovation is much better than the maintenance of the status quo.

Chapter 12

A case study of sentencing and recidivism

Inference on treatment effect, whether from observational or experimental data, must confront the fact that the data are inherently incomplete: each person being observed can experience at most only one treatment. Identification of treatment effects necessarily requires assumptions about the process determining treatment selection. The prevalence practice can be summarized by saying that among persons with specified observable covariates, treatment selection is statistically independent of outcomes: this is the so-called strong ignorability condition or exogenous treatment selection. The specified covariates are often said to "control for" treatment assignment.

Random treatment selection is appropriate in the analysis of data from classical randomized experiments, but it is suspect when observed treatment are self-selected or otherwise chosen purposefully. A solution to this problem is to move away from exact identification of treatment effect. In this case, inferences predicated on weak assumptions can achieve wide consensus, whereas inferences that require strong assumptions almost inevitably are subject to sharp disagreements.

12.1 Context

The problem[1] we will analyze in this chapter can be summarized through the following question: How judges should sentence convicted juvenile offenders? The idea of this chapter is to show how conclusions about treatment effects vary depending on the assumptions made. More specifically, we will present identification regions obtained under two alternative models of judicial decision making:

- The *outcome optimization model* assumes that judges make sentencing decisions that minimize the chance of recidivism.
- The *skimming model* assumes that judges classify offenders as "high risk" or "lower risk", sentencing former to residential treatment and the latter to nonresidential treatment.

[1]This chapter is based on Manski and Nagin (1998).

Juvenile justice system was (is) predicated on a "medical model" of deviance. This lead to make a distinction between criminal responsability and deviance, which in turn is viewed as an underlying pathology that requires treatment. We consider the National Juvenile Count Data Archive 1992, particularly male offenders born between 1970 through 1974, extracted from the state of Utah's extensive records on youth contacts with its juvenile justice system. The data for each case include the dates of referral and disposition and the reasons for and disposition of the referral. Also recorded are the age, race, sex, and number of prior referrals of the youth who is the subject of the case.

Attention is restricted to referrals that are not ultimately dismissed and in which the most serious charge involves an act that would be criminal if committed by an adult. For this segment of the population, we aggregate the many sentencing options available to juvenile court judges into two broad categories of treatments: placement in a residential facility or assignment to nonresidential treatment such as probation, restitution or counselling. The residential treatment alternative included confinement in a secure facility but more generally involved assignment to a nonsecure group home located in the individual's community.

In 1980, Utah adopted a policy of assigning juvenile offenders to the least intrusive treatment alternative. As part of this policy, Utah closed its large centralized detention facility, replaced it with a much smaller secure facility and, as a second alternative, opened many more local nonsecure group house to provide residential treatment.

Our target is to use the Utah data to bound the impact on recidivism of these two modalities of treatment. Recidivism is defined as the youth generating a subsequent referred that meets the above criteria for inclusion in the sample within 24-months period following the date of treatment. Referral is a filtered indicator of a return to offending. A referral occurs only when a youth is apprehended for an offense and the authorities deem it appropriate to refer the apprehended youth to the Utah juvenile justice system.

The date of treatment is defined as the date of disposition, and the date of recidivism as the date of referral of the subsequent qualifying case. Because the age of majority in Utah is 18, we consider only those cases in which the offender was less than age 16 at the date of treatment.

The 24-month exposure period includes the time that an offender sentenced to residential treatment is detained in the residential facility. I principle, residential treatment affects recidivism through two channels:

1. by incapacitating the offender while detained;
2. by deterring or stigmatizing the offender after the sentence is completed.

In practice, the incapacitative effect is limited by the nonsecure nature of some residential facilities and by the short duration of juvenile residential sentences, which usually are less than 6 months in length.

12.1.1 Judge's sentences

In practice, a judge sentences some offenders to a residential program and others to non-residential treatment. Definition of treatment effects requires specification of a population of interest. Here this is the

population of male offenders who are age 16 or less at the time of treatment. For various reasons we might be interested in treatment effects within an identifiable subset of the population, characterized by specified covariates, rather than the entire population. Our use of covariates to specify a subpopulation of interest differs fundamentally from the common use of covariates to control for treatment assignment. When we present estimates of, or bounds on, classical treatment effects conditioning on specific covariates, we seek to answer this question:

> Among persons with the specified covariates, what would be the difference in recidivism rate if all such persons were assigned one treatment rather than the other?

This question is well-posed however the covariates are specified. Researchers commonly assert that there is some minimal set of "correct" covariates to use in the analysis of treatment effects and that "omitted variable bias" may occur if one conditions on only a subset of these covariates. These statements relate to the use of covariates to control for treatment assignment. A set of covariates is said to be "correct" if treatment assignment is random conditional on these covariates; "omitted variable bias" is said to occur if one conditions only on a subset of these covariates and if treatment assignment is not random conditional on this subset. We do not assume that treatment assignment is random conditional on (gender, age, prior referrals) or any other set of covariates. Hence, the concepts of "correct" covariates and "omitted variable bias" are not germane to our analysis.

12.1.2 Structure of the problem

What quantities do the data reveal? For each juvenile offenders, we know

1. the treatment received, residential or nonresidential;
2. whether the offender recidivated and, if so, how long after treatment.

These features of the population are revealed by the data:

- The probability that an offender is sentenced to residential treatment.
- The probability of recidivism among offenders who are residentially treated.
- The probability of recidivism among offenders who are nonresidentially treated.

Two vital counterfactual are missing:

- One is the probability of recidivism among offenders sentenced to residential treatment had they instead been sentenced to nonresidential treatment.
- The other is the probability of recidivism among offenders sentenced to nonresidential treatment had they instead been sentenced to residential treatment.

Let $t = 1$ denote residential treatment and $t = 0$ nonresidential treatment. Let $y(t) = 1$ if an offender would recidivate during the exposure period if that offender were to receive treatment t, and $y(t)0 =$ if the offender would not recidivate during the exposure period if that offender were to receive treatment t. Let $z = 1$ if an offender is actually sentenced to residential treatment and $z = 0$ if an offender is actually sentenced to nonresidential treatment.

12.1.3 Identified and unidentified parameters

Quantities revealed by the data are the following:

- $P(z = 1)$.
- $P(y(1) = 1 \mid z = 1)$: the probability of recidivism under treatment 1 among those offenders who actually receive treatment 1.
- $P(y(0) = 1 \mid z = 0)$: the probability of recidivism under treatment 0 among those offenders who actually receive treatment 0.

Counterfactual quantities not revealed by the data:

- $P(y(1) = 1 \mid z = 0)$: the probability of recidivism under treatment 1 among those offenders who actually received treatment 0.
- $P(y(0) = 1 \mid z = 1)$: the probability of recidivism under treatment 0 among those offenders who actually received treatment 1.

12.2 Impact evaluation of the treatments

There are two treatment effect typically used in the literature: the classical treatment effect (CTE) and the status quo treatment effect (STE), which are defined as follows:

- $CTE \doteq P[y(1) = 1] - P[y(0) = 1]$, where $P[y(1) = 1]$ is the probability of recidivism if all offenders were sentenced to residential treatment, and $P[y(0) = 1]$ is the probability of recidivism if all offenders were sentenced to nonresidential treatment.
- $STE \doteq P[y(1) = 1] - P[y(z) = 1]$, where $y(z)$ is the recidivism outcome experienced by an offender under the treatment that this offender actually received. Thus, $P[y(z) = 1]$ measures the probability of recidivism under the status quo sentencing rules used by judges.

Now, the recidivism probability under mandatory residential treatment is given by

$$P[y(1) = 1] = P[y(1) = 1 \mid z = 1]P[z = 1] + P[y(1) = 1 \mid z = 0]P[z = 0]$$

and it is therefore unidentified. Similarly, the recidivism probability under mandatory nonresidential treatment is given by

$$P[y(0) = 1] = P[y(0) = 1 \mid z = 1]P[z = 1] + P[y(0) = 1 \mid z = 0]P[z = 0]$$

and it is also unidentified. The recidivism probability under status quo is given by

$$P[y(z) = 1] = P[y(1) = 1 \mid z = 1]P[z = 1] + P[y(0) = 1 \mid z = 0]P[z = 0],$$

which is identified.

The previous unidentified probabilities can be partially identified. It is enough to take the following two polar cases:

- $P[y(1) = 1 \mid z = 0] = 0$, that is, none of the offenders who receive nonresidential treatment would have recidivated had they received residential treatment.
- $P[y(1) = 1 \mid z = 0] = 1$, that is, all of the offenders who receive nonresidential treatment would have recidivated had they received residential treatment.

Consequently, the plausible recidivism probabilities under mandatory residential treatment are given by the following region:

$$P[y(1) = 1 \mid z = 1]P[z = 1] \leqslant P[y(1) = 1] \leqslant P[y(1) = 1 \mid z = 1]P[z = 1] + P[z = 0].$$

Similarly, the plausible recidivism probabilities under mandatory nonresidetial treatment are given by the following region:

$$P[y(0) = 1 \mid z = 1]P[z = 1] \leqslant P[y(0) = 1] \leqslant P[y(0) = 1 \mid z = 1]P[z = 1] + P[z = 0].$$

Consequently, the CTE is bounded as follows:

$$\begin{aligned} &P[y(1) = 1 \mid z = 1]P[z = 1] - P[y(0) = 1 \mid z = 0]P[z = 0] + P[z = 1] \leqslant \\ &\quad CTE \leqslant \\ &\quad P[y(1) = 1 \mid z = 1]P[z = 1] + P[z = 0] - P[y(0) = 1 \mid z = 0]P[z = 0]. \end{aligned}$$

It should be remarked that the width of this interval is $P[z = 0] + P[z = 1] = 1$.

Similarly, the STE is bounded as follows:

$$-P[y(0) = 1 \mid z = 0]P[z = 0] \leqslant STE \leqslant P[y(0) = 0 \mid z = 0]P[z = 0].$$

The width of this interval is $P[z = 0]$.

12.2.1 Application

From the data detailed above, the following figures follow:

- Probability of residential treatment: $P[z = 1] = 0.11$.
- Status quo recidivism probability: $P[y(z) = 1] = 0.61$.
- Recidivism probability in subpopulation receiving residential treatment: $P[y(1) = 1 \mid z = 1] = 0.77$.
- Recidivism probability in subpopulation receiving nonresidential treatment: $P[y(0) = 1 \mid z = 0] = 0.59$.

Using this information, we get the following results:

- CTE assuming random treatment selection: $0.77 - 0.59 = 0.18$.
- Bound on recidivism probability under mandatory residential treatment: $0.08 \leqslant P[y(1) = 1] \leqslant 0.97$.
- Bound on recidivism probability under mandatory nonresidential treatment: $0.53 \leqslant P[y(0) = 1] \leqslant 0.64$.
- Bound on CTE: $-0.56 \leqslant CTE \leqslant 0.44$.
- Bound on STE: $-0.53 \leqslant CTE \leqslant 0.36$.

12.3 Two models of treatment selection

12.3.1 Motivation

Although the logical starting point for the analysis of treatment effects is to assume nothing about the counterfactual probabilities, researchers ordinarily want to make sharper inferences than those implied by the data alone. The price of sharper inferences is the loss of credibility associated with the imposition of untestable assumptions.

In the case of sentencing and recidivism there are two models of treatment selection:

- The *outcome optimization model* operationalizes the economist's standard assumption that decisions under uncertainty are made to maximize expected utility. Given the available information, the decision maker chooses the treatment that yields the better expected outcome. This model is consistent with sentencing strategies designed to reduce crime through incapacitation of high rate offenders rather than by their rehabilitation.

- The *skimming model* operationalizes the idea that program administrations may act to maximize the apparent rather than the actual effectiveness of their programs. Judges assign only high risk offenders to residential treatment and only low risk offenders to nonresidential treatment, regardless of which treatment is more effective in reducing recidivism. In educational and training programs it is sometimes hypothesized that administrators select participant who they expect will be most successful following completion of the program, regardless of how this success is influenced by the program. In our juvenile justice application, the skimming model assumes that the population of offenders can be decomposed into "high risk" and "lower risk" individuals. High risk individual (type A) have higher probabilities of recidivism under both residential and nonresidential treatment than do lower risk individual (type B).

Why might judges sentence in accord with the skimming model? One possibility is that judges, like the administrators of educational and training programs, act to maximize the apparent risk rather than the actual effectiveness of their sentencing decision. Alternatively, if the public blames the judicial system when offenders given "light" sentences commit subsequent crimes, then judges have an incentive to give light sentences only to type B persons, regardless of how sentencing actually affects recidivism.

12.3.2 Outcome optimization model

Let s be all the characteristic of the offender and environment that may be associated with treatment selection in a given case. Therefore, z is a function of s.

Consider the following assumptions:

Assumption 1: the judge knows the probabilities $P[y(1) = 1 \mid s]$ and $P[y(0) = 1 \mid s]$ of recidivism under the two posisble treatments.

Assumption 2: the judge selects the sentence yielding the smaller probability of recidivism. Thus,

$$\begin{aligned} P[y(1) = 1 \mid s] \leqslant P[y(0) = 1 \mid s] \quad &\Longrightarrow \quad z = 1; \\ P[y(0) = 1 \mid s] \leqslant P[y(1) = 1 \mid s] \quad &\Longrightarrow \quad z = 0 \end{aligned} \tag{12.3.1}$$

Remark 12.1 Formally, $z = 1$ implies that $P[y(1) = 1 \mid s] \leqslant P[y(0) = 1 \mid s]$; and $z = 0$ implies that $P[y(0) = 1 \mid s] \leqslant P[y(1) = 1 \mid s]$. Taking into account that z is a function of s, it follows that

$$P[y(1) = 1 \mid s, z = 1] \leqslant P[y(0) = 1 \mid s, z = 1],$$

$$P[y(0) = 1 \mid s, z = 0] \leqslant P[y(1) = 1 \mid s, z = 0].$$

Taking $E(\cdot \mid z)$, it follows (12.3.1).

■

Therefore, using the data alone,

$$P[y(0) = 1 \mid z = 1] \in [0, 1],$$

but under the outcome optimization assumption the interval decrease:

$$P[y(0) = 1 \mid z = 1] \in [P[y(1) = 1 \mid z = 1], 1] \,.$$

Similarly,

$$P[y(1) = 1 \mid z = 0] \in [P[y(0) = 1 \mid = 0], 1] \,.$$

It follows that

$$P[y(z) = 1] \leqslant P[y(1) = 1] \leqslant P[y(1) = 1 \mid z = 1]P[z = 1] + P[z = 0],$$

$$P[y(z) = 1] \leqslant P[y(0) = 1] \leqslant P[y(0) = 1 \mid z = 0]P[z = 0] + P[z = 1].$$

Consequently,

$$-P[y(1) = 0 \mid z = 1]P[z = 1] \leqslant CTE \leqslant P[y(0) = 0 \mid z = 0]P[z = 0]$$

and

$$0 \leqslant STE \leqslant P[y(0) = 0 \mid z = 0]P[z = 0].$$

12.3.3 Skimming model

Type A and type B offenders are distinguished by their values of $P[y(1) = 1 \mid s]$ and $P[y(0) = 1 \mid s]$. In particular, there exist thresholds Π_1 and Π_0 such that

$$\text{Type A offenders } P[y(1) = 1 \mid s] \geqslant \Pi_1 \text{ and } P[y(0) = 1 \mid s] \geqslant \Pi_0.$$

$$\text{Type B offenders } P[y(1) = 1 \mid s] < \Pi_1 \text{ and } P[y(0) = 1 \mid s] < \Pi_0.$$

Thus, for each treatment, type A offenders have higher recidivism probabilities than type B offenders. It follows that

$$z = 1 \Longrightarrow P[y(1) = 1 \mid s] \geqslant \Pi_1 \text{ and } P[y(0) = 1 \mid s] \geqslant \Pi_0.$$

$$z = 0 \Longrightarrow P[y(1) = 1 \mid s] < \Pi_1 \text{ and } P[y(0) = 1 \mid s] < \Pi_0.$$

Using the Law of Total Probability (because z i a function of s), it follows that

$$P[y(1) = 1 \mid z = 1] \geqslant \Pi_1 \text{and } P[y(0) = 1 \mid z = 1] \geqslant \Pi_0.$$

$$P[y(1) = 1 \mid z = 0] < \Pi_1 \text{and } P[y(0) = 1 \mid z = 1] < \Pi_0.$$

It follows that

$$P[y(1) = 1 \mid z = 0] < P[y(1) = 1 \mid z = 1], \qquad P[y(0) = 1 \mid z = 0] < P[y(0) = 1 \mid z = 1],$$

where

- $P[y(1) = 1 \mid z = 0]$ corresponds to the recidivism probability that offenders sentenced to non-residential treatment would have if they were sentenced to residential treatment, which under the skimming model
$$P[y(1) = 1 \mid z = 0] \in [0, P[y(1) = 1 \mid z = 1]].$$
- $P[y(0) = 1 \mid z = 1]$ corresponds to the recidivism probability that offenders sentenced to residential treatment would have if they were sentenced to nonresidential treatment, which under the skimming model
$$P[y(0) = 1 \mid z = 1] \in [P[y(0) = 1 \mid z = 0], 1].$$

Consequently,

$$P[y(1) = 1 \mid z = 1]P[z = 1] \leqslant P[y(1) = 1] \leqslant P[y(1) = 1 \mid z = 1],$$

$$P[y(0) = 1 \mid z = 0] \leqslant P[y(0) = 1] \leqslant P[y(0) = 1 \mid z = 0]P[z = 0] + P[z = 1],$$

and thus

$$\begin{aligned} &P[y(1) = 1 \mid z = 1]P[z = 1] - \{P[y(0) = 1 \mid z = 0]P[z = 0] + P[z = 1]\} \leqslant \\ &\quad CTE \leqslant \\ &\quad P[y(1) = 1 \mid z = 1] - P[y(0) = 1 \mid z = 0]; \end{aligned}$$

and

$$-P[y(0) = 1 \mid z = 0]P[z = 0] \leqslant STE \leqslant \{P[y(1) = 1 \mid z = 1] - P[y(0) = 1 \mid z = 0]\}P[z = 0].$$

Following with the Application of Section 12.2.1, we have the following identification region:

- Under the data alone, we have that $-0.48 \leqslant CTE \leqslant 0.52$.
- Under the outcome optimization model, we have $-0.02 \leqslant CTE \leqslant 0.50$.
- Under the skimming model, we have $-0.48 \leqslant TE \leqslant 0.09$.

It can be seen that the choice of the treatment highly depends on the assumption that we are ready to believe!

Chapter 13

Identification Problems underlying Randomized Control Trials

13.1 How to define a Randomized Trial?

Broadly speaking, a controlled experiment is characterized by the fact that a researcher purposefully *randomizes* treatment assignments, whereas a natural experiment is characterized by the fact that *randomization* is a consequence of some process eternal to the research project. In both cases, *randomization* is the key aspect. However, how randomization is defined?

13.1.1 Kolmogorov's axioms of probability theory

One of the fundamental works that introduced both probability theory and statistical inference as tools for econometric research. was Haavelmo (1944):

> This study is intended as a contribution to econometrics. It represents an attempt to supply a theoretical foundation for the analysis of interrelations between economic variables. It is based upon modern theory of probability and statistical inference (p. iii).

Haavelmo motivates his contribution by criticizing the practice of economic research of his time: statistics are used as a tool without accepting the consequences that this tool has on the empirical research itself:

> The application of such simple "statistics" has been considered legitimate, while, at the same time, the adoption of definite probability models has been deemed a crime in economic research, a violation of the very nature of economic data. That is to say, it has been considered legitimate to use some of the *tools* developed in statistical theory *without* accepting the very *foundation* upon which statistical theory is built. For *no tool developed in the theory of statistics has any meaning* -except, perhaps, for descriptive purposes- *without being referred to some stochastic scheme* (p.iii, the italics in the text).

This caveat is still relevant when attempting to answer the question of what *randomization* means.

The foundation of both Statistics and Econometrics is the axiomatic theory of probability as proposed by Kolmogorov (1950). In the International Congress of Mathematicians held in Paris in 1900, Hilbert introduced a series of problems, one of them, the sixth, related to the theory of probability:

> 6. Mathematical treatment of the axioms of physics.
>
> The investigations on the foundations of geometry suggest the problem: *To treat in the same manner, by means of axioms, those physical sciences in which mathematics plays an important part; in the first rank are the theory of probabilities and mechanics.*
>
> As to the axioms of the theory of probabilities, it seems to me desirable that their logical investigation should be accompanied by a rigorous and satisfactory development of the method of mean values in mathematical physics, and in particular in the kinetic theory of gases.

Kolmogorov explicitly solves this problem, keeping in mind that the axioms are mere syntactic rules and that their logical consequences cannot be based on their eventual meaning:

> The theory of probability, as a mathematical discipline, can and should be developed from axioms in exactly the same way as Geometry and Algebra. This means that after we have defined the elements to be studied and their basic relations, and have stated the axioms by which these relations are to be governed, all further exposition must be based exclusively on these axioms, independent of the usual concrete meaning of these elements and their relations (Kolmogorov, 1950, p.1).

This independence of the axioms with respect to their eventual meaning is what allows them to be used in different fields: it is enough to propose a certain semantics to the syntactic rules:

> Every axiomatic (abstract) theory admits, as is well known, of an unlimited number of concrete interpretations besides those from which it was derived. Thus we find applications in fields of science which have no relation to the concepts of random event and of probability in the precise meaning of these words (p.1).

Let us review what is the "precise meaning of [the] words" *random* and *probability* in the axiomatic treatment of the theory of probability.

Kolmogorov introduces the axioms in section 1 of chapter 1 of his contribution. Let us quote *in extenso*:

> Let E be a collection of elements $\xi, \eta, \zeta, \ldots$, which we shall call *elementary events*, and $\mathfrak{F}$ a set of subsets of E; the elements of the set $\mathfrak{F}$ will be called *random events*.
>
> > *I. $\mathfrak{F}$ is a field of sets.*
> >
> > *II. $\mathfrak{F}$ contains E.*
> >
> > *III. To each set A in $\mathfrak{F}$ is assigned a non-negative real number $P(A)$. This number $P(A)$ is called the probability of event A.*

IV. $P(E)$ equals 1.

V. If A and B have no element in common, then

$$P(A + B) = P(A) + P(B)$$

.

A system of sets, $\mathfrak{F}$, together with a definite assignment of numbers $P(A)$, satisfying Axioms I-V, is called a *field of probability*.

Some precisions of Kolmogorov's terminology: a system of sets is called a field if the union, intersection and difference of two sets of the system also belong to the same system. Currently, the system $\mathfrak{F}$ is viewed as an algebra, which is equivalent to the field $\mathfrak{F}$, along with the axiom II. On the other hand, in axiom V Kolmogorov denotes the union of two disjoint sets as the sum $A + B$.

As can be seen, the words *random* and *probability* are mere terms characterized by the axioms. Thus, when we ask *what a random event is*, the answer is the following: a random event is a set that belongs to the system $\mathfrak{F}$ characterized by the axioms I and II. Similarly, when we ask *what a probability is*, the answer is the following: a probability is a function defined from the field $\mathfrak{F}$ to $[0, 1]$ satisfying the axioms III, IV and V.

When one reviews the notes on terminology that Kolmogorov develops in section 3 of chapter 1, one can conclude with greater certainty that the word *random* and the expression *random event* are only a mere change of terminology with respect to that used in set theory:

We have defined the objects of our future study, *random events, as sets*. However, in the theory of probability many set-theoretic concepts are designated by *other terms*[1]. We hall give here a brief list of concepts.

[1]These italics are mine.

Theory of Sets	*Random Events*
1. A and B do not intersect, i.e. $AB = 0$.	1. Events A and B are incompatible.
2. $AB \ldots N = 0$.	2. Events $A, B, \ldots, N$ are incompatible.
3. $AB \ldots N = X$.	3. Events X is defined as the simultaneous occurrence of events $A, B, \ldots, N$.
4. $A \dot{+} B \dot{+} \ldots \dot{+} N = X$	4. Event X is defined as the occurrence of at least one of the events $A, B, \ldots, N$.
5. The complementary set $\overline{A}$.	5. The opposite event $\overline{A}$ consisting of the non-occurrence of event A.
6. $A = 0$.	6. Event A is impossible.
7. $A = E$.	7. Event A must occur.
8. The system $\mathfrak{U}$ of the sets $A_1, A_2, \ldots, A_n$ forms a *decomposition* of the set E if $A_1 + A_2 + \cdots + A_n = E$ (This assumes that the sets A_i do not intersect, in pairs).	8. *Experiment* $\mathfrak{U}$ consists of determining which of the events $A_1, A_2, \ldots, A_n$ occurs. We therefore call $A_1, A_2, \ldots . A_n$ the possible results of experiment $\mathfrak{U}$.
9. B is a subset of A: $B \subset A$.	From the occurrence of event B follows the inevitable occurrence of A.

From all this, we can draw some conclusions:

1. *Random event* does not imply any underlying mechanism that generates it. It is simply an element of the field $\mathfrak{F}$.
2. Any set function fdefined from $\mathfrak{F}$ into $[0, 1]$ that satisfies axioms III to V deserves to be called *probability*. In particular, a relative frequency satisfies these axioms, so it is, axiomatically speaking, a probability.
3. It is instructive to emphasize that an experiment for Kolmogorov is a partition of a system of sets. This idea is fundamental when constructing the conditional probability as a random variable.

13.1.2 What intuition underlies the expression *randomized control trial*?

Stephenson and Imrie (1998) point out that a randomized control trial (RCT) is the best way of measuring the efficacy of intervention "because of its ability to minimise bias and avoid false conclusions. Random assignment of individuals to different treatment groups is the best way of achieving a balance between

groups for the known and unknown factors that influence outcome" (p.611). Odgaard-Jensen et al. (2011) are even more explicit in affirming that bias is reduced thanks to random assignment:

> Randomised trials use the play of chance to assign participants to comparison groups. The unpredictability of the process, if not subverted, should prevent systematic differences between comparison groups (selection bias). Differences due to chance will still occur and these are minimised by randomising a sufficiently large number of people.

Bhide et al. (2018) contrasts a RCT with an observational study: the evidence based on RCTs is the highest one because "evidence based on observational data is prone to bias". *Bias* is understood as "the systematic tendency of any factors associated with the design, conduct, analysis, evaluation and interpretation of the results of a study to make the estimate of the effect of a treatment or intervention deviate from its true value". The presence of bias in an observational study is described in the following terms: "If two or more groups are being compared in an observational study, there are often systematic differences between the groups, so much so that the outcome of the groups may be different because of these differences rather than actual exposure or intervention" (p.381). The way to overcome this serious drawback is through a random assignment:

> The only way to eliminate these differences is to allocate each individual to one or the other intervention at random. Therefore, the probability of any individual receiving one intervention or the other is decided solely by chance (p.381).

All these statements provide us with an intuition, but they fail to be formal in probabilistic terms because *random* is not a probabilistic concept, just a designation, as we discussed previously. However, they provide us with a term that can indeed be defined in probabilistic terms: *bias*. We propose a *structural definition* of bias, that is, one that makes explicit the concept of bias and, therefore, is applicable to many concrete situations.

13.1.3 A structural definition of bias

This definition arise from the Law of Total Probability because it allows us to make explicit the counterfactual aspect underlying bias selection. More specifically, let M be the sample space representing the population of interest. Let V be a random variable defined on M representing the outcome of interest (for instance, the consequence of a treatment), and C a random vector defined on M describing characteristics of the statistical units. The target is to learn about $P(V \leqslant v \mid C)$ from $P(V \leqslant v \mid C, Z = 1)$, where Z denotes the random assignment variable, namely

$$Z(m) = \begin{cases} 1, & \text{if the statistical unit } m \in M \text{ is selected;} \\ 0, & \text{if a statistical unit } m \in M \text{ is not selected.} \end{cases}$$

Since Z induces a partition on M, it is natural to decompose $P(V \leqslant v \mid C)$ through the Law of Total Probability, namely

$$P(V \leqslant v \mid C) = P(V \leqslant v \mid C, Z = 1)P(Z = 1 \mid C) + P(V \leqslant v \mid C, Z = 0)P(Z = 0 \mid C);$$

here $P(V \leqslant v \mid C, Z = 0)$ corresponds to the conditional probability of the outcome that the statistical units would have experienced if they had been assigned to the treatment. Consequently, to learn about $P(V \leqslant v \mid C)$, it is possible to ignore this last conditional probability only if

$$P(V \leqslant v \mid C, Z = 1) = P(V \leqslant v \mid C, Z = 0), \tag{13.1.1}$$

or equivalently, if $V \perp\!\!\!\perp Z \mid C$, which means *absence of bias*.

Additionally, if the statistical units in $Z^{-1}\{1\}$ were chosen without taking into account their characteristics captured by C, then

$$Z \perp\!\!\!\perp C. \tag{13.1.2}$$

A RCT can thus accordingly be defined through the structural conditions (13.1.1) and (13.1.2). Furthermore, *bias* should be defined as the logical negation of (13.1.1). Let us remark that we call these conditions *structural* because we emphasize the "form" or "anatomy" of the concept, which need to be revealed for specific problems. Moreover, up to the best of our knowledge, there not exist a formal proof deriving conditions (13.1.1) and (13.1.2) from "random assignment mechanism": if the proof exists, these conditions would be a consequence; if not, they would be a formal definition.

Thee considerations were published in San Martín and González (2022).

13.2 Compliance with Treatment Assignment

A participant in an experiment is said to comply with his assigned treatment if the treatment received coincides with the treatment assigned. In practice, noncompliance occurs often as a consequence of choices made by the participants in experiments.

Example: the Illinois Unemployment Insurance (UI) Experiment carried out in 1984-1985. Newly unemployed persons were randomly assigned to conventional Ui or to UI augmented by a wage subsidy paid to the employer if the unemployed person should find a full-time job within eleven weeks. Dubin and Rivers (1993) report that 32 percent of those assigned to UI with the wage subsidy did not comply. These subjects chose instead to receive conventional UI.

Let us assume that we have a randomized experiment and let us consider the case where some subjects may not comply with their assigned treatment. We also assume that a person's outcome depends only on the treatment that he receives, not on the one he is assigned. To study noncompliance requires the introduction of a notation allowing to distinguish the treatment received from the treatment assigned.

Let z be a random variable denoting the received treatment , and let ζ be denote the assigned treatment. Noncompliance occurs when $z(m) \neq \zeta(m)$ for $m \in M$.

Let us consider an experiment involving two treatments. One, labeled $t = b$, is an *innovation* that is not available to the general public. The other, labeled $t = a$, is a *status quo* treatment thta it is available outside of the experimental setting. Subjects who are randomly assigned to the innovation can refuse to comply and can choose the status quo treatment instead. However, subjects who are randomly assigned to the status quo treatment are not permitted to *cross over* to receive the innovation. These subjects must comply with their assignment.

An experiment without crossover point identifies the distribution $P[y(a) \mid x]$ of outcomes that would occur if all person were to receive the status quo treatment. It is the case that

$$P[y(a) \mid x] = P[y(a) \mid x, \zeta = a] = P(y \mid x, \zeta = a).$$

Randomization of treatment assignment gives the first equality. The samping process reveals $P(y \mid x, \zeta = a)$.

If there is partial compliance with assignment to the innovation, the experiment partially identifies the distribution $P[y(b) \mid x]$ of outcomes thta would occur if all persons were to receive the innovation. In the absence of crossover, outcomes under treatment b are observed only when $\zeta = b$ and $z = b$. It is the case that

$$\begin{aligned} P[y(b) \mid x] &= P[y(b) \mid x, \zeta = b] \\ &= P[y(b) \mid x, \zeta = b, z = b]P[z = b \mid x, \zeta = b] + P[y(b) \mid x, \zeta = b, z \neq b]P(z \neq b \mid x, \zeta = b) \\ &= P(y \mid x, \zeta = b, z = b)P(z = b \mid x, \zeta = b) + P[y(b) \mid x, \zeta = b, z \neq b]P(z \neq b \mid x, \zeta = b). \end{aligned}$$

Randomization of treatment assignment gives the first equality, the Law of Total Probability gives the second, and the observability of $y(b)$ when $z = b$ gives the third. The empirical evidence reveals $P(y \mid x, \zeta = b, z = b)$ and the compliance probbilities $P(z \mid x, \zeta = b)$. The evidence is uninformative about the outcome distribution $P[y(b) \mid x, \zeta = b, z \neq b]$ for those who do not comply with asisgnment to treatment b. Hence

$$P[y(b) \mid x] \in \{P(y \mid x, \zeta = b, z = b)P(z = b \mid x, \zeta = b) + \gamma P(z \neq b \mid x, \zeta = b) : \gamma \in [0, 1]\}.$$

13.2.1 Intention-to-Treat

Researchers analyzing experimental data sometimes define treatment in a way that makes noncompliance logically impossible. Suppose one defines t to be receipt of an offer of a specified treatment rather than receipt of this treatment. Randomized assignment in an experiment without compulsion essentially gives a subject an offer of treatment. Hence z necessarily equals ζ. The term *intention-to-treat* is used to describe an experimental offer of a treatment.

It is important to understand that redefining t to be an offer of treatment does not solve the selection problem but rather dismisses it by redefining the objective of research. The original objective was to learn the outcome distribution that would occur if all members of the population were to actually receive a treatment. The redefined objective is to learn the outcome distribution that would occur if all members of the population were to receive an offer of the treatment, which they may either accept or reject.

Example: Consider again the Illinois Unemployment Insurance Experiment. Woodbury and Spiegelman (1987) describe the experiment as randomly assigning newly unemployed persons to conventional UI or to an offer of UI with a wage subsidy, allowing the unemployed persons to choose between conventional UI and the wage-subsidy-augmented UI. Hence they view all subjects as complying with their assigned treatments, and they analyze the experimental data in the classical manner. This contrast with the Dubin and Rivers (1993) study, which pay considerable attention to noncompliance.

The findings reported by Woodbury and Spiegelman (1987) differ from those reported by Dubin and Rivers (1993). This is because the two sets of authors have different objectives. Woodbury and Spiegelman (1987) want to predict the outcomes that would occur if unemployed persons were permitted to choose between conventional UI and a program with a wage subsidy. Dubin and Rivers (1993) want to predict the outcomes that would occur if conventional UI were replaced by a program with a wage subsidy. From a policy perspective it is important to emphasize that, in the first case, the policy os to offer both programs to all unemployed persons; in the second case, the policy is either to offer the conventional UI to all persons or the program with wage subsidy: which one will depend on the results of the analysis.

Additional remark: intention to treat is an identification retsriction, not a property of the statistical units in a randomized experiment set-up.

13.2.2 The Average Effect of Treatment on Compliers

Consider again an experiment without crossover, treatment b being the innovation and treatment a being the status quo. We proved that $P[y(b) \mid x]$ is partially identified, hence the ATE $E[y(b) \mid x] - E[y(a) \mid x]$ is also partially identified.

Some researchers have studied inference on the average treatment effect within the subpopulation of persons who comply with the assignment to the innovation; that is, the persons for whom $z = b$ and $\zeta = b$; see Bloom (1984). This *average effect of treatment on compliers* is

$$E[y(b) \mid z = b, \zeta = b] - E[y(a) \mid z = b, \zeta = b].$$

In the absence of crossover, the subpopulation that complies with assignment b is the same as the group that actually receives b. Hence the average effect of treatment on compliers is the same here as $E[y(b) \mid \zeta = b] - E[y(a) \mid \zeta = b]$, which is called *average effect of treatment on the treated*.

Remark 13.1 From the perspective of welfare economics, the relevant population for analysis of treatment response is the population that will be subjected to treatment. Nevertheless, researchers sometimes study treatment response *within the subset of the study population that has received a specified tretamen*. These researchers often report the *average effect of treatment on the treated*:

$$ATT(t) \doteq E[y(b) \mid z = t] - E[y(a) \mid z = t], \qquad t \in \{a, b\}.$$

Example: In Gueron and Pauly (1991), treatment a was the then-current American social program of Aid to Families with Dependent Children AFDC). The analysis focused on

$$E[y(b) \mid z = a] - E[y(a) \mid z = a],$$

namely, the average effect on current AFDC recipients if these persons were to participate in tje alternative instead.

Following Björklund and Moffitt (1987), it can be proved that, in general, the effect on the treated *does not coincide with the average treatment effect for the population of welfare-economic interest*. As a matter of fact, by the Law of Iterated Expectation, we have

$$\begin{aligned} E[y(b)] - E[y(a)] &= \{E[y(b) \mid z = a] - E[y(a) \mid z = a]\} P(z = a) + \\ &\quad \{E[y(b) \mid z = b] - E[y(a) \mid z = b]\} P(z = b) \\ &= ATT(a)P(z = a) + ATT(b)P(z = b). \end{aligned}$$

According to Manski (2007a), "it is rare to find an explicit rationale" explaining why reserachers analyze the effect of treatment on the treated. Manski (2007a) refers to Angrist (1990) who seems to provides a motivation. Here, treatment b denoted being drafted into American military service during the period of the Vietnam War draft lottery, and treatment a denoted not being drafted. Taking the outcome to be person's lifetime earnings, Angrist estimate $E[y(b) \mid z = b] - E[y(a) \mid z = b]$, the average effect of the draft on the lifetime earnings of those who were drafted. He motivates interest in this quantity by writing: " A central question in th edebate over military manpower policy is whether veterans are adequately compensated by their service".

■

Let us come back to the average effect of treatment on compliers. It turns out that this treatment is point-identified. Moreover, it is given by

$$E[y(b) \mid z = b, \zeta = b] - E[y(a) \mid z = b, \zeta = b] = \frac{E(y \mid \zeta = b) - E(y \mid \zeta = a)}{P(z = b \mid \zeta = b)}. \tag{13.2.3}$$

The quantity $E(y \mid \zeta = b) - E(y \mid \zeta = a)$ is the average effect of intention-to-treat. Thus, the average effect of treatment on compliers is proportional to the average effect of intention-to-treat, the factor of proportionality being the reciprocal of the compliance probability $P(z = b \mid \zeta = b)$. It follows that

$$E[y(b) \mid z = b, \zeta = b] - E[y(a) \mid z = b, \zeta = b] > E(y \mid \zeta = b) - E(y \mid \zeta = a).$$

Let prove identity (13.2.3).

First,

$$E[y(b) \mid z = b, \zeta = b] = E(y \mid z = b, \zeta = b).$$

Second, $E[y(a) \mid z = b, \zeta = b]$ is point-identify. As a matter of act,

$$\begin{aligned} E[y(a) \mid \zeta = b] &= E[y(a) \mid z = a, \zeta = b]P(z = a \mid \zeta = b) + E[y(a) \mid z = b, \zeta = b]P(z = b \mid \zeta = b) \\ &= E(y \mid z = a, \zeta = b)P(z = a \mid \zeta = b) + E[y(a) \mid z = b, \zeta = b]P(z = b \mid \zeta = b). \end{aligned}$$

The first equality is the Law of Iterate Expectations (or, equivalently, the Law of Total Probability), and the second holds because

$$\{m \in M : z(m) = a, \zeta(m) = b\} \neq \varnothing$$

since it is the set of all persons who were assigned to teratment b but follow treatment a; hence $y(a)$ is observed. Solving for $E[y(a) \mid z = b, \zeta = b]$ yields

$$E[y(a) \mid z = b, \zeta = b] = \frac{E[y(a) \mid \zeta = b] - E(y \mid z = a, \zeta = b)P(z = a \mid \zeta = b)}{P(z = b \mid \zeta = b)}. \tag{13.2.4}$$

Inspect the right-hand of (13.2.4). The experiment reveals $E(y \mid z = a, \zeta = b)$a d $P(z = b \mid \zeta)b)$. Moreover,

$$E[y(a) \mid \zeta = b] = E[y(a) \mid \zeta = a] = E(Y \mid \zeta = a).$$

The first equality holds because treatment asisgnment is randomized., and the second because there are no crossover from the status quo treatment to innovation. Thus $E[y(a) \mid z = b, \zeta = b]$ is point-identified, with

$$E[y(a) \mid z = b, \zeta = b] = \frac{E(y \mid \zeta = b) - E(y \mid z = a, \zeta = b)P(z = a \mid \zeta = b)}{P(z = b \mid \zeta = b)}. \tag{13.2.5}$$

Application of (13.2.5) gives this expression for the effect of treatment on compliers:

$$E[y(b) \mid z = b, \zeta = b] - E[y(a) \mid z = b, \zeta = b] =$$

$$= E(y \mid z = b, \zeta = b) - \frac{E(y \mid \zeta = b) - E(y \mid z = a, \zeta = b)P(z = a \mid \zeta = b)}{P(z = b \mid \zeta = b)}.$$

Application of the Law of Iterated Expectation gives (13.2.3).

■

Note that the average treatment effect on the subpopulation of compliers is point-identified even though he average treatment effect on the full population is only partially identified.

The original objective was to compare the outcomes of alternative treatments on the full population of interest. The average treatment effect on compliers compare outcomes only on the subpopulation of compliers. This is a special subpopulation whose response to treatment may, in principle, differ considerably from that of the complementary subpopulation of noncompliers. An experiment with partial compliers reveals nothing about response to treatment b within the latter subpopulation.

Chapter 14

Identification problems in the extrapolation of experimental results

In this chapter, which is based on Manski (1997), we focus our attention on the following problem:

> To learn the distribution of outcomes when a specified treatment policy, or assignment rule, determines the treatment received by each member of a specified population.

14.1 Extrapolation of the Perry Preschool Project

Early 1960s, intensive and social services were provided to a random sample of about sixty black children, aged 3 on 4, living in a low-income neighborhood of Ypslilanti, Utah. Two treatments were administered:

- No special services were provided to a second random sample of such children drawn to serve as a control group.
- The treatment and control groups were subsequently followed into adulthood and some of the relevant results are the following:

The two groups were followed across the time. At the age of 19 years old, it was observed that 67% of the treated group were High School (HS) graduated, whereas 49% of the control group were High School (HS) graduated. The Perry Preschool Project is an example of a classical randomized experiment and therefore the difference of outcomes has been attributed to the treatment:

> In 1980 in the United States, 66 percent of all blacks aged 19 and 20 graduated from high school or received a graduate equivalency diploma, while for the study's control group, the rate was 49 percent and for the experimental group, 67 percent. Preschool thus raised the rate of graduation to the national level for blacks (Berrueta-Clement et al., 1984).

> After 25 years we don't know much about how early childhood intervention programs work, but current research suggests they should be extended beyond early childhood (Holden, 1990).

The cohorts has been followed by many years. By the age of 40, the cohorts were still followed. The conclusions of the differeces of outcomes in favor of the treatment group were still attributed to the randomized experiment:

> Because of the random assignment strategy, children's preschool experience remains the best explanation for subsequent group differences in their performance over the years (Schweinhart et al., 2005).

Let us formalize the experimental situation: let $t = 1$ be the educational and social services provided to children participating in the project, and $t = 0$ be the services available to children in the control group. Let $y(t)$ be a binary variable indicating HS graduation by age 19. Taking into account that the Perry Preschool Project is a classical randomized experiment, we can state that the data revealed that

$$P[y(1) = 1] = 0,67, \qquad P[y(0) = 1] = 0.49.$$

Now, the problem can be stated as follows:

> What does the experiment reveal about the HS graduation rate that would occur under a treatment rule in which *some children receive the PP services and the rest do not*?

In order to answer this question, remark that each member of the population has one of these four values for $(y(1), y(0))$:

$$(y(1) = 0, y(0) = 0), \quad (y(1) = 0, y(0) = 1)$$

$$(y(1) = 1, y(0) = 0), \quad (y(1) = 1, y(0) = 1)$$

Treatment assignment has no impact on persons for whom $y(1) = y(0)$, but determines the outcome of persons for whom $y(1) \neq y(0)$. Thus, the *highest feasible graduation rate* is attained by a treatment rule that always selects *the treatment with the better graduation outcome*, and so gives treatment $t = 1$ to each person with

$$(y(1) = 1, y(0) = 0)$$

and treatment $t = 0$ to each person with

$$(y(1) = 0, y(0) = 1).$$

Then the only persons who do not graduate are those with $(y(1) = 0, y(0) = 0)$, so the graduation rate is

$$1 - P[y(1) = 0, y(0) = 0].$$

Similarly, the *lowest feasible graduation rate* is attained by a treatment rule that, by design or error, gives treatment $t = 0$ to persons with

$$(y(1) = 1, y(0) = 0)$$

and treatment $t = 1$ to each person with

$$(y(1) = 0, y(0) = 1).$$

So the graduation rate is

$$P[y(1) = 1, y(0) = 1].$$

However, the experiment does not reveal the joint probability $P[y(1) = 0, y(0) = 0]$ and $P[y(1) = 1, y(0) = 1]$, because treatments $t = 1$ and $t = 0$ are mutually exclusive. Consequently, the experiment does reveal the marginal probabilities

$$P[y(1) = 1] = 0.67, \qquad P[y(0) = 1] = 0.49.$$

It can be shown that among all joint distributions $P[y(1), y(0)]$ that are consistent with the marginals, there is one that *maximize both* $P[y(1) = 0, y(0) = 0]$ *and* $P[y(1) = 1, y(0) = 1]$. This is given by

$$P[y(1) = 0, y(0) = 0] = 0, \quad P[y(1) = 1, y(0) = 0] = 0.51$$

$$P[y(1) = 0, y(0) = 1] = 0.33, \quad P[y(1) = 1, y(0) = 1] = 0.16$$

Thus, from the randomized experiment, we can conclude that *he highest graduation rate consistent with the experimental evidence is 1 and the lowest is 0.16.*

14.2 The mixing problem

Let each member of the population be described by values for

$$[y(1), y(0), z_m, x]$$

, where x is a vector of covariates. There are two feasible treatments, labelled 1 and 0. The letter m denotes the treatment policy of interest. A treatment policy determines which treatment each person receives. The indicator variable z_m denotes the treatment that a given person receives under policy m: $z_m = 1$ if the person receives treatment 1 and $z_m = 0$ otherwise. The outcome a person realizes under policy m is a probability mixture of $y(1)$ and $y(0)$, namely

$$y_m \doteq y(1)z_m + y(0)(1 - z_m).$$

The distribution of outcomes realized by those persons sharing the same value of x is

$$P(y_m \mid x) = P(y(1) \mid x, z_m = 1)P(z_m = 1 \mid x) + P(y(0) \mid x, z_m = 0)P(z_m = 0 \mid x).$$

For example, a welfare recipient might be treated by job-specific training or by basic education. The relevant outcome might be earned income following treatment. One treatment policy might mandate job training for all welfare recipients and enforce the mandate. A second policy might attempt to mandate basic education but not be able to enforce compliance. A third policy might permit a person's case worker to select the treatment expected to yield the larger net benefit, measured as earned income minus treatment costs.

The problem can be stated in the following terms: What does the knowledge of $[P(y(1) \mid x), P(y(0) \mid x)]$ imply about $P(y_m \mid x)$?

14.2.1 Partial identification without prior information

What can we learn on $P(y_m \in B)$ that y_m falls in any specified set B without prior information? The following proposition, which is based on the inequality of Frechet (Fréchet, 1960b), provides a general answer:

Proposition 14.1 *Let $P(y(1))$ and $P(y(0))$ be known. Let $C = P(y(1) \in B) + P(y(0) \in B)$. Then*

$$\max\{0, C - 1\} \leqslant P(y_m \in B) \leqslant \min\{C, 1\}.$$

Before proving this proposition, let us remark its nature:

- Suppose $C \geqslant 1$. Then $C - 1 \leqslant P(y_m \in B) \leqslant 1$, which means that the identification region is informative at the lower bound.
- Suppose $C \leqslant 1$. Then $0 \leqslant P(y_m \in B) \leqslant C$, which means that the identification region is informative at the upper bound.

The information provided by the randomized experiment is contained in C, which is the with of the identification region.

Proof of Proposition 14.1: The largest possible value of $P(y_m \in B)$ is $P(y(1) \in B \cup y(0) \in B)$: this is achieved by a treatment policy that always chooses a treatment whose outcome lies in B, when such a treatment exist; that is, it is ensured that the expected outcome be observed either among persons under treatment $t = 1$ or $t = 0$. The smallest possible value of $P(y_m \in B)$ is $P(y(1) \in B \cap y(0) \in B)$: this is achieved by a treatment policy that always chooses a treatment whose outcome lies in the complement of B, when such a treatment exists. This is due to the fact that

$$P(y(1) \in B \cap y(0) \in B) = 1 - P(y(1) \notin B \cup y(0) \notin B).$$

Therefore

$$P(y(1) \in B \cap y(0) \in B) \leqslant P(y_m \in B) \leqslant P(y(1) \in B \cup y(0) \in B). \tag{14.2.1}$$

However, as it was remarked before, $P(y(1), y(0))$ is not identified. The problem now is to relate the marginals $P(y(1))$ and $P(y(0))$ with the joint distribution. This can be done using the Frechet inequality:

$$\max\{0, C-1\} \leqslant P(y(1) \in B \cap y(0) \in B) \leqslant \min\{P(y(1) \in B), P(y(0) \in B)\}.$$

It follows that $\max\{0, C-1\}$ corresponds to the lower bound of $P(y_m \in B)$. To obtain the upper bound, note that

$$P(y(1) \in B \cup y(0) \in B) = C - P(y(1) \in B \cap y(0) \in B).$$

It is enough to apply the Frechet inequality to $P(y(1) \in B \cap y(0) \in B)$.

■

14.2.2 Ordered outcomes

Outcomes $y(1)$ and $y(0)$ are said to be ordered with respect to a given set B if

$$y(1) \in B \Longrightarrow y(0) \in B.$$

For example, let the outcomes be binary and suppose that

$$y(1) = 0 \Longrightarrow y(0) = 0.$$

The the outcomes are ordered with respect to the set $B = \{0\}$. This assumption means that one may believe that receiving the intervention cannot diminish a child's prospects for graduation. Then any child who receives the intervention and does not graduate would not graduate in the absence of the intervention.

As another example, suppose that the outcomes are real-valued and that

$$y(1) \geqslant y(0).$$

Then $y(1)$ and $y(0)$ are ordered with respect to all sets of the form $B = (-\infty, u]$, because

$$y(1) \leqslant u \Longrightarrow y(0) \leqslant u.$$

Proposition 14.2 *Let $P(y(1))$ and $P(y(0))$ be known. Let it be known that $y(1) \in B \Longrightarrow y(0) \in B$. Then*

$$P(y(1) \in B) \leqslant P(y_m \in B) \leqslant P(y(0) \in B).$$

The proof follows from relation (14.2.1) because under ordered outcome with respect to B we have

$$P(y(1) \in B \cap y(0) \in B) = P(y(1) \in B), \quad P(y(1) \in B \cup y(0) \in B) = P(y(0) \in B).$$

14.2.3 Treatment independent from outcomes

Suppose it is known that the treatment z_m received by each person is statistically independent of the person's outcomes $(y(1), y(0))$, that is,

$$(y(1), y(0)) \perp\!\!\!\perp z_m.$$

Then

$$P(y_m) = P(y(1))P(z_m = 1) + P(y(0))P(z_m = 0).$$

In the absence of other information, the only restriction on the treatment distribution is that $P(z_m = 1)$ and $P(z_m = 0)$ must lie in $[0, 1]$ and added up one. Hence the following proposition follows immediately:

Proposition 14.3 *Let $P(y(1))$ and $P(y(0))$ be known. Let it be known that $(y(1), y(0)) \perp\!\!\!\perp z_m$. Then*

$$\min\{P(y(1)), P(y(0))\} \leqslant P(y_m \in B) \leqslant \max\{P(y(1)), P(y(0))\}.$$

14.2.4 Optimization policies

To prove Proposition 14.1, we examined two extreme treatment policies, one minimizing $P(y_m \in B)$ and the other maximizing it. Suppose that one of these policies is actually implemented. What can be learned about $P(y_m \in B)$ in the absence of prior information about the outcome distribution?

The policy minimizing $P(y_m \in B)$ makes

$$P(y_m \in B) = P(y(1) \in B \cap y(0) \in B).$$

The policy maximizing $P(y_m \in B)$ makes

$$P(y_m \in B) = P(y(1) \in B \cup y(0) \in B).$$

Applying the Frechet inequality, we obtain the following proposition:

Proposition 14.4 *Let $P(y(1))$ and $P(y(0))$ be known.*

1. *If $P(y_m \in B) = P(y(1) \in B \cap y(0) \in B)$, then*

$$\max\{0, C - 1\} \leqslant P(y_m \in B) \leqslant \min\{P(y(1) \in B), P(y(0) \in B)\}.$$

2. *If $P(y_m \in B) = P(y(1) \in B \cup y(0) \in B)$, then*

$$\max\{P(y(1) \in B), P(y(0) \in B)\} \leqslant P(y_m \in B) \leqslant \min(C, 1).$$

14.2.5 Known treatment distribution

The implementation of a public policy is typically subjected to both economical and logistic constrains. Therefore, a treatment policy assignment can be performed progressively, or even applied to a specific proportion of the population. Thus, we will assume that, under policy m, a known fraction p of the population receive treatment $t = 0$ and the remaining fraction $1 - p$ receive treatment $t = 1$. So

$$P(z_m = 0) = p, \tag{14.2.2}$$

where p is known. No information is available on the rule used to make treatment assignments that satisfy (14.2.2).

Now, $P(y_m)$ can be decomposed as follows:

$$P(y_m) = P(y(1) \mid z_m = 1)(1 - p) + P(y(0) \mid z_m = 0)p. \tag{14.2.3}$$

But $P(y(1) \mid z_m = 1)$ and $P(y(0) \mid z_m = 0)$ can not be arbitrarily chosen because of the following decompositions:

$$P(y(1)) = P(y(1) \mid z_m = 1)(1 - p) + P(y(1) \mid z_m = 0)p; \tag{14.2.4}$$

and

$$P(y(0)) = P(y(0) \mid z_m = 1)(1 - p) + P(y(0) \mid z_m = 0)p. \tag{14.2.5}$$

That is, knowledge of $P(y(1))$ and p restricts $P(y(1) \mid z_m = 1)$ and $P(y(1) \mid z_m = 0)$ to satisfy (14.2.4); and knowledge of $P(y(0))$ and p restricts $P(y(0) \mid z_m = 1)$ and $P(y(0) \mid z_m = 0)$ to satisfy (14.2.5). More specifically,

$$P(y(1) \mid z_m = 1) \in \Gamma_{11}(p) \doteq \Gamma_Y \cap \left\{ \frac{P(y(1)) - p\,\gamma}{1 - p} : \gamma \in \Gamma_Y \right\} \tag{14.2.6}$$

and

$$P(y(0) \mid z_m = 1) \in \Gamma_{00}(p) \doteq \Gamma_Y \cap \left\{ \frac{P(y(0)) - (1 - p)\,\gamma}{p} : \gamma \in \Gamma_Y \right\}. \tag{14.2.7}$$

It follows that $P(y_m)$ is a mixture of a distribution in $\Gamma_{11}(p)$ and one in $\Psi_{00}(p)$, where the mixing distribution is given by p. That is,

$$P(y_m) \in \{(1 - p)\gamma_{ii} + p\gamma_{00} : (\gamma_{11}, \gamma_{00}) \in \Gamma_{11}(p) \times \Gamma_{00}(p)\}.$$

More precisely, from (14.2.6) it follows that

$$\max\left\{0, \frac{P(y(1)) - p}{1 - p}\right\} \leqslant P(y(1) \mid z_m = 1) \leqslant \min\left\{1, \frac{P(y(1))}{1 - p}\right\},$$

and

$$\max\left\{0, \frac{P(y(0)) - (1 - p)}{p}\right\} \leqslant P(y(0) \mid z_m = 0) \leqslant \min\left\{0, \frac{P(y(0))}{p}\right\}.$$

It is enough to consider the polar cases $\gamma = 1$ and $\gamma = 0$, respectively.

These regions and (14.2.3) implies the following proposition:

Proposition 14.5 *Let $P(y(1))$ and $P(y(0))$ be known. Let $P(z_m = 0) = p$, for known p. Then*

$$\max\{0, P(y(1) \in B) - p\} + \max\{0, P(y(0) \in B) - (1-p)\} \leqslant$$

$$P(y_m \in B) \leqslant \min\{1-p, P(y(1) \in B)\} + \min\{p, P(y(0) \in B)\}.$$

It can be noticed that if $p = 0$, that is, if treatment $t = 1$ is mandatory, then

$$P(y_m \in B) = P(y(1) \in B),$$

which means to "reproduce" the results of the experiment due to the fact that all the persons now are under treatment $t = 1$. Similarly, if $p = 1$, then treatment $t = 0$ is mandatory and it follows that

$$P(y_m \in B) = P(y(0) \in B).$$

14.2.6 Application

Considering the results from the Perry Preschool experiment, we have the following results:

	$P(y_m = 1)$
no prior information	[0.16;1]
ordered outcomes	[0.49;0.67]
treatment indep. of outcomes	[0.49;0.67]
treatment maximizing HS	[0.67;1]
$1-p = 1/10$ receiving $t = 1$	[0.39;0.59]
$1-p = 9/10$ receiving $t = 1$	[0.57;0.77]
$1-p = 5/10$ receiving $t = 1$	[0.17;0.94]

Appendix A

Appendix

A.1 Fréchet inequality

If you want to know what is proved, look at the proof.
Wittgenstein (*Philosophical Grammar*)

In this section, we revisit the arguments developed by Fréchet (1960a) in his article *Les tableaux dont les marges sont données*.

Let us consider a contingency table of dimension $r \times q$ given by

$$\mathbf{N} \doteq \begin{pmatrix} n_{11} & n_{12} & \cdots & n_{1r} \\ n_{21} & n_{22} & \cdots & n_{2r} \\ \vdots & \vdots & \vdots & \vdots \\ n_{q1} & n_{q2} & \cdots & n_{qr} \end{pmatrix}.$$

The rows marginal are denoted as $n_{1\bullet},\ n_{2\bullet},\ \cdots,\ n_{q\bullet}$, where

$$n_{i\bullet} = \sum_{j=1}^{r} n_{ij}, \qquad i = 1, \ldots, q. \tag{A.1.1}$$

Similarly, the columns marginal are denoted as $n_{\bullet 1},\ n_{\bullet 2},\ \cdots,\ n_{\bullet r}$, where

$$n_{\bullet j} = \sum_{i=1}^{q} n_{ij}, \qquad j = 1, \ldots, r. \tag{A.1.2}$$

We have that

$$\sum_{i=1}^{q} n_{i\bullet} = \sum_{j=1}^{r} n_{\bullet j} \doteq n. \tag{A.1.3}$$

A.1.1 Statement of the problem

The problem to be solved is the following: Given non-negative integer numbers

$$\{n_{1\bullet},\ n_{2\bullet},\ \cdots,\ n_{q\bullet}, n_{\bullet 1},\ n_{\bullet 2},\ \cdots,\ n_{q\bullet}\},$$

1. There exists a contingency table $\mathbf{N}$ such that these numbers correspond to its margins?
2. If there exists at least one contingency table $\mathbf{N}$, there exists other contingency table $\mathbf{N}'$?
3. How to compute the elements of the contingency table $\mathbf{N}$ that it is a solution of the problem?

A.1.2 Solution

The solution of the problem should satisfy condition (A.1.3). If this is the case, we will show that there exists at least one solution: the proof is constructive, so it will indicate how to compute it.

Step 1

Let us start by the first column and to determine the possible values for the number n_{i1}: by definition of $n_{\bullet 1}$ we have

$$\underbrace{n_{11} + \cdots + n_{i-1,1}}_{\text{already determined}} + n_{i1} + \underbrace{n_{i+1,1} + \cdots + n_{q1}}_{\text{not yet determined}} = n_{\bullet 1};$$

it follows that

$$n_{i1} \leqslant n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}).$$

On the other hand, $n_{i1} \leqslant n_{i\bullet}$. Consequently,

$$\begin{aligned} \text{(i) } n_{11} &\leqslant \min\{n_{1\bullet}, n_{\bullet 1}\}; \\ \text{(ii) } n_{i1} &\leqslant \min\{n_{i\bullet}, n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1})\}, \quad i = 2, \ldots, q. \end{aligned} \tag{A.1.4}$$

Now, in the initial equality the not yet determined numbers are not fully free, but bounded by the rows margin:

$$\begin{aligned} n_{\bullet 1} &= \underbrace{n_{11} + \cdots + n_{i-1,1}}_{\text{already determined}} + n_{i1} + \underbrace{n_{i+1,1} + \cdots + n_{q1}}_{\text{not yet determined}} \\ &\leqslant n_{11} + \cdots + n_{i-1,1} + n_{i1} + n_{i+1,\bullet} + \cdots + n_{q\bullet}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{(i)}\ \max\{0, n_{\bullet 1} - (n_{2\bullet} + \cdots + n_{q\bullet}\} &\leqslant n_{11}; \\ \text{(ii)}\ \max\{0, n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) - (n_{i+1,\bullet} + \cdots + n_{q\bullet})\} &\leqslant n_{i1}, \quad i = 2, \ldots, q. \end{aligned} \tag{A.1.5}$$

Summarizing, (A.1.4) and (A.1.5) can be written as

$$\max\{0, n_{\bullet 1} - (n_{2\bullet} + \cdots + n_{q\bullet})\} \leqslant n_{11} \leqslant \min\{n_{1\bullet}, n_{\bullet 1}\};$$

and for $i = 2, \ldots, q$,

$$\max\{0, n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) - (n_{i+1,\bullet} + \cdots + n_{q\bullet})\} \leqslant n_{i1} \leqslant \min\{n_{i\bullet}, n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1})\}.$$

Step 1 recursively viewed

1. To choose n_{11}:

   - If $n_{\bullet 1} < n_{1\bullet}$ then
   $$n_{\bullet 1} - (n_{2\bullet} + \cdots + n_{q\bullet}) < n_{1\bullet}$$
   and, therefore, it is possible to choose one n_{11}.
   - If $n_{1\bullet} < n_{\bullet 1}$ then
   $$\begin{aligned} n_{\bullet 1} - (n_{2\bullet} + \cdots + n_{q\bullet}) < n_{1\bullet} &\iff n_{\bullet 1} - (n_{1\bullet} + n_{2\bullet} + \cdots + n_{q\bullet}) < 0 \\ &\iff n_{\bullet 1} - n < 0, \end{aligned}$$
   which is true. Therefore, it is possible to choose one n_{11}.

2. To choose n_{21}, the set of possible solutions is given by

   $$\max\{0, n_{\bullet 1} - n_{11} - (n_{3\bullet} + \cdots + n_{q\bullet})\} \leqslant n_{21} \leqslant \min\{n_{2\bullet}, n_{\bullet 1} - n_{11}\}.$$

   - If $n_{\bullet 1} - n_{11} < n_{2\bullet}$ then
   $$n_{\bullet 1} - n_{11} - (n_{3\bullet} + \cdots + n_{q\bullet}) \leqslant n_{\bullet 1} - n_{11},$$
   which is true. Therefore, it is possible to choose one n_{21}.
   - If $n_{2\bullet} < n_{\bullet 1} - n_{11}$ then
   $$\begin{aligned} n_{\bullet 1} - n_{11} - (n_{3\bullet} + \cdots + n_{q\bullet}) \leqslant n_{2\bullet} &\iff n_{\bullet 1} - n_{11} - (n_{2\bullet} + n_{3\bullet} + \cdots + n_{q\bullet}) \leqslant 0 \\ &\iff n_{\bullet 1} - (n_{2\bullet} + n_{3\bullet} + \cdots + n_{q\bullet}) \leqslant n_{11}, \end{aligned}$$
   which is true by the previous step. Therefore, it is possible to choose one n_{21}.

3. In general, to choose n_{i1}:

   - If $n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) < n_{i\bullet}$ then

     $$n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) - (n_{i+1,\bullet} + \cdots + n_{q\bullet}) < n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1})$$

     and, therefore, it is possible to choose one n_{i1}.
   - If $n_{i\bullet} < n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1})$, then

     $$\begin{aligned} n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) - (n_{i+1,\bullet} + \cdots + n_{q\bullet}) < n_{i\bullet} \quad &\Longleftrightarrow \\ n_{\bullet 1} - (n_{11} + \cdots + n_{i-1,1}) - (n_{i\bullet} + n_{i+1,\bullet} + \cdots + n_{q\bullet}) < 0 \quad &\Longleftrightarrow \\ n_{\bullet 1} - (n_{11} + \cdots + n_{i-2,1}) - (n_{i\bullet} + n_{i+1,\bullet} + \cdots + n_{q\bullet}) < n_{i-1,1}, \end{aligned}$$

     which is true by the previous step. Therefore, it is possible to choose one n_{i1}.

4. Finally,

   $$\begin{aligned} n_{\bullet 1} = n_{11} + \cdots + n_{q-1,1} + n_{q1} \quad &\Longleftrightarrow \\ n_{\bullet 1} - (n_{11} + \cdots + n_{q-1,1}) = n_{q1}, \end{aligned}$$

   and, therefore, it is possible to choose one n_{q1}. Thus,

   $$\sum_{i=1}^{q} n_{i1} = n_{\bullet 1}.$$

A.1.3 Step 2

We now proceed by recurrence. Suppose we obtained the elements of the $h = 1$ column. To obtain the elements of the h-th column, it is enough to observe that it is the first column of the table obtained by suppressing the $h - 1$ first columns already obtained and replacing the $n_{i\bullet}$ by the quantities

$$\widetilde{n}_{i\bullet} \doteq n_{i\bullet} - (n_{i1} + n_{i2} + \cdots + n_{i,h-1}).$$

The new table is the following:

$$\begin{array}{c|cccc|c} & j = h & j = h+1 & \cdots & j = r & \\ \hline i = 1 & n_{1h} & n_{1,h+1} & \cdots & n_{1r} & \widetilde{n}_{1\bullet} \\ i = 2 & n_{2h} & n_{2,h+1} & \cdots & n_{2r} & \widetilde{n}_{2\bullet} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ i = q & n_{qh} & n_{q,h+1} & \cdots & n_{qr} & \widetilde{n}_{q\bullet} \\ \hline \end{array}$$

After applying step 1 to this new table, we obtain

$$\max\{0, n_{\bullet h} - (n_{1h} + \cdots + n_{i-1,h}) - (\widetilde{n}_{i+1,\bullet} + \cdots + \widetilde{n}_{q\bullet})\} \leqslant n_{ih}$$

$$\leqslant \min\{n_{i\bullet}, n_{\bullet h} - (n_{1h} + \cdots + n_{i-1,h})\},$$

which can equivalently be rewritten as (changing the index)

$$\begin{aligned}
\max\{0, n_{\bullet j} - (n_{1j} + \cdots + n_{i-1,j}) & \\
-(n_{i+1,\bullet} - n_{i1} - \cdots - n_{i,j-1}) & \\
\vdots & \\
-(n_{q\bullet} - n_{q1} - \cdots - n_{q,j-1})\} \quad \leqslant \quad & n_{ij} \leqslant \min\{n_{i\bullet}, n_{\bullet j} - (n_{1j} + \cdots + n_{i-1,j})\}.
\end{aligned} \tag{A.1.6}$$

References

ACE. (2016). *Sistema de Aseguramiento de la Calidad. Resultados Categoría de Desempeño 2016*. Santiago de Chile: Author.

Alarcón-Bustamante, E., San Martín, E., & González, J. (2020). Predictive validity under partial observability. In M. Wiberg, D. Molenaar, J. González, U. Böckenholt, & J.-S. Kim (Eds.), *Quantitative psychology* (pp. 135–145). Cham: Springer International Publishing.

Allais, M. (1953). Le comportement de l'homme rationnel devant le risque: critique des postulats et axiomes de l'école américaine. *Econometrica*, *21*, 503–546.

Andereggen, S., Vischer, M., & Boutellier, R. (2012). Honest but broke: The dilemma of universities acting as honest brokers. *Technology in Society*, *34*, 118–126.

Andrade, M. J. (2019). La lucha por el territorio mapuche en Chile: una cuestión de pobreza y medio ambiente. *L'Ordinaire des Amériques*(225).

Angoff, W. H. (1984). *Scales, Norms, and Equivalent Scores*. New Jersey: Educational Testing Service, Princeton.

Angrist, J. D. (1990). Lifetime earnings and the vietnam era draft lottery: evidence from social security administrative records. *The American Economic Review*, 313–336.

Angrist, J. D., & Pischke, J.-S. (2008). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton University Press.

Arendt, H. (1958). The Modern Concept of History. *The Review of Politics*, *20*, 570–590.

Arendt, H. (2013). *Hannah Arendt: The Last Interview and Other Conversations*. Melville House.

Austin, J. L. (1962). *How to Do Things with Words*. Clarendon Press.

Barber, M., & Mourshed, M. (2007). *How the world's best-performing schools systems come out on top*. McKinsey & Company.

Beaulieu, E., & Hyde, S. D. (2009). In the shadow of democracy promotion: Strategic manipulation, international observers, and election boycotts. *Comparative Political Studies*, *42*(3), 392–415.

Bellei, C., Vanni, X., Valenzuela, J., & Contreras, D. (2016). School improvement trajectories: an empirical typology. *School Effectiveness and School Improvement*, *27*(3), 275–292.

Bengoa, J. (2007). *El Tratado de Quilín*. Santiago, Chile: Catalonia.

Bengoa, J. (2016). Sarmiento y sarmientadas. In Domingo Faustino Sarmiento (Ed.), *Conflicto y Armonía de las Razas en América* (pp. 5–14). Santiago, Chile: Akal-Inter Pares.

Berinsky, A. J. (2017). Measuring public opinion with surveys. *Annual Review of Political Science*, *20*, 309–329.

Berrueta-Clement, J. R., et al. (1984). *Changed Lives: The Effects of the Perry Preschool Program on Youths through Age 19. Monographs of the High/Scope Educational Research Foundation, Number Eight.* ERIC.

Bhide, A., Shah, P. S., & Acharya, G. (2018). A simplified guide to randomized controlled trials. *Acta Obstetricia et Gynecologica Scandinavica*, *97*(4), 380–387.

Birch, S. (2010). Perceptions of electoral fairness and voter turnout. *Comparative political studies*, *43*(12), 1601–1622.

Björklund, A., & Moffitt, R. (1987). The estimation of wage gains and welfare gains in self-selection models. *The Review of Economics and Statistics*, 42–49.

Bloom, H. S. (1984). Accounting for no-shows in experimental evaluation designs. *Evaluation review*, *8*(2), 225–246.

Blundell, R., Gosling, A., Ichimura, H., & Meghir, C. (2007). Changes in the Distribution of Male and Female Wages Accounting for Emplyement Composition Using Bounds. *Econometrica*, *75*, 323–363.

Bolsinova, M., & Maris, G. (2016). Can irt solve the missing data problem in test equating? *Frontiers in Psychology*, *6*, 1956.

Braun, H., & Holland, P. W. (1982). Observed-score test equating: a mathematical analysis of some ETS equating procedures. In P. W. Holland & D. Rubin (Eds.), *Test Equating* (pp. 9–49). New York: Academic Press.

Brennan, R. L., & Kolen, M. J. (1987). A reply to angoff. *Applied Psychological Measurement*, *11*(3), 301-306.

Burns, P. B., Rohrich, R. J., & Chung, K. C. (2011). The levels of evidence and their role in evidence-based medicine. *Plastic and Reconstructive Surgery*, *128*, 305.

Bush, T., & Glover, D. (2003). *School Leadership: Concepts and Evidence*. National College for School Leadership.

Bush, V. (1945). *Science, the endless frontier. Report to the President on a Program for Postwar Scientific Research* (Tech. Rep.). Office of Scientific Research and Development, Washington DC.

CADEM. (2018). *Diseño metodológico de Plaza Pública Cadem 2018.* Retrieved 2021-12-30, from `https://cadem.cl/plaza-publica/`

CADEM. (2022). *Encuesta Plaza Pública. Quinta Semana de Diciembre.* Retrieved 2022-01-03, from `https://cadem.cl/wp-content/uploads/2022/01/Plaza-Publica-416-63-tiene-una-imagen-positiva-o-muy-positiva-del-presidente-electo-Gabriel-Boric-su-mejor-evaluacion-historica-1.pdf`

Cairney, P. (2016). *The Politics of Evidence-Based Policy Making*. Palgrave Macmillan.

Cairney, P. (2017). Evidence-based best practice is more political than it looks: a case study of the "Scottish Approach". *Evidence & Policy: A Journal of Research, Debate and Practice*, *13*, 499–515.

Cairney, P., Oliver, K., & Wellstead, A. (2016). To bridge the divide between evidence and policy: reduce ambiguity as much as uncertainty. *Public Administration Review*, *76*, 399–402.

Carnap, R. (1962). *Logical Foundations of Probability*. Chicago: The University of Chicago Press.

Cayul, P., Durán, E., & Jaimovich, D. (2021). *¿Es representativa la consulta ciudadana en la Araucanía? (Is the citizen consultation in Araucanía representative?).* Retrieved from http://www.mipp.cl/miradas/2021/11/11/representatividad-consulta-araucania/

Christensen, R., & Schultz, T. J. (2014). Identifying election fraud using orphan and low propensity voters. *American Politics Research*, *42*(2), 311–337.

Claro, M., Preiss, D., San Martín, E., Jara, I., Hinostroza, J. E., Valenzuela, S., ... Nussbaun, M. (2012). Assessment of 21st century ICT skills in Chile: Test design and results from high school level students. *Computers and Education*, *59*, 1042–1053.

Clearinghouse, W. W. (2007). Welcome to WWC. *Retrieved April*, *5*, 2019.

Clifford, P. (1982). Some General Comments on Nonidentifiability. In L. Le Cam & J. Neyman (Eds.), *Probability models and cancer* (pp. 81–83). North Holland, Amsterdam, The Netherlands.

Cochran, W. G., & Chambers, S. (1965). The Planning of Observational Studies of Human Populations. *Journal of the Royal Statistical Society, Series A*, *128*(2), 234–266.

Cochrane, A. L. (1972). *Effectiveness and Efficiency: Random Reflections on Health Services*. Nuffield Provincial Hospitals Trust London.

Constitución de la República de Chile. (2005, Sep. 11). *Decreto 100. fija el texto refundido, coordinado y sistematizado de la Constitución Política de la República de Chile.*

Dacey, R. (2003). The S-Shaped Utility Function. *Synthese*, *135*(2), 243–272.

Davies, P., & Boruch, R. (2001). The Campbell Collaboration: Does for Public Policy what Cochrane Does for Health. *British Medical Journal Publishing Group*, *323*, 294-295.

Decree 17. (2014). *Aprueba la metodología de ordenación de todos los establecimientos educacionales reconocidos por el estado conforme a lo dispuesto en el inciso cuarto del artículo 17.* (https://www.leychile.cl/Navegar?idNorma=1060182&idVersion=2014-12-01)

DESUC-CEPPE. (2015). *Encuesta Nacional de Percepciones, Opiniones y Actitudes de Sostenedores, Equipos Directivos y Docentes sobre la Macha Blanca del Sistema de Aseguramiento de Calidad de la Educación: Fase Implementación Ordenación* (Tech. Rep.). Santigo de Chile: Instituto de Sociología, Pontificia Universidad Católica de Chile & Centro de Estudios de Políticas y Prácticas en Educación.

Diario Oficial de la República de Chile. (2021, Oct. 13). *Declara estado de excepción constitucional de emergenia en las zonas del territorio nacional que indica.*

Dreisbach, S., & Guevara, D. (2017). The Asian Disease Problem and the Ethical Implications of Prospect Theory. *Noûs*, *53*, 613-638.

Dubin, J. A., & Rivers, D. (1993). Experimental estimates of the impact of wage subsidies. *Journal of econometrics*, *56*(1-2), 219–242.

Embrechts, P., & Hofert, M. (2013). A Note on Generalized Inverses. *Mathematical Methods of Operations Research*, *77*(3), 423–432.

European Commission. (2017). *Mechanism for Evidence Based Policy-Making in Education. Eurydice Report*. Luxembourg: Publications Office if the European Union.

Fariña, P., González, J., & San Martín, E. (2019). The Use of an Identification-Based Strategy for the Interpretation of Parameters in the 1PL-G and Rasch Models. *Psychometrika*, *84*, 511–528.

Felline, L. (2018). Mechanisms Meet Structural Explanation. *Synthese*, *195*, 99–114.

Ferrari, D., McAlister, K., & Mebane Jr, W. R. (2018). Developments in positive empirical models of election frauds: Dimensions and decisions. In *Summer meeting of the political methodology society, provo, ut, july* (pp. 16–18).

Fisher, R. A. (1922). On the Mathematical Foundations of Theoretical Statistics. *Philosophical Transactions of the Royal Society of London, Series A*, *222*, 309–368.

Fisher, R. A. (1935). *The Design of Experiments*. Edinburgh: Oliver and Boyd.

Fisher, R. A. (1973). *Statistical Methods for Research Workers*. Hafner Publishhing, New York, USA.

Florens, J. P., & Mouchart, M. (1982). A Note on Noncausality. *Econometrica*, *50*(3), 583–592.

Florens, J.-P., & Mouchart, M. (1985). A linear theory for noncausality. *Econometrica*, 157–175.

Florens, J. P., Mouchart, M., & Rolin, J.-M. (1990). *Elements of Bayesian Statistics*. New York: Marcel Dekker.

Fréchet, M. (1960a). Les tableaux dont les marges sont données. *Trabajos de Estadística*, *11*, 1–18.

Fréchet, M. (1960b). Sur les tableaux dont les marges et des bornes sont données. *Revue de l'Institut International de Statistique*, *28*, 10–32.

Gigerenzer, G. (2004). Mindless Statistics. *The Journal of Socio-Economics*, *33*, 587–606.

Gobierno de Datos. (2021a). *Resultados por mesa de las elecciones presidenciales 2021 primera vuelta - [oficina nacional de procesos electorales (onpe)].*

Gobierno de Datos. (2021b). *Resultados por mesa de las elecciones presidenciales 2021 primera vuelta - [oficina nacional de procesos electorales (onpe)].*

Godin, B. (2006). The Linear Model of Innovation: The Historical Construction of an Analytical Framework. *Science, Technology, & Human Values*, *31*(6), 639–667.

Godin, B. (2011). The Linear Model of Innovation: Maurice Holland and the Research Cycle. *Social Science Information*, *50*(3-4), 569–581.

González, J., & Wiberg, M. (2017). *Applying Test Equating Methods Using R*. New Yok: Springer.

Gonzalez, R. M. (2021). Cell phone access and election fraud: Evidence from a spatial regression discontinuity design in afghanistan. *American Economic Journal: Applied Economics*, *13*(2), 1–51.

Gueron, J. M., & Pauly, E. (1991). *From welfare to work*. Russell Sage Foundation.

Gundersen, C., & Kreider, B. (2009). Bounding the effects of food insecurity on children's health outcomes. *Journal of Health Economics*, *28*, 971–983.

Haavelmo, T. (1944). The probability approach in econometrics. *Econometrica*, iii–115.

Haavelmo, T. (1997). Econometrics and the Welfare State. *The American Economic Review*, *87*, 13–15.

Hallinger, P. (2003). Leading educational change: Reflections on the practice of instructional and transformational leadership. *Cambridge Journal of Education*, *33*(3), 329–352.

Hausmann, R., & Rigobon, R. (2011). In search of the black swan: analysis of the statistical evidence of electoral fraud in Venezuela. *Statistical Science*, *26*(4), 543-563.

Head, B. W. (2010). Reconsidering evidence-based policy: Key issues and challenges. *Policy and Society*, *29*, 77-94.

Hirano, K., & Imbens, G. W. (2004). The propensity score with continuous treatments. In W. A. Shewhart & S. S. Wilks (Eds.), *Applied bayesian modeling and causal inference from incomplete-data*

perspectives: An essential journey with donald rubin's statistical family (p. 73-84). Wiley Series in Probability and Statistics.

Holden, C. (1990). Head start enters adulthood. *Science*, *247*(4949), 1400–1403.

Holland, P. W., Sinharay, S., von Davier, A. A., & Han, N. (2008). An approach to evaluating the missing data assumptions of the chain and post-stratification equating methods for the neat design. *Journal of Educational Measurement*, *45*(1), 17-43.

Horn, A., & Marfán, J. (2010). Relación entre liderazgo educativo y desempeño escolar: Revisión de la investigación en chile. *Psicoperspectivas*, *9*(2), 82–104.

Horowitz, J., & Manski, C. F. (1995). Identification and Robustness with Contaminated and Corrupted Data. *Econometrica*, *63*, 281–302.

Horowitz, J. L., & Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica: Journal of the Econometric Society*, 281–302.

Hurwicz, L. (1950). Generalization of the Concept of Identification. In T. Koopmans (Ed.), *Statistical inference in dynamic economic models, cowles commission research monograph 11* (p. 238-257). Wiley, New York.

Hyde, S. D. (2007). The observer effect in international politics: Evidence from a natural experiment. *World politics*, *60*(1), 37–63.

Hyndman, R. J., & Fan, Y. (1996). Sample Quantiles in Statistical Packages. *The American Statistician*, *50*(4), 361–365.

Ichino, N., & Schündeln, M. (2012). Deterring or displacing electoral irregularities? spillover effects of observers in a randomized field experiment in ghana. *The Journal of Politics*, *74*(1), 292–307.

Imbens, G. W. (2000). The role of the propensity score in estimating dose-response functions. *Biometrika*, *87*(3), 706–710.

Instituto Forestal. (2021). *Anuario Forestal. Boletín Estadístico Número 174*. Santiago, Chile: Instituto Forestal.

Itô, K. (1984). *An Introduction to Probability Theory*. Cambridge: Cambridge University Press.

Jullien, D. (2013). *Asian Disease-type of Framing of Outcomes as an Historical Curiosity* (Tech. Rep.). Groupe de Recherche en Droit, Economie, Gestion (GREDEG CNRS), Université Nice Sophie Antipolis.

Kahneman, D., & Tversky, A. (1979). Prospect Theory: An Analysis of Decision Under Risk. *Econometrica*, *47*, 263–291.

Kahneman, D., & Tversky, A. (1984). Choices, values, and frames. *American Psychologist*, *39*, 341–350.

Kahneman, D., & Tversky, A. (2013). Prospect Theory: An Analysis of Decision Under Risk. In *Handbook of the fundamentals of financial decision making: Part i* (pp. 99–127). World Scientific.

Kalton, G., & Flores-Cervantes, I. (2003). Weighting methods. *Journal of Official Statistics*, *19*(2), 81-97.

King, T. (2010). *Statistics in Society: three case studies in the uk* (Tech. Rep.). Southampton Statistical Sciences Research Institute.

Klimek, P., Yegorov, Y., Hanel, R., & Thurner, S. (2012). Statistical detection of systematic election irregularities. *Proceedings of the National Academy of Sciences of the United States of America*, *109*(41), 16469-16473.

Kolen, M. J. (2007). Data collection designs and linking procedures. In N. J. Dorans, M. Pommerich, & P. W. Holland (Eds.), *Linking and aligning scores and scales* (pp. 31–55). Springer.

Kolen, M. J., & Brennan, R. L. (2014). *Test equating, scaling, and linking: Methods and practices* (3rd ed.). New York: Springer.

Kolmogorov, A. N. (1950). *Foundations of the Theoty of Probability*. New Yok: Chelsea Publishing Company.

Koopmans, T. (1949). Identification Problems in Economic Model Construction. *Econometrica*, *17*, 125-144.

Koopmans, T., & Reiersøl, O. (1950). The Identification of Structural Characteristics. *The Annals of Mathematical Statistics*, *21*, 165–181.

Laplace, P. S. (1820). *Théorie Analytique des Probabilités*. Paris: Mme. Ve. Courcier, Imprimeur-Libraire pour les Mathématiques.

Leemann, L., & Bochsler, D. (2014). A systematic approach to study electoral fraud. *Electoral Studies*, *35*, 33–47.

Lehmann, E. L. (1995). Neyman's Statistical Philosophy. *Probability and Mathematical Statistics*, *15*, 29–36.

Lehmann, E. L. (2011). *Fisher, Neyman, and the Creation of Classical Statistics*. Springer Science & Business Media.

Lehoucq, F. (2003). Electoral fraud: Causes, types, and consequences. *Annual review of political science*, *6*(1), 233–256.

Leithwood, K., Day, C., Sammons, P., Harris, A., & Hopkins, D. (2006). *Successful school leadership: What it is and how it influences pupil learning*. Nottingham: National College for School Leadership.

Leithwood, K., & Jantzi, D. (2006). Transformational school leadership for large-scale reform: Effects on students, teachers, and their classroom practices. *School Effectiveness and School Improvement*, *17*(2), 201–227.

Levy, J. S. (1992). An Introduction to Prospect Theory. *Political Psychology*, *13*, 171–186.

Levy, J. S. (2003). Applications of Prospect Theory to Political Science. *Synthese*, *135*, 215–241.

Li, S., & Xie, X. (2006). A new look at the "asian disease" problem: A choice between the best possible outcomes or between the worst possible outcomes? *Thinking & reasoning*, *12*, 129–143.

Linde, J., & Vis, B. (2017). Do Politicians Take Risks Like the Rest of Us? An Experimental Test of Prospect Theory under MPs. *Political Psychology*, *38*, 101–117.

Lindley, D. V. (1972). *Bayesian Statistics: A Review* (Vol. 2). SIAM.

Liou, M., & Cheng, P. E. (1995). Equipercentile Equating Via Data-Imputation Technique. *Psychometrika*, *60*, 119-136.

Little, A. T., et al. (2012). Elections, fraud, and election monitoring in the shadow of revolution. *Quarterly Journal of Political Science*, *7*(3), 249–283.

Little, R. J., & Rubin, D. B. (1994). Test equating from biased samples, with application to the Armed Services Vocational Aptitude Battery. *Journal of Educational and Behavioral Statistics*, 309–335.

Little, R. J., & Rubin, D. B. (2019). *Statistical Analysis with Missing Data* (Vol. 793). John Wiley & Sons.

López-Pintor, R. (2010). Assessing electoral fraud in new democracies: a basic conceptual framework. *Washington DC International Foundation for Electoral Systems: White Paper Series Electoral Fraud*.

Lord, F. M. (1950). *Notes on Comparable Scales for Test Scores* (Tech. Rep.). Educational Testing Service.

Lord, F. M., & Novick, M. (1968). *Statistical theories of mental test scores.* Reading, MA: Addison-Wesley.

Lorenzen, P. (1987). *Constructive Philosophy*. Univ of Massachusetts Press.

Maddala, G. (1983). *Qualitative and limited dependent variable models in econometrics*. Cambridge: Cambridge University Press.

Manski, C. F. (1992). Educational Choice (Vouchers) and Social Mobility. *Economics of Education Review*, *11*, 351–369.

Manski, C. F. (1995). *Identification Problems in the Social Sciences*. New York: Harvard University Press.

Manski, C. F. (1997). The Mixing Problem in Programme Evaluation. *Review of Economics Studies*, *64*, 537–553.

Manski, C. F. (2003). *Partial identification of probability distributions*. New York: Springer.

Manski, C. F. (2005a). *Fractional Treatment Rules for Social Diversification of Indivisible Private Risks* (Tech. Rep.). National Bureau of Economic Research.

Manski, C. F. (2005b). Partial identification with missing data: concepts and findings. *International Journal of Approximate Reasoning*, *39*, 151–165.

Manski, C. F. (2005c). *Social Choice with Partial Knowledge of Treatment Response*. Princeton University Press.

Manski, C. F. (2007a). *Identification for Prediction and Decision*. Cambridge: Harvard University Press.

Manski, C. F. (2007b). Minimax-regret treatment choice with missing outcome data. *Journal of Econometrics*, *139*, 105–115.

Manski, C. F. (2013a). Diagnostic testing and treatment under ambiguity: using decision analysis to inform clinical practice. *Proceedings of the National Academy of Sciences*, *150*, 2064–2069.

Manski, C. F. (2013b). Diagnostic testing and treatment under ambiguity: using decision analysis to inform clinical practice. *Proceedings of the National Academy of Sciences*, *110*, 2064–2069.

Manski, C. F. (2013c). *Public Policy in an Uncertain World: Analysis and Decision*. Cambridge: Harvard University Press.

Manski, C. F. (2016). Credible interval estimates for official statistics with survey nonresponse. *Journal of Econometrics*, *191*, 293-301.

Manski, C. F. (2019). *Patient Care Under Uncertainty*. Princeton University Press.

Manski, C. F., & Nagin, D. S. (1998). Bounding disagreements about treatment effects: A case study of sentencing and recidivism. *Sociological Methodology*, *28*(1), 99–137.

Manzi, J., & Preiss, D. (2013). Educational assessment and educational achievement in South America. *International guide to student achievement*, 472–474.

Marks, H. M., & Printy, S. M. (2003). Principal leadership and school performance: An integration of transformational and instructional leadership. *Educational Administration Quarterly*, *39*(3),

370–397.

Mayo, D. G., & Spanos, A. (2006). Severe testing as a basic concept in a Neyman–Pearson philosophy of induction. *The British Journal for the Philosophy of Science*, *57*, 323–357.

McHugh, R. B. (1956). Efficient Estimation and Local Identification in Latent Class Analysis. *Psychometrika*, *21*(4), 331–347.

Mebane Jr, W. R. (2016). Election forensics: Frauds tests and observation-level frauds probabilities. In *Annual meeting of the midwest political science association, chicago* (pp. 7–10).

Meckes, L., & Carrasco, R. (2010). Two decades of SIMCE: an overview of the National Assessment System in Chile. *Assessment in Education: Principles, Policy & Practice*, *17*(2), 233–248.

MINEDUC. (2005). *Marco para la Buena Dirección. Criteris para el Desarrollo Profesional y Evaluación de Desempeño*. Santiago de Chile: Unidad de Gestión y Mejoramiento Educativo. División de Educación General. Ministerio de Educación. República de Chile.

MINEDUC. (2014b). *Estándares indicativos de desempeño para los establecimientos educacionales y sus sostenedores* (Unidad de Currículum y Evaluacíon ed.). Santiago de Chile.

MINEDUC. (2015). *Marco para la Buena Dirección y el Liderazgo Escolar*. Santiago de Chile: Centro de Perfeccionamiento, Experimentación e Investigaciones Pedagógicas, CPEIP. Ministerio de Educación. República de Chile.

Miyazaki, K., Hoshino, T., Mayekawa, S.-i., & Shigemasu, K. (2009). A new concurrent calibration method for nonequivalent group design under nonrandom assignment. *Psychometrika*, *74*(1), 1.

Molinari, F. (2010). Missing Treatments. *Journal of Business & Economic Statistics*, *28*, 82–95.

Mouchart, M., & Russo, F. (2011). Causal explanation: Recursive decompositions and mechanisms. In P. M. Illari, F. Russo, & J. Williamson (Eds.), *Causality in the Sciences* (pp. 317–337). Oxford University Press.

Müller, S. M., Garcia-Retamero, R., Cokely, E., & Maldonado, A. (2011). Causal beliefs and empirical evidence: Decision-making processes in two-alternative forced-choice tasks. *Experimental Psychology*, *58*, 324–332.

Newman, J. (2017). Deconstructing the debate over evidence-based policy. *Critical Policy Studies*, *11*, 211–226.

Oaks, M. (1986). *Statistical Inference: A Commentary for the Social and Behavioral Sciences.* Wiley.

Odgaard-Jensen, J., Vist, G. E., Timmer, A., Kunz, R., Akl, E. A., Schünemann, H., ... Oxman, A. D. (2011). Randomisation to protect against selection bias in healthcare trials. *Cochrane Database of Systematic Reviews*(4).

OECD. (2007). *Evidence in Education: Linking Research and Policy*. Paris: OECD.

Parkhurst, J. (2017). *The Politics of Evidence: From Evidence-Based Policy to the Good Governance of Evidence*. Routledge.

Parmigiani, G., & Inoue, L. (2009). *Decision Theory: Principles and Approaches* (Vol. 812). John Wiley & Sons.

Pedersen, D. B. (2014). The political epistemology of science-based policy-making. *Society*, *51*, 547–551.

Pepper, J. (2000). The Intergenerational Transmission of Welfare Receipt: A Nonparametric Bounds Analysis. *The Review of Economics and Statistics*, *82*, 472–488.

Peterson, M. (2017). *An Introduction to Decision Theory. Second Edition*. Cambridge University Press.

Pielke Jr, R. A. (2007). *The Honest Broker: Making Sense of Science in Policy and Politics*. Cambridge University Press.

Pielke Jr, R. A. (2012). "Basic Research" as a Political Symbol. *Minerva*, *50*(3), 339–361.

Pont, B., Nusche, D., & Moorman, H. (2008). *Improving School Leadership, Volume 1 Policy and Practice: Policy and Practice* (Vol. 1). OECD Publishing.

R Core Team. (2020). R: A language and environment for statistical computing [Computer software manual]. Vienna, Austria. Retrieved from `https://www.R-project.org/`

Rao, M. (2005). *Conditional Measures and Applications. Second Edition*. New Yok: Chapman & Hall/CRC.

Robinson, V., Lloyd, C., & Rowe, K. J. (2008). The impact of leadership on student outcomes: An analysis of the differential effects of leadership types. *Educational Administration Quarterly*, *44*(5), 635-674.

Rosenbaum, P. R., & Rubin, D. B. (1983). The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika*, *70*(1), 41–55.

Rousseau, D. M., & Gunia, B. C. (2016). Evidence-based Practice: The Psychology of EBP Implementation. *Annual Review of Psychology*, *67*, 667–692.

Rundlett, A., & Svolik, M. W. (2016). Deliver the vote! micromotives and macrobehavior in electoral fraud. *American Political Science Review*, *110*(1), 180–197.

Ryan, R., & Hill, S. (2016). *How to GRADE the Quality of the Evidence* (Tech. Rep.). Cochrane Consumers and Communication Group.

Rycroft-Malone, J., & Bucknall, T. (2011). *Models and Frameworks for Implementing Evidence-Based Practice: Linking Evidence to Action (Vol. 2)*. John Wiley & Sons.

Sackett, D. L., Straus, S. E., Richardson, W., Rosenberg, W., & Haynes, R. B. (2000). *Evidence Based Medicine: How to Practice and Teach EBM (2nd ed.)*. Churchill Livingstone.

San Martín, E. (2015). Identification of Item Resposne Theory Models. In W. J. van der Linden (Ed.), *Handbook of Item Response Theory. Volumen Two. Statistical Tools* (pp. 127–150). CRC Press.

San Martín, E. (2018). Identifiability of Structural Characteristics: How Relevant is it for the Bayesian Approach? *Brazilian Journal of Probability and Statistics*, *32*, 346–373.

San Martín, E., & Alarcón-Bustamante, E. (2022, 1). *Dissecting Chilean Surveys R-Code*. Retrieved from `https://github.com/edalarconb/DissectingSurveys`

San Martín, E., & De Boeck, P. (2015). What Do You Mean by a Difficult Item? On the Interpretation of the Difficulty Parameter in a Rasch Model. In R. E. Millsap, D. M. Bolt, L. A. van der Ark, & W.-C. Wang (Eds.), *Quantitative Psychology Research, Springer Proceedings in Mathematics & Statistics 89* (pp. 1–14). Springer.

San Martín, E., & González, J. (2022). A critical view on the neat equating design: Statistical modeling and identifiability problems. *Journal of Educational and Behavioral Statistics*, *47*(4), 406–437.

San Martín, E., & González, J. (2022). A Critical View on the NEAT Equating Design: Statistical Modelling and Identifiability Problems. *Journal of Educational and Behavioral Statistics*, *Accepted for publication*.

San Martín, E., González, J., & Tuerlinckx, F. (2009). Identified Parameters, Parameters of Interest and Their Relationships. *Measurement: Interdisciplinary Research and Perspective*, *7*, 97–105.

San Martín, E., González, J., & Tuerlinckx, F. (2015). On the unidentifiability of the fixed-effects 3pl model. *Psychometrika*, *80*(2), 450–467.

San Martín, E., & González-Larrondo, T. (2020). *How to Broker the Evaluation of Public Policies? A Strategy to Reveal Inherent Uncertainties in Policy-Decision Making* (Tech. Rep. No. DP2020—02). Interdisciplinary Laboratory of Social Statistics, Pontificia Universidad Católica de Chile, Chile.

Savage, L. J. (1951). The Theory of Statistical Decision. *Journal of the American Statistical Association*, *46*, 55–67.

Schünemann, H. J. (2016). Interpreting GRADE's levels of certainty or quality of the evidence: GRADE for statisticians, considering review information size or less emphasis on imprecision? *Journal of Clinical Epidemiology*, *75*, 6–15.

Schweinhart, L. J., Montie, J., Xiang, Z., Barnett, W. S., Belfield, C., & Nores, M. (2005). The High/Scope Perry Preschool Study Through Age 40. *Ypsilanti, MI: High Scope Educational Research Foundation*.

Seashore, K., Leithwood, K., Wahlstrom, K., & Anderson, S. (2010). *Investigating the links to improved student learning: Final report of research findings*. The Wallace Foundation.

Sinharay, S., & Holland, P. W. (2010). The Missing Data Assumptions of the NEAT Design and Their Implications for Test Equating. *Psychometrika*, *75*, 309–327.

Stephenson, J., & Imrie, J. (1998). Why do we need randomised controlled trials to assess behavioural interventions? *BMJ*, *316*(7131), 611–613.

Strassheim, H. (2017). Bringing the political back in: reconstructing the debate over evidence-based policy. a response to Newman. *Critical Policy Studies*, *11*, 235–245.

Tamer, E. (2010a). Partial identification in econometrics. *Annual Review of Economics*, *2*(1), 167–195.

Tamer, E. (2010b). Partial Identification in Econometrics. *Annual Review of Economics*, *2*, 167–195.

Toland, J. (1702). *Christianity not Mysterious, or, A Treatise Shewing, That there is nothing in the Gospel Contrary to Reason, nor Above it: And that no Christian Doctrine can be properly call'd a Mystery*. Printed for Sam. Buckley at the Dolphin over against St. Dunstans Church in Fleetstreet.

Turnhout, E., Stuiver, M., Klostermann, J., Harms, B., & Leeuwis, C. (2013). New Roles of Science in Society: Different Repertoires of Knowledge Brokering. *Science and Public Policy*, *40*, 354–365.

Tversky, A., & Kahneman, D. (1981). The Framing of Decisions and the Psychology of Choice. *Science*, *211*(4481), 453–458.

Van Woensel, L. (2020). *A Bias Radar for Responsible Policy-Making: Foresight-Based Scientific Advice*. Palgrave Macmillan, Springer Nature.

von Davier, A. A., Holland, P., & Thayer, D. (2004). *The kernel method of test equating*. New York: Springer.

Von Neumann, J., Morgenstern, O., & Kuhn, H. W. (2007). *Theory of Games and Economic Behavior (Commemorative Edition)*. Princeton University Press.

Wald, A. (1950). *Statistical Decision Functions*. John Wiley and Sons, Inc.

Waters, T., Marzano, R., & McNulty, B. (2003). *Balanced Leadership: What 30 Years of Research Tells Us about the Effect of Leadership on Student Achievement. A Working Paper*. ERIC.

Weinstein, J., & Hernández, M. (2014). Políticas hacia el liderazgo directivo escolar en chile: Una mirada comparada con otros sistemas escolares de américa latina. *Psicoperspectivas*, *13*(3), 52–68.

Wiseman, A. W. (2010). The Uses of Evidence for Educational Policymaking: Global Contexts and International Trends. *Review of Research in Education*, *34*, 1–24.

Wolpin, K. I. (2013). *The Limits of Inference without Theory*. The MIT Press.

Woodbury, S. A., & Spiegelman, R. G. (1987). Bonuses to workers and employers to reduce unemployment: Randomized trials in illinois. *The American Economic Review*, 513–530.

Wunsch, G., Mouchart, M., & Russo, F. (2014). Functions and Mechanisms in Structural-Modelling Explanations. *Journal for General Philosophy of Science*, *45*, 187–208.